EGA IV, SECTIONS 19–21: SOURCE-ALIGNED ENGLISH READER

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CONTENTS

§19. REGULAR IMMERSIONS AND NORMAL FLATNESS

This section is devoted, on the one hand, to a study of regular immersions (16.9.2) $Y \to X$ of preschemes, especially when X and Y are flat over the same prescheme S; on the other hand, it gives supplements concerning M-regular sequences and normal flatness, notably generalizing flatness results established by H. Hironaka in his theory of resolution of singularities [35].

19.1. Properties of regular immersions **IV-4** | 185

Proposition (19.1.1). — *Let X be a locally Noetherian prescheme, $j : Y \to X$ an immersion, and y a regular point of Y. For j to be a regular immersion at the point y, it is necessary and sufficient that X be regular at the point $j(y)$.*

Proof. Taking (16.9.10) into account, we are reduced to proving the analogous result for local rings, which follows from (**0**, 17.3.3). □

Corollary (19.1.2). — *Let A be a Noetherian local ring and $\mathfrak{J}$ an ideal of A; suppose that the quotient ring $B = A/\mathfrak{J}$ is regular. For A to be regular, it is necessary and sufficient that the ideal $\mathfrak{J}$ be regular.*

Proof. Indeed, if A is regular, then $\mathfrak{J}$ is generated by part of a regular system of parameters of A (**0**, 17.1.9), and hence $\mathfrak{J}$ is regular. Conversely, if $\mathfrak{J}$ is regular and $(x_i)_{1\leqslant i\leqslant r}$ is an A-regular sequence generating $\mathfrak{J}$, then (x_i) is part of a system of parameters of A (**0**, 16.4.1); it therefore follows from (**0**, 17.1.7) that A is regular. □

Definition (19.1.3). — Let X be a prescheme, Y a subprescheme of X, and U an open subset of X containing Y such that Y is defined by an ideal $\mathscr{J}$ of $\mathscr{O}_U$; suppose that $\mathscr{J}/\mathscr{J}^2$ is a locally free $(\mathscr{O}_U/\mathscr{J})$-module (as is the case when Y is quasi-regularly immersed in X). For every $y \in Y$, the *transverse codimension* of Y in X at y, denoted by $\operatorname{codim}^*_y(Y, X)$, is the rank of the free $(\mathscr{O}_{U,y}/\mathscr{J}_y)$-module $\mathscr{J}_y/\mathscr{J}_y^2$. If $f : Z \to X$ is an immersion with image Y, the transverse codimension of f at $z \in Z$ is likewise the transverse codimension in X, at $f(z)$, of the subprescheme Y.

Proposition (19.1.4). — *Let X be a locally Noetherian prescheme and Y a subprescheme regularly immersed in X. For every $y \in Y$, one has*

$$\operatorname{codim}^*_y(Y, X) = \operatorname{codim}_y(Y, X). \tag{19.1.4.1}$$

Proof. By virtue of (5.1.3.2), $\operatorname{codim}_y(Y, X)$ is equal to the least value of the dimensions of the local rings $A = \mathscr{O}_{X,z}$, where z runs through the generic points of the irreducible components of Y containing y. Such a point z belongs to every neighbourhood of y in X, and $\mathscr{J}_z/\mathscr{J}_z^2$ is a free $(A/\mathscr{J}_z)$-module of rank

$$n = \operatorname{codim}^*_y(Y, X).$$

It therefore remains to prove that $\dim(A) = n$.

Since z is a maximal point of the subprescheme Y defined by $\mathscr{J}$, one has $\dim(A/\mathscr{J}_z) = 0$; hence **IV-4** | 186 $\mathscr{J}_z$ is an ideal of definition of the Noetherian local ring A. Moreover, the quasi-regularity of $\mathscr{J}$ implies that $\mathscr{J}_z^m/\mathscr{J}_z^{m+1}$ is a free $(A/\mathscr{J}_z)$-module of rank $\binom{m+n-1}{n-1}$. If r denotes the length of the Artinian ring $A/\mathscr{J}_z$, the length of $\mathscr{J}_z^m/\mathscr{J}_z^{m+1}$ is therefore

$$r\binom{m+n-1}{n-1}.$$

Consequently, the Hilbert–Samuel polynomial of A for the $\mathscr{J}_z$-adic filtration has degree n, which proves the proposition by (**0**, 16.2.3). □

By virtue of (19.1.4), henceforth we shall say "codimension" instead of "transverse codimension" when speaking of a subprescheme Y regularly immersed in X, even when X is not locally Noetherian.

Proposition (19.1.5). —

(i) *For an immersion $j : Y \to X$ to be open, it is necessary and sufficient that it be regular and of codimension 0 at every point.*

(ii) *Let $f : Y \to X$ be a regular (resp. quasi-regular) immersion, $g : X' \to X$ a flat morphism, $Y' = Y \times_X X'$; then the morphism $f' = f_{(X')} : Y' \to X'$ is a regular (resp. quasi-regular) immersion, and for every $y' \in Y'$, if y is the projection of y' in Y, the codimension of f' at the point y' is equal to the codimension of f at the point y. Conversely, if f is an immersion and $g : X' \to X$ a faithfully flat and quasi-compact morphism, then, if f' is a quasi-regular immersion, so is f.*

(iii) *If $f : Y \to X$ and $g : Z \to Y$ are regular immersions, then $f \circ g : Z \to X$ is a regular immersion, and for every $z \in Z$, the codimension of $f \circ g$ at the point z is the sum of the codimension of g at the point z and of the codimension of f at the point $g(z)$. Moreover, the sequence of canonical homomorphisms* (16.2.7.1)

$$0 \longrightarrow g^*(\mathscr{N}_{Y/X}) \longrightarrow \mathscr{N}_{Z/X} \longrightarrow \mathscr{N}_{Z/Y} \longrightarrow 0$$

(19.1.5.1) *is exact, and for every $z \in Z$, there exists a neighbourhood of z in Z in which the restrictions of the homomorphisms of this sequence form a split exact sequence.*

(iv) *Let X be a locally Noetherian prescheme, $f : Y \to X$ and $g : Z \to Y$ two immersions, z a point of Z. The following conditions are equivalent:*

a) *g is a regular immersion at the point z and f is a regular immersion at the point $g(z)$.*

b) *g and $f \circ g$ are regular immersions at the point z.*

c) *$f \circ g$ is a regular immersion at the point z, f is a regular immersion at the point $g(z)$, and the sequence of canonical homomorphisms* (19.1.5.1)*, restricted to a sufficiently small open neighbourhood of z in Z, is exact and split.*

(v) *Let X be a locally Noetherian prescheme, $f : Y \to X$, $g : Z \to Y$ two immersions, z a point of Z; suppose that $y = g(z)$ is a regular point of Y and $x = f(g(z))$ a regular point of X (so that, by* (19.1.1)*, f is a regular immersion at the point $g(z)$); then, for $f \circ g$ to be a regular immersion at the point z, it is necessary and sufficient that g be a regular immersion at the point z.* **IV-4** | 187

Proof. Assertion (i) is another way of expressing (16.1.10). To prove the first assertion of (ii), we can restrict ourselves to the case where Y is a closed sub-prescheme of X defined by a regular (resp. quasi-regular) ideal $\mathscr{J}$; then Y' is the closed sub-prescheme of X' defined by the ideal $g^*(\mathscr{J})\mathscr{O}_{X'}$, which is identified here with $g^*(\mathscr{J})$ since g is flat; by replacing X by a sufficiently small affine open, we can suppose that $\mathscr{J}$ admits a regular (resp. quasi-regular) sequence of generators f_i (sections of $\mathscr{O}_X$ over X). If the sequence (f_i) is regular, so is the sequence $(f_i \otimes 1)$ (which generates $g^*(\mathscr{J})\mathscr{O}_{X'}$), by virtue of (**0**, 15.2.5); the analogous result for quasi-regular sequences follows immediately from the flatness of g and from the criterion (16.9.4). Similarly, if g is faithfully flat and quasi-compact, and if $g^*(\mathscr{J})$ is quasi-regular, $\mathscr{J}$ is of finite type by virtue of (2.5.2); by flatness, we have $g^*(\mathscr{J})/g^*(\mathscr{J})^2 = g^*(\mathscr{J}/\mathscr{J}^2)$, hence, since $g^*(\mathscr{J}/\mathscr{J}^2)$ is locally free by hypothesis (16.9.4), it follows from (2.5.2) that $\mathscr{J}/\mathscr{J}^2$ is locally free. Finally, condition (iii) of (16.9.4) relative to $g^*(\mathscr{J})$ implies the same condition for $\mathscr{J}$ by virtue of the faithful flatness of g (2.2.7). We conclude from (16.9.4) that $\mathscr{J}$ is quasi-regular.

To prove (iii), we can likewise suppose that Y and Z are closed sub-preschemes of X, defined respectively by ideals $\mathscr{J}$, $\mathscr{K}$ of $\mathscr{O}_X$ such that $\mathscr{J} \subset \mathscr{K}$; moreover, we can suppose that there exist sections f_i ($1 \leqslant i \leqslant m$) of $\mathscr{J}$ over X generating $\mathscr{J}$ and such that for $1 \leqslant i \leqslant m$, the endomorphism of $\mathscr{O}_X/(\sum_{j=1}^{i-1} f_j\mathscr{O}_X)$ defined by f_i is injective, and sections $\bar{f}_i$ ($m+1 \leqslant i \leqslant n$) of $\mathscr{K}/\mathscr{J}$ over X generating $\mathscr{K}/\mathscr{J}$ and such that for $m+1 \leqslant i \leqslant n$, the endomorphism of

$$(\mathscr{O}_X/\mathscr{J})/\Big(\sum_{j=m+1}^{i-1} \bar{f}_j(\mathscr{O}_X/\mathscr{J})\Big)$$

defined by $\bar{f}_i$ is injective. For every $z \in Z$, let U be an open neighbourhood of z in X such that there exist sections f_i of $\mathscr{K}$ over U, whose classes in $\Gamma(U, \mathscr{K}/\mathscr{J})$ are $\bar{f}_i|U$ for $m+1 \leqslant i \leqslant n$; then the restriction to U of

$$(\mathscr{O}_X/\mathscr{J})/\Big(\sum_{j=m+1}^{i-1} \bar{f}_j(\mathscr{O}_X/\mathscr{J})\Big)$$

is isomorphic to

$$\mathscr{O}_U/\Big((\mathscr{J}|U) + \sum_{j=m+1}^{i-1} f_j\mathscr{O}_U\Big) = \mathscr{O}_U/\Big(\sum_{j=1}^{i-1} f_j\mathscr{O}_U\Big),$$

and we see therefore that the sequence $(f_i)_{1 \leqslant i \leqslant n}$ is regular in $\mathscr{O}_U$; as it generates $\mathscr{K}|U$, this proves the first assertion of (iii). To prove the last assertion of (iii), we can restrict ourselves (with the same notation) to the case where the images t_i of the f_i in $\mathscr{K}/\mathscr{K}^2$ (for $1 \leqslant i \leqslant n$) form a basis of this $(\mathscr{O}_X/\mathscr{K})$-module (16.9.5); for the same reason, we can suppose that the t_i for $1 \leqslant i \leqslant m$ are the canonical images of elements of a basis of the $(\mathscr{O}_X/\mathscr{K})$-module $g^*(\mathscr{J}/\mathscr{J}^2)$, and that the canonical images $\bar{t}_i$ of the t_i in $(\mathscr{K}/\mathscr{J})/(\mathscr{K}/\mathscr{J})^2$ for $m+1 \leqslant i \leqslant n$ form a basis of this $(\mathscr{O}_X/\mathscr{K})$-module. This completes the proof of (iii).

Let us pass to the proof of (iv). The fact that *a)* implies *c)* follows from (iii). Let us prove that *c)* implies *a)*. Set again $\mathrm{A} = \mathscr{O}_{X,f(g(z))}$, so that $\mathscr{O}_{Y,g(z)} = \mathrm{A}/\mathfrak{J}$, $\mathscr{O}_{Z,z} = \mathrm{A}/\mathfrak{K}$ with $\mathfrak{J} \subset \mathfrak{K}$. By hypothesis, **IV-4** | 188
$\mathfrak{J}$ and $\mathfrak{K}$ are regular ideals of A and we have a split exact sequence

$$0 \longrightarrow \mathfrak{J}/\mathfrak{J}\mathfrak{K} \longrightarrow \mathfrak{K}/\mathfrak{K}^2 \longrightarrow (\mathfrak{K}/\mathfrak{J})/(\mathfrak{K}/\mathfrak{J})^2 \longrightarrow 0$$

(cf. (16.2.7)). This implies that there exist elements f_j $(1 \leqslant j \leqslant r+s)$ of $\mathfrak{K}$ whose images f'_j in $\mathfrak{K}/\mathfrak{K}^2$ form a basis of this $(\mathrm{A}/\mathfrak{K})$-module, and such that, in addition, the f_j of index $j \leqslant r$ belong to $\mathfrak{J}$ and are such that their images in $\mathfrak{J}/\mathfrak{J}\mathfrak{K}$ form a basis of this $(\mathrm{A}/\mathfrak{K})$-module. The f_j for $1 \leqslant j \leqslant r+s$ generate $\mathfrak{K}$, and the f_j for $1 \leqslant j \leqslant r$ generate $\mathfrak{J}$ since A is Noetherian (Bourbaki, *Alg. comm.*, chap. II, § 3, n° 2, prop. 5); as $\mathfrak{K}$ is a regular ideal, it follows from (16.9.5) that the sequence $(f_j)_{1 \leqslant j \leqslant r+s}$ is regular. By definition, the images of $f_{r+1}, \ldots, f_{r+s}$ in $\mathfrak{K}/\mathfrak{J}$ form a regular sequence in this $(\mathrm{A}/\mathfrak{J})$-module, which completes the proof that *c)* implies *a)* in (iv).

Finally, taking into account (16.9.10), it is immediate that (v), as well as the equivalence of *a)* and *b)* in (iv), follow from the following corollary. □

Corollary (19.1.6). — *Let* A *be a local Noetherian ring,* $\mathfrak{J}$, $\mathfrak{K}$ *two ideals of* A *contained in its maximal ideal* $\mathfrak{m}$ *and such that* $\mathfrak{J} \subset \mathfrak{K}$, $\mathrm{A}' = \mathrm{A}/\mathfrak{J}$, $\mathfrak{K}' = \mathfrak{K}/\mathfrak{J}$. *Then:*

(i) If $\mathfrak{J}$ *and* $\mathfrak{K}'$ *are regular ideals, so is* $\mathfrak{K}$.
(ii) If $\mathfrak{K}$ *and* $\mathfrak{K}'$ *are regular ideals, so is* $\mathfrak{J}$.
(iii) Suppose the rings A *and* $\mathrm{A}' = \mathrm{A}/\mathfrak{J}$ *are regular. For the ideal* $\mathfrak{K}$ *to be regular, it is necessary and sufficient that* $\mathfrak{K}'$ *be so.*

Proof. Assertion (i) is a particular case of (19.1.5), (iii). To prove (ii), consider the canonical surjective homomorphism $u : \mathfrak{K}/\mathfrak{K}^2 \to \mathfrak{K}'/\mathfrak{K}'^2 = \mathfrak{K}/(\mathfrak{K}^2 + \mathfrak{J})$. Since by hypothesis $\mathfrak{K}/\mathfrak{K}^2$ and $\mathfrak{K}'/\mathfrak{K}'^2$ are free $(\mathrm{A}/\mathfrak{K})$-modules, whose ranks we denote respectively by $p+q$ and q, the kernel of u is an $(\mathrm{A}/\mathfrak{K})$-module *projective* (Bourbaki, *Alg.*, chap. II, 3e ed., § 2, n° 2, prop. 4), hence *free* of rank p since $\mathrm{A}/\mathfrak{K}$ is a local Noetherian ring ((**0**, 10.1.3)); in other words, there exists a basis of $\mathfrak{K}/\mathfrak{K}^2$ of which the first p elements form a basis of $(\mathfrak{K}^2 + \mathfrak{J})/\mathfrak{K}^2$. This means (by virtue of Nakayama's lemma) that there exists a system of generators $(f_i)_{1 \leqslant i \leqslant p+q}$ of $\mathfrak{K}$, forming a regular sequence in A, such that the sequence $(f_i)_{1 \leqslant i \leqslant p}$ consists of elements of $\mathfrak{J}$; it remains to show that this last sequence generates $\mathfrak{J}$. For this, denote by $\mathfrak{J}' \subset \mathfrak{J}$ the ideal it generates; considering the ring $\mathrm{A}/\mathfrak{J}'$, we see that we are reduced to proving the following lemma: □

Lemma (19.1.6.1). — *Let* B *be a Noetherian local ring with maximal ideal* $\mathfrak{n}$, $(g_j)_{1 \leqslant j \leqslant q}$ *a regular sequence in* B *consisting of elements of* $\mathfrak{n}$, $\mathfrak{b}$ *the ideal it generates, and* $\mathfrak{c} \subset \mathfrak{b}$ *an ideal such that the canonical images of the* g_j *in* $\mathrm{B}/\mathfrak{c}$ *still form a regular sequence in this ring. Then* $\mathfrak{c} = 0$.

Proof. The lemma being trivial for $q = 0$, we reason by induction on q. The induction hypothesis, applied to the ring $\mathrm{B}/g_1\mathrm{B}$ and to the canonical images of the g_j $(2 \leqslant j \leqslant q)$ and of $\mathfrak{c}$ in this quotient **IV-4** | 189
ring, shows that $\mathfrak{c} \subset g_1\mathrm{B}$. Let $\mathfrak{d}$ be the ideal of B consisting of the $x \in \mathrm{B}$ such that $g_1 x \in \mathfrak{c}$; we thus have $\mathfrak{c} = g_1\mathfrak{d}$; but since by hypothesis g_1 is regular in $\mathrm{B}/\mathfrak{c}$, we necessarily have $\mathfrak{d} = \mathfrak{c}$, hence $\mathfrak{c} = g_1\mathfrak{c}$. Since $\mathfrak{c}$ is of finite type and g_1 is contained in the radical of B, Nakayama's lemma shows that $\mathfrak{c} = 0$.

Let us now prove assertion (iii) of (19.1.6). By virtue of (i) and of (19.1.2), it suffices to show that if $\mathfrak{K}$ is regular, so is $\mathfrak{K}'$. Note that $\mathfrak{J}$ is regular (19.1.2), and reason by induction on the rank q of the $(\mathrm{A}/\mathfrak{J})$-module $\mathfrak{J}/\mathfrak{J}^2$. If $q = 0$, we have $\mathfrak{J} = 0$ by Nakayama's lemma and the assertion is trivial. Since A and $\mathrm{A}/\mathfrak{J}$ are regular, we know ((**0**, 17.1.9)) that $\mathfrak{J}$ is generated by a subset of q elements of a regular system of parameters of A, hence, if f is one of the elements of this system of generators of $\mathfrak{J}$, $\mathrm{A}_1 = \mathrm{A}/f\mathrm{A}$ is regular and $f \notin \mathfrak{m}^2$ ((**0**, 17.1.8)). Let $\mathfrak{J}_1, \mathfrak{K}_1$ be the images of $\mathfrak{J}, \mathfrak{K}$ in A_1; we have $\mathfrak{J}_1 \subset \mathfrak{K}_1$, $\mathrm{A}_1/\mathfrak{J}_1 = \mathrm{A}/\mathfrak{J}$, hence $\mathrm{A}_1/\mathfrak{J}_1$ is regular and consequently $\mathfrak{J}_1$ is a regular ideal (19.1.2). Let us show that $\mathfrak{K}_1$ is a regular ideal in A_1; since $\mathfrak{K}$ is a regular ideal in A, it suffices to show that f is part of a regular system of generators of $\mathfrak{K}$, and for this (16.9.5) it suffices to show that the image of f in $\mathfrak{K}/(\mathfrak{K}^2 + \mathfrak{m}\mathfrak{K}) = \mathfrak{K}/\mathfrak{m}\mathfrak{K}$ is part of a basis of this $(\mathrm{A}/\mathfrak{m})$-vector space, in other words that $f \notin \mathfrak{m}\mathfrak{K}$, which indeed follows from $f \notin \mathfrak{m}^2$. Then, as $\mathfrak{J}_1/\mathfrak{J}_1^2$ is a free $(\mathrm{A}_1/\mathfrak{J}_1)$-module of rank

$q-1$, the induction hypothesis shows that $\mathfrak{R}'_1 = \mathfrak{R}_1/\mathfrak{J}_1$ is a regular ideal of $\mathrm{A}_1/\mathfrak{J}_1$, and since $\mathfrak{R}'$ is identified with $\mathfrak{R}'_1$ in the canonical isomorphism between $\mathrm{A}/\mathfrak{J}$ and $\mathrm{A}_1/\mathfrak{J}_1$, $\mathfrak{R}'$ is regular. □

Remarks (19.1.7). —

(i) We do not know whether the composite of two quasi-regular immersions is quasi-regular.

(ii) In a ringed space with local rings X, for every ideal $\mathscr{J}$ of $\mathscr{O}_X$, $\mathscr{O}_X/\mathscr{J}$ is of finite type and therefore has closed support in X ((**0**, 5.2.2)). We can therefore define, as in (**I**, 4.1.3) and (4.2.1), the notions of a (locally closed) ringed subspace of X, locally defined by an ideal $\mathscr{J}$ of a sheaf $\mathscr{O}_U$, where U is an open subset of X, and of an immersion. The definitions of quasi-regular and regular immersions and of transverse codimension then apply without change, and the results of (19.1.5), (i) to (iv) are still valid, [1] provided that, in the second assertion of (i) and in (iv), the sheaf $\mathscr{O}_X$ is coherent and the local rings $\mathscr{O}_{X,x}$ are Noetherian; as for (ii), it is necessary to *define* Y' as the ringed subspace defined by the $\mathscr{O}_{X'}$-ideal $g^*(\mathscr{J})$; the proofs are unchanged.

(19.1.8). We have already noted (in particular (16.9.6), (ii)) that in non-Noetherian rings, the notion of "regular sequence" of elements of the ring does not possess the properties that would make it usable. Recent research seems to show that in this case the notion that should be substituted for that of regular sequence is that of the sequence $\mathbf{f} = (f_1, \ldots, f_n)$ having the property that the complex of the exterior algebra $\mathrm{K}_0(\mathbf{f}, \mathrm{M})$ ((**III**, 1.1.3)) has *zero homology in positive degrees* (cf. (**III**, 1.1.4)); see in particular A. Grothendieck, Séminaire de géométrie algébrique, 1966, to appear.

19.2. Transversally regular immersions

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Definition (19.2.1). — Let $u : X \to S$ be a morphism of preschemes and $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module. We say that a sequence $(f_i)_{1\leqslant i\leqslant n}$ of sections of $\mathscr{O}_X$ over X is *transversally $\mathscr{F}$-regular relative to S* (or relative to u) if the sequence (f_i) is $\mathscr{F}$-regular and the $\mathscr{O}_X$-modules

$$\mathscr{F}\Big/\Big(\sum_{1\leqslant j\leqslant i} f_j\mathscr{F}\Big)$$

are flat over S for $0 \leqslant i \leqslant n$. We say that a quasi-coherent ideal $\mathscr{J}$ of $\mathscr{O}_X$ is *transversally regular relative to S* (or relative to u) at a point $x \in \mathrm{Supp}(\mathscr{O}_X/\mathscr{J})$ if there exists an open neighbourhood U of x and a finite sequence (f_i) of sections of $\mathscr{J}$ over U that is transversally $\mathscr{O}_U$-regular relative to S and generates $\mathscr{J}|U$.

In an A-algebra B, we say that an ideal $\mathfrak{J}$ is *transversally regular relative to A* if $\widetilde{\mathfrak{J}}$ is transversally regular relative to $S = \mathrm{Spec}(A)$ at every point of $V(\mathfrak{J})$ in $\mathrm{Spec}(B)$.

Definition (19.2.2). — Let S be a prescheme, X and Y two S-preschemes, and $u : Y \to X$ an S-morphism which is an immersion. We say that u is a *transversally regular immersion relative to S* at a point $y \in Y$ if there exists a neighbourhood V of $u(y)$ in X such that the sub-prescheme induced on $u(Y) \cap V$ by the sub-prescheme of X associated to u is defined by an ideal of $\mathscr{O}_V$ transversally regular relative to S at the point $u(y)$.

It is clear that the set of points of Y where an immersion is transversally regular relative to S is *open* in Y; if it is equal to Y, we say simply that u is a *transversally regular immersion relative to S*.

If $S \to T$ is a flat morphism and $\mathscr{J}$ is an ideal of $\mathscr{O}_X$ transversally regular relative to S at $x \in X$, then it is also transversally regular relative to T ((**0**, 6.2.1)). An S-immersion $Y \to X$ transversally regular relative to S at a point $y \in Y$ is therefore also a T-immersion transversally regular relative to T at this point.

Proposition (19.2.3). — *Let S be a prescheme, X and Y two S-preschemes, and $u : Y \to X$ an S-morphism that is a transversally regular immersion relative to S at a point $y \in Y$. Then, for every base change $S' \to S$, if we set $X' = X \times_S S'$ and $Y' = Y \times_S S'$, the immersion $u' = u_{(S')} : Y' \to X'$ is transversally regular relative to S' at every point $y' \in Y'$ above y.*

Proof. This follows immediately from the definition and from (**0**, 15.1.15). □

[1] The printed text reads "(19.1.4), (i) to (iv)"; the reference must be to Proposition 19.1.5, the itemized proposition immediately above.

Proposition (19.2.4). — *Let S be a prescheme, X and Y two S-preschemes, and $u : Y \to X$ an S-morphism that is an immersion. Suppose either that S and X are locally Noetherian, or that the structure morphisms $g : X \to S$ and $h : Y \to S$ are locally of finite presentation. Let y be a point of Y, s its image in S. The following conditions are equivalent:*

a) *u is transversally regular relative to S at the point y.*
b) *The morphisms g and h are flat in a neighbourhood of the points $u(y)$ and y respectively, and if we set $X_s = g^{-1}(s)$, $Y_s = h^{-1}(s)$, the immersion $u_s : Y_s \to X_s$ is regular at the point y.*
b′) *The morphisms g and h are flat in a neighbourhood of the points $u(y)$ and y respectively, and, for every change of base $S' \to S$, if we set $X' = X \times_S S'$, $Y' = Y \times_S S'$, the immersion $u' : Y' \to X'$ is regular at every point of Y' above y.*
c) *The morphism h is flat in a neighbourhood of the point y, and the immersion u is regular in a neighbourhood of the point y.* **IV-4** | 191
c′) *The morphism h is flat in a neighbourhood of the point y, and the immersion u is quasi-regular in a neighbourhood of the point y.*

Proof. We have already proved in (19.2.3) that *a)* implies *b′)* without the hypothesis of finiteness, and it is trivial that *b′)* implies *b)*. Let us show that *b)* implies *c)*. For this, since the question is local on X and on S, we may suppose that Y is a closed sub-prescheme of X defined by a quasi-coherent ideal $\mathscr{J}$ of $\mathscr{O}_X$, and that u is the canonical injection. The hypothesis that $\mathscr{O}_X/\mathscr{J}$ is g-flat implies that the sequence

$$0 \longrightarrow \mathscr{J} \otimes_{\mathscr{O}_S} \mathbf{k}(s) \longrightarrow \mathscr{O}_X \otimes_{\mathscr{O}_S} \mathbf{k}(s) \longrightarrow (\mathscr{O}_X/\mathscr{J}) \otimes_{\mathscr{O}_S} \mathbf{k}(s) \longrightarrow 0$$

is exact ((**0**, 6.1.2)); hence Y_s is the closed sub-prescheme of X_s defined by the ideal $\mathscr{J}_s$ of $\mathscr{O}_{X_s} = \mathscr{O}_X \otimes_{\mathscr{O}_S} \mathbf{k}(s)$ that is identified with $\mathscr{J} \otimes_{\mathscr{O}_S} \mathbf{k}(s) = \mathscr{J}/\mathfrak{m}_s\mathscr{J}$. Consider a finite sequence (f_i) of sections of $\mathscr{J}$ over a neighbourhood of y in X whose images in

$$\mathscr{J}_y/(\mathfrak{m}_s\mathscr{O}_{X,y}\mathscr{J}_y + \mathscr{J}_y^2)$$

form a basis of this $\big(\mathscr{O}_{X,y}/(\mathfrak{m}_s\mathscr{O}_{X,y} + \mathscr{J}_y)\big)$-module, which is free by hypothesis *b)*. Since X_s is locally Noetherian, it follows from (16.9.11), (16.9.5), and hypothesis *b)* that the images $(\overline{f}_i)_y = (f_i)_y \otimes 1$, sections of $\mathscr{J}_s = \mathscr{J} \otimes_{\mathscr{O}_S} \mathbf{k}(s)$ over a neighbourhood of y in X_s, form a regular sequence generating the restriction of $\mathscr{J}_s$ to this neighbourhood. Since $\mathfrak{m}_s\mathscr{O}_{X,y}$ is contained in the radical of $\mathscr{O}_{X,y}$, and since in both cases under consideration $\mathscr{J}_y$ is an ideal of finite type of $\mathscr{O}_{X,y}$, Nakayama's lemma shows that the $(f_i)_y$ also generate $\mathscr{J}_y$; since $\mathscr{J}$ is an ideal of finite type of $\mathscr{O}_X$ in both cases, there is a neighbourhood U of y in X such that the $f_i|U$ generate $\mathscr{J}|U$ ((**0**, 5.2.2)). The implication *b)⇒c)* now follows from (**0**, 15.1.16) when S and X are locally Noetherian, and from (11.3.8) when g and h are locally of finite presentation; in the latter case, (11.3.8) also proves the equivalence of *c)* and *c′)*, which is immediate when X and S are locally Noetherian ((**0**, 15.1.11)). Finally, if *c)* holds, then to prove *a)* we may again reduce to the case where Y is a closed sub-prescheme of X defined by $\mathscr{J}$. By hypothesis there is a regular sequence (f_i) of sections of $\mathscr{J}$ over a neighbourhood U of y in X such that the f_i generate $\mathscr{J}|U$ and h is flat on U; to see that *c)* implies *a)*, it again suffices to apply (**0**, 15.1.6) when S and X are locally Noetherian, and (11.3.8) when g and h are locally of finite presentation. □

Remarks (19.2.5). —

(i) If the hypothesis that h is flat in a neighbourhood of y is removed from condition *b)*, the resulting condition no longer implies *a)*. It suffices to take for S the spectrum of a local ring that is not a field, for s the closed point of S, and $Y = X_s$.

(ii) Suppose that Y is a closed sub-prescheme of X defined by a quasi-coherent ideal $\mathscr{J}$ of $\mathscr{O}_X$. The proof of (19.2.4) shows that, when the equivalent conditions of that proposition hold, then for every system (f_i) of sections of $\mathscr{J}$ over a neighbourhood of y in X whose images in $\mathscr{J}_y/(\mathfrak{m}_s\mathscr{O}_{X,y}\mathscr{J}_y + \mathscr{J}_y^2)$ form a basis of this $\big(\mathscr{O}_{X,y}/(\mathfrak{m}_s\mathscr{O}_{X,y} + \mathscr{J}_y)\big)$-module, there is an open neighbourhood U of y in X such that the $f_i|U$ satisfy the conditions of Definition (19.2.1).

Corollary (19.2.6). — *Suppose either that S and X are locally Noetherian, or that g and h are locally of finite presentation. Suppose moreover that the fibres X_s and Y_s are regular at the points $u(y)$ and y respectively, and that the morphisms g and h are flat in a neighbourhood of $u(y)$ and y respectively. Then the immersion $Y \to X$ is transversally regular at the point y.*[2] **IV-4** | 192

Proof. Indeed, (19.1.1) shows that the immersion $Y_s \to X_s$ is regular at the point y, and it suffices to apply (19.2.4), *b)*. □

Proposition (19.2.7). —

(i) Every open S-immersion is transversally regular relative to S.
(ii) Let $f : Y \to X$ be an S-immersion, $g : S' \to S$ a morphism and set $X' = X_{(S')}$, $Y' = Y_{(S')}$, $f' = f_{(S')} : Y' \to X'$. If f is transversally regular relative to S, then f' is transversally regular relative to S'. The converse holds if g is faithfully flat and, moreover, X and Y are locally of finite presentation over S.
(iii) If $f : Y \to X$ and $g : Z \to Y$ are transversally regular immersions relative to S, so is $f \circ g : Z \to X$.
(iv) Let X be an S-prescheme, $f : Y \to X$, $g : Z \to Y$ two S-immersions. Suppose either that S and X are locally Noetherian, or that X, Y, and Z are locally of finite presentation over S. Then, if g and $f \circ g$ are transversally regular at the point $z \in Z$ relative to S, f is transversally regular at the point $g(z)$ relative to S.
(v) Under the general conditions of (iv), suppose that X and Y are S-flat, Y_s is regular at the point $g(z)$, and X_s is regular at the point $f(g(z))$; then, if $f \circ g$ is transversally regular at the point z, so is g.

Proof. Assertion (i) is immediate, and the first assertion of (ii) follows immediately from (19.2.3). To prove the second assertion of (ii), apply (19.2.4). Since X' and Y' are locally of finite presentation over S', this criterion first shows that X' and Y' are flat over S'; hence X and Y are flat over S ((2.5.1)). On the other hand, for every $s \in S$, if $s' \in S'$ lies above s, then $Y'_{s'} \to X'_{s'}$ is a regular immersion by hypothesis; (19.1.5), (ii), therefore shows that the same is true of $Y_s \to X_s$, since these preschemes are locally Noetherian and, in that case, regularity and quasi-regularity of the immersion $Y_s \to X_s$ are equivalent. To prove (iii), reduce to the case where Z is a closed sub-prescheme of Y and Y a closed sub-prescheme of X, and form, as in (19.1.5), (iii), a system of generators of the ideal of $\mathscr{O}_X$ defining Z. For (iv), the hypothesis together with (19.2.4) first shows that X, Y, and Z are S-flat in a neighbourhood of z. If s is the image of z in S, it remains only to prove that $f_s : Y_s \to X_s$ is regular at the point $g(z)$, knowing that $g_s : Z_s \to Y_s$ and $f_s \circ g_s$ are regular at z; this follows from (19.1.5), (iv). Finally, for (v), argue in the same way as for (iv). The hypothesis on $f \circ g$ already implies by (19.2.4), *b)*, that Z is S-flat in a neighbourhood of z, and the same is true of X and Y by hypothesis. We are therefore reduced to proving that g_s is regular at z, knowing that $f_s \circ g_s$ is regular there; this follows from (19.1.5), (v), under the stated hypotheses on X_s and Y_s. □

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Proposition (19.2.8). — *Let X be an S-prescheme, let Y be a closed sub-prescheme of X defined by a quasi-coherent ideal $\mathscr{J}$ of $\mathscr{O}_X$, and let $u : Y \to X$ be the canonical injection. If u is transversally regular relative to S, the $\mathscr{O}_X$-modules $\mathscr{J}^n$ and $\mathscr{J}^n/\mathscr{J}^m$ $(0 \leqslant n < m)$ are S-flat at the points of Y. For every base change $S' \to S$, if $X' = X_{(S')}$, $Y' = Y_{(S')}$, and $u' = u_{(S')}$, then, up to canonical isomorphism,*

$$\mathscr{G}r_n(u') = \mathscr{G}r_n(u) \otimes_{\mathscr{O}_S} \mathscr{O}_{S'}$$

for every $n > 0$.

Proof. Indeed, $\mathscr{O}_X$ and $\mathscr{O}_X/\mathscr{J}$ are, by hypothesis, S-flat at the points of Y, while $\mathscr{J}^n/\mathscr{J}^{n+1}$ is a locally free $(\mathscr{O}_X/\mathscr{J})$-module ((16.9.3)), and hence is S-flat at those points. The first assertion then follows from (**0**, 6.1.2), and the second from (11.2.9.2). □

The following proposition generalizes and sharpens (17.16.1).

Proposition (19.2.9). — *Let $f : X \to S$ be a morphism, x a point of X, $s = f(x)$. Suppose that f is of finite presentation at x and is a Cohen–Macaulay morphism at x ((6.8.1)). Then there exists a sub-prescheme Y of X, containing x, with the following properties:*

[2] The printed text has "x" here and again in the proof; the point introduced in Proposition 19.2.4 is y.

(i) The canonical injection $j : Y \to X$ is transversally regular relative to S.
(ii) The morphism $g = f \circ j : Y \to S$ is Cohen–Macaulay; equivalently, in view of (i), all the fibres of g are Cohen–Macaulay preschemes.
(iii) The point x is a maximal point of $Y_s = g^{-1}(s)$.

Proof. Set $X_s = f^{-1}(s)$. By hypothesis, the ring $\mathscr{O}_{X_s,x}$ is Cohen–Macaulay; let $(t_i)_{1 \leqslant i \leqslant n}$ be a system of parameters of this ring, and hence a regular sequence ((**0**, 16.5.7)). There is an open neighbourhood U of x in X and sections h_i ($1 \leqslant i \leqslant n$) of $\mathscr{O}_X$ over U such that the t_i are the canonical images of the germs $(h_i)_x$ in the quotient ring $\mathscr{O}_{X_s,x}$ of $\mathscr{O}_{X,x}$. Let $\mathscr{J}$ be the ideal of $\mathscr{O}_U$ generated by the h_i, and let Y be the closed sub-prescheme of U defined by $\mathscr{J}$. We may suppose that $f|U$ is locally of finite presentation; by the definition of Y, the same is then true of g. The equivalence of *c')* and *a)* in (11.3.8) shows that the sequence $((h_i)_x)$ is regular in $\mathscr{O}_{X,x}$ and that $\mathscr{O}_{Y,x}$ is flat over $\mathscr{O}_{S,s}$. By (11.3.1), f and g are therefore flat in a neighbourhood of x, and (19.2.4) shows that j is a transversally regular immersion relative to S, after replacing Y, if necessary, by $Y \cap V$ for an open neighbourhood V of x in X. Thus (i) is proved, and we may suppose that $g : Y \to S$ is flat. Since $\mathscr{O}_{Y_s,x}$, the quotient of $\mathscr{O}_{X_s,x}$ by the ideal generated by the t_i, is Artinian by construction ((**0**, 16.3.6)), x is a maximal point of Y_s. Finally, every Artinian ring is Cohen–Macaulay ((**0**, 16.5.1)); replacing Y, if necessary, by its intersection with an open neighbourhood of x in X, we may therefore suppose, by (12.1.7), (iii), that g is a Cohen–Macaulay morphism.

In particular, we recover the existence of flat "quasi-sections" Y of a morphism $f : X \to S$ at a point $x \in X_s$ that is closed in X_s, under the hypotheses that f is flat at x and $\mathscr{O}_{X_s,x}$ is Cohen–Macaulay ((17.16.1)); moreover, the immersion $Y \to X$ is transversally regular relative to S. This last refinement also applies to the *étale* quasi-section defined in (17.16.3), (i). ☐

19.3. Relative complete intersections (flat case) **IV-4** | 194

Definition (19.3.1). — Let A be a Noetherian local ring. We say that A is a *complete-intersection ring* (or an *absolute complete-intersection ring*, when there is danger of confusion) if its completion $\widehat{\mathrm{A}}$ is isomorphic to the quotient of a complete regular Noetherian local ring B by a regular ideal, that is, by an ideal generated by a regular sequence of elements of B ((16.9.7)).

Every regular local ring is a complete intersection ring.

We say that a locally Noetherian prescheme X is an *(absolute) complete intersection* at a point $x \in X$ if the ring $\mathscr{O}_{X,x}$ is a complete-intersection ring.

Proposition (19.3.2). — *Let* B *be a regular Noetherian local ring and* $\mathfrak{J}$ *an ideal of* B*. For* $\mathrm{A} = \mathrm{B}/\mathfrak{J}$ *to be a complete-intersection ring, it is necessary and sufficient that* $\mathfrak{J}$ *be a regular ideal, that is, be generated by a regular sequence of elements of* B *((16.9.7)).*

Proof. We may write $\widehat{\mathrm{A}} = \widehat{\mathrm{B}}/\mathfrak{J}\widehat{\mathrm{B}}$. Since $\widehat{\mathrm{B}}$ is faithfully flat over B and is Noetherian, $\mathfrak{J}\widehat{\mathrm{B}}$ is regular if and only if $\mathfrak{J}$ is regular ((19.1.5), (ii)). This shows, by (**0**, 17.1.5), both that the condition in the statement is sufficient and that, in proving its necessity, we may reduce to the case where A and B are complete. By hypothesis, there is then a complete regular Noetherian local ring B' such that A is isomorphic to a quotient $\mathrm{B}'/\mathfrak{J}'$, where $\mathfrak{J}'$ is a regular ideal of B'. Consider the fibre product $\mathrm{B}'' = \mathrm{B} \times_{\mathrm{A}} \mathrm{B}'$ for the surjective homomorphisms $\mathrm{B} \to \mathrm{A}$ and $\mathrm{B}' \to \mathrm{A}$ ((**0**, 18.1.2)). We first prove the following lemma.

Lemma (19.3.2.1). — *Let* A, E, F *be three rings,* $f : \mathrm{E} \to \mathrm{A}$*,* $g : \mathrm{F} \to \mathrm{A}$ *two homomorphisms, and set* $\mathrm{G} = \mathrm{E} \times_{\mathrm{A}} \mathrm{F}$ *((**0**, 18.1.2)).*

(i) If f is surjective and if E *and* F *are Noetherian rings,* G *is Noetherian.*
(ii) If A, E, *and* F *are local rings and f and g are local homomorphisms, then* G *is a local ring and the canonical homomorphisms* $\mathrm{G} \to \mathrm{E}$ *and* $\mathrm{G} \to \mathrm{F}$ *are local.*
(iii) If the conditions of (i) and (ii) are satisfied, if g is surjective, if E *and* F *are complete Noetherian local rings,* G *is complete.*

Proof. (i) In view of (**0**, 18.1.3) and (18.1.5), the projection $\mathrm{G} \to \mathrm{F}$ is surjective. Its kernel $\mathfrak{J}'$ is canonically identified with the kernel $\mathfrak{J}$ of f, and the G-module structure on $\mathfrak{J}'$ corresponds to the E-module structure on $\mathfrak{J}$ via the canonical homomorphism $\mathrm{G} \to \mathrm{E}$. If $\mathfrak{a}$ is any ideal of G, then $\mathfrak{a}/(\mathfrak{a} \cap \mathfrak{J}')$ is isomorphic to $(\mathfrak{a} + \mathfrak{J}')/\mathfrak{J}'$, and hence to an ideal of F, so it is finitely generated. On the other hand, $\mathfrak{a} \cap \mathfrak{J}'$ is isomorphic to an ideal of E contained in $\mathfrak{J}$, and is therefore also finitely generated. Thus $\mathfrak{a}$ is finitely generated and G is Noetherian.

(ii) Let $\mathfrak{r}$ and $\mathfrak{s}$ be the maximal ideals of E and F respectively. The set $\mathfrak{m} = \mathfrak{r} \times_{\mathrm{A}} \mathfrak{s}$ is an ideal of G, the inverse image of $\mathfrak{r}$ under $\mathrm{G} \to \mathrm{E}$ and of $\mathfrak{s}$ under $\mathrm{G} \to \mathrm{F}$. Its elements are therefore non-invertible in G. On the other hand, if $(x, y) \notin \mathfrak{m}$ and, for example, $x \notin \mathfrak{r}$, then $f(x)$ does not belong to the maximal ideal of A. Since $g(y) = f(x)$ and g is local, we also have $y \notin \mathfrak{s}$; we conclude that x and y are invertible, hence (x, y) is invertible, which completes the proof of (ii). **IV-4** | 195

(iii) The image of $\mathfrak{m}$ under the surjective homomorphism $\mathrm{G} \to \mathrm{E}$ is the maximal ideal $\mathfrak{r}$ of E; hence the image of $\mathfrak{m}^n \cap \mathfrak{J}'$ is $\mathfrak{r}^n \cap \mathfrak{J}$. Since $\mathfrak{J}$ is closed, and therefore complete, for the topology induced by the $\mathfrak{r}$-adic topology of E ((**0**, 7.3.5)), $\mathfrak{J}'$ is complete for the topology induced by the $\mathfrak{m}$-preadic topology of G. On the other hand, $\mathrm{G}/\mathfrak{J}'$, endowed with the $\mathfrak{m}$-preadic topology, is isomorphic to F endowed with the $\mathfrak{s}$-adic topology, and is therefore complete for the quotient topology of the $\mathfrak{m}$-preadic topology of G. It follows that G is complete for the $\mathfrak{m}$-preadic topology.

□

The lemma established, Cohen's theorem ((**0**, 19.8.8)) shows that B″ is isomorphic to a quotient of a complete regular Noetherian local ring C. It follows from (19.1.2) that the immersion $\mathrm{Spec}(\mathrm{B}') \to \mathrm{Spec}(\mathrm{C})$ is regular. Since, by hypothesis, the immersion $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B}')$ is regular, (19.1.5), (iii), shows that $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{C})$ is regular. But this immersion is also the composite

$$\mathrm{Spec}(\mathrm{A}) \longrightarrow \mathrm{Spec}(\mathrm{B}) \longrightarrow \mathrm{Spec}(\mathrm{C}).$$

Since B is regular, the immersion $\mathrm{Spec}(\mathrm{B}) \to \mathrm{Spec}(\mathrm{C})$ is regular ((19.1.2)); hence $\mathrm{Spec}(\mathrm{A}) \to \mathrm{Spec}(\mathrm{B})$ is regular by (19.1.5), (iv). □

Corollary (19.3.3). — *Let X be a locally Noetherian prescheme locally immersible in a regular prescheme. Then the set of $x \in X$ where X is a complete intersection is open in X and contains the set of regular points of X.*

Proof. We can indeed restrict ourselves to the case where $X = \mathrm{Spec}(\mathrm{B}/\mathfrak{J})$, where B is a regular ring and $\mathfrak{J}$ an ideal of B. The set of $\mathfrak{p} \supset \mathfrak{J}$ in Spec(B) such that $\mathfrak{J}_{\mathfrak{p}}$ is regular in $\mathrm{B}_{\mathfrak{p}}$ is open ((16.9.5)), and, by (19.3.2), it is the set of points of X at which X is a complete intersection. □

Corollary (19.3.4). — *Let k be a field, X a prescheme locally of finite type over k, k' an extension of k, $X' = X \otimes_k k'$. Let x' be a point of X', x its projection in X. For X to be a complete intersection at x, it is necessary and sufficient that X' be a complete intersection at x'.*

Proof. The question being local on X, we can suppose that $X = \mathrm{Spec}(\mathrm{A})$, where A is a k-algebra of finite type, hence a quotient of a polynomial algebra $k[\mathrm{T}_1, \ldots, \mathrm{T}_n]$, so X is a closed sub-prescheme of the regular scheme $Y = \mathrm{Spec}(k[\mathrm{T}_1, \ldots, \mathrm{T}_n])$ ((**0**, 17.3.7)). If we set $Y' = Y \otimes_k k' = \mathrm{Spec}(k'[\mathrm{T}_1, \ldots, \mathrm{T}_n])$, Y' is also a regular scheme. Now, since $\mathrm{Spec}(k')$ is faithfully flat over $\mathrm{Spec}(k)$, it follows from (19.1.5), (ii), and the fact that these preschemes are Noetherian that the immersion $X \to Y$ is regular at x if and only if the immersion $X' \to Y'$ is regular at x'. The conclusion then follows from the criterion (19.3.2). □

Corollary (19.3.5). — *Let* A *and* C *be Noetherian local rings, let $\rho : \mathrm{A} \to \mathrm{C}$ be a local homomorphism, let $\mathrm{B} = \mathrm{C}/\mathfrak{J}$ be a quotient ring of* C*, and let k be the residue field of* A*. Suppose that ρ makes* C *a flat* A*-module and that $\mathrm{C} \otimes_{\mathrm{A}} k$ is regular. Then the following conditions are equivalent:*

a) *The ideal $\mathfrak{J}$ is transversally regular relatively to* A.
b) B *is a flat* A*-module, and $\mathrm{B} \otimes_{\mathrm{A}} k$ is a complete intersection ring.*

Proof. By virtue of (19.2.4), condition *a)* amounts to saying that B is a flat A-module and that $\mathfrak{J} \otimes_{\mathrm{A}} k$ (which is identified with an ideal of $\mathrm{C} \otimes_{\mathrm{A}} k$ by virtue of the flatness of B over A ((**0**, 6.1.2))) is regular in this ring. Since the ring $\mathrm{C} \otimes_{\mathrm{A}} k$ is regular, it amounts to the same, by virtue of (19.3.2), to say (when B is a flat A-module) that $\mathfrak{J} \otimes_{\mathrm{A}} k$ is a regular ideal of $\mathrm{C} \otimes_{\mathrm{A}} k$, or that $\mathrm{B} \otimes_{\mathrm{A}} k = (\mathrm{C} \otimes_{\mathrm{A}} k)/(\mathfrak{J} \otimes_{\mathrm{A}} k)$ is a complete intersection ring; whence the corollary. □ **IV-4** | 196

Definition (19.3.6). — Let $f : X \to S$ be a flat morphism locally of finite presentation. We say that X is a *complete intersection relatively to* S at a point x, if the fibre $f^{-1}(f(x))$ is a complete intersection (absolute) at the point x. We say that X is a *complete intersection relatively to* S, or that the morphism f is a *morphism of complete intersection*, if X is a complete intersection relatively to S at each of its points.

Proposition (19.3.7). — *Let $g : X \to S$, $h : Y \to S$ be two flat morphisms locally of finite presentation, $f : Y \to X$ an S-immersion, y a point of Y, $x = f(y)$, $s = g(x) = h(y)$. Suppose that the fibre $X_s = g^{-1}(s)$ is regular at the point x. Then, for f to be a transversally regular immersion at the point y, it is necessary and sufficient that Y be a complete intersection relatively to S at the point y.*

Proof. To say that f is transversally regular at the point y amounts, by virtue of (19.2.4), to saying that $f_s : Y_s \to X_s$ is a regular immersion at the point y. But since X_s is regular at the point $x = f(y)$, this is equivalent, by virtue of (19.3.2), to saying that Y_s is a complete intersection (absolute) at the point y, whence the proposition. □

Corollary (19.3.8). — *Let $f : X \to S$ be a flat morphism locally of finite presentation. The set U of $x \in X$ for which X is a complete intersection relatively to S is open in X. If moreover f is proper, the set E of $s \in S$ such that $X_s = f^{-1}(s)$ is a complete intersection at each of its points is an open subset of S.*

Proof. Since $E = S - f(X - U)$,[3] the second assertion follows from the first and from the fact that f is a closed map. The first assertion is local on X, hence we can suppose that X is a closed subprescheme of a prescheme of the form $Z = S[\mathrm{T}_1, \dots, \mathrm{T}_n]$ (T_i indeterminate). Since Z is smooth over S ((17.3.8)), its fibres Z_s are regular, and, by (19.3.7), the assertion that $x \in U$ is equivalent to the assertion that the immersion $X \to Z$ is transversally regular at x; the result follows from (19.2.2). □

Proposition (19.3.9). —

(i) *Every open immersion is a complete-intersection morphism.*

(ii) *Let $f : X \to S$ be a flat morphism locally of finite presentation that is a complete-intersection morphism. For every change of base $g : S' \to S$, $f' = f_{(S')} : X_{(S')} \to S'$ is a morphism of complete intersection. The converse is true if g is faithfully flat and quasi-compact.*

(iii) *If $f : X \to Y$ and $g : Y \to Z$ are flat morphisms locally of finite presentation which are complete intersection morphisms, so is $g \circ f : X \to Z$.*

Proof. Assertion (i) follows immediately from the definition (19.3.1). To prove (ii), note first that **IV-4** | 197
if f is flat and locally of finite presentation, then so is f' ((2.1.4)); conversely, these two properties descend when g is faithfully flat and quasi-compact ((2.5.1) and (2.7.1)). Moreover, for every $s' \in S'$, if $s = g(s')$, the fibre $f^{-1}(s)$ is a complete intersection if and only if $f'^{-1}(s')$ is one ((19.3.4)). This proves (ii), since a faithfully flat morphism is surjective.

Finally, under the hypotheses of (iii), $g \circ f$ is flat and locally of finite presentation. By definition (19.3.6), we are reduced to the case where Z is the spectrum of a field, and must then prove that the local rings $\mathscr{O}_{X,x}$ are complete-intersection rings. This is contained in the following more general result. □

Corollary (19.3.10). — *Let $f : X \to S$ be a flat morphism locally of finite presentation that is a complete-intersection morphism, let x be a point of X, and set $s = f(x)$. If $\mathscr{O}_{S,s}$ is a Noetherian complete-intersection ring, then so is $\mathscr{O}_{X,x}$.*

Proof. We may plainly reduce to the case where $S = \mathrm{Spec}(\mathrm{A})$ with $\mathrm{A} = \mathscr{O}_{S,s}$. We first show that we may further reduce to the case where A is complete. Set $\mathrm{A}' = \widehat{\mathrm{A}}$, $X' = X \otimes_{\mathrm{A}} \mathrm{A}'$, and let s' be the unique closed point of $S' = \mathrm{Spec}(\mathrm{A}')$, hence the unique point of S' above s. If $f' = f_{(S')} : X' \to S'$, then $f'^{-1}(s')$ is canonically isomorphic to $f^{-1}(s)$ because A and A′ have the same residue field ((**I**, 3.6.4)). Thus there is a unique point $x' \in X'$ above x and s', and $\mathscr{O}_{X',x'} = \mathscr{O}_{X,x} \otimes_{\mathrm{A}} \mathrm{A}'$. The local rings $\mathscr{O}_{X,x}$ and $\mathscr{O}_{X',x'}$ have isomorphic completions. Indeed, if E is a local ring that is a local A-algebra, then the separated completion $(\mathrm{E} \otimes_{\mathrm{A}} \widehat{\mathrm{A}})^{\wedge}$ of $\mathrm{E} \otimes_{\mathrm{A}} \widehat{\mathrm{A}}$, endowed with the tensor-product

[3] The printed text has $Y - f(X - U)$; the ambient set in which E was just defined is S.

topology, is isomorphic to the separated completion of $\widehat{E}\widehat{\otimes}_A\widehat{A} \cong \widehat{E}$, and hence to $\widehat{E}$ itself ((**0**, 7.7.1)). By (19.3.1), it is therefore equivalent for $\mathscr{O}_{X,x}$ or $\mathscr{O}_{X',x'}$ to be a complete-intersection ring.

Suppose, then, that A is complete. We may also suppose that $X = \mathrm{Spec}(B)$, where B is a quotient of the polynomial algebra $C = A[T_1, \dots, T_r]$. Since polynomial rings over a field are regular ((**0**, 17.3.7)), (19.3.7) shows that the immersion $X \to Z = \mathrm{Spec}(C)$ may be taken transversally regular relative to S. Thus $B = C/\mathfrak{J}$, where $\mathfrak{J}$ is generated by a regular sequence $(g_i)_{1\leqslant i\leqslant n}$ in C. On the other hand, since A is complete, Cohen's theorem allows us to write $A = A'/\mathfrak{R}$, where A' is a regular local ring and $\mathfrak{R}$ an ideal of A' ((**0**, 19.8.8)). Since A is a complete-intersection ring, $\mathfrak{R}$ is generated by a regular sequence $(f_j)_{1\leqslant j\leqslant m}$ ((19.3.2)). In $C' = A'[T_1, \dots, T_r]$, the elements f_j still form a regular sequence, generating $\mathfrak{R}' = \mathfrak{R}[T_1, \dots, T_r]$ ((**0**, 15.1.4)). Since $C'/\mathfrak{R}' = C$, choose, for each i, an element $g'_i \in C'$ whose image in C is g_i. The sequence consisting of the f_j and the g'_i is regular in C'. If $\mathfrak{J}'$ is the ideal it generates, then $B = C'/\mathfrak{J}'$; the conclusion follows from (19.3.2), since C' is regular ((**0**, 17.3.7)). □

19.4. Application: criteria of regularity and smoothness for blown-up preschemes **IV-4** | 198

(19.4.1). Let X be a prescheme and Y a closed sub-prescheme of X, defined by a quasi-coherent ideal $\mathscr{J}$ of $\mathscr{O}_X$. Recall ((**II**, 8.1.3)) that the X-scheme X' obtained by blowing up Y is the prescheme

$$X' = \mathrm{Proj}(\mathscr{S}), \quad \text{where} \quad \mathscr{S} = \bigoplus_{n\geqslant 0} \mathscr{J}^n.$$

If $\mathscr{J}$ is of finite type, the structural morphism $f : X' \to X$ is projective. Without hypothesis on $\mathscr{J}$, the closed sub-prescheme

$$Y' = f^{-1}(Y)$$

of X' is defined by the quasi-coherent Ideal $\mathscr{J}\mathscr{O}_{X'}$ of $\mathscr{O}_{X'}$, canonically isomorphic to $\mathscr{O}_{X'}(1)$; more precisely, we have an exact sequence

$$0 \longrightarrow \mathscr{O}_{X'}(1) \xrightarrow{\widetilde{u}_0} \mathscr{O}_{X'} \longrightarrow \mathscr{O}_{Y'} \longrightarrow 0$$

(19.4.1.1) where $\mathscr{J}\mathscr{O}_{X'}$ is the image of $\widetilde{u}_0$ ((**II**, 8.1.7) and (8.1.8)). Since $\mathscr{O}_{X'}(1)$ is free of rank 1 in a neighbourhood of every point of X', $\mathscr{J}\mathscr{O}_{X'}$ is locally generated by a non-zero-divisor of $\mathscr{O}_{X'}$, and $\mathscr{J}\mathscr{O}_{X'}/\mathscr{J}^2\mathscr{O}_{X'}$ is consequently a free $(\mathscr{O}_{X'}/\mathscr{J}\mathscr{O}_{X'})$-module of rank 1. This proves the first assertion of the following lemma:

Lemma (19.4.2). — *The canonical immersion $Y' \to X'$ is regular and of codimension* 1, *and Y' is canonically isomorphic to the Y-scheme* $\mathrm{Proj}(\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X))$.

Proof. The second assertion follows from (**II**, 3.5.3), applied to the canonical injection $Y \to X$, because

$$\mathscr{S} \otimes_{\mathscr{O}_X} \mathscr{O}_Y = \mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$$

by definition: $\mathscr{O}_Y \cong \mathscr{O}_X/\mathscr{J}$ and $\mathscr{J}^n \otimes_{\mathscr{O}_X} (\mathscr{O}_X/\mathscr{J}) \cong \mathscr{J}^n/\mathscr{J}^{n+1}$. □

Corollary (19.4.3). — *If the canonical immersion $Y \to X$ is quasi-regular, the restriction $g : Y' \to Y$ of the morphism f is smooth.*

Proof. Indeed, $\mathscr{N}_{Y/X} = \mathscr{J}/\mathscr{J}^2$ is then an $\mathscr{O}_Y$-Module locally free of finite rank and $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is isomorphic to $\mathbf{S}^*_{\mathscr{O}_Y}(\mathscr{N}_{Y/X})$ (16.9.8), hence Y' is Y-isomorphic to $\mathbf{P}(\mathscr{N}_{Y/X})$ and is therefore smooth over Y (17.3.9). □

We shall see later (chap. V) that g is smooth also when $Y \to X$ "has only ordinary singularities".

Proposition (19.4.4). — *With the notations of* (19.4.1), *suppose that X is locally Noetherian and that the morphism $g : Y' \to Y$, restriction of f, is smooth. Let x' be a point of Y', $x = g(x')$. For Y to be regular at the point x, it is necessary and sufficient that Y' be regular at the point x'; when this is so, X' is then regular at the point x'.*

Proof. The first assertion follows from (17.5.8) and the second from (19.1.1) and (19.4.2). □

Corollary (19.4.5). — *Under the general hypotheses of (19.4.4), suppose that Y is regular at the point x. Then, for every generization x_1 of x in X not belonging to Y, X is regular at the point x_1. If $\mathrm{Reg}(X)$ is open (((6.12)), for example if X is excellent ((7.8.6), (iii))) and if Y is regular, then there exists an open neighbourhood U of Y in X such that X is regular at the points of $U - Y$.* **IV-4** | 199

Proof. To prove the first assertion, we may reduce to the case where X is a local scheme with closed point x. The hypothesis then implies that Y is regular ((**0**, 17.3.2)), and hence so is Y' by (19.4.4). The morphism $f : X' \to X$ is projective, hence proper. If x_1' denotes the unique point of X' above x_1 ((**II**, 8.1.3)), the valuative specialization property ((**II**, 7.2.1)) gives a specialization of x_1' lying in Y'. Applying (19.4.4) and (**0**, 17.3.2), we conclude that X' is regular at x_1', and therefore that X is regular at x_1, since $X' - Y'$ is isomorphic to $X - Y$ ((**II**, 8.1.3)). If Y is regular, every point of $X - Y$ which is a generization of a point of Y belongs to $\mathrm{Reg}(X)$; if $\mathrm{Reg}(X)$ is open, $U = Y \cup \mathrm{Reg}(X)$ is then locally constructible and stable under generization, hence open ((**0**, 9.2.5)), and responds to the question. □

Proposition (19.4.6). — *Let $g : X \to S$ be a morphism, Y a closed sub-prescheme of X defined by a quasi-coherent Ideal $\mathscr{J}$, X' the X-scheme obtained by blowing up Y, Y' the inverse image of Y in X'.*

- *(i) The following conditions are equivalent:*
  - a) $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ *is a g-flat $\mathscr{O}_X$-algebra.*
  - b) *For every $n \geqslant 0$, the $\mathscr{O}_X$-Module $\mathscr{O}_X/\mathscr{J}^{n+1}$ is g-flat (in other words, the n-th infinitesimal neighbourhood $Y^{(n)}$ of Y in X (16.1.2) is S-flat).*
  - c) *For every base change $S_1 \to S$ and every integer $n \geqslant 1$, if $X_1 = X \times_S S_1$, the canonical homomorphism $\mathscr{J}^n \otimes_{\mathscr{O}_S} \mathscr{O}_{S_1} \to \mathscr{J}^n \mathscr{O}_{X_1}$ is bijective.*

  When this is so, Y' is S-flat, and for every change of base $S_1 \to S$, if Y_1 is the inverse image of Y in X_1, the prescheme X_1' obtained by blowing up Y_1 in X_1 is canonically isomorphic to $X' \times_S S_1$.
- *(ii) Suppose that the equivalent conditions of (i) are satisfied and moreover that the morphisms $X \to S$ and $Y \to S$ are locally of finite presentation. Then $\mathscr{S} = \bigoplus_{n \geqslant 0} \mathscr{J}^n$ is a $\mathscr{O}_X$-Algebra of finite presentation, the morphism $X' \to X$ is of finite presentation, the canonical immersion $Y' \to X'$ is transversally regular of codimension 1 relatively to S, and the set of points of X (resp. X') where X (resp. X') is S-flat, is a neighbourhood of Y (resp. Y').*

Proof. **IV-4** | 200

- (i) The equivalence of *a)* and *b)* follows immediately from (**0**, 6.1.2). To prove the equivalence of *b)* and *c)*, we may suppose that $S = \mathrm{Spec}(\mathrm{A})$ and $X = \mathrm{Spec}(\mathrm{B})$ are affine, with $\mathscr{J} = \widetilde{\mathfrak{J}}$. The Tor exact sequence shows that *b)* implies $\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{B}/\mathfrak{J}^{n+1}, \mathrm{A}') = 0$ for every $n \geqslant 0$ and every A-algebra A'. Taking $\mathrm{A}' = \mathrm{A} \oplus \mathrm{M}$, for an arbitrary A-module M and multiplication $(a, m)(a', m') = (aa', am' + a'm)$, shows that this is equivalent to $\mathrm{Tor}_1^{\mathrm{A}}(\mathrm{B}/\mathfrak{J}^{n+1}, \mathrm{M}) = 0$, and hence to the flatness of $\mathrm{B}/\mathfrak{J}^{n+1}$ over A. If $\mathfrak{J}_1 = \mathfrak{J} \otimes_{\mathrm{A}} \mathrm{A}_1$, then

  $$\bigoplus_{n \geqslant 0} \mathfrak{J}_1^n = \left(\bigoplus_{n \geqslant 0} \mathfrak{J}^n \right) \otimes_{\mathrm{A}} \mathrm{A}_1,$$

  and the assertion concerning the blown-up prescheme X_1' follows immediately. Finally, to prove that Y' is S-flat, we may again suppose that S and X are affine. Put $\mathrm{C} = \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B})$. By (19.4.2) and (**II**, 2.3.6), every point of Y' has an affine neighbourhood whose ring is $\mathrm{C}_{(t)}$, with t homogeneous of degree 1. Since C is flat over A, so is the localization C_t, and hence its degree-zero direct summand $\mathrm{C}_{(t)}$. Thus Y' is S-flat.
- (ii) Again suppose that S and X are affine, so that B is a finitely presented A-algebra and $\mathfrak{J}$ is a finitely generated ideal of B. By (8.9.1), (8.6.3), and (11.2.9), there exist a Noetherian subring $\mathrm{A}_0 \subset \mathrm{A}$, a finite-type A_0-algebra B_0, and a finitely generated ideal $\mathfrak{J}_0 \subset \mathrm{B}_0$ such that

  $$\mathrm{B} = \mathrm{B}_0 \otimes_{\mathrm{A}_0} \mathrm{A}, \qquad \mathfrak{J} = \mathfrak{J}_0 \mathrm{B}, \qquad \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{B}) = \mathrm{gr}^*_{\mathfrak{J}_0}(\mathrm{B}_0) \otimes_{\mathrm{A}_0} \mathrm{A},$$

  with $\mathrm{gr}^*_{\mathfrak{J}_0}(\mathrm{B}_0)$ flat over A_0. By (i),

  $$\bigoplus_{n \geqslant 0} \mathfrak{J}^n = \left(\bigoplus_{n \geqslant 0} \mathfrak{J}_0^n \right) \otimes_{\mathrm{A}_0} \mathrm{A} = \left(\bigoplus_{n \geqslant 0} \mathfrak{J}_0^n \right) \otimes_{\mathrm{B}_0} \mathrm{B}.$$

The algebra $\bigoplus_{n\geqslant 0}\mathfrak{J}_0^n$ is generated over B_0 by the finitely generated ideal $\mathfrak{J}_0$; it is therefore of finite type, and hence of finite presentation because B_0 is Noetherian. Thus $\bigoplus_{n\geqslant 0}\mathfrak{J}^n$ is a finitely presented B-algebra. Likewise, $\mathrm{Proj}(\bigoplus_{n\geqslant 0}\mathfrak{J}_0^n) \to \mathrm{Spec}(\mathrm{B}_0)$ is of finite type ((**II**, 2.7.1)), hence of finite presentation, and its base change $X' \to X$ is of finite presentation. By (19.4.2), the canonical immersion $Y' \to X'$ is regular. Since Y' is S-flat by (i), (19.2.4) shows that $Y' \to X'$ is transversally regular relative to S and that X' is S-flat at the points of Y', hence on an open neighbourhood of Y' by (11.3.1). Finally, since $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is an S-flat $\mathscr{O}_X$-algebra, (11.3.4) shows that X is S-flat at the points of Y, and hence on a neighbourhood of Y by (11.3.1).

□

Corollary (19.4.7). — *Let $g : X \to S$ be a morphism locally of finite presentation, Y a closed sub-prescheme of X such that the composite morphism $Y \to X \to S$ is locally of finite presentation; suppose moreover that Y is S-flat and that $\mathscr{O}_X$ is normally flat along Y ((11.3.4)). Then (with the notations of (19.4.1)), the morphism $X' \to X$ is of finite presentation, the canonical immersion $Y' \to X'$ is transversally regular of codimension* 1 *relative to S, Y' is flat over Y (and hence over S), and the set of points of X (resp. X') where X (resp. X') is S-flat is a neighbourhood of Y (resp. Y').*

Proof. Indeed, by hypothesis $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is a flat $(\mathscr{O}_X/\mathscr{J})$-module, and hence is S-flat because Y is S-flat. We can therefore apply the results of (19.4.6), (i) and (ii). Since Y' is Y-isomorphic to $\mathrm{Proj}(\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X))$ ((19.4.2)), the same argument as in (19.4.6) shows that Y' is flat over Y. □

Proposition (19.4.8). — *Suppose the hypotheses of (19.4.7) satisfied and moreover that the morphism $Y' \to Y$ is smooth. Let x' be a point of Y', x its image in Y. For Y to be smooth over S at the point x, it is necessary and sufficient that Y' be smooth over S at the point x'; when this is so, X' is then smooth over S at the point x'. The preceding conclusions hold in particular when $X \to S$ is locally of finite presentation and the immersion $Y \to X$ is transversally regular relative to S.*

Proof. Taking into account the compatibility of the formation of a blown-up prescheme with changes of base in the case envisaged (19.4.6), (i) and of ((17.5.1)) and (17.7.1), it suffices to consider **IV-4** | 201
the case where S is the spectrum of an algebraically closed field. But then it amounts to the same to say that an S-prescheme locally of finite type is smooth or that it is regular at this point ((17.15.1)), hence the conclusions follow from (19.4.4).

When $X \to S$ is locally of finite presentation and the canonical immersion $Y \to X$ is quasi-regular, that immersion is, by definition, of finite presentation, since the Ideal defining it is of finite type; hence the composite $Y \to X \to S$ is locally of finite presentation. We have already seen in (19.4.3) that under these conditions $Y' \to Y$ is smooth; moreover, $\mathscr{N}_{Y/X}$ is a locally free $\mathscr{O}_Y$-Module and $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{O}_X)$ is isomorphic to $\mathbf{S}^*_{\mathscr{O}_Y}(\mathscr{N}_{Y/X})$, hence is a flat $\mathscr{O}_Y$-Module. The hypotheses of (19.4.7) are thus all satisfied and we can apply the conclusions of (19.4.8). □

Corollary (19.4.9). — *Under the general hypotheses of (19.4.8), suppose moreover that Y is smooth over S at the point x. Then, for every generization x_1 of x in X not belonging to Y, X is smooth over S at the point x_1. If Y is smooth over S, then there exists an open neighbourhood U of Y in X such that X is smooth over S at the points of $U - Y$.*

Proof. To prove the first assertion, we can restrict ourselves to the case where X is a local scheme with closed point x; the hypothesis then implies that Y is smooth over S, since every neighbourhood in Y of its closed point is necessarily all of Y. We conclude by the same reasoning as in (19.4.5), using the fact that the set of points where X' is smooth over S is open in X'. To prove the second assertion, we can reduce to the case where S is Noetherian, and X of finite type over S, using ((8.9.1)), ((8.6.3)), ((11.2.9)) and (17.7.6); we then conclude as in (19.4.5), using the fact that the set of points where X is smooth over S is open in X. □

Remark (19.4.10). — In (19.4.6), (ii), replace the hypothesis that X and Y are locally of finite presentation over S by the hypothesis that S and X (hence also Y) are *locally Noetherian*. Then it is clear that $\bigoplus_{n\geqslant 0}\mathscr{J}^n$ is an $\mathscr{O}_X$-Algebra of finite type, that the morphism $X' \to X$ is of finite type, and it follows again from (19.2.4) applied to locally Noetherian preschemes, that the immersion $Y' \to X'$ is transversally regular relatively to S and that X' is S-flat at the points of Y'.

Proposition (19.4.11). — *Let X be a prescheme, $\mathscr{E}$ a quasi-coherent $\mathscr{O}_X$-Module, $u : \mathscr{E} \to \mathscr{O}_X$ a homomorphism of $\mathscr{O}_X$-Modules, $\mathscr{J} = u(\mathscr{E})$ the quasi-coherent Ideal of $\mathscr{O}_X$ which is the image of u, and Y the closed sub-prescheme of X defined by $\mathscr{J}$. Let X' be the X-scheme obtained by blowing up Y. On the other hand, let $P = \mathbf{P}(\mathscr{E})$ be the projective bundle over X defined by $\mathscr{E}$ ((**II**, 4.1.1)), $p : P \to X$ the structural morphism, $\alpha_1^\# : p^*(\mathscr{E}) \to \mathscr{O}_P(1)$ the canonical homomorphism ((**II**, 4.1.5.1)) and $\mathscr{H}$ its kernel, so that we have an exact sequence*

$$0 \longrightarrow \mathscr{H} \longrightarrow p^*(\mathscr{E}) \xrightarrow{\alpha_1^\#} \mathscr{O}_P(1) \longrightarrow 0.$$

Finally, let $\mathscr{K}$ be the quasi-coherent Ideal of $\mathscr{O}_P$ which is the image of the restriction $v : \mathscr{H} \to \mathscr{O}_P$ of $p^*(u)$, and let Z be the closed sub-prescheme of P defined by $\mathscr{K}$. **IV-4** | 202

(i) *The image of X' by the closed immersion $j : X' \to \mathbf{P}(\mathscr{E})$ corresponding to the surjective homomorphism of graded $\mathscr{O}_X$-Algebras*

$$\mathbf{S}_{\mathscr{O}_X}(\mathscr{E}) \longrightarrow \mathbf{S}_{\mathscr{O}_X}(\mathscr{J}) \longrightarrow \bigoplus_{n \geqslant 0} \mathscr{J}^n$$

((**II**, 3.6.2)) *is a closed sub-prescheme of Z.*

(ii) *If $\mathscr{E}$ is an $\mathscr{O}_X$-Module locally free of rank m, $\mathscr{H}$ is an $\mathscr{O}_P$-Module locally free of rank $m-1$.*

(iii) *Suppose X is locally Noetherian, that $\mathscr{E}$ is locally free of finite type, that the sub-prescheme Y is regular, and that at every point $x \in Y$, the canonical immersion $i : Y \to X$ is regular and of codimension equal to $\mathrm{rg}_{\mathscr{O}_{X,x}}(\mathscr{E}_x)$ (which we shall express by saying that the homomorphism u is regular at the point x (cf. chap. V)). Then the image of X' by the closed immersion $j : X' \to P$ is the sub-prescheme Z; at all points of $X' - Y'$ (Y' being the inverse image of Y in X') the immersion j is transversally regular relatively to X; and finally, at all points x' of Y', X' is regular, the immersion j is regular and of codimension $\mathrm{rg}_{\mathscr{O}_{P,z}}(\mathscr{H}_z)$ where $z = j(x')$ (in other words the homomorphism $v : \mathscr{H} \to \mathscr{O}_P$ is regular in z).*

Proof. All the questions being local on X, we can restrict ourselves to the case where $X = \mathrm{Spec}(A)$ is affine and $\mathscr{E} = \widetilde{E}$, where E is an A-module. To verify (i), we can in addition examine what happens in an affine open of P of the form $D_+(t)$, where $t \in E = \mathbf{S}^1_A(E)$ ((**II**, 2.3.14)). If we set $S = \mathbf{S}_A(E)$, then $\Gamma(D_+(t), \mathscr{O}_P) = S_{(t)}$ ((**II**, 2.4.1)) and $p^*(\mathscr{E})|_{D_+(t)} = (E \otimes_A S_{(t)})^\sim$. Referring to the definition of $\alpha_1^\#$ ((**II**, 2.6.2.2)), we see that to every element

$$a = \sum_i x^{(i)} \otimes \frac{x_1^{(i)} x_2^{(i)} \cdots x_n^{(i)}}{t^n}$$

of $E \otimes_A S_{(t)}$, where the $x^{(i)}$ and $x_k^{(i)}$ belong to E, it assigns

$$\sum_i \frac{x^{(i)} x_1^{(i)} x_2^{(i)} \cdots x_n^{(i)}}{t^{n+1}} \quad \text{in } (S(1))_{(t)}.$$

To prove (i), it is enough to show that, if this last element is zero, then the image of

$$a' = v(a) = \sum_i u(x^{(i)}) \frac{x_1^{(i)} x_2^{(i)} \cdots x_n^{(i)}}{t^n}$$

under the canonical homomorphism $S_{(t)} \to \left(\bigoplus_{n \geqslant 0} u(E)^n\right)_{(u(t))}$ ((**II**, 3.6.2)) is also zero. Its image in $A_{u(t)}$ is

$$\sum_i \frac{u(x^{(i)}) u(x_1^{(i)}) \cdots u(x_n^{(i)})}{u(t)^n},$$

that is, the product of $u(t)$ and the canonical image of $\sum_i x^{(i)} x_1^{(i)} \cdots x_n^{(i)} / t^{n+1}$ under the algebra homomorphism extending $u : E \to A$ from $S_{(t)}$. This proves (i).

To establish (ii), we may suppose that E is a free A-module of rank m. Since $\alpha_1^\# : E \otimes_A S_{(t)} \to (S(1))_{(t)}$ is surjective and $(S(1))_{(t)}$ is a free $S_{(t)}$-module of rank 1, the exact sequence

$$0 \longrightarrow H_{(t)} \longrightarrow E \otimes_A S_{(t)} \xrightarrow{\alpha_1^\#} (S(1))_{(t)} \longrightarrow 0$$

splits, and $\mathrm{H}_{(t)}$ is therefore a projective $\mathrm{S}_{(t)}$-module of rank $m-1$, whence (ii).

To prove (iii), let us examine first what happens above a point $x \in X - Y$. The homomorphism $u : \mathscr{E} \to \mathscr{O}_X$, restricted to a neighbourhood of x, is then surjective, so we may restrict to the case $\mathscr{J} = \mathscr{O}_X$, in which X' is canonically identified with X. To see that the closed sub-prescheme of P which is the image of X' is then identical with Z, we may, as at the beginning, suppose $X = \mathrm{Spec}(A)$ and $\mathrm{E} = \mathrm{A}^m$, so that $\mathrm{S} = \mathrm{A}[\mathrm{T}_0, \dots, \mathrm{T}_{m-1}]$. With the notation above we may suppose $u(t) \neq 0$; the elements $t \in \mathrm{E}$ having this property generate the symmetric algebra $\mathbf{S}_\mathrm{A}(\mathrm{E})$. After changing the basis of E if necessary, we may therefore suppose $t = \mathrm{T}_0$ and $u(\mathrm{T}_i) = 0$ for $1 \leqslant i \leqslant m-1$, so that $\mathrm{S}_{(t)}$ is canonically identified with $\mathrm{A}[\mathrm{T}_1, \dots, \mathrm{T}_{m-1}]$. The elements of H are then those of the form

$$\sum_\alpha \mathrm{L}_\alpha(\mathbf{T}) \otimes \mathbf{T}^\alpha$$

in $\mathrm{E} \otimes_\mathrm{A} \mathrm{A}[\mathrm{T}_1, \dots, \mathrm{T}_{m-1}]$, where $\mathrm{L}_\alpha(\mathbf{T}) = \lambda_{\alpha 0} + \lambda_{\alpha 1}\mathrm{T}_1 + \dots + \lambda_{\alpha, m-1}\mathrm{T}_{m-1}$ and $\sum_\alpha \mathrm{L}_\alpha(\mathbf{T})\mathbf{T}^\alpha = 0$. One checks readily that, subject to $\lambda_{\mathbf{0},0} = 0$, the values of $\lambda_{\alpha 0}$ may be chosen arbitrarily for $|\alpha| \geqslant 1$, after which the $\lambda_{\alpha i}$ for $i \geqslant 1$ may be chosen so as to satisfy the displayed relation. The corresponding elements of the Ideal $\mathscr{K}$ are therefore precisely the polynomials $\sum_\alpha \lambda_{\alpha 0}\mathbf{T}^\alpha$ without constant term. Since $(\mathbf{S}_\mathrm{A}(\mathrm{A}))_{(t)}$ is identified with A, these polynomials are exactly the kernel of $(\mathbf{S}_\mathrm{A}(\mathrm{E}))_{(t)} \to (\mathbf{S}_\mathrm{A}(\mathrm{A}))_{(t)}$. We have thus proved, by (19.2.1), that $j : X' \to \mathrm{P}$ is transversally regular relative to X at every point of $X' - Y'$ and that $j(X') = Z$ at those points.

It remains to examine what happens above a point $x \in Y$. The regularity hypothesis on u at x **IV-4** | 203
in (iii) is equivalent, by (19.1.1) and ((**0**, 17.1.7)), to saying that X is regular at x and that

$$u_x \otimes 1 : \mathscr{E}_x \otimes_{\mathscr{O}_{X,x}} \mathbf{k}(x) \longrightarrow \mathfrak{m}_x/\mathfrak{m}_x^2$$

is injective. Since $p : \mathrm{P} \to X$ is smooth ((17.3.9)), for every $z \in \mathrm{P}$ above x, P is regular at z ((17.5.8)); moreover, by ((**0**, 17.3.3)), the canonical homomorphism

$$(\mathfrak{m}_x/\mathfrak{m}_x^2) \otimes_{\mathbf{k}(x)} \mathbf{k}(z) \longrightarrow \mathfrak{m}_z/\mathfrak{m}_z^2 \tag{19.4.11.1}$$

is injective. It follows that the homomorphism

$$(\mathscr{E}_x \otimes_{\mathscr{O}_{X,x}} \mathscr{O}_{\mathrm{P},z}) \otimes_{\mathscr{O}_{\mathrm{P},z}} \mathbf{k}(z) \longrightarrow \mathfrak{m}_z/\mathfrak{m}_z^2$$

induced by $p^*(u)$ is also injective, since it is the composite

$$(\mathscr{E}_x \otimes_{\mathscr{O}_{X,x}} \mathbf{k}(x)) \otimes_{\mathbf{k}(x)} \mathbf{k}(z) \longrightarrow (\mathfrak{m}_x/\mathfrak{m}_x^2) \otimes_{\mathbf{k}(x)} \mathbf{k}(z) \longrightarrow \mathfrak{m}_z/\mathfrak{m}_z^2,$$

where the first arrow is injective by flatness and the second is the injective homomorphism (19.4.11.1). Since $\mathscr{H}$ is, in a neighbourhood of z in P, a direct factor of $p^*(\mathscr{E})$, the homomorphism $v_z \otimes 1 : \mathscr{H}_z \otimes_{\mathscr{O}_{\mathrm{P},z}} \mathbf{k}(z) \to \mathfrak{m}_z/\mathfrak{m}_z^2$ is also injective; this establishes the regularity of v at z. It remains to see that the image by j of X' is identical to the closed sub-prescheme Z. It will suffice for this to show that the image by j of the points of $q^{-1}(x)$ (where $q : X' \to X$ is the structural morphism) are exactly those of $\mathrm{Z} \cap p^{-1}(x)$ and that on the other hand at each of these points $z = j(x')$, the surjective homomorphism $\mathscr{O}_{Z,z} \to \mathscr{O}_{X',x'}$ is bijective. By the hypothesis and (19.4.3), the restriction $Y' \to Y$ of q is smooth; since $\mathscr{O}_{Y,x}$ is regular, so is $\mathscr{O}_{Y',x'}$ ((17.5.8)). Moreover, $Y' \to X'$ is a regular immersion ((19.4.2)), and hence $\mathscr{O}_{X',x'}$ is regular ((19.1.1)). The regularity of v and of P at z similarly shows that $\mathscr{O}_{Z,z}$ is regular. To prove that $\mathscr{O}_{Z,z} \to \mathscr{O}_{X',x'}$ is bijective, it therefore suffices to show that the dimensions of these two regular rings are equal ((**0**, 17.1.9)). By ((**0**, 17.3.3)), this is equivalent to **IV-4** | 204
equality of the dimensions of $\mathscr{O}_{X',x'} \otimes_{\mathscr{O}_{X,x}} \mathbf{k}(x)$ and $\mathscr{O}_{Z,z} \otimes_{\mathscr{O}_{X,x}} \mathbf{k}(x)$. We may therefore replace X' and Z by the fibres $q^{-1}(x)$ and $\mathrm{Z} \cap p^{-1}(x)$ respectively, and reduce to proving that the morphism $q^{-1}(x) \to \mathrm{Z} \cap p^{-1}(x)$ induced by j is an isomorphism. We may now suppose that $X = \mathrm{Spec}(\mathrm{B})$, where $\mathrm{B} = \mathscr{O}_{X,x}$, and that $\mathrm{E} = \mathrm{B}^m$, so that $\mathrm{S} = \mathrm{B}[\mathrm{T}_1, \dots, \mathrm{T}_m]$. The hypothesis means that there is a regular system of parameters $(t_i)_{1 \leqslant i \leqslant n}$ of B such that $u(\mathrm{T}_i) = t_i$ for $1 \leqslant i \leqslant m$ ((**0**, 17.1.7)). Let $\mathfrak{J}$ be the image of u, and let $\mathfrak{m}$ be the maximal ideal of B. Then

$$\mathfrak{J}^n \otimes_\mathrm{B} (\mathrm{B}/\mathfrak{m}) = \mathfrak{J}^n/\mathfrak{m}\mathfrak{J}^n = (\mathfrak{J}^n/\mathfrak{J}^{n+1}) \otimes_\mathrm{B} (\mathrm{B}/\mathfrak{m}),$$

and consequently, by (16.9.4.1), $(\bigoplus_{n \geqslant 0} \mathfrak{J}^n) \otimes_\mathrm{B} \mathbf{k}(x)$ is identified with $\mathbf{k}(x)[\mathrm{T}_1, \dots, \mathrm{T}_m]$. It follows that the immersion $q^{-1}(x) \to p^{-1}(x)$ is an isomorphism; this implies *a fortiori* that the closed sub-prescheme $\mathrm{Z} \cap p^{-1}(x)$ is identical with $j(q^{-1}(x))$, and completes the proof of (19.4.11). □

19.5. Criteria for M-regularity We resume here, in completed form, the criteria for a sequence of elements of a ring A to be M-regular or M-quasi-regular, where M is an A-module, already studied in ((**0**, 15.1)).

Theorem (19.5.1). — *Let* A *be a ring,* $\mathbf{f} = (f_i)_{1\leqslant i\leqslant n}$ *a sequence of elements of* A, *and* M *an* A*-module. Consider the following properties:*

a) *The sequence* **f** *is M-regular.*
b) $\mathrm{H}_i(\mathbf{f}, \mathrm{M}) = 0$ *for every* $i > 0$ *((**III**, 1.1.3)).*
b') $\mathrm{H}_1(\mathbf{f}, \mathrm{M}) = 0$.
c) *The sequence* **f** *is M-quasi-regular.*

Then we have the implications

$$a) \Longrightarrow b) \Longrightarrow b') \qquad and \qquad a) \Longrightarrow c).$$

Put $\mathfrak{J} = \sum_{i=1}^{n} f_i\mathrm{A}$. *If the modules* $\mathrm{M}_i = \mathrm{M}/(\sum_{j=1}^{i} f_j\mathrm{M})$ *are separated for the* $\mathfrak{J}$*-preadic topology (respectively, if every quotient module of a submodule of* M *is separated for the* $\mathfrak{J}$*-preadic topology), then we also have the implication c)⇒a) (respectively, b')⇒a)).*

Proof. The implications a)⇒c), and c)⇒a) when the M_i are separated for the $\mathfrak{J}$-preadic topology, were already proved in ((**0**, 15.1.9)) and are recalled only for completeness. We have also shown in ((**III**, 1.1.4)) and ((**III**, 1.1.3.3)) that a) implies b), and b) trivially implies b'). It therefore remains to prove that, when every quotient module of a submodule of M is separated for the $\mathfrak{J}$-preadic topology, b') implies a). We argue by induction on n. For $n = 1$, the assertion follows immediately from the definitions, since $\mathrm{H}_1(\mathbf{f}, \mathrm{M})$ is precisely the kernel in M of multiplication $z \mapsto f_1 z$ ((**III**, 1.1.1), (**III**, 1.1.2)). Suppose then that $n \geqslant 2$, and put $\mathbf{f}' = (f_i)_{1\leqslant i\leqslant n-1}$. With the notation of ((**III**, 1.1.2)), $\mathrm{K}_\bullet(\mathbf{f}, \mathrm{M}) = \mathrm{K}_\bullet(f_n) \otimes_\mathrm{A} \mathrm{K}_\bullet(\mathbf{f}', \mathrm{M})$. We therefore have ((**III**, 1.1.4.1)) the exact sequence

$$0 \longrightarrow \mathrm{H}_0(f_n, \mathrm{H}_1(\mathbf{f}', \mathrm{M})) \longrightarrow \mathrm{H}_1(\mathbf{f}, \mathrm{M}) \longrightarrow \mathrm{H}_1(f_n, \mathrm{H}_0(\mathbf{f}', \mathrm{M})) \longrightarrow 0. \tag{19.5.1.1}$$

The relation $\mathrm{H}_1(\mathbf{f}, \mathrm{M}) = 0$ implies therefore $\mathrm{H}_0(f_n, \mathrm{H}_1(\mathbf{f}', \mathrm{M})) = 0$ and $\mathrm{H}_1(f_n, \mathrm{H}_0(\mathbf{f}', \mathrm{M})) = 0$. **IV-4** | 205
The first of these two relations means that we have $f_n\mathrm{H}_1(\mathbf{f}', \mathrm{M}) = \mathrm{H}_1(\mathbf{f}', \mathrm{M})$ (**III**, 1.1.3.5). Now, by definition (**III**, 1.1.1), $\mathrm{H}_1(\mathbf{f}', \mathrm{M})$ is isomorphic to a quotient of a submodule N of M^{n-1}; taking on M^{n-1} the filtration of the M^j for $j \leqslant n-1$, on N the induced filtration and on $\mathrm{H}_1(\mathbf{f}', \mathrm{M})$ the quotient filtration from that of N, we obtain on $\mathrm{H}_1(\mathbf{f}', \mathrm{M})$ a finite filtration whose quotients are isomorphic to quotients of submodules of M, hence are separated for the $\mathfrak{J}$-preadic topology by hypothesis. As the relation $f_n\mathrm{H}_1(\mathbf{f}', \mathrm{M}) = \mathrm{H}_1(\mathbf{f}', \mathrm{M})$ implies *a fortiori* $\mathfrak{J}\mathrm{H}_1(\mathbf{f}', \mathrm{M}) = \mathrm{H}_1(\mathbf{f}', \mathrm{M})$, we deduce from the hypothesis and the preceding remarks that $\mathrm{H}_1(\mathbf{f}', \mathrm{M}) = 0$. The induction hypothesis proves then that the sequence $\mathbf{f}'$ is M-regular. On the other hand, the relation $\mathrm{H}_1(f_n, \mathrm{H}_0(\mathbf{f}', \mathrm{M})) = 0$ means that f_n is $(\mathrm{M}/(\sum_{i=1}^{n-1} f_i\mathrm{M}))$-regular, hence the sequence **f** is M-regular. □

Corollary (19.5.2). — *Suppose that* A *is a Noetherian ring, that the* f_i $(1 \leqslant i \leqslant n)$ *belong to the radical of* A, *and that* M *is a finitely generated* A*-module. Then the four conditions* a), b), b'), c) *of* ((19.5.1)) *are equivalent.*

Proof. We know indeed from $(\mathbf{0}_\mathrm{I}, 7.3.5)$ that every finitely generated A-module is separated for the $\mathfrak{J}$-preadic topology. □

Proposition (19.5.3). — *Let*

$$0 \longrightarrow \mathrm{M}' \longrightarrow \mathrm{M} \longrightarrow \mathrm{M}'' \longrightarrow 0$$

be an exact sequence of A*-modules,* $\mathbf{f} = (f_i)_{1\leqslant i\leqslant n}$ *an M-regular sequence,* $\mathfrak{J} = \sum_{i=1}^{n} f_i\mathrm{A}$. *Consider the following properties:*

a) *The sequence* **f** *is* M''*-regular.*
b) *The* $\mathfrak{J}$*-preadic filtration of* M' *is induced on* M' *by the* $\mathfrak{J}$*-preadic filtration of* M *(in other words, the canonical homomorphism* $\mathrm{gr}^*_\mathfrak{J}(\mathrm{M}') \to \mathrm{gr}^*_\mathfrak{J}(\mathrm{M})$ *is injective).*
c) *The canonical homomorphism* $\mathrm{M}'/(\sum_{i=1}^{n} f_i\mathrm{M}') \to \mathrm{M}/(\sum_{i=1}^{n} f_i\mathrm{M})$ *is injective.*

Then we have the implications a) ⇒ b) ⇒ c), *and* a) *implies moreover that the sequence* **f** *is* M'*-regular. If every quotient of a submodule of* M'' *is separated for the* $\mathfrak{J}$*-preadic topology, the conditions* a), b) *and* c) *are equivalent.*

Proof. It is clear that $c)$ is a consequence of $b)$. Let us prove that under all the hypotheses of the last assertion, $c)$ implies $a)$: since $\mathbf{f}$ is M-regular, by (19.5.1), $\mathrm{H}_1(\mathbf{f}, \mathrm{M}) = 0$, whence an exact sequence, portion of the exact homology sequence

$$0 \longrightarrow \mathrm{H}_1(\mathbf{f}, \mathrm{M}'') \longrightarrow \mathrm{H}_0(\mathbf{f}, \mathrm{M}') \longrightarrow \mathrm{H}_0(\mathbf{f}, \mathrm{M}).$$

Now, condition $c)$ expresses that the homomorphism $\mathrm{H}_0(\mathbf{f}, \mathrm{M}') \to \mathrm{H}_0(\mathbf{f}, \mathrm{M})$ is injective (**III**, 1.1.3.5); it is therefore equivalent to $\mathrm{H}_1(\mathbf{f}, \mathrm{M}'') = 0$, whence, by virtue of the separation hypothesis and (19.5.1), the fact that $\mathbf{f}$ is M″-regular.

Let us show next that $a)$ implies $c)$, and also that the sequence $\mathbf{f}$ is M′-regular, proceeding by induction on n. For $n = 1$, f_1 being M-regular is also M′-regular and property $c)$ is precisely the lemma ((3.4.1.4)). For $n > 1$, the induction hypothesis shows that if we set $\mathbf{f}' = (f_1, \ldots, f_{n-1})$, the sequence $\mathbf{f}'$ is M′-regular and, by virtue of $c)$, we have an exact sequence **IV-4** | 206

$$0 \longrightarrow \mathrm{M}'/(\textstyle\sum_{i=1}^{n-1} f_i\mathrm{M}') \to \mathrm{M}/(\sum_{i=1}^{n-1} f_i\mathrm{M}) \to \mathrm{M}''/(\sum_{i=1}^{n-1} f_i\mathrm{M}'') \to 0.$$

By hypothesis, f_n is $(\mathrm{M}/(\sum_{i=1}^{n-1} f_i\mathrm{M}))$-regular and $(\mathrm{M}''/(\sum_{i=1}^{n-1} f_i\mathrm{M}''))$-regular. The same reasoning shows, on the one hand, that f_n is also $(\mathrm{M}'/(\sum_{i=1}^{n-1} f_i\mathrm{M}'))$-regular and, on the other hand, by ((3.4.1.4)), that the sequence

$$0 \longrightarrow \mathrm{M}'/(\textstyle\sum_{i=1}^{n} f_i\mathrm{M}') \to \mathrm{M}/(\sum_{i=1}^{n} f_i\mathrm{M}) \to \mathrm{M}''/(\sum_{i=1}^{n} f_i\mathrm{M}'') \to 0$$

is exact, whence $c)$.

Let us show finally that $a)$ implies $b)$. As the sequence $\mathbf{f}$ is then M-regular, M-quasi-regular, and M′-quasi-regular, we have canonical isomorphisms

$$\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}') \xrightarrow{\sim} \mathrm{gr}^0_{\mathfrak{J}}(\mathrm{M}')[\mathrm{T}_1, \ldots, \mathrm{T}_n], \qquad \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}) \xrightarrow{\sim} \mathrm{gr}^0_{\mathfrak{J}}(\mathrm{M})[\mathrm{T}_1, \ldots, \mathrm{T}_n]$$

$(\mathbf{0}, 15.1.7)$, and as we saw above that $\mathrm{gr}^0_{\mathfrak{J}}(\mathrm{M}') \to \mathrm{gr}^0_{\mathfrak{J}}(\mathrm{M})$ is injective, the same holds for $\mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M}') \to \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{M})$.

We note moreover that the equivalence of conditions $a)$, $b)$, $c)$ of (19.5.3) holds in particular when A is Noetherian, M″ is a finitely generated A-module, and the f_i belong to the radical of A. ☐

(19.5.4). In the rest of this number, we preserve the notations A, $\mathbf{f}$, M, $\mathfrak{J}$, and we suppose moreover M endowed with a decreasing filtration (M_k) formed of sub-A-modules, such that $\mathrm{M}_0 = \mathrm{M}$. Recall $(\mathbf{0}, 15.1.5)$ that we then define on M a second decreasing filtration formed of submodules

$$\mathrm{M}'_k = \mathrm{M}_k + \mathfrak{J}\mathrm{M}_{k-1} + \ldots + \mathfrak{J}^{k-1}\mathrm{M}_1 + \mathfrak{J}^k\mathrm{M}_0$$

and that, if $\mathrm{gr}_\bullet(\mathrm{M})$ and $\mathrm{gr}'_\bullet(\mathrm{M})$ are the graded A-modules associated to the filtrations (M_k) and (M'_k) respectively, we define a surjective graded homomorphism of degree 0 $(\mathbf{0}, 15.1.5.2)$

$$\psi_{\mathbf{M}} : (\mathrm{gr}_\bullet(\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}'_\bullet(\mathrm{M})$$

Recall also that we have the canonical surjective homomorphism $(\mathbf{0}, 15.1.1.1)$

$$\varphi_{\mathrm{gr}_\bullet(\mathrm{M})} : (\mathrm{gr}_\bullet(\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}^*_{\mathfrak{J}}(\mathrm{gr}_\bullet(\mathrm{M}))$$

Theorem (19.5.5). — *With the notations of ((19.5.4)), consider the following properties:*

a) *The sequence* $\mathbf{f}$ *is* $\mathrm{gr}_\bullet(\mathrm{M})$*-regular.*
b) *The canonical homomorphism* $\psi_{\mathbf{M}}$ *is bijective.*
c) *The canonical homomorphism* $\varphi_{\mathrm{gr}_\bullet(\mathrm{M})}$ *is bijective (in other words* $(\mathbf{0}, 15.1.7)$*, the sequence* $\mathbf{f}$ *is* $\mathrm{gr}_\bullet(\mathrm{M})$*-quasi-regular).*

Then we have the implications[4] **IV-4** | 207

$$a) \Rightarrow b) \Rightarrow c).$$

Moreover, if for every integer $k \geqslant 0$*, the quotient modules of submodules of* $\mathrm{gr}_k(\mathrm{M})$ *are separated for the* $\mathfrak{J}$*-preadic topology (which is the case when* A *is Noetherian, the* f_i *belong to the radical of* A*, and the* $\mathrm{gr}_k(\mathrm{M})$ *are finitely generated* A*-modules),* then all conditions a), b) *and* c) are equivalent.

[4]The implication $b) \Rightarrow c)$, without the separation hypothesis on the $\mathrm{gr}_k(\mathrm{M})$, is due to P. Deligne, whose demonstration we reproduce below.

Proof. The implication $a) \Rightarrow b)$ has been demonstrated in $(\mathbf{0}, 15.1.8)$; the implication $c) \Rightarrow a)$ under the separation hypothesis is a particular case of $(\mathbf{0}, 15.1.9)$. It remains to prove that $b)$ implies $c)$, which will be done in several steps.

We denote by $\mathfrak{J}$ the ideal $\sum_{i=1}^{n} f_i \mathrm{A}$ generated by the f_i; for every sequence $\mathbf{q} = (q_1, \ldots, q_n)$ of n integers $\geqslant 0$, we set $|\mathbf{q}| = \sum_{i=1}^{n} q_i$ and $\mathbf{f}^{\mathbf{q}} = f_1^{q_1} f_2^{q_2} \cdots f_n^{q_n}$.

(19.5.5.1). For the map $\psi_{\mathbf{M}}$ to be injective (and hence bijective), it is necessary and sufficient that the following condition be satisfied:

$(\mathrm{I}_{q,p})$ *For every family* $(x_{\mathbf{q}}^{p})_{|\mathbf{q}|=q}$ *of elements of* M_p*, the relation*

$$\textstyle\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x_{\mathbf{q}}^{p} \in \sum_{i=1}^{q} \mathfrak{J}^{q-i} \mathrm{M}_{p+i} + \mathrm{M}'_{p+q+1}$$

implies $x_{\mathbf{q}}^{p} \in \mathrm{M}_{p+1} + \mathfrak{J}\mathrm{M}_p$ *for every* $\mathbf{q}$ *such that* $|\mathbf{q}| = q$.

We proceed as in $(\mathbf{0}, 15.1.6)$ by considering the sub-A-module Q_k (resp. Q'_k) as a term of degree k in the first (resp. second) member of $(\mathbf{0}, 15.1.5.2)$. We endow Q_k with the filtration

$$(\mathrm{Q}_k)_i = \textstyle\sum_{j \leqslant k-i} \big(\sum_{|\mathbf{l}|=j} (\mathrm{gr}_{k-j}(\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}))\big) \mathrm{T}^{\mathbf{l}}$$

and Q'_k the image filtration formed by $\psi_{\mathbf{M}}((\mathrm{Q}_k)_i) = (\mathrm{Q}'_k)_i$; it suffices then to prove that the homomorphisms $(\psi_{\mathbf{M}})_{ki} : \mathrm{gr}_i(\mathrm{Q}_k) \to \mathrm{gr}_i(\mathrm{Q}'_k)$ are injective. Now, we have

$$\mathrm{gr}_i(\mathrm{Q}_k) = \textstyle\sum_{|\mathbf{l}|=k-i} \big((\mathrm{M}_i/\mathrm{M}_{i+1}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J})\big) \mathrm{T}^{\mathbf{l}} = \sum_{|\mathbf{l}|=k-i} \big(\mathrm{M}_i/(\mathfrak{J}\mathrm{M}_i + \mathrm{M}_{i+1})\big) \mathrm{T}^{\mathbf{l}}$$

on the other hand, $(\mathrm{Q}'_k)_{i+1}$ is the image of $\sum_{h=0}^{k-i-1} \mathfrak{J}^{k-i-h-1} \mathrm{M}_{i+h+1}$ in $\mathrm{M}'_k/\mathrm{M}'_{k+1}$. To write that **IV-4** | 208
$(\psi_{\mathbf{M}})_{ki}$ is injective is therefore equivalent to writing the condition $(\mathrm{I}_{k-i,i})$; whence our assertion.

(19.5.5.2). *For the sequence* $\mathbf{f}$ *to be* $\mathrm{gr}_\bullet(\mathrm{M})$*-quasi-regular, it suffices that, for* $p \geqslant 0$ *and* $q \geqslant 0$*, the following condition be satisfied:*

$(\mathrm{S}_{q,p})$ *For every family* $(x_{\mathbf{q}}^{p})_{|\mathbf{q}|=q}$ *of elements of* M_p*, the relation*

$$\textstyle\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x_{\mathbf{q}}^{p} \in \mathrm{M}_{p+1}$$

implies $x_{\mathbf{q}}^{p} \in \mathrm{M}_{p+1} + \mathfrak{J}\mathrm{M}_p$ *for every* $\mathbf{q}$ *such that* $|\mathbf{q}| = q$.

By definition, saying that $\mathbf{f}$ is $\mathrm{gr}_\bullet(\mathrm{M})$-quasi-regular means that, for every $p \geqslant 0$ and every $q \geqslant 0$, the relation

$$\textstyle\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x_{\mathbf{q}}^{p} \in \mathrm{M}_{p+1} + \mathfrak{J}^{q+1}\mathrm{M}_p$$

for a family $(x_{\mathbf{q}}^{p})_{|\mathbf{q}|=q}$ of elements of M_p implies $x_{\mathbf{q}}^{p} \in \mathrm{M}_{p+1} + \mathfrak{J}\mathrm{M}_p$ for every $\mathbf{q}$ such that $|\mathbf{q}| = q$. The condition $(\mathrm{S}_{q,p})$ means that there exists a family $(y_{\mathbf{q}}^{p})_{|\mathbf{q}|=q}$ of elements of $\mathfrak{J}\mathrm{M}_p$ such that $\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} (x_{\mathbf{q}}^{p} - y_{\mathbf{q}}^{p}) \in \mathrm{M}_{p+1}$; it then implies $x_{\mathbf{q}}^{p} - y_{\mathbf{q}}^{p} \in \mathrm{M}_{p+1} + \mathfrak{J}\mathrm{M}_p$, whence $x_{\mathbf{q}}^{p} \in \mathrm{M}_{p+1} + \mathfrak{J}\mathrm{M}_p$; hence one sees that the conditions $(\mathrm{S}_{q,p})$ imply that the sequence $\mathbf{f}$ is $\mathrm{gr}_\bullet(\mathrm{M})$-quasi-regular.

It suffices therefore to prove the following proposition:

(19.5.5.3). *For every* $r \geqslant 0$*, if the conditions* $(\mathrm{I}_{q,p})$ *are satisfied for* $p + q < r$*, then the conditions* $(\mathrm{S}_{q,p})$ *are satisfied for* $p + q < r$*.*

Let us introduce the conditions

$(\mathrm{A}_{q,p})$ *For every family* $(x_{\mathbf{q}}^{p})_{|\mathbf{q}|=q}$ *of elements of* M_p*, the relation*

$$\textstyle\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x_{\mathbf{q}}^{p} \in \mathrm{M}_{p+1}$$

implies

$$\textstyle\sum_{|\mathbf{q}|=q} \mathbf{f}^{\mathbf{q}} x_{\mathbf{q}}^{p} \in \sum_{i=1}^{q} \mathfrak{J}^{q-i} \mathrm{M}_{p+i}.$$

It is clear that the conjunction of $(\mathrm{A}_{q,p})$ and $(\mathrm{I}_{q,p})$ implies $(\mathrm{S}_{q,p})$. On the other hand, the conditions $(\mathrm{A}_{1,p})$ and $(\mathrm{S}_{0,p})$ are trivial. Consequently, (19.5.5.3) will follow from the following proposition:

(19.5.5.4). *For all* $p \geqslant 0$ *and* $q \geqslant 1$*, the condition* $(\mathrm{A}_{q+1,p})$ *is implied by the conjunction of the conditions* $(\mathrm{A}_{q,p})$*,* $(\mathrm{I}_{q-1,p+1})$*,* $(\mathrm{I}_{q-2,p+2})$*, …,* $(\mathrm{I}_{0,p+q})$*.*

Indeed, once this proposition is proved, the conditions $(\mathrm{I}_{q,p})$ supposed satisfied for $p + q < r$ will imply, for each $p \leqslant r$, the condition $(\mathrm{A}_{q,p})$ for $1 \leqslant q < r - p$, by induction on q.

Let us prove (19.5.5.4). Let us note that the condition $(\mathrm{A}_{q,p})$ is also equivalent to

(19.5.5.5).

$$\mathfrak{J}^q\mathrm{M}_p \cap \mathrm{M}_{p+1} \subset \sum_{i=1}^q \mathfrak{J}^{q-i}\mathrm{M}_{p+i}.$$

Let therefore $m \in \mathfrak{J}^{q+1}\mathrm{M}_p \cap \mathrm{M}_{p+1} \subset \mathfrak{J}^q\mathrm{M}_p \cap \mathrm{M}_{p+1}$. Applying condition $(\mathrm{A}_{q,p})$, we see that for each i such that $1 \leqslant i \leqslant q$, there exists a family $(y_{\mathbf{q}}^{p+i})_{|\mathbf{q}|=q-i}$ of elements of M_{p+i} such that

$$m = \sum_{i=1}^q \big(\sum_{|\mathbf{q}|=q-i} \mathbf{f}^{\mathbf{q}} y_{\mathbf{q}}^{p+i}\big).$$

Suppose that for $1 \leqslant i < j$ ($j \geqslant 1$), we have proved that

$$y_{\mathbf{q}}^{p+i} \in \mathfrak{J}\mathrm{M}_{p+i} + \mathrm{M}_{p+i+1} \qquad (\text{for } |\mathbf{q}| = q-i).$$

We deduce by definition (19.5.4)

$$m - \sum_{i<j}\big(\sum_{|\mathbf{q}|=q-i} \mathbf{f}^{\mathbf{q}} y_{\mathbf{q}}^{p+i}\big) \in \mathrm{M}'_{p+q+1},$$

Since by hypothesis $(\mathrm{I}_{q-j,p+j})$ is satisfied, we deduce

$$y_{\mathbf{q}}^{p+j} \in \mathfrak{J}\mathrm{M}_{p+j} + \mathrm{M}_{p+j+1}$$

and, by induction on j, one sees therefore that this condition is satisfied for *every* j such that $1 \leqslant j \leqslant q$. We have therefore **IV-4** | 209

$$m \in \sum_{i=1}^q \mathfrak{J}^{q-i}(\mathfrak{J}\mathrm{M}_{p+i} + \mathrm{M}_{p+i+1}) = \sum_{i=1}^{q+1} \mathfrak{J}^{q-i+1}\mathrm{M}_{p+i}$$

and we have thus proved $(\mathrm{A}_{q+1,p})$ and completed the demonstration of (19.5.5).

□

Remarks (19.5.6). —

(i) The results of this number remain valid when, instead of an A-module M and elements $f_i \in \mathrm{A}$, one takes an object M of any abelian category, and n endomorphisms f_i of M that mutually commute; the demonstrations adapt without difficulty to this more general case. The hypothesis of separation for an A-module N relative to the $\mathfrak{J}$-preadic topology must here be replaced by the hypothesis that every sub-object of N contained in all the $\sum f_i^r(\mathrm{N})$ (r an arbitrary integer) is necessarily zero. The same remark applies to the results of (19.7).

(ii) Suppose that A is a *graded* ring in degrees $\geqslant 0$. If M (resp. each $\mathrm{gr}_p(\mathrm{M})$) is a graded A-module with degrees bounded below, and if the f_i contain no homogeneous component of degree 0, the separation hypothesis of $(\mathbf{0}, 15.1.9)$ is *ipso facto* satisfied for M (resp. each $\mathrm{gr}_p(\mathrm{M})$), and one sees that in this case we have the implication $c) \Rightarrow a)$ in (19.5.1) (resp. (19.5.5)).

(iii) Recall that, without the separation hypothesis, the implication $c) \Rightarrow b)$ is no longer valid even for $n = 1$ $(\mathbf{0}, 15.1.12, (\text{iii}))$.

19.6. Regular sequences relative to a filtered quotient module

(19.6.1). Let A be a ring, $\mathbf{f} = (f_1, \ldots, f_n)$ a finite sequence of elements of A, $\mathfrak{J} = f_1\mathrm{A} + \ldots + f_n\mathrm{A}$ the ideal it generates, and consider an exact sequence of A-modules

$$0 \longrightarrow \mathrm{R} \xrightarrow{j} \mathrm{N} \xrightarrow{p} \mathrm{M} \longrightarrow 0$$

where one supposes that N is endowed with a decreasing filtration (N_k) of sub-A-modules, with $\mathrm{N}_0 = \mathrm{N}$. We set $\mathrm{R}_k = \mathrm{R} \cap \mathrm{N}_k$ (filtration induced by (N_k)), $\mathrm{M}_k = p(\mathrm{N}_k)$ (quotient filtration of (N_k)) and we denote by $\mathrm{gr}_\bullet(\mathrm{N})$, $\mathrm{gr}_\bullet(\mathrm{R})$ and $\mathrm{gr}_\bullet(\mathrm{M})$ the graded A-modules associated with these three filtrations. We set on the other hand for all k,

$$\mathrm{N}'_k = \mathrm{N}_k + \mathfrak{J}\mathrm{N}_{k-1} + \ldots + \mathfrak{J}^{k-1}\mathrm{N}_1 + \mathfrak{J}^k\mathrm{N}_0$$

so that $\mathrm{M}'_k = p(\mathrm{N}'_k) = \mathrm{M}_k + \mathfrak{J}\mathrm{M}_{k-1} + \ldots + \mathfrak{J}^{k-1}\mathrm{M}_1 + \mathfrak{J}^k\mathrm{M}_0$; we set on the other hand $\mathrm{R}'_k = \mathrm{R} \cap \mathrm{N}'_k$ (filtration *induced* by (N'_k)), and we denote by $\mathrm{gr}'_\bullet(\mathrm{N})$, $\mathrm{gr}'_\bullet(\mathrm{M})$ and $\mathrm{gr}'_\bullet(\mathrm{R})$ the graded A-modules associated with these three filtrations; we thus have a commutative diagram of exact sequences

(19.6.1.1)
$$\begin{tikzcd} 0 \arrow[r] & \mathrm{gr}_\bullet(\mathrm{R}) \arrow[r, "\mathrm{gr}(j)"] \arrow[d] & \mathrm{gr}_\bullet(\mathrm{N}) \arrow[r, "\mathrm{gr}(p)"] \arrow[d] & \mathrm{gr}_\bullet(\mathrm{M}) \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & \mathrm{gr}'_\bullet(\mathrm{R}) \arrow[r, "\mathrm{gr}'(j)"] & \mathrm{gr}'_\bullet(\mathrm{N}) \arrow[r, "\mathrm{gr}'(p)"] & \mathrm{gr}'_\bullet(\mathrm{M}) \arrow[r] & 0 \end{tikzcd}$$

where the vertical arrows are the canonical homomorphisms from a graded module associated with **IV-4** | 210
a filtration to a graded module associated with a coarser filtration.

We note that if one sets

$$\mathrm{R}''_k = \mathrm{R}_k + \mathfrak{J}\mathrm{R}_{k-1} + \ldots + \mathfrak{J}^{k-1}\mathrm{R}_1 + \mathfrak{J}^k\mathrm{R}_0$$

we evidently have $\mathrm{R}''_k \subset \mathrm{R}'_k$, but the filtrations (R''_k) and (R'_k) are in general distinct; we denote by $\mathrm{gr}''_\bullet(\mathrm{R})$ the graded A-module associated with the filtration (R''_k).

(19.6.2). We defined in $(\mathbf{0}, 15.1.5.2)$ the surjective graded homomorphisms of degree 0

$$\psi_\mathrm{N} : (\mathrm{gr}_\bullet(\mathrm{N}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}'_\bullet(\mathrm{N})$$
$$\psi_\mathrm{M} : (\mathrm{gr}_\bullet(\mathrm{M}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}'_\bullet(\mathrm{M})$$
$$\psi_\mathrm{R} : (\mathrm{gr}_\bullet(\mathrm{R}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}''_\bullet(\mathrm{R})$$

the first two of which are deduced from the last two vertical arrows of (19.6.1.1).

As the filtration (R''_k) is finer than (R'_k), on R we have a canonical homomorphism $\gamma : \mathrm{gr}''_\bullet(\mathrm{R}) \to \mathrm{gr}'_\bullet(\mathrm{R})$; we will denote by ψ'_R the composite homomorphism $\gamma \circ \psi_\mathrm{R}$. It then follows immediately from the definitions that we have a commutative diagram

(19.6.2.1)
$$\begin{tikzcd} & 0 \arrow[d] \\ (\mathrm{gr}_\bullet(\mathrm{R}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \arrow[r, "\psi'_\mathrm{R}"] \arrow[d, "j'"] & \mathrm{gr}'_\bullet(\mathrm{R}) \arrow[d, "\mathrm{gr}'(j)"] \\ (\mathrm{gr}_\bullet(\mathrm{N}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \arrow[r, "\psi_\mathrm{N}"] \arrow[d, "p'"] & \mathrm{gr}'_\bullet(\mathrm{N}) \arrow[d, "\mathrm{gr}'(p)"] \\ (\mathrm{gr}_\bullet(\mathrm{M}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \arrow[r, "\psi_\mathrm{M}"] \arrow[d] & \mathrm{gr}'_\bullet(\mathrm{M}) \arrow[d] \\ 0 & 0 \end{tikzcd}$$

where the columns are exact.

Proposition (19.6.3). — *With the preceding notations, suppose that the sequence* $\mathbf{f}$ *is* $\mathrm{gr}_\bullet(\mathrm{N})$*-regular. Consider the following properties:*

a) $\mathbf{f}$ *is* $\mathrm{gr}_\bullet(\mathrm{M})$*-regular.*
b) ψ_M *is bijective.*
c) ψ'_R *is surjective (in other words* $\mathrm{gr}'_\bullet(\mathrm{R})$ *is generated as* $\mathrm{gr}^*_\mathfrak{J}(\mathrm{A})$*-module by* $\mathrm{Im}(\mathrm{gr}_\bullet(\mathrm{R}) \to \mathrm{gr}'_\bullet(\mathrm{R}))$*).*
d) ψ'_R *is injective.*
d′) *The homomorphism* $(\psi'_\mathrm{R})_0 : \mathrm{gr}_\bullet(\mathrm{R}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}) \to \mathrm{gr}'_\bullet(\mathrm{R})$ *obtained by restricting* ψ'_R *to polynomials of degree* 0*, is injective.*
e) ψ'_R *is bijective.*
f) j' *is injective.* **IV-4** | 211
f′) *The homomorphism* $\mathrm{gr}_\bullet(\mathrm{R}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}) \to \mathrm{gr}_\bullet(\mathrm{N}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J})$ *obtained by restricting* j' *to polynomials of degree* 0*, is injective.*
g) *The* $\mathfrak{J}$*-preadic filtration of* $\mathrm{gr}_\bullet(\mathrm{R})$ *is induced by that of* $\mathrm{gr}_\bullet(\mathrm{N})$*.*
h) *The filtrations* (R'_k) *and* (R''_k) *on* R *are identical.*

Then we have the following implications:

$$\begin{tikzcd}[column sep=small]
a) \arrow[r, Rightarrow] \arrow[dr, Rightarrow] & e) \arrow[r, Rightarrow] \arrow[dr, Rightarrow] & b) \arrow[r, Leftrightarrow] & c) \arrow[r, Leftrightarrow] & h) & \\
& g) \arrow[r, Rightarrow] & d) \arrow[r, Leftrightarrow] & d') \arrow[r, Leftrightarrow] & f) \arrow[r, Leftrightarrow] & f')
\end{tikzcd}$$

Moreover, when every quotient module of a submodule of $\mathrm{gr}_k(\mathrm{M})$ *is separated for the* $\mathfrak{J}$*-preadic topology (which is the case when* A *is Noetherian, the* f_i *belong to the radical of* A*, and the* $\mathrm{gr}_k(\mathrm{M})$ *are finitely generated* A*-modules),* then all conditions a) *to* h) are equivalent*.*

Proof. Let us note that, from (19.5.5), the hypothesis that $\mathbf{f}$ is $\mathrm{gr}_\bullet(\mathrm{N})$-regular implies that ψ_N is *bijective*; as $\mathrm{gr}'(j)$ is injective, the equivalence of $d)$ and $f)$ results from the diagram (19.6.2.1). Likewise $d')$ and $f')$ are equivalent, while $f)$ and $f')$ are trivially equivalent; this proves the equivalence of $d)$, $d')$, $f)$ and $f')$.

As one already knows that ψ_M is surjective and ψ_N bijective, the equivalence of $b)$ and $c)$ also follows from the diagram (19.6.2.1) by a particular case of the five lemma.

The relation $\psi'_\mathrm{R} = \gamma \circ \psi_\mathrm{R}$ (19.6.2) and the fact that ψ_R is surjective show that condition $c)$ is equivalent to saying that γ is surjective, which is also equivalent to condition $h)$; hence we have proved the equivalence of $b)$, $c)$ and $h)$.

It is trivial that $e)$ is equivalent to the conjunction of $c)$ and $d)$, and hence also of $b)$ and $d)$. Let us note now that $a)$ implies $b)$ and that $b)$ implies $a)$ under the separation hypothesis (19.5.5). On the other hand, the implications $a) \Rightarrow g) \Rightarrow f')$, and the implication $f') \Rightarrow a)$ under the separation hypothesis, result from (19.5.3). Hence we have proved $a)$ implies $e)$ (equivalent to the conjunction of $b)$ and $f')$), and also that all conditions are equivalent under the separation hypothesis. □

Corollary (19.6.4). — *Let* $\mathfrak{K}$ *be an ideal of* A, $\mathfrak{L} = \mathfrak{J} + \mathfrak{K}$*. Suppose that the filtration (hence also* (M_k)*) is the* $\mathfrak{K}$*-preadic filtration, so that the filtrations* (M'_k) *and* (N'_k) *are* $\mathfrak{L}$*-preadic filtrations. Consider* $\mathrm{gr}_\bullet(\mathrm{R})$ *(resp.* $\mathrm{gr}'_\bullet(\mathrm{R})$*) as a graded module over* $\mathrm{gr}^*_{\mathfrak{K}}(\mathrm{A})$ *(resp.* $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$*).*

- (i) *Let* S *be a part of* $\mathrm{gr}_\bullet(\mathrm{R})$ *that generates* $\mathrm{gr}_\bullet(\mathrm{R})$ *as a* $\mathrm{gr}^*_{\mathfrak{K}}(\mathrm{A})$*-module. Then condition* c) *of* ((19.6.3)) *is equivalent to the following condition:*
- c′) *The image of* S *in* $\mathrm{gr}'_\bullet(\mathrm{R})$ *by* ψ'_R *generates* $\mathrm{gr}'_\bullet(\mathrm{R})$ *as a* $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$*-module.*
- (ii) *Suppose condition* e) *of* ((19.6.3)) *is satisfied, and also that the quotient modules of submodules of* $\mathrm{gr}_k(\mathrm{R})$ *are separated for the* $\mathfrak{J}$*-preadic topology (which is the case when* A *is Noetherian, the* f_i *belong to the radical of* A*, and the* $\mathrm{gr}_k(\mathrm{R})$ *are finitely generated* A*-modules). Then,* for a part S of $\mathrm{gr}_\bullet(\mathrm{R})$ formed of homogeneous elements to generate $\mathrm{gr}_\bullet(\mathrm{R})$ as a $\mathrm{gr}^*_{\mathfrak{K}}(\mathrm{A})$-module, it is necessary and sufficient that its image by ψ'_R generates $\mathrm{gr}'_\bullet(\mathrm{R})$ as a $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$-module.

Proof. **IV-4** | 212

- (i) Consider the homomorphism of degree 0

$$\psi_\mathrm{A} : (\mathrm{gr}^*_{\mathfrak{K}}(\mathrm{A}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}'_\bullet(\mathrm{A}) = \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$$

  ($\mathbf{0}$, 15.1.5.2). The fact that this homomorphism is *surjective* implies that the sub-$\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$-module of $\mathrm{gr}'_\bullet(\mathrm{R})$ generated by $\mathrm{S}' = \psi'_\mathrm{R}(\mathrm{S})$ is none other than $\mathrm{Im}(\psi'_\mathrm{R})$, whence the equivalence of $c)$ and $c')$.
- (ii) Since condition $e)$ is satisfied, it follows by (i) that it remains to prove the sufficiency of the condition in the statement. Now, as ψ'_R is bijective, to say that S' generates $\mathrm{gr}'_\bullet(\mathrm{R})$ considered as a $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$-module amounts to saying that S generates $(\mathrm{gr}_\bullet(\mathrm{R}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ as a $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$-module, and by the surjectivity of ψ_A, we can in this statement replace the ring $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{A})$ by $(\mathrm{gr}^*_{\mathfrak{K}}(\mathrm{A}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J}))[\mathrm{T}_1, \ldots, \mathrm{T}_n]$. It amounts to saying that S generates $\mathrm{gr}_\bullet(\mathrm{R}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J})$ as a module over the ring $\mathrm{gr}^*_{\mathfrak{K}}(\mathrm{A}) \otimes_\mathrm{A} (\mathrm{A}/\mathfrak{J})$, or also as a $\mathrm{gr}^*_{\mathfrak{K}}(\mathrm{A})$-module. Now, if T is the graded sub-$\mathrm{gr}^*_{\mathfrak{K}}(\mathrm{A})$-module of $\mathrm{gr}_\bullet(\mathrm{R})$ generated by S, and if we set $\mathrm{T}' = \mathrm{gr}_\bullet(\mathrm{R})/\mathrm{T}$, then by hypothesis $\mathrm{T}'/\mathfrak{J}\mathrm{T}' = 0$, or $\mathfrak{J}\mathrm{T}' = \mathrm{T}'$; but T' is a graded module, and each T'_p is a quotient of a submodule of $\mathrm{gr}_p(\mathrm{R})$, hence separated for the $\mathfrak{J}$-preadic topology. The hypothesis therefore implies $\mathrm{T}' = 0$, which completes the proof of the corollary.

□

19.7. Hironaka's normal flatness criterion

Theorem (19.7.1). — (Hironaka). *Let* A *be a ring,* $\mathfrak{R}$ *an ideal of* A, $\mathbf{f} = (f_1, \ldots, f_n)$ *a sequence of elements of* A *that is* $(\mathrm{A}/\mathfrak{R})$*-regular,* $\mathfrak{J} = f_1\mathrm{A} + \ldots + f_n\mathrm{A}$ *the ideal generated by* $\mathbf{f}$, $\mathfrak{L} = \mathfrak{J} + \mathfrak{R}$, M *an* A*-module. In this case* $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M})) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}) = (\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M})) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{L}) = (\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M})) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{L})$*, since* $(\mathfrak{R}^n\mathrm{M}/\mathfrak{R}^{n+1}\mathrm{M}) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{J}) = \mathfrak{R}^n\mathrm{M}/(\mathfrak{R}^{n+1} + \mathfrak{J}\mathfrak{R}^n)\mathrm{M} = \mathfrak{R}^n\mathrm{M}/(\mathfrak{R}^{n+1} + \mathfrak{L}\mathfrak{R}^n)\mathrm{M}$*, and the last equality is a result of Bourbaki,* Alg.*, chap. II,* 3^e *ed., § 3, n° 7, cor. 2 of prop. 6.*

Consider the following conditions:

a) $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M})$ *is a flat* $(\mathrm{A}/\mathfrak{R})$*-module.*
b) $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$ *is a flat* $(\mathrm{A}/\mathfrak{L})$*-module and the canonical homomorphism* $(\mathbf{0}, 15.1.5.2)$

$$\psi_{\mathrm{M}} : ((\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M})) \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \ldots, \mathrm{T}_n] \longrightarrow \mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$$

is bijective.
c) $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M})) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{L})$ *is a flat* $(\mathrm{A}/\mathfrak{L})$*-module and the sequence* $\mathbf{f}$ *is* $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M}))$*-regular.*
d) $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$ *is a flat* $(\mathrm{A}/\mathfrak{L})$*-module, and for every* $\mathfrak{p} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{R}}(\mathrm{A}/\mathfrak{L})$, $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M}))_{\mathfrak{p}}$ *is a flat* $(\mathrm{A}/\mathfrak{R})_{\mathfrak{p}}$*-module.*

Then we have the implications

$$\begin{tikzcd} a) \arrow[r, Rightarrow] \arrow[dr, Rightarrow] & c) \arrow[r, Rightarrow] & b) \\ & d) & \end{tikzcd}$$

When every quotient of a submodule of $\mathrm{gr}^k_{\mathfrak{R}}(\mathrm{M})$ $(k \geqslant 0)$ *is ideally separated for* $\mathfrak{J}$ *(Bourbaki,* Alg. comm.*, chap. III, § 5, n° 1), the conditions* a), b), c) *are equivalent.*

Finally, when A *is Noetherian, the* f_i *belong to the radical of* A*, the sequence* $\mathbf{f}$ *is* $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{A})$*-regular, and* M **IV-4** | 213
is a finitely generated A*-module, the conditions* a), b), c), d) *are equivalent.*

Proof. The implication $a) \Rightarrow c)$ is immediate $(\mathbf{0}, 15.1.13)$. If $c)$ is satisfied, it results from (19.5.5) that ψ_{M} is bijective; moreover, $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ is a flat $(\mathrm{A}/\mathfrak{L})$-module, and hence so is $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$, since ψ_{M} is an $(\mathrm{A}/\mathfrak{L})$-isomorphism; thus $c)$ implies $b)$. The fact that $a)$ also implies $d)$ is immediate.

When every quotient of a submodule of $\mathrm{gr}^k_{\mathfrak{R}}(\mathrm{M})$ is ideally separated for $\mathfrak{J}$, it results from (19.5.5) that condition $b)$ implies that $\mathbf{f}$ is $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M}))$-regular; moreover, $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \ldots, \mathrm{T}_n]$ is isomorphic to $\mathrm{gr}^*_{\mathfrak{L}}(\mathrm{M})$ and hence is flat over $\mathrm{A}/\mathfrak{L}$. Its degree-zero direct summand $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{L})$ is therefore flat; this proves that $b)$ implies $c)$. On the other hand, under the same hypothesis, $c)$ implies $a)$ by virtue of $(\mathbf{0}, 15.1.21)$ (where we can replace the Noetherian hypothesis by the hypothesis that M is ideally separated for $\mathfrak{J}$, by virtue of $(\mathbf{0}_{\mathrm{III}}, 10.2.1)$).

It remains therefore to prove that when A is Noetherian, M of finite type, and the f_i belong to the radical of A, $d)$ implies $a)$.

There exists a free A-module N of *finite type* and an exact sequence $0 \to \mathrm{R} \to \mathrm{N} \to \mathrm{M} \to 0$. It suffices to prove that $d)$ implies $b)$, since every quotient of a sub-$\mathrm{gr}^k_{\mathfrak{R}}(\mathrm{M})$ is then ideally separated for $\mathfrak{J}$, by virtue of the fact that $\mathrm{gr}^k_{\mathfrak{R}}(\mathrm{M})$ is a finitely generated A-module, that $\mathfrak{J}$ is contained in the radical of A, and that A is Noetherian (Bourbaki, *Alg. comm.*, chap. III, § 5, n° 1); in other words, it suffices to show that ψ_{M} is *bijective*.

As N is free and the sequence $\mathbf{f}$ is $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{A})$-regular by hypothesis, it is also $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{N})$-regular. One can therefore apply (19.6.3), since every quotient module of a sub-$\mathrm{gr}_k(\mathrm{M})$ is separated for the $\mathfrak{J}$-preadic topology, the f_i belonging to the radical of A by hypothesis. The diagram (19.6.2.1) here reads

(19.7.1.1)
$$\begin{tikzcd}
 & 0 \arrow[d] \\
(\mathrm{gr}^*_{\mathfrak{R}}(R,N) \otimes_{A/\mathfrak{R}} (A/\mathfrak{L}))[T_1,\dots,T_n] \arrow[r, "\psi'_R"] \arrow[d] & \mathrm{gr}^*_{\mathfrak{L}}(R,N) \arrow[d] \\
(\mathrm{gr}^*_{\mathfrak{R}}(N) \otimes_{A/\mathfrak{R}} (A/\mathfrak{L}))[T_1,\dots,T_n] \arrow[r, "\psi_N"] \arrow[d] & \mathrm{gr}^*_{\mathfrak{L}}(N) \arrow[d] \\
(\mathrm{gr}^*_{\mathfrak{R}}(M) \otimes_{A/\mathfrak{R}} (A/\mathfrak{L}))[T_1,\dots,T_n] \arrow[r, "\psi_M"] \arrow[d] & \mathrm{gr}^*_{\mathfrak{L}}(M) \arrow[d] \\
0 & 0
\end{tikzcd}$$

where $\mathrm{gr}^*_{\mathfrak{R}}(R,N)$ (resp. $\mathrm{gr}^*_{\mathfrak{L}}(R,N)$) is the graded associated to R for the filtration $R \cap \mathfrak{R}^m N$ (resp. $R \cap \mathfrak{L}^m N$); recall that in this diagram the columns are exact and that ψ_N is *bijective*. **IV-4** | 214

(19.7.1.2). Set $B = \mathrm{gr}^*_{\mathfrak{R}} A \otimes_{A/\mathfrak{R}} (A/\mathfrak{L})$, $P = \mathrm{gr}^*_{\mathfrak{R}}(N) \otimes_{A/\mathfrak{R}} (A/\mathfrak{L})$, $Q = \mathrm{gr}^*_{\mathfrak{L}}(R,N)$; since ψ_N is bijective, $P[T_1,\dots,T_n]$ identifies with $\mathrm{gr}^*_{\mathfrak{L}}(N)$, and Q is a sub-$B[T_1,\dots,T_n]$-module of $P[T_1,\dots,T_n]$; we will first show that we have

(19.7.1.3).
$$Q = (Q \cap P)[T_1,\dots,T_n].$$

For this, set $Z = Q/((Q\cap P)[T_1,\dots,T_n])$; this is a sub-$B[T_1,\dots,T_n]$-module of $P[T_1,\dots,T_n]/((Q\cap P)[T_1,\dots,T_n]) = (P/(Q\cap P))[T_1,\dots,T_n]$. As an $(A/\mathfrak{L})$-module, it is therefore isomorphic to a submodule of a direct sum of $(A/\mathfrak{L})$-modules isomorphic to $P/(Q\cap P) = (P+Q)/Q$. But $(P+Q)/Q$ is an $(A/\mathfrak{L})$-module isomorphic to a submodule of $P[T_1,\dots,T_n]/Q$, which is none other than $\mathrm{gr}^*_{\mathfrak{L}}(M)$ by the diagram (19.7.1.1).

But each $\mathrm{gr}^m_{\mathfrak{L}}(M)$ is by hypothesis a *flat* $(A/\mathfrak{L})$-module of finite type, hence projective, and we have the inclusion $\mathrm{Ass}_{A/\mathfrak{L}}(\mathrm{gr}^*_{\mathfrak{L}}(M)) \subset \mathrm{Ass}_{A/\mathfrak{L}}(A/\mathfrak{L})$. To see that $Z = 0$, it suffices by (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 1, cor. 1 of prop. 2 and n° 3, prop. 7) to see that for every $\mathfrak{q} \in \mathrm{Ass}_{A/\mathfrak{L}}(A/\mathfrak{L})$, $Z_{\mathfrak{q}} = 0$. But the ideals of $\mathrm{Ass}_{A/\mathfrak{L}}(A/\mathfrak{L})$ are of the form $\mathfrak{p}/(\mathfrak{L}/\mathfrak{R})$, where $\mathfrak{p} \in \mathrm{Ass}_{A/\mathfrak{R}}(A/\mathfrak{L})$, and it comes down to the same to see that $Z_{\mathfrak{p}} = 0$ for every $\mathfrak{p} \in \mathrm{Ass}_{A/\mathfrak{R}}(A/\mathfrak{L})$. Now, if $\mathfrak{p} = \mathfrak{r}/\mathfrak{R}$, where $\mathfrak{r}$ is a prime ideal of A, then $(A/\mathfrak{L})_{\mathfrak{p}} = A_{\mathfrak{r}}/\mathfrak{L}_{\mathfrak{r}}$, and the images of the f_i in $A_{\mathfrak{r}}$ form a $\mathrm{gr}^*_{\mathfrak{R}_{\mathfrak{r}}}(M_{\mathfrak{r}})$-regular sequence $(\mathbf{0}, 15.1.14)$; applying to $A_{\mathfrak{r}}$, $\mathfrak{R}_{\mathfrak{r}}$ and $M_{\mathfrak{r}}$ the implication *a*) ⇒ *b*) of the statement, we see from the hypothesis *d*) that $\psi_{M_{\mathfrak{r}}}$ is bijective, hence also $\psi'_{R_{\mathfrak{r}}}$ by virtue of (19.6.3); in other words, $Q_{\mathfrak{r}} = Q'_{\mathfrak{r}}[T_1,\dots,T_n]$, where we have set $Q' = \mathrm{gr}^*_{\mathfrak{R}}(R,N) \otimes_{A/\mathfrak{R}} (A/\mathfrak{L})$, which is here a sub-B-module of P; in particular $Q'_{\mathfrak{r}} \subset Q_{\mathfrak{r}} \cap P_{\mathfrak{r}}$; hence finally $Z_{\mathfrak{r}} = Z_{\mathfrak{p}} = 0$, which completes the demonstration of (19.7.1.3).

(19.7.1.5). By virtue of (19.7.1.3), to prove that ψ'_R is surjective, it suffices to show that every homogeneous element of degree m of $Q \cap P$ is the image of an element of the same degree of $Q' = \mathrm{gr}^*_{\mathfrak{R}}(R,N) \otimes_{A/\mathfrak{R}} (A/\mathfrak{L})$. Now, the elements of degree m of P identify (by ψ_N) with the elements of $(\mathfrak{R}^m N + \mathfrak{L}^{m+1}N)/\mathfrak{L}^{m+1}N$; those which belong to Q identify with the elements of $((R\cap\mathfrak{L}^m N) + \mathfrak{L}^{m+1}N)/\mathfrak{L}^{m+1}N$; and the images of the elements of degree m of Q' identify with the elements of $((R\cap\mathfrak{R}^m N) + \mathfrak{L}^{m+1}N)/\mathfrak{L}^{m+1}N$. It will therefore suffice to prove, for $m > 0$, the inclusion

$$((R\cap\mathfrak{L}^m N) + \mathfrak{L}^{m+1}N) \cap (\mathfrak{R}^m N + \mathfrak{L}^{m+1}N) \subset (R\cap\mathfrak{R}^m N) + \mathfrak{L}^{m+1}N.$$

As $\mathfrak{R} \subset \mathfrak{L}$, we see immediately that the first member of this relation is equal to $(R \cap (\mathfrak{R}^m N + \mathfrak{L}^{m+1}N)) + \mathfrak{L}^{m+1}N$, so that everything reduces to proving

(19.7.1.6).
$$R \cap (\mathfrak{R}^m N + \mathfrak{L}^{m+1}N) \subset (R\cap\mathfrak{R}^m N) + \mathfrak{L}^{m+1}N.$$

We will proceed by induction on m, assuming therefore (19.7.1.6) verified when m is replaced by an integer $m' < m$. We will consider on the other hand, for m fixed and for an integer $d \geqslant 0$, the relation **IV-4 | 215**

$$(*_d) \qquad \mathrm{R} \cap (\mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N}) \subset (\mathrm{R} \cap \mathfrak{R}^d\mathrm{N} \cap \mathfrak{L}^m\mathrm{N}) + \mathfrak{L}^{m+1}\mathrm{N}.$$

It is clear that it suffices to demonstrate $(*_d)$ for $d = m$, and on the other hand $(*_0)$ is trivially true. We will prove $(*_d)$ by *induction on d*; in other words, we suppose, for a $d \geqslant 0$ fixed (with $d < m$), that $(*_d)$ is true, and we wish to prove

$$(*_{d+1}) \qquad \mathrm{R} \cap (\mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N}) \subset (\mathrm{R} \cap \mathfrak{R}^{d+1}\mathrm{N} \cap \mathfrak{L}^m\mathrm{N}) + \mathfrak{L}^{m+1}\mathrm{N}.$$

Consider for this, for an $h \geqslant 0$, the relation

$$(**_{d+1,h}) \qquad \mathrm{R} \cap (\mathfrak{R}^m\mathrm{N} + \mathfrak{R}^d\mathfrak{L}^h\mathrm{N}) \subset (\mathrm{R} \cap (\mathfrak{R}^{d+1}\mathrm{N} + \mathfrak{R}^d\mathfrak{L}^h\mathrm{N}) \cap \mathfrak{L}^m\mathrm{N}) + \mathfrak{L}^{m+1}\mathrm{N}.$$

The hypothesis $(*_d)$ implies that $(**_{d+1,0})$ is true. We will prove by *induction on h* that $(**_{d+1,h})$ is true for every $h \geqslant 0$. But we have

Lemma (19.7.1.7). — *Let* A *be a Noetherian ring,* N *a finitely generated* A*-module,* $\mathfrak{L}$ *an ideal of* A, E, F *two submodules of* N*. For every $k > 0$ there exists $h > 0$ such that* $\mathrm{E} \cap (\mathrm{F} + \mathfrak{L}^h\mathrm{N}) \subset (\mathrm{E} \cap \mathrm{F}) + \mathfrak{L}^k\mathrm{N}$.

Proof. Indeed, let $\varphi : \mathrm{N} \to \mathrm{N}/(\mathrm{E} \cap \mathrm{F}) = \mathrm{N}_1$ be the canonical homomorphism, and set $\mathrm{E}_1 = \varphi(\mathrm{E})$, $\mathrm{F}_1 = \varphi(\mathrm{F})$, so that $\mathrm{E}_1 \cap \mathrm{F}_1 = 0$ and $\varphi(\mathrm{E} \cap (\mathrm{F} + \mathfrak{L}^h\mathrm{N})) = \mathrm{E}_1 \cap (\mathrm{F}_1 + \mathfrak{L}^h\mathrm{N}_1)$. We know $(\mathbf{0}_{\mathrm{I}}, 7.3.2)$ that there exists h such that $(\mathrm{E}_1 + \mathrm{F}_1) \cap \mathfrak{L}^h\mathrm{N}_1 \subset \mathfrak{L}^k(\mathrm{E}_1 + \mathrm{F}_1)$; if h is thus chosen and if $x_1 \in \mathrm{E}_1$ is such that there exists $y_1 \in \mathrm{F}_1$ for which $x_1 - y_1 \in \mathfrak{L}^h\mathrm{N}_1$, we deduce $x_1 - y_1 \in \mathfrak{L}^k\mathrm{N}_1$, hence $\mathrm{E}_1 \cap (\mathrm{F}_1 + \mathfrak{L}^h\mathrm{N}_1) \subset \mathfrak{L}^k\mathrm{N}_1$; therefore $\mathrm{E} \cap (\mathrm{F} + \mathfrak{L}^h\mathrm{N}) \subset \mathrm{E} \cap \mathrm{F} + \mathfrak{L}^k\mathrm{N}$, which proves the lemma. □

It suffices then to apply this lemma with $\mathrm{E} = \mathrm{R} \cap \mathfrak{L}^m\mathrm{N}$, $\mathrm{F} = \mathfrak{R}^{d+1}\mathrm{N}$ and to take $k = m + 1$ to deduce that $(**_{d+1,h})$ implies $(*_{d+1})$.

We suppose therefore in all that follows that $(**_{d+1,h})$ *is satisfied.*

(19.7.1.8). *Demonstration of $(**_{d+1,h+1})$; First case: $h \leqslant m - d$.*

Start with an element

(19.7.1.9).
$$g \in \mathrm{R} \cap (\mathfrak{R}^{d+1}\mathrm{N} + \mathfrak{R}^d\mathfrak{L}^h\mathrm{N}) \cap \mathfrak{L}^m\mathrm{N}$$

congruent modulo $\mathfrak{L}^{m+1}\mathrm{N}$ to an element $g_1 \in \mathrm{R} \cap (\mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N})$; we therefore have

(19.7.1.10).
$$g \in \mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N}.$$

It will suffice to prove that g is also congruent modulo $\mathfrak{L}^{m+1}\mathrm{N}$ to an element $g_2 \in \mathrm{R} \cap (\mathfrak{R}^{d+1}\mathrm{N} + \mathfrak{R}^d\mathfrak{L}^{h+1}\mathrm{N})$, for this will imply $g_2 \in \mathfrak{L}^m\mathrm{N}$, since $g \in \mathfrak{L}^m\mathrm{N}$ and $g_2 - g \in \mathfrak{L}^{m+1}\mathrm{N}$. In the case $h \leqslant m - d$ under consideration, it is enough to prove

(19.7.1.11).
$$(\mathfrak{R}^m\mathrm{N} + \mathfrak{L}^{m+1}\mathrm{N}) \cap \mathfrak{R}^d\mathrm{N} \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d\mathrm{N} + \mathfrak{R}^{d+1}\mathrm{N}.$$

Indeed, relations (19.7.1.9) and (19.7.1.10) imply that g belongs to the left-hand side of (19.7.1.11); and since $h \leqslant m - d$, we have $\mathfrak{L}^{m-d+1} \subset \mathfrak{L}^{h+1}$, which will establish $(**_{d+1,h+1})$. Moreover $\mathfrak{R}^m\mathrm{N} \subset \mathfrak{R}^d\mathrm{N}$, and the relation **IV-4 | 216**

(19.7.1.12).
$$\mathfrak{L}^{m+1}\mathrm{N} \cap \mathfrak{R}^d\mathrm{N} \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d\mathrm{N}$$

will imply (19.7.1.11). Since N is a free A-module of finite type, it is therefore enough to prove

(19.7.1.13).
$$\mathfrak{L}^{m+1} \cap \mathfrak{R}^d \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d.$$

It will suffice to prove that, in general, for $d < m$, one has

(19.7.1.14).
$$\mathfrak{L}^{m+1} \cap \mathfrak{R}^d \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d + \mathfrak{R}^{d+1}.$$

Indeed, since $\mathfrak{R} \subset \mathfrak{L}$, hence $\mathfrak{L}^{m-d+1}\mathfrak{R}^d \subset \mathfrak{L}^{m+1}$, the preceding relation, applied successively with $d, d+1, \dots, m$, implies

$$\mathfrak{L}^{m+1} \cap \mathfrak{R}^d \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d + \mathfrak{R}^{m+1},$$

whence (19.7.1.13), since $\mathfrak{R}^{m+1} \subset \mathfrak{L}^{m-d+1}\mathfrak{R}^d$.

(19.7.1.15). As $\mathfrak{L} = \mathfrak{J} + \mathfrak{R}$, the relation (19.7.1.14) also reads

$$\begin{aligned}&(\mathfrak{J}^{m+1} + \mathfrak{J}^m\mathfrak{R} + \cdots + \mathfrak{J}\mathfrak{R}^m + \mathfrak{R}^{m+1}) \cap \mathfrak{R}^d \\ &\quad \subset \mathfrak{J}^{m-d+1}\mathfrak{R}^d + \mathfrak{J}^{m-d}\mathfrak{R}^{d+1} + \cdots + \mathfrak{J}\mathfrak{R}^m + \mathfrak{R}^{m+1} + \mathfrak{R}^{d+1} \\ &\quad = \mathfrak{J}^{m-d+1}\mathfrak{R}^d + \mathfrak{R}^{d+1}.\end{aligned}$$

We will show that this inclusion is itself a consequence of

(19.7.1.16).

$$\mathfrak{R}^a\mathfrak{J}^b \cap \mathfrak{R}^{a+1} \subset \mathfrak{R}^{a+1}\mathfrak{J}^{b-1}$$

valid for $a \geqslant 0$, $b \geqslant 1$. Indeed, it suffices, to prove (19.7.1.15), to show that for $0 < q \leqslant d$,

(19.7.1.17).

$$(\mathfrak{J}^{m+1} + \mathfrak{J}^m\mathfrak{R} + \ldots + \mathfrak{J}\mathfrak{R}^m + \mathfrak{R}^{m+1}) \cap \mathfrak{R}^d \subset (\mathfrak{J}^{m+1-q}\mathfrak{R}^q + \mathfrak{J}^{m-q}\mathfrak{R}^{q+1} + \ldots + \mathfrak{R}^{m+1}) \cap \mathfrak{R}^d$$

since for $q = d$, this will imply (19.7.1.15). To prove (19.7.1.17), proceed by induction on q, supposing the relation true for $q < d$. An element of the right-hand side corresponding to q can be written $y + z$, with $y \in \mathfrak{J}^{m+1-q}\mathfrak{R}^q$ and $z \in \mathfrak{J}^{m-q}\mathfrak{R}^{q+1} + \cdots + \mathfrak{R}^{m+1}$. Since $y + z \in \mathfrak{R}^d \subset \mathfrak{R}^{q+1}$ and $z \in \mathfrak{R}^{q+1}$, we have $y \in \mathfrak{R}^{q+1}$; by (19.7.1.16), $y \in \mathfrak{J}^{m-q}\mathfrak{R}^{q+1}$, and hence

$$y + z \in \mathfrak{J}^{m-q}\mathfrak{R}^{q+1} + \ldots + \mathfrak{R}^{m+1}, \quad \text{and} \quad y + z \in \mathfrak{R}^d$$

by hypothesis, which proves (19.7.1.17) where q has been replaced by $q + 1$.

It remains therefore to establish (19.7.1.16). An element of the left-hand side is written $a_1 f_1 + \ldots + a_n f_n$, where the a_i belong to $\mathfrak{R}^a\mathfrak{J}^{b-1}$. Since the sequence $\mathbf{f}$ is $(\mathfrak{R}^a/\mathfrak{R}^{a+1})$-regular, hence also $(\mathfrak{R}^a/\mathfrak{R}^{a+1})$-quasi-regular $(\mathbf{0}, 15.1.9)$, it follows from the definition $(\mathbf{0}, 15.1.7)$ that the relation $a_1 f_1 + \ldots + a_n f_n \in \mathfrak{R}^{a+1}$, with $a_i \in \mathfrak{R}^a$, necessarily implies $a_i \in \mathfrak{R}^{a+1}$, and hence $a_i \in \mathfrak{R}^a\mathfrak{J}^{b-1} \cap \mathfrak{R}^{a+1}$. It suffices to argue by induction on b for $b \geqslant 1$ (the case $b = 1$ being trivial) to establish (19.7.1.16) and by extension also $(**_{d+1,h+1})$ for $h \leqslant m - d$. **IV-4** | 217

(19.7.1.18). *Demonstration of* $(**_{d+1,h+1})$*; Second case:* $h > m - d$.

We will first prove that, for every $h \geqslant 0$, we have

(19.7.1.19).

$$\mathrm{R} \cap (\mathfrak{R}^{d+1}\mathrm{N} + \mathfrak{R}^d\mathfrak{L}^h\mathrm{N}) \subset \mathfrak{J}^h(\mathrm{R} \cap \mathfrak{R}^d\mathrm{N}) + \mathfrak{L}^{h+1}\mathfrak{R}^d\mathrm{N} + \mathfrak{R}^{d+1}\mathrm{N}.$$

Consider an element z of the left-hand side of (19.7.1.19). We note that $\mathfrak{R}^{d+1}\mathrm{N} + \mathfrak{R}^d\mathfrak{L}^h\mathrm{N} = \mathfrak{R}^{d+1}\mathrm{N} + \mathfrak{R}^d\mathfrak{J}^h\mathrm{N}$, since $\mathfrak{L} = \mathfrak{J} + \mathfrak{R}$. We consider the class $\bar{z}$ of z in $\mathrm{gr}^h_{\mathfrak{J}}(\mathrm{gr}^d_{\mathfrak{R}}(\mathrm{N}))$, which, by virtue of (19.5.5) and the hypothesis on $\mathbf{f}$, identifies with a sub-$(\mathrm{A}/\mathfrak{L})$-module of $\mathrm{gr}^{h+d}_{\mathfrak{L}}(\mathrm{N})$. Since $z \in \mathrm{R}$, we will show that, under this identification, one even has

(19.7.1.20).

$$\bar{z} \in \mathrm{gr}^h_{\mathfrak{J}}(\mathrm{gr}^d_{\mathfrak{R}}(\mathrm{R}, \mathrm{N})).$$

Indeed, we noted in the proof of (19.7.1.3) that, for every prime ideal $\mathfrak{r}$ of A such that $\mathfrak{p} = \mathfrak{r}/\mathfrak{R} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{R}}(\mathrm{A}/\mathfrak{L})$, the map $\psi'_{\mathrm{R}_{\mathfrak{r}}}$ is bijective, and consequently relation (19.7.1.6), with all modules replaced by their localisations at $\mathfrak{r}$, is true. Since z belongs to $\mathrm{R} \cap (\mathfrak{R}^d\mathrm{N} + \mathfrak{L}^{d+1}\mathrm{N})$, its image $z/1$ in $\mathrm{R}_{\mathfrak{r}}$ belongs to $(\mathrm{R}_{\mathfrak{r}} \cap \mathfrak{R}^d_{\mathfrak{r}}\mathrm{N}_{\mathfrak{r}}) + \mathfrak{L}^{d+1}_{\mathfrak{r}}\mathrm{N}_{\mathfrak{r}}$. Hence the image $\bar{z}/1$ in $\mathrm{gr}^h_{\mathfrak{J}}(\mathrm{gr}^d_{\mathfrak{R}}(\mathrm{N}_{\mathfrak{r}}))$ belongs to $\mathrm{gr}^h_{\mathfrak{J}}(\mathrm{gr}^d_{\mathfrak{R}}(\mathrm{R}_{\mathfrak{r}}, \mathrm{N}_{\mathfrak{r}}))$, the localisation at $\mathfrak{q} = \mathfrak{r}/\mathfrak{L}$ of $\mathrm{gr}^h_{\mathfrak{J}}(\mathrm{gr}^d_{\mathfrak{R}}(\mathrm{R}, \mathrm{N}))$. In other words, the image of $\bar{z}$ by the canonical map

$$(\mathrm{gr}^h_{\mathfrak{J}}(\mathrm{gr}^d_{\mathfrak{R}}(\mathrm{N})))_{\mathfrak{q}} \longrightarrow (\mathrm{gr}^h_{\mathfrak{J}}(\mathrm{gr}^d_{\mathfrak{R}}(\mathrm{M})))_{\mathfrak{q}}$$

is zero for every $\mathfrak{q} \in \mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{A}/\mathfrak{L})$. As in the proof of (19.7.1.3), the associated primes of the target are contained in $\mathrm{Ass}_{\mathrm{A}/\mathfrak{L}}(\mathrm{A}/\mathfrak{L})$; it follows (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 1, cor. 1 of prop. 2 and n° 3, prop. 7) that the image of $\bar{z}$ is zero, which is precisely (19.7.1.20).

(19.7.1.21). This being so, we can write by definition

$$z \equiv \sum_{|\mathbf{p}|=h} c_{\mathbf{p}} \mathbf{f}^{\mathbf{p}} \bmod \mathfrak{R}^{d+1}\mathrm{N}$$

where one sets as usual $\mathbf{p} = (p_1, \dots, p_n)$, $\mathbf{f}^{\mathbf{p}} = f_1^{p_1} f_2^{p_2} \cdots f_n^{p_n}$, $|\mathbf{p}| = p_1 + \ldots + p_n$ and $c_{\mathbf{p}} \in \mathfrak{R}^d\mathrm{N}$. Moreover, as ψ_{N} identifies the relevant graded module with $(\mathrm{gr}^d_{\mathfrak{R}}(\mathrm{N}) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{L}))[\mathrm{T}_1, \dots, \mathrm{T}_n]$, the $c_{\mathbf{p}}$ are determined modulo $\mathfrak{R}^d\mathfrak{L}\mathrm{N}$; if $\bar{c}_{\mathbf{p}}$ denotes the class of $c_{\mathbf{p}}$ modulo $\mathfrak{R}^d\mathfrak{L}\mathrm{N}$, then, under the preceding identification,

(19.7.1.22).

$$\bar{z} = \sum_{|\mathbf{p}|=h} \bar{c}_{\mathbf{p}} \mathrm{T}^{\mathbf{p}}.$$

Using (19.7.1.3), we deduce that necessarily, for every $\mathbf{p}$ such that $|\mathbf{p}| = h$ ($\bar{c}_{\mathbf{p}}$ being this time identified by ψ_{N} with an element of $\mathrm{gr}^d_{\mathfrak{R}}(\mathrm{N})$),

(19.7.1.23).

$$\bar{c}_{\mathbf{p}} \in \mathrm{gr}^d_{\mathfrak{R}}(\mathrm{R}, \mathrm{N}).$$

But since $d < m$, the hypothesis that $(*_d)$ is true implies that $\bar{c}_{\mathbf{p}}$ belongs to the image of ψ'_{R}, hence we can suppose that $c_{\mathbf{p}} \in (\mathrm{R} \cap \mathfrak{R}^d\mathrm{N}) + \mathfrak{R}^d\mathfrak{L}\mathrm{N}$. The relation (19.7.1.21) then gives

$$z \in \mathfrak{J}^h(\mathrm{R} \cap \mathfrak{R}^d\mathrm{N}) + \mathfrak{J}^h\mathfrak{R}^d\mathfrak{L}\mathrm{N} + \mathfrak{R}^{d+1}\mathrm{N}$$

and as $\mathfrak{J} \subset \mathfrak{L}$, this proves (19.7.1.19).

In particular, we can therefore write **IV-4** | 218

$$g = t + g_2$$

with $t \in \mathfrak{J}^h(\mathrm{R} \cap \mathfrak{R}^d\mathrm{N})$ and $g_2 \in \mathfrak{R}^{d+1}\mathrm{N} + \mathfrak{R}^d\mathfrak{L}^{h+1}\mathrm{N}$; moreover, as $t \in \mathrm{R}$ and $g \in \mathrm{R}$, we have $g_2 \in \mathrm{R}$. On the other hand, using the hypothesis $h \geqslant m - d + 1$, one sees that $t \in \mathfrak{L}^{m+1}\mathrm{N}$. The element g_2 hence satisfies all the conditions stated at the start of (19.7.1.8), which completes the demonstration of (19.7.1). □

Remarks (19.7.2). —

(i) When one supposes that $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{A})$ is an $(\mathrm{A}/\mathfrak{R})$-module *flat*, the hypothesis that the sequence $\mathbf{f}$ is $(\mathrm{A}/\mathfrak{R})$-regular implies that it is also $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{A})$-regular $(\mathbf{0}, 15.1.14)$. We note that this will be the case when A and $\mathrm{A}/\mathfrak{R}$ *are regular rings* $(\mathbf{0}, 17.3.6)$: indeed, $\mathfrak{R}$ is then a regular ideal of A (19.1.2), hence quasi-regular, and localising at the maximal ideals of A containing $\mathfrak{R}$ and using $(\mathbf{0}, 15.1.7)$, one sees that $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{A})$ is a *projective* $(\mathrm{A}/\mathfrak{R})$-module.

(ii) It would be interesting to be able to prove the last conclusion of (19.7.1) under the sole hypothesis that there exists an A-algebra B which is a Noetherian ring, for which $\mathfrak{J}\mathrm{B}$ is contained in the radical of B, and that M is a finitely generated B-module.

Corollary (19.7.3). — *Let* A *be a local Noetherian ring,* $\mathfrak{m}$ *its maximal ideal,* $\mathfrak{R}$ *an ideal of* A *such that the ring* $\mathrm{A}/\mathfrak{R}$ *is regular and of dimension* n, M *a finitely generated* A*-module. The following conditions are equivalent:*

a) $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M})$ *is a flat* $(\mathrm{A}/\mathfrak{R})$*-module.*

b) *If* $\mathrm{P}(\mathrm{T})$ *and* $\mathrm{Q}(\mathrm{T})$ *are the Poincaré series of the* $(\mathrm{A}/\mathfrak{m})$*-modules graded* $\mathrm{gr}^*_{\mathfrak{m}}(\mathrm{M})$ *and* $\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{m})$*, we have*

$$\mathrm{Q}(\mathrm{T}) = \mathrm{P}(\mathrm{T})(1 - \mathrm{T})^n. \tag{19.7.3.1}$$

Let $\mathbf{f} = (f_1, \dots, f_n)$ *be a sequence of elements of* A *such that the images of the* f_i *in* $\mathrm{A}/\mathfrak{R}$ *form a regular system of parameters of* $\mathrm{A}/\mathfrak{R}$ $(\mathbf{0}, 17.1.6)$*, so that* $\mathbf{f}$ *is an* $(\mathrm{A}/\mathfrak{R})$*-regular sequence; moreover, if* $\mathfrak{J}$ *is the ideal generated by* $\mathbf{f}$, $\mathfrak{J} + \mathfrak{R} = \mathfrak{m}$*, and as* $\mathrm{A}/\mathfrak{m}$ *is a field,* $\mathrm{gr}^*_{\mathfrak{m}}(\mathrm{M})$ *is a flat* $(\mathrm{A}/\mathfrak{m})$*-module. As* A *is Noetherian and* M *a finitely generated* A*-module, one can apply the equivalence of a) and b) in* (19.7.1)*. Moreover,* $\mathrm{A}/\mathfrak{m}$ *being a field, the fact that* ψ_{M} *is bijective is equivalent to saying that for every* $h \geqslant 0$, $\mathrm{gr}^h_{\mathfrak{m}}(\mathrm{M})$ *and the submodule of elements of degree* h *of* $(\mathrm{gr}^*_{\mathfrak{R}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{m}))[\mathrm{T}_1, \dots, \mathrm{T}_n]$ *have the same rank over* $\mathrm{A}/\mathfrak{m}$*; but for the second of these modules the rank in question is clearly equal to*

$\sum_{r=0}^{h} \binom{r+n-1}{n-1} \mathrm{rg}(\mathrm{gr}^{h-r}_{\mathfrak{R}}(\mathrm{M}) \otimes_{\mathrm{A}/\mathfrak{R}} (\mathrm{A}/\mathfrak{m}))$. *As* $(1 - \mathrm{T})^{-n} = \sum_{r=0}^{\infty} \binom{r+n-1}{n-1} \mathrm{T}^r$, *this proves that the relation* (19.7.3.1) *is necessary and sufficient for* ψ_{M} *to be bijective.*

Corollary (19.7.4). — *Let* X *be a locally Noetherian prescheme,* Y *a closed sub-prescheme of* X, Y *regular and connected,* $\mathscr{F}$ *a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module normally flat* ((6.10.1)) *along* Y. *Then there exists a formal power series* $\mathrm{R} \in \mathbf{Q}[[\mathrm{T}]]$ *(independent of* $x \in \mathrm{Y}$*) such that, for every* $x \in \mathrm{Y}$*, the Poincaré series of* $\mathrm{gr}^*_{\mathfrak{m}_x}(\mathscr{F}_x)$ *is given by the formula*

$$\mathrm{P}_x(\mathscr{F})(\mathrm{T}) = \mathrm{R}(\mathrm{T})(1-\mathrm{T})^{-n} \qquad \textit{where } n = \dim(\mathscr{O}_{\mathrm{Y},x}). \tag{19.7.4.1}$$

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Let $\mathscr{J}$ *be the coherent Ideal of* $\mathscr{O}_{\mathrm{X}}$ *defining* Y*; the hypothesis on* $\mathscr{F}$ *means that* $\mathrm{gr}^*_{\mathscr{J}}(\mathscr{F})$ *is a flat* $\mathscr{O}_{\mathrm{Y}}$*-Module. We can therefore, for every* $x \in \mathrm{Y}$*, apply* (19.7.3) *taking* $\mathrm{A} = \mathscr{O}_{\mathrm{X},x}$, $\mathfrak{R} = \mathscr{J}_x$, $\mathrm{M} = \mathscr{F}_x$*; condition* a) *being satisfied, the Poincaré series* $\mathrm{P}_x(\mathscr{F})(\mathrm{T})$ *is equal to* $\mathrm{R}_x(\mathrm{T})(1-\mathrm{T})^{-n}$*, where* $\mathrm{R}_x(\mathrm{T})$ *is the Poincaré series of* $\mathrm{gr}^*_{\mathscr{J}_x}(\mathscr{F}_x) \otimes_{\mathscr{O}_{\mathrm{Y},x}} k(x)$*. Each* $\mathscr{O}_{\mathrm{Y}}$*-Module* $\mathscr{J}^h\mathscr{F}/\mathscr{J}^{h+1}\mathscr{F}$ *is flat and coherent by hypothesis, hence locally free* (2.1.12)*; and since* Y *is connected, its rank is constant on* Y*. This proves that* $\mathrm{R}_x(\mathrm{T})$ *is independent of* $x \in \mathrm{Y}$.

Corollary (19.7.5). — *Let* X *be a locally Noetherian prescheme,* Y *a closed sub-prescheme of* X, Z *a closed sub-prescheme of* Y*; one supposes* Y *and* Z *regular. Let* $\mathscr{F}$ *be a coherent* $\mathscr{O}_{\mathrm{X}}$*-Module.*

(i) *If* $\mathscr{F}$ *is normally flat along* Y *at the points of* Z *(cf.* (11.3.4) *for the definition of normal flatness at a point), then* $\mathscr{F}$ *is normally flat along* Z.

(ii) *Suppose moreover that* Z *is connected and that* $\mathscr{O}_{\mathrm{X}}$ *is normally flat along* Y *at the points of* Z *(which is the case in particular if* X *is regular at the points of* Z*). If* $\mathscr{F}$ *is normally flat along* Z *and if there exists a point* $z \in \mathrm{Z}$ *at which* $\mathscr{F}$ *is normally flat along* Y*, then* $\mathscr{F}$ *is normally flat along* Y *at all the points of* Z.

Let $\mathscr{K}$ *and* $\mathscr{L} \supset \mathscr{K}$ *be the coherent Ideals of* $\mathscr{O}_{\mathrm{X}}$ *defining respectively* Y *and* Z*. For every point* $z \in \mathrm{Z}$*, the canonical immersion* $\mathrm{Z} \to \mathrm{Y}$ *is regular at* z*, since* Y *and* Z *are regular at that point* (19.1.1)*; in other words, there exists a sequence* $(f_i)_{1\leqslant i\leqslant n}$ *of elements of* $\mathscr{O}_{\mathrm{X},z}$ *which is* $\mathscr{K}_z$*-regular and such that* $\mathscr{L}_z = \mathscr{K}_z + \sum_i f_i\mathscr{O}_{\mathrm{X},z}$.

(i) *The hypothesis means that* $\mathrm{gr}^*_{\mathscr{K}_z}(\mathscr{F}_z)$ *is a flat* $\mathscr{O}_{\mathrm{Y},z}$*-module; it therefore implies that* $\mathrm{gr}^*_{\mathscr{L}_z}(\mathscr{F}_z)$ *is a flat* $\mathscr{O}_{\mathrm{Z},z}$*-module, by the implication* a) $\Rightarrow$ b) *of* (19.7.1).

(ii) *The supplementary hypothesis on* X *implies that the sequence* (f_i) *is* $\mathrm{gr}^*_{\mathscr{K}_z}(\mathscr{O}_{\mathrm{X},z})$*-regular at every point* $z \in \mathrm{Z}$ (19.7.2)*. On the other hand,* Z*, being regular and connected, is integral; if* $\mathrm{gr}^*_{\mathscr{K}_z}(\mathscr{F}_z)$ *is flat over* $\mathscr{O}_{\mathrm{Y},z}$ *at one point* $z \in \mathrm{Z}$*, then, since the generic point* ζ *of* Z *is a generization of* z, $\mathrm{gr}^*_{\mathscr{K}_\zeta}(\mathscr{F}_\zeta)$ *is flat over* $\mathscr{O}_{\mathrm{Y},\zeta}$ ($\mathbf{0}_{\mathrm{I}}$, 6.3.1)*. If, at every point* z' *of* Z, $\mathrm{gr}^*_{\mathscr{L}_{z'}}(\mathscr{F}_{z'})$ *is flat over* $\mathscr{O}_{\mathrm{Z},z'}$*, we then conclude that* $\mathrm{gr}^*_{\mathscr{K}_{z'}}(\mathscr{F}_{z'})$ *is flat over* $\mathscr{O}_{\mathrm{Y},z'}$*, by applying the equivalence of* a) *and* d) *in* (19.7.1).

19.8. Properties of passage to the projective limit In this number, the notations and conventions on projective limits are those of (8.5.1) and (8.8.1).

Proposition (19.8.1). — *Suppose that the transition morphisms* $\mathrm{S}_\mu \to \mathrm{S}_\lambda$ ($\lambda \leqslant \mu$) *are flat, and moreover that one of the two following hypotheses is satisfied:*

1° *The preschemes* S_λ *are locally Noetherian.*

2° *The transition morphisms* $\mathrm{S}_\mu \to \mathrm{S}_\lambda$ *are surjective (and hence faithfully flat).*

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(i) *Suppose* S_α *quasi-compact. Let* $\mathscr{F}_\alpha$ *be a quasi-coherent* $\mathscr{O}_{\mathrm{S}_\alpha}$*-Module, which one supposes moreover of finite type when hypothesis* 1° *is satisfied. Let* $(f_{i\alpha})_{1\leqslant i\leqslant n}$ *be a finite sequence of sections of* $\mathscr{F}_\alpha$ *above* S_α*, the sequence of sections* f_i *of the* $\mathscr{O}_{\mathrm{S}}$*-Module* $\mathscr{F}$ *above* S*, corresponding to the* $f_{i\alpha}$*. For the sequence of sections* f_i ($1 \leqslant i \leqslant n$) *to be* $\mathscr{F}$*-regular, it is necessary and sufficient that there exist* $\lambda \geqslant \alpha$ *such that the sequence of sections* $f_{i\lambda}$ *of* $\mathscr{F}_\lambda$ *above* S_λ *is* $\mathscr{F}_\lambda$*-regular.*

(ii) *Suppose* X_α *quasi-compact, and let* $j_\alpha : \mathrm{X}_\alpha \to \mathrm{S}_\alpha$ *be an immersion, which one supposes locally of finite presentation when hypothesis* 2° *is satisfied. Then, for the corresponding immersion* $j : \mathrm{X} \to \mathrm{S}$ *to be regular, it is necessary and sufficient that there exist* $\lambda \geqslant \alpha$ *such that* $j_\lambda : \mathrm{X}_\lambda \to \mathrm{S}_\lambda$ *is regular.*

Proof. (i) If $p_\lambda : \mathrm{S} \to \mathrm{S}_\lambda$ is the canonical projection, we have $\mathscr{F}/(\sum_{j=1}^{i-1} f_j\mathscr{F}) = p_\lambda^*(\mathscr{F}_\lambda/(\sum_{j=1}^{i-1} f_{j\lambda}\mathscr{F}_\lambda))$, hence we are reduced to the case $n = 1$, in which case we omit the index i. Sufficiency follows from the flatness of p_λ (8.3.8) and ($\mathbf{0}$, 15.1.5). In case 2°, p_λ is faithfully flat (8.3.8) and necessity follows from $(\mathbf{0}, 15.2.5)$, with $\lambda = \alpha$. In case 1°, let $\mathscr{N}_\lambda$ (resp. $\mathscr{N}$) be the kernel of multiplication by f_λ on $\mathscr{F}_\lambda$ (resp. by f on $\mathscr{F}$); since $\mathscr{F}_\lambda$ is coherent, so is $\mathscr{N}_\lambda$, and the hypothesis $\mathscr{N} = 0$ implies $\mathscr{N}_\lambda = 0$ for some $\lambda \geqslant \alpha$ by (8.5.8), (ii).

(ii) As p_λ is flat, sufficiency follows from (19.1.5), (ii). For necessity, since $j_\alpha(\mathrm{X}_\alpha)$ is quasi-compact, it is contained in a quasi-compact open subset of S_α; we may therefore reduce to the case in which S_α is quasi-compact and j_α is a closed immersion. Its image is then defined by a quasi-coherent Ideal $\mathscr{J}_\alpha$ of $\mathscr{O}_{\mathrm{S}_\alpha}$, which may moreover be supposed of finite type in cases 1° and 2°. The image of X_λ (resp. X) in S_λ (resp. S) is defined by $\mathscr{J}_\lambda = \mathscr{J}_\alpha \mathscr{O}_{\mathrm{S}_\lambda}$ (resp. $\mathscr{J} = \mathscr{J}_\alpha \mathscr{O}_{\mathrm{S}}$), again of finite type. By (8.2.11), we may suppose that $\mathscr{J}$ is generated by a regular sequence $(f_i)_{1 \leqslant i \leqslant n}$ of sections over S, and hence by a surjection $u : \mathscr{O}_{\mathrm{S}}^n \to \mathscr{J}$. By (8.5.2), (i), and (8.5.7), there exist $\lambda \geqslant \alpha$ and a surjection $u_\lambda : \mathscr{O}_{\mathrm{S}_\lambda}^n \to \mathscr{J}_\lambda$ such that $u = p_\lambda^*(u_\lambda)$; thus the f_i are the canonical images of sections $f_{i\lambda}$ of $\mathscr{J}_\lambda$ generating this Ideal. By (i), there is then $\mu \geqslant \lambda$ for which $(f_{i\mu})$ is $\mathscr{O}_{\mathrm{S}_\mu}$-regular, and hence j_μ is a regular immersion. □

Proposition (19.8.2). — *Let X_α be an S_α-prescheme quasi-compact and locally of finite presentation.*

(i) *Let $\mathscr{F}_\alpha$ be a quasi-coherent $\mathscr{O}_{\mathrm{X}_\alpha}$-Module of finite presentation, and $(f_{i\alpha})_{1 \leqslant i \leqslant n}$ a finite sequence of sections of $\mathscr{O}_{\mathrm{X}_\alpha}$ over X_α, corresponding to sections f_i of $\mathscr{O}_{\mathrm{X}}$ over X. For the sequence (f_i) to be $\mathscr{F}$-transversally regular relative to* S *((19.2.1)), it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that the sequence $(f_{i\lambda})$ over X_λ is $\mathscr{F}_\lambda$-transversally regular relative to S_λ.*

(ii) *Let Y_α be an S_α-prescheme quasi-compact, $j_\alpha : \mathrm{Y}_\alpha \to \mathrm{X}_\alpha$ an S_α-immersion locally of finite presentation. For the corresponding* S*-immersion $j : \mathrm{Y} \to \mathrm{X}$ to be transversally regular relative to* S*, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that $j_\lambda : \mathrm{Y}_\lambda \to \mathrm{X}_\lambda$ is transversally regular relative to S_λ.*

Proof. (i) The sufficiency of the condition follows from $(\mathbf{0}, 15.1.15)$. To prove necessity, it is enough to show that every point $x \in \mathrm{X}$ has an open neighbourhood $\mathrm{V}(x)$ for which there exist an index $\lambda = \lambda(x) \geqslant \alpha$ and an open neighbourhood V_λ of the image x_λ of x in X_λ, such that $\mathrm{V}(x)$ is the inverse image of V_λ and the restrictions of the $f_{i\lambda}$ to V_λ form an $(\mathscr{F}_\lambda|\mathrm{V}_\lambda)$-transversally regular sequence relative to S_λ. Indeed, one then applies (8.3.4) to the open subset U_λ of X_λ defined as the largest open on which the restrictions of the $f_{i\lambda}$ form such a sequence. **IV-4** | 221

Let s be the image of x in S and, for every $\lambda \geqslant \alpha$, let s_λ be the image of s in S_λ. The fibre X_s is the projective limit of the fibres $(\mathrm{X}_\lambda)_{s_\lambda}$ and is locally Noetherian, since $\mathrm{X} \to \mathrm{S}$ is locally of finite presentation. Moreover $(\mathrm{X}_\alpha)_{s_\alpha}$ is quasi-compact and the transition morphisms $(\mathrm{X}_\mu)_{s_\mu} \to (\mathrm{X}_\lambda)_{s_\lambda}$ are flat, since this is so for $\mathrm{Spec}(k(s_\mu)) \to \mathrm{Spec}(k(s_\lambda))$. We may therefore apply (19.8.1), (i), to the sections $f_i \otimes 1$ of $\mathscr{F}_s = \mathscr{F} \otimes_{\mathscr{O}_{\mathrm{S}}} k(s)$ over X_s. Thus there exists $\lambda \geqslant \alpha$ such that the sections $f_{i\lambda} \otimes 1$ of $(\mathscr{F}_\lambda)_{s_\lambda} = \mathscr{F}_\lambda \otimes_{\mathscr{O}_{\mathrm{S}_\lambda}} k(s_\lambda)$ form an $(\mathscr{F}_\lambda)_{s_\lambda}$-regular sequence. By (11.2.6), one may moreover suppose that $\mathscr{F}_\lambda$ is flat over S_λ at x_λ. It then follows from (11.3.8) that there exists an open neighbourhood V_λ of x_λ in X_λ satisfying the required conditions.

(ii) Sufficiency follows from (19.2.7), (ii). For necessity, it again suffices to show that every point $x \in j(\mathrm{Y})$ has an open neighbourhood $\mathrm{V}(x)$ for which there exist an index $\lambda = \lambda(x) \geqslant \alpha$ and an open neighbourhood V_λ of the image x_λ of x in X_λ, such that $\mathrm{V}(x)$ is the inverse image of V_λ and the restriction $j_\lambda^{-1}(\mathrm{V}_\lambda) \to \mathrm{V}_\lambda$ is transversally regular relative to S_λ.

Let s be the image of x in S and, for every $\lambda \geqslant \alpha$, let s_λ be the image of s in S_λ. Since X_s (resp. Y_s) is the projective limit of the fibres $(\mathrm{X}_\lambda)_{s_\lambda}$ (resp. $(\mathrm{Y}_\lambda)_{s_\lambda}$), we may use (19.8.1), (ii), as in (i); and since the immersion $\mathrm{Y}_s \to \mathrm{X}_s$ is regular by hypothesis, there exists an index $\lambda(s)$ such that, for $\lambda \geqslant \lambda(s)$, the immersion $(\mathrm{Y}_\lambda)_{s_\lambda} \to (\mathrm{X}_\lambda)_{s_\lambda}$ is regular. Moreover, by (19.2.4), there exists a quasi-compact open neighbourhood $\mathrm{W}(x)$ of x in X such that the structural morphisms $\mathrm{W}(x) \to \mathrm{S}$ and $\mathrm{Y} \cap \mathrm{W}(x) \to \mathrm{S}$ are flat. Applying (11.2.6) and (8.2.11), one may suppose that there exist $\lambda \geqslant \lambda(s)$ and a quasi-compact open neighbourhood $\mathrm{W}(x_\lambda)$ of x_λ in X_λ, such that $\mathrm{W}(x)$ is the inverse image of $\mathrm{W}(x_\lambda)$ and the restrictions to $\mathrm{W}(x_\lambda)$ and $\mathrm{Y}_\lambda \cap \mathrm{W}(x_\lambda)$ of the structural morphisms $\mathrm{X}_\lambda \to \mathrm{S}_\lambda$ and $\mathrm{Y}_\lambda \to \mathrm{S}_\lambda$ are flat. We conclude from (19.2.4) that there exists an open neighbourhood V_λ of x_λ satisfying the required conditions.

□

Remark (19.8.3). — The preceding demonstrations show that the statements of (19.8.1) and (19.8.2) subsist when one replaces the conditions concerning regularity or transversal regularity throughout X (resp. X_λ) by these conditions in a neighbourhood of a point $x \in \mathrm{X}$ (resp. in a neighbourhood of x_λ, projection of x).

19.9. $\mathscr{F}$-regular sequences and depth IV-4 | 222

(19.9.1). Recall that if X is a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_{\mathrm{X}}$-Module, T a part of X, one sets (5.10.1)

$$\operatorname{prof}_{\mathrm{T}}(\mathscr{F}) = \inf_{x\in\mathrm{T}} \operatorname{prof}(\mathscr{F}_x). \tag{19.9.1.1}$$

We set on the other hand, for every point $t \in \mathrm{T}$,

$$\operatorname{prof}_{\mathrm{T},t}(\mathscr{F}) = \inf_{x\in\mathrm{T}\cap\operatorname{Spec}(\mathscr{O}_{\mathrm{X},t})} \operatorname{prof}(\mathscr{F}_x), \tag{19.9.1.2}$$

and we say that $\operatorname{prof}_{\mathrm{T},t}(\mathscr{F})$ is the *T-depth of $\mathscr{F}$ at the point t*; it is clear that we have

$$\operatorname{prof}_{\mathrm{T}}(\mathscr{F}) = \inf_{t\in\mathrm{T}} \operatorname{prof}_{\mathrm{T},t}(\mathscr{F}). \tag{19.9.1.3}$$

Lemma (19.9.2). — *Let A be a Noetherian ring, M a finitely generated A-module, $\mathfrak{J}$ an ideal of A. In order that, for every prime ideal $\mathfrak{p} \supset \mathfrak{J}$, we have $\operatorname{prof}_{\mathrm{A}_\mathfrak{p}}(\mathrm{M}_\mathfrak{p}) \geqslant r$, it is necessary and sufficient that there exist an M-regular sequence of r elements belonging to $\mathfrak{J}$.*

Proof. The sufficiency of the condition follows immediately from $(\mathbf{0}, 16.4.5)$; let us prove its necessity. We argue by induction on r (there is nothing to prove if $r = 0$): since $\operatorname{prof}_{\mathrm{A}_\mathfrak{p}}(\mathrm{M}_\mathfrak{p}) \geqslant r-1$ for every $\mathfrak{p} \supset \mathfrak{J}$, there exists by the induction hypothesis an M-regular sequence $(g_i)_{1\leqslant i\leqslant r-1}$ of elements of $\mathfrak{J}$. Set $\mathrm{N} = \mathrm{M}/(g_1\mathrm{M} + \ldots + g_{r-1}\mathrm{M})$; for every $\mathfrak{p} \supset \mathfrak{J}$, the images of the g_i in $\mathrm{A}_\mathfrak{p}$ belong to the maximal ideal $\mathfrak{p}\mathrm{A}_\mathfrak{p}$ and form an $\mathrm{M}_\mathfrak{p}$-regular sequence $(\mathbf{0}, 15.1.14)$. It follows from $(\mathbf{0}, 16.4.6)$ that $\operatorname{prof}_{\mathrm{A}_\mathfrak{p}}(\mathrm{N}_\mathfrak{p}) = \operatorname{prof}_{\mathrm{A}_\mathfrak{p}}(\mathrm{M}_\mathfrak{p}) - (r-1) \geqslant 1$, so everything reduces to proving the lemma for $r = 1$. The hypothesis then means that, for every $\mathfrak{p} \supset \mathfrak{J}$, $\mathfrak{p}\mathrm{A}_\mathfrak{p}$ is not associated with $\mathrm{M}_\mathfrak{p}$ $(\mathbf{0}, 16.4.6)$; hence (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 2, prop. 5), $\mathfrak{p}$ is not associated with M. In other words, $\mathfrak{J}$ is contained in none of the prime ideals of $\operatorname{Ass}(\mathrm{M})$, and therefore is not contained in their union (Bourbaki, *Alg. comm.*, chap. II, § 1, n° 1, prop. 2). Since this union is the set of elements that are not M-regular (Bourbaki, *Alg. comm.*, chap. IV, § 1, cor. 2 of prop. 2), this proves the lemma. □

Proposition (19.9.3). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a coherent $\mathscr{O}_{\mathrm{X}}$-Module, Y a closed subset of X, y a point of Y, n an integer > 0. The following conditions are equivalent:*

a) *There exists an open neighbourhood U of y in X such that $\operatorname{prof}_{\mathrm{Y}\cap\mathrm{U}}(\mathscr{F}|\mathrm{U}) \geqslant n$.*
b) *There exists an open neighbourhood U of y in X and an $(\mathscr{F}|\mathrm{U})$-regular sequence of n sections f_i of $\mathscr{O}_{\mathrm{U}}$ above U, such that $f_i(x) = 0$ $(\mathbf{0}_{\mathrm{I}}, 5.5.1)$ for $1 \leqslant i \leqslant n$ at every point $x \in \mathrm{U}\cap\mathrm{Y}$.*
c) *$\operatorname{prof}_{\mathrm{Y},y}(\mathscr{F}) \geqslant n$.*

Moreover, the set of $y \in \mathrm{Y}$ satisfying these conditions is open in Y.

Proof. The last assertion follows trivially from a). The equivalence of a) and b) follows from (19.9.2), applied to an affine open neighbourhood of y in X, and from $(\mathbf{0}, 15.1.14)$. It is clear that a) implies c), since every open neighbourhood of y in X contains $\operatorname{Spec}(\mathscr{O}_{\mathrm{X},y})$; let us prove conversely that c) implies b). It follows from c) and the definition (19.9.1) that we can apply (19.9.2) to the $\mathscr{O}_{\mathrm{X},y}$- IV-4 | 223
module $\mathscr{F}_y$ and the ideal $\mathscr{J}_y$ of $\mathscr{O}_{\mathrm{X},y}$, where $\mathscr{J}$ is an ideal defining Y near y. Hence there exists an $\mathscr{F}_y$-regular sequence $(t_i)_{1\leqslant i\leqslant n}$ of n elements of $\mathscr{J}_y$. There is therefore an open neighbourhood U of y in X and a sequence $(f_i)_{1\leqslant i\leqslant n}$ of sections of $\mathscr{J}$ above U such that the t_i are the germs of the f_i at y; using $(\mathbf{0}, 15.2.4)$, one deduces that there exists an open neighbourhood $\mathrm{U}' \subset \mathrm{U}$ of y in X such that the $f_i|\mathrm{U}'$ form an $(\mathscr{F}|\mathrm{U}')$-regular sequence. □

Corollary (19.9.4). — *With the notations of ((19.9.3)), the function $y \mapsto \operatorname{prof}_{\mathrm{Y},y}(\mathscr{F})$ is lower semi-continuous on Y.*

Proposition (19.9.5). — *With the notations of ((19.9.3)), let X' be a locally Noetherian prescheme, $g : X' \to X$ a flat morphism, $Y' = g^{-1}(Y)$, $\mathscr{F}' = g^*(\mathscr{F})$. Then, for every point $y' \in Y'$ above y, we have $\mathrm{prof}_{Y',y'}(\mathscr{F}') = \mathrm{prof}_{Y,y}(\mathscr{F})$.*

Proof. Indeed, for every $z' \in \mathrm{Spec}(\mathscr{O}_{X',y'})$, it follows from (6.3.1) that $\mathrm{prof}(\mathscr{F}'_{z'}) \geqslant \mathrm{prof}(\mathscr{F}_{g(z')})$, and if z'' is a maximal point of the fibre $g^{-1}(g(z'))$ that is a generization of z' (and hence also belongs to $\mathrm{Spec}(\mathscr{O}_{X',y'})$), we have (*loc. cit.*) $\mathrm{prof}(\mathscr{F}'_{z''}) = \mathrm{prof}(\mathscr{F}_{g(z')})$. The proposition then follows from the fact that $g(\mathrm{Spec}(\mathscr{O}_{X',y'})) = \mathrm{Spec}(\mathscr{O}_{X,g(y')})$, since g is flat (2.3.4). □

Proposition (19.9.6). — *Let $f : X \to S$ be a morphism locally of finite presentation, Y a closed subset of X, $\mathscr{F}$ a coherent $\mathscr{O}_X$-Module of finite presentation and f-flat, n an integer > 0. For every point $y \in Y$, denoting by $X_{f(y)}$ the fibre of f at the point $f(y)$, by $\mathscr{F}_{f(y)}$ the inverse image of $\mathscr{F}$ in $X_{f(y)}$, and setting $Y_{f(y)} = X_{f(y)} \cap Y$:*

a) $\mathrm{prof}_{Y_{f(y)},y}(\mathscr{F}_{f(y)}) \geqslant n$.
b) *There exists an open neighbourhood U of y in X and a sequence $(g_i)_{1\leqslant i\leqslant n}$ of n sections of $\mathscr{O}_X$ above U, that is transversally $\mathscr{F}$-regular relative to S (cf. (19.2.1) for the terminology) and such that $g_i(x) = 0$ for $1 \leqslant i \leqslant n$ at every point $x \in U \cap Y$.*

The set V_n of $y \in Y$ satisfying these conditions is open in Y; if moreover Y is locally constructible in X, V_n is retrocompact in X ($\mathbf{0}_{\mathrm{III}}$, 9.1.1).

Proof. By virtue of (19.9.3), condition a) is equivalent to the existence of an open neighbourhood U of y in X and of an $(\mathscr{F}_{f(y)}|(U \cap X_{f(y)}))$-regular sequence $(h_i)_{1\leqslant i\leqslant n}$ of sections of $\mathscr{O}_{X_{f(y)}}$ above $U \cap X_{f(y)}$, such that $h_i(z) = 0$ for $1 \leqslant i \leqslant n$ at every point $z \in U \cap Y_{f(y)}$. If we regard Y as a closed sub-prescheme of X having the given closed subset as its underlying space, and denote by $\mathscr{J}$ the quasi-coherent ideal of $\mathscr{O}_X$ defining it, one can further require that $h_i \in \mathscr{J}_{f(y)}$ for $1 \leqslant i \leqslant n$. Replacing U by a smaller affine open neighbourhood of y, one can suppose that the h_i are the images of a sequence $(g_i)_{1\leqslant i\leqslant n}$ of sections of $\mathscr{J}$ above U, so that the germs $(g_i)_y$ belong to the maximal ideal $\mathfrak{m}_y$ of $\mathscr{O}_{X,y}$. The equivalence of a) and b) then follows from (11.3.8), which also proves that the set V_n is open in Y.

It remains to prove the last assertion. Since this again concerns properties local on X, we may suppose that $S = \mathrm{Spec}(A)$ is affine and that X is of finite presentation over S. There then exist a Noetherian subring A_0 of A, a prescheme X_0 of finite type over $S_0 = \mathrm{Spec}(A_0)$, and a coherent $\mathscr{O}_{X_0}$-module $\mathscr{F}_0$ such that $X = X_0 \times_{S_0} S$ and $\mathscr{F} = \mathscr{F}_0 \otimes_{\mathscr{O}_{X_0}} \mathscr{O}_X$ (8.9.1). We may moreover suppose that, if $f_0 : X_0 \to S_0$ is the structural morphism, $\mathscr{F}_0$ is f_0-flat ((11.2.6)). Since Y is constructible, we may further suppose (8.3.11) that $Y = p^{-1}(Y_0)$, where $p : X \to X_0$ is the canonical projection. Then, for every $y \in Y$, (19.9.5) and the transitivity of fibres give

$$\mathrm{prof}_{Y_{f(y)},y}(\mathscr{F}_{f(y)}) = \mathrm{prof}_{(Y_0)_{f_0(y_0)},y_0}((\mathscr{F}_0)_{f_0(y_0)}),$$

where $y_0 = p(y)$. If V_n^0 is the set of $y_0 \in Y_0$ for which the right-hand side is at least n, then $V_n = p^{-1}(V_n^0)$. Since X_0 is Noetherian, V_n^0 is a retrocompact open subset of X_0, and consequently (cf. the proof of (I, 8.2)) V_n is retrocompact in X. □

Corollary (19.9.7). — *Under the general hypotheses of (19.9.6), the function $y \mapsto \mathrm{prof}_{Y_{f(y)},y}(\mathscr{F}_{f(y)})$ is lower semi-continuous in Y. If moreover Y is locally constructible, this function is locally constructible.*

Proposition (19.9.8). — *Let $f : X \to S$ be a morphism locally of finite presentation, Z a closed subset of X, $\mathscr{F}$ an $\mathscr{O}_X$-module of finite presentation and f-flat, and $\mathscr{G}$ a quasi-coherent $\mathscr{O}_S$-module. Suppose that, for every $x \in Z$, we have $\mathrm{prof}((\mathscr{F}_{f(x)})_x) \geqslant 1$ (resp. $\mathrm{prof}((\mathscr{F}_{f(x)})_x) \geqslant 2$). Then, if $j : X - Z \to X$ is the canonical injection and if we set $\mathscr{H} = \mathscr{F} \otimes_{\mathscr{O}_X} f^*(\mathscr{G})$, the canonical homomorphism $\rho_{\mathscr{H}} : \mathscr{H} \to j_*(j^*(\mathscr{H}))$ relative to j is injective (resp. bijective).*

Proof. The question is evidently local on X and S, and it suffices to prove the proposition in a neighbourhood of a point $x \in Z$. In other words, we may restrict to the case where X and S are affine. By (19.9.6), we may suppose that there exists a section g_1 of $\mathscr{O}_X$ over X that is transversally $\mathscr{F}$-regular relative to S (resp. a transversally $\mathscr{F}$-regular sequence (g_1, g_2) of two sections of $\mathscr{O}_X$ over

X relative to S) such that, if Z' is the set of $x' \in X$ for which $g_1(x') = 0$ (resp. $g_1(x') = g_2(x') = 0$), then $Z \subset Z'$.

This being so, return to the proof of (19.9.8). By (19.9.6), we still have $\mathrm{prof}((\mathscr{F}_{f(x)})_x) \geqslant 1$ (resp. $\mathrm{prof}((\mathscr{F}_{f(x)})_x) \geqslant 2$) for every $x \in Z'$. If the proposition is proved after replacing the pair (X, Z) by each of (X, Z') and $(X - Z, Z' \cap (X - Z))$, it is immediate that it also holds for (X, Z); we may therefore restrict to proving it for X and Z'. In other words, we may restrict to proving (19.9.8) when $S = \mathrm{Spec}(A)$ and $X = \mathrm{Spec}(B)$, where B is an A-algebra of finite presentation, $Z = V(\mathfrak{J})$ for an ideal $\mathfrak{J}$ of finite type of B, $\mathscr{F} = \widetilde{M}$, and $\mathscr{G} = \widetilde{N}$, where N is an A-module and M is a finitely presented B-module that is flat as an A-module. We may moreover reduce to the case where N is a finitely presented A-module. Indeed, N is the filtered inductive limit of a system of finitely presented A-modules N_α (by taking a double inductive limit). If $\mathscr{G}_\alpha = \widetilde{N_\alpha}$, then $\mathscr{H}$ is the inductive limit of the $\mathscr{H}_\alpha = \mathscr{F} \otimes_{\mathscr{O}_X} f^*(\mathscr{G}_\alpha)$; since j_* and j^* commute with inductive limits and $\varinjlim$ is an exact functor in the category of quasi-coherent modules, the proposition for every $\mathscr{H}_\alpha$ implies it for $\mathscr{H}$.

It further suffices to prove that the canonical homomorphism

$$\Gamma(X, \mathscr{H}) \longrightarrow \Gamma(X - Z, \mathscr{H})$$

is injective (resp. bijective). There then exist a Noetherian subring A_0 of A, a finite-type A_0-algebra B_0, an ideal $\mathfrak{J}_0$ of B_0, a finitely generated B_0-module M_0 that is flat as an A_0-module, and an A_0-module N_0, such that $B = B_0 \otimes_{A_0} A$, $\mathfrak{J} = \mathfrak{J}_0 B$, $M = M_0 \otimes_{A_0} A$, and $N = N_0 \otimes_{A_0} A$ (8.9.1), (8.5.11), **IV-4** | 225
and ((11.2.7)). Let (A_λ) be the family of finite-type A_0-subalgebras of A, so that $A = \varinjlim A_\lambda$; set

$$B_\lambda = B_0 \otimes_{A_0} A_\lambda, \qquad \mathfrak{J}_\lambda = \mathfrak{J}_0 B_\lambda, \qquad M_\lambda = M_0 \otimes_{A_0} A_\lambda = M_0 \otimes_{B_0} B_\lambda,$$
$$N_\lambda = N_0 \otimes_{A_0} A_\lambda, \qquad S_\lambda = \mathrm{Spec}(A_\lambda), \quad X_\lambda = \mathrm{Spec}(B_\lambda), \quad Z_\lambda = V(\mathfrak{J}_\lambda).$$

If $p_\lambda : X \to X_\lambda$ is the canonical projection, then $Z = p_\lambda^{-1}(Z_\lambda)$ and $X - Z = p_\lambda^{-1}(X_\lambda - Z_\lambda)$. Since $X_\lambda - Z_\lambda$ is quasi-compact and quasi-separated, it follows from (8.5.2) that, on setting $\mathscr{F}_\lambda = \widetilde{M_\lambda}$ and $\mathscr{H}_\lambda = (M_\lambda \otimes_{A_\lambda} N_\lambda)^\sim$, the homomorphisms

$$\varinjlim \Gamma(X_\lambda, \mathscr{H}_\lambda) \longrightarrow \Gamma(X, \mathscr{H}), \qquad \varinjlim \Gamma(X_\lambda - Z_\lambda, \mathscr{H}_\lambda) \longrightarrow \Gamma(X - Z, \mathscr{H})$$

are bijective. By exactness of $\varinjlim$, it therefore suffices to prove that, for every sufficiently large λ, the canonical homomorphism $\Gamma(X_\lambda, \mathscr{H}_\lambda) \to \Gamma(X_\lambda - Z_\lambda, \mathscr{H}_\lambda)$ is injective (resp. bijective).

For every $x \in Z$, if x_λ is the projection of x in Z_λ, then (4.2.7) and (6.7.1) give

$$\mathrm{prof}((\mathscr{F}_{f(x)})_x) = \mathrm{prof}(((\mathscr{F}_\lambda)_{f_\lambda(x_\lambda)})_{x_\lambda}).$$

Thus, to reduce to proving (19.9.8) when A is Noetherian, it remains to prove that, for every sufficiently large λ, $\mathrm{prof}(((\mathscr{F}_\lambda)_{f_\lambda(x_\lambda)})_{x_\lambda}) \geqslant 1$ (resp. $\geqslant 2$) for every $x_\lambda \in Z_\lambda$. Let Z'_λ be the set of points $x_\lambda \in Z_\lambda$ such that, for every generization z_λ of x_λ in $Z_\lambda \cap (X_\lambda)_{f_\lambda(x_\lambda)}$, one has $\mathrm{prof}(((\mathscr{F}_\lambda)_{f_\lambda(x_\lambda)})_{z_\lambda}) \geqslant 1$ (resp. $\geqslant 2$). If $p_{\lambda\mu} : Z_\mu \to Z_\lambda$ is the canonical projection for $\lambda \leqslant \mu$, the hypothesis and (19.9.5) give $Z'_\mu = p_{\lambda\mu}^{-1}(Z'_\lambda)$ and $Z = p_\lambda^{-1}(Z'_\lambda)$. On the other hand, (19.9.6) shows that Z'_λ is open in Z_λ; the assertion therefore follows from (8.3.4).

Finally, when A is Noetherian, the flatness hypothesis and (6.3.1) give $\mathrm{prof}_Z(\mathscr{H}) \geqslant 1$ (resp. $\mathrm{prof}_Z(\mathscr{H}) \geqslant 2$); in this case the conclusion of (19.9.8) follows from (5.10.2) (resp. (5.10.4)). ☐

Remark (19.9.9). — The same proof, combined with the results of chap. III, 3rd part on depth and local cohomology, allows one to establish the following generalisation of (19.9.8):

Under the general conditions of (19.9.8), suppose that, for every $x \in Z$, we have $\mathrm{prof}((\mathscr{F}_{f(x)})_x) \geqslant k$; then the canonical homomorphism

$$H^i(X, \mathscr{H}) \longrightarrow H^i(X - Z, \mathscr{H}|(X - Z))$$

is bijective for $i \leqslant k - 2$ and injective for $i = k - 1$ (which, expressed using the general cohomological notion of depth introduced in chap. III, is also written $\mathrm{prof}_Z(\mathscr{H}) \geqslant k$).

§20. Meromorphic functions and pseudo-morphisms

20.0. Introduction Most of the concepts and results of §§20 and 21 relate directly to Chapter I and hardly depend on Chapters II to IV, except for the occasional use of the notion of depth and of a regular local ring (in (20.6), (21.11), (21.13), and (21.15)), of Zariski's "Main Theorem" in (20.4) and (21.12), and of the properties of transversally regular immersions in (20.6) and (21.15).

In §20, we introduce several variants of the concept of a rational map, already studied in (**I**, 7) **IV-4** | 226
from a point of view still fairly close to the classical point of view, and for this reason quite ill-suited to the case of not necessarily reduced preschemes. The notions and results of §20 are used in §21 (n^{os}(21.1) and (21.7)) to develop the general notion of a divisor and its most elementary properties. This notion is especially convenient when the local rings of the preschemes considered are Noetherian and integrally closed, and especially when they are also *factorial* ((21.6) and (21.7)), because in the latter case it is identified with the notion of a *codimension*-1 *cycle* (a linear combination of irreducible subpreschemes of codimension 1). In (21.9), we determine the divisors on a Noetherian prescheme of dimension 1 but not necessarily normal, which is useful for various applications. Numbers (21.11) and (21.12) give two important theorems, due respectively to Auslander–Buchsbaum and Van der Waerden, and relating to the notion of a factorial ring (numbers (21.9), (21.11), and (21.12) are independent of one another). In numbers (21.13) and (21.14), which are likewise independent of the preceding three, we study a useful variant of the notion of a factorial local ring, namely that of a *parafactorial local ring*; this notion is introduced in particular in [**?**] in the development of comparison theorems for the Picard group of a projective prescheme X over a field k and of a "hyperplane section". We will see in (21.14.1) (Ramanujam–Samuel theorem) that the local parafactorial rings are much more numerous than one would have expected *a priori*.

In (20.5), (20.6), and (21.15), we review the previous notions but from a point of view "relative" to a fixed base prescheme. For the moment these notions are used only rather rarely; in particular, the notion of a relative divisor is hardly used except for positive divisors, and in that case it can be described advantageously without recourse to relative meromorphic functions, by using the notion of a transversally regular immersion of codimension 1. It is therefore advisable to omit these sections on a first reading.

20.1. Meromorphic functions

(20.1.1). Let $(X, \mathscr{O}_X)$ be a ringed space, and let $\mathscr{S}$ be a subsheaf *of sets* of $\mathscr{O}_X$. For every open U of X, consider the *ring of fractions* $\Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}]$ (Bourbaki, *Alg. comm.*, chap. II, §2, $\mathrm{n}^{\mathrm{o}}1$). It is immediate that the map $U \mapsto \Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}]$ is a *presheaf of rings* ((**0**, 1.5.1) and (**0**, 1.5.7)). We denote by $\mathscr{O}_X[\mathscr{S}^{-1}]$ the *sheaf of rings* associated to this presheaf and we say that this is the *sheaf of rings of fractions of $\mathscr{O}_X$ with denominators in $\mathscr{S}$*; this is a *flat* $\mathscr{O}_X$-module. It is immediate that for every $x \in X$, we have a canonical isomorphism

$$(\mathscr{O}_X[\mathscr{S}^{-1}])_x \simeq \mathscr{O}_{X,x}[\mathscr{S}_x^{-1}], \tag{20.1.1.1}$$

since the reasoning of (**0**, 1.4.5) generalizes immediately in the case where we have an inductive system $(A_\alpha, \phi_{\beta\alpha})$ of rings, and for each index α a subset S_α of A_α such that $\phi_{\beta\alpha}(S_\alpha) \subset S_\beta$ for $\alpha \leqslant \beta$; **IV-4** | 227
we then take S to be the inductive limit, in $A = \varinjlim A_\alpha$, of the inductive system of subsets (S_α).

(20.1.2). Now let $\mathscr{F}$ be an $\mathscr{O}_X$-module. We then set

$$\mathscr{F}[\mathscr{S}^{-1}] = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{O}_X[\mathscr{S}^{-1}], \tag{20.1.2.1}$$

and we say that this is the *sheaf of modules of fractions of $\mathscr{F}$ with denominators in $\mathscr{S}$*; it is immediate that it is associated to the presheaf of modules $U \mapsto \Gamma(U, \mathscr{F})[\Gamma(U, \mathscr{S})^{-1}]$, and that for every $x \in X$, we have a canonical isomorphism

$$(\mathscr{F}[\mathscr{S}^{-1}])_x \simeq \mathscr{F}_x[\mathscr{S}_x^{-1}]. \tag{20.1.2.2}$$

(20.1.3). We will focus here on the case where $\mathscr{S}$ is the subsheaf $\mathscr{S}(\mathscr{O}_X)$ of $\mathscr{O}_X$ such that for every open U, $\Gamma(U, \mathscr{S})$ is the *set of regular elements* of the ring $\Gamma(U, \mathscr{O}_X)$; it is immediate that it is a sheaf (and not merely a presheaf), since regularity of a section of $\mathscr{O}_X$ over U can be checked "fibre by fibre" (that is, its germ at x is regular in $\mathscr{O}_{X,x}$ for every $x \in U$); in other words, $\mathscr{S}(\mathscr{O}_X)_x$ is precisely the set of regular elements of $\mathscr{O}_{X,x}$. The corresponding sheaf of rings

$$\mathscr{M}_X = \mathscr{O}_X[\mathscr{S}^{-1}]$$

is called the *sheaf of germs of meromorphic functions on* X, and the sections of $\mathscr{M}_X$ over X are called the *meromorphic functions on* X; they form a ring which we denote by $M(X)$. For every $\mathscr{O}_X$*-module* $\mathscr{F}$,

$$\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_X = \mathscr{F}[\mathscr{S}^{-1}]$$

is also denoted $\mathscr{M}_X(\mathscr{F})$ and called the *sheaf of germs of meromorphic sections of* $\mathscr{F}$; its sections over X form an $M(X)$-module denoted by $M(X, \mathscr{F})$, whose elements are called *meromorphic sections of* $\mathscr{F}$ *over* X. These definitions imply that for every open U of X, we have a canonical isomorphism $\mathscr{M}_X(\mathscr{F})|U \simeq \mathscr{M}_U(\mathscr{F}|U)$, in particular $\mathscr{M}_X|U \simeq \mathscr{M}_U$.

(20.1.3.1). If X is a *reduced prescheme*, then we note that if an element $s \in \Gamma(U, \mathscr{O}_X)$ is such that $s_\xi \neq 0$ for every maximal point ξ of U, then s is *regular*. Indeed, if $st = 0$ for a $t \in \Gamma(U, \mathscr{O}_X)$, then we have $s_\xi t_\xi = 0$, so $t_\xi = 0$ since $\mathscr{O}_{X,\xi}$ is a field, and say that $t_\xi = 0$ for every maximal point ξ of X means that $t = 0$: we are immediately reduced to the case where U is affine, and an element of a reduced ring which belongs to all the minimal prime ideals is zero by definition. The converse is true if the set of irreducible components of X is *locally finite*. We immediately reduce to the case where $X = \operatorname{Spec}(A)$ is affine; if $\mathfrak{p}_i$ $(1 \leqslant i \leqslant n)$ are the minimal prime ideals of A and if $s \in \mathfrak{p}_i$ for an index i, then there exists $t \in A$ such that $t \in \mathfrak{p}_j$ for $j \neq i$ and $t \notin \mathfrak{p}_i$ (Bourbaki, *Alg. comm.*, chap. II, §1, n°1, Prop. 1); so we have $st \in \mathfrak{p}_i$ for all i, and as a result $st = 0$ since A is reduced; s is therefore nonregular.

(20.1.4). For every open U of X, the homomorphism $t \mapsto t/1$ from $\Gamma(U, \mathscr{O}_X)$ to $\Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}]$ (which is none other than the *total ring of fractions* of $\Gamma(U, \mathscr{O}_X)$) is injective; these homomorphisms thus define a *canonical injective homomorphism* **IV-4** | 228

(20.1.4.1) $$i : \mathscr{O}_X \longrightarrow \mathscr{M}_X$$

which allows us to identify $\mathscr{O}_X$ with a subsheaf of $\mathscr{M}_X$. Given a meromorphic function $\phi \in M(X)$, we say that ϕ is *defined* on an open U of X if $\phi|U$ is a *section of* $\mathscr{O}_U$ over U; the axioms of sheaves show that there is, for a given section ϕ, a *largest* open on which ϕ is defined; we call this the *domain of definition* of ϕ and denote it by $\operatorname{dom}(\phi)$.

(20.1.5). For every $\mathscr{O}_X$-module $\mathscr{F}$, we obtain from (20.1.4.1) a di-homomorphism consisting of i and the homomorphism of sheaves of additive groups

(20.1.5.1) $$1_{\mathscr{F}} \otimes i : \mathscr{F} \longrightarrow \mathscr{M}_X(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_X.$$

We note that the latter is not injective in general; when it is injective, we say that $\mathscr{F}$ is *strictly torsion-free*: this means that for every open U of X and every section $s \in \Gamma(U, \mathscr{O}_X)$ which is a regular element in this ring the homothety $z \mapsto sz$ of $\Gamma(U, \mathscr{F})$ is injective; this condition is evidently satisfied if $\mathscr{F}$ is *locally free*.

Proposition (20.1.6). — *Let X be a locally Noetherian prescheme, $\mathscr{F}$ a quasi-coherent $\mathscr{O}_X$-module. For $\mathscr{F}$ to be strictly torsion-free, it is necessary and sufficient that* $\operatorname{Ass}(\mathscr{F}) \subset \operatorname{Ass}(\mathscr{O}_X)$.

Proof. We immediately reduce to the case where $X = \operatorname{Spec}(A)$ is affine, $\mathscr{F} = \widetilde{M}$, and we know that the elements s of A belonging to an ideal of $\operatorname{Ass}(M)$ are exactly those for which the homothety $z \mapsto sz$ is not injective (Bourbaki, *Alg. comm.*, chap. IV, §1, n°1, cor. 2 of prop. 2). ☐

(20.1.7). If u is a section of $\mathscr{M}_X(\mathscr{F})$ over X, then we say that u is *defined* at a point $x \in X$ if there exists an open neighbourhood V of x in X such that $u|V$ is the image of a section of $\mathscr{F}$ over V under the di-homomorphism (20.1.5.1). We say that u is *defined* on an open U of X if it is defined at every point in U; there is still a larger open in which u is defined, called the *domain of definition* of u and denoted $\operatorname{dom}(u)$. When $\mathscr{F}$ is strictly torsion-free, so that $\mathscr{F}$ is identified by (20.1.5.1) with a subsheaf of $\mathscr{M}_X(\mathscr{F})$, saying that u is defined on U means that $u|U$ is a *section of* $\mathscr{F}$ *over* U.

(20.1.8). In accordance with the general notation of ($\mathbf{0_I}$, 5.4.7), we denote by $\mathscr{M}_X^*$ the sheaf of multiplicative groups such that $\Gamma(U, \mathscr{M}_X^*)$ is (for every open U of X) the group of *invertible elements* of $\Gamma(U, \mathscr{M}_X)$. This sheaf is none other than the sheaf $\mathscr{S}(\mathscr{M}_X)$ defined in (20.1.3): indeed, if $s \in \Gamma(U, \mathscr{S}(\mathscr{M}_X))$, then for every $x \in U$, there exists an open neighbourhood $V \subset U$ of x such that $s|V$ is a regular element in the *total ring of fractions* of $\Gamma(V, \mathscr{O}_X)$, and we know that such an element is necessarily invertible in this ring of fractions. We say that the sections of $\mathscr{M}_X^*$ over X are the *regular meromorphic functions* (note that we are deviating here from the terminology followed by certain authors, who call "regular" meromorphic functions those which are *sections of* $\mathscr{O}_X$, identified with a subsheaf of $\mathscr{M}_X$).

Let $\mathscr{L}$ be an *invertible* $\mathscr{O}_X$*-module* ($\mathbf{0_I}$, 5.4.1); then it is clear that $\mathscr{M}_X(\mathscr{L}) = \mathscr{L} \otimes_{\mathscr{O}_X} \mathscr{M}_X$ is an *invertible* $\mathscr{M}_X$*-module*. Let U be an open such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_U$; as every automorphism of $\mathscr{M}_U$ is multiplication by an invertible element of $\Gamma(U, \mathscr{M}_X)$ ($\mathbf{0_I}$, 5.4.7), it amounts to the same thing to say that a section $s \in \Gamma(U, \mathscr{M}_X(\mathscr{L}))$ has an invertible image in $\Gamma(U, \mathscr{M}_X)$ under *some* isomorphism or under *every* isomorphism onto $\Gamma(U, \mathscr{M}_X)$; we say in this case that s is a *regular meromorphic section of* $\mathscr{L}$ over U; a section s of $\mathscr{L}$ over X is called a *regular meromorphic section of* $\mathscr{L}$ if, for every open U such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_U$, $s|U$ is a regular meromorphic section of $\mathscr{L}$ over U. We denote by $(\mathscr{M}_X(\mathscr{L}))^*$ the subsheaf of $\mathscr{M}_X(\mathscr{L})$ such that for every open U, $\Gamma(U, (\mathscr{M}_X(\mathscr{L}))^*)$ is the set of regular meromorphic sections of $\mathscr{L}$ over U. Let s be a meromorphic section of $\mathscr{L}$ over X (i.e. a section of $\mathscr{M}_X(\mathscr{L})$); it defines a homomorphism $h_s : \mathscr{M}_X \to \mathscr{M}_X(\mathscr{L})$ which sends every section t of $\mathscr{M}_X$ over an open U to $(s|U)t$. It follows immediately from the above that for s to be *regular*, it is necessary and sufficient for h_s to be *injective*, and in fact h_s is then a *bijective* homomorphism from $\mathscr{M}_X$ to $\mathscr{M}_X(\mathscr{L})$, and its restriction to $\mathscr{M}_X^*$ is a bijection to $(\mathscr{M}_X(\mathscr{L}))^*$. We conclude that the homothety $t \mapsto ts$ is an isomorphism from $M(X)$ to $M(X, \mathscr{L})$. **IV-4** | 229

(20.1.9). Let s be a regular meromorphic section of an invertible $\mathscr{O}_X$-module $\mathscr{L}$ over X; then for every $\mathscr{O}_X$-module $\mathscr{F}$, s similarly defines a homomorphism $h_s \otimes 1_{\mathscr{F}} : \mathscr{M}_X(\mathscr{F}) \to \mathscr{M}_X(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L})$, which is again *bijective*.

(20.1.10). Let s be a meromorphic section of an invertible $\mathscr{O}_X$-module $\mathscr{L}$ over X; for s to be regular, it is necessary and sufficient for there to exist a meromorphic section s' of $\mathscr{L}^{-1}$ over X such that the canonical image of $s \otimes s'$ in $\mathscr{M}_X$ ($\mathbf{0_I}$, 5.4.3) is the unit section, and this section s' is then unique: indeed, the necessity of the local existence of such a section is evident, and its local uniqueness implies its global (and unique) existence; moreover, the existence of s' is trivially sufficient for s to be regular. We will take $s' = s^{-1}$.

Finally, if $\mathscr{L}'$ is a second invertible $\mathscr{O}_X$-module, s (resp. s') a regular meromorphic section of $\mathscr{L}$ (resp. $\mathscr{L}'$) over X, then $s \otimes s'$ is evidently a regular meromorphic section of $\mathscr{L} \otimes \mathscr{L}'$ over X.

(20.1.11). If $f : X' \to X$ is a morphism of ringed spaces, then there is in general no natural map sending a meromorphic function on X to a meromorphic function on X'. For example, if X is the spectrum of a local integral domain A and X' is the spectrum of its residue field k, then there is no natural homomorphism from the field of fractions K of A to k, and we can only send an element of K to an element of k if it is already in A.

In general, if $f = (\psi, \theta)$, then for every open U of X, denote by $\mathscr{S}_f(U)$ the set of *regular* sections $s \in \Gamma(U, \mathscr{O}_X)$ such that the image of s under

$$\Gamma(\theta^\#) : \Gamma(U, \mathscr{O}_X) \longrightarrow \Gamma(f^{-1}(U), \mathscr{O}_{X'})$$

is a *regular* section. It is immediate that $U \mapsto \mathscr{S}_f(U)$ is a *subsheaf* of the sheaf of sets $\mathscr{S}(\mathscr{O}_X)$, which we denote by $\mathscr{S}_f$. We set $\mathscr{M}_f = \mathscr{O}_X[\mathscr{S}_f^{-1}]$; this is a subsheaf of rings of $\mathscr{M}_X$, and we canonically obtain from $\theta^\# : \psi^*(\mathscr{O}_X) \to \mathscr{O}_{X'}$ a homomorphism of sheaves of rings $\theta'^\# : \psi^*(\mathscr{M}_f) \to \mathscr{M}_{X'}$ extending $\theta^\#$ (Bourbaki, *Alg. comm.*, chap. II, §2, n°1, prop. 2); hence, recalling that $f^*(\mathscr{M}_f) = \psi^*(\mathscr{M}_f) \otimes_{\psi^*(\mathscr{O}_X)} \mathscr{O}_{X'}$, we get a canonical homomorphism of $\mathscr{O}_{X'}$-algebras **IV-4** | 230

$$f^*(\mathscr{M}_f) \longrightarrow \mathscr{M}_{X'}. \tag{20.1.11.1}$$

For every meromorphic function ϕ on X that *is a section of* $\mathscr{M}_f$, $\Gamma(\theta'^\#)(\phi)$ is a meromorphic function on X', called the *inverse image of* ϕ *under* f, and denoted by $\phi \circ f$ if there is no cause for confusion.

Similarly, if $\mathscr{F}$ is an $\mathscr{O}_X$-module, then we set $\mathscr{M}_f(\mathscr{F}) = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_f$, and we immediately obtain from $\theta'^{\#}$ a canonical homomorphism (which is also written as $u \mapsto u \circ f$)

$$\Gamma(X, \mathscr{M}_f(\mathscr{F})) \longrightarrow \Gamma(X', \mathscr{M}_{X'}(f^*(\mathscr{F}))).$$

In addition, if $u \in \Gamma(X, \mathscr{M}_f(\mathscr{F}))$ is defined (20.1.7) at a point x, then u coincides, on a neighbourhood U of x, with a section of the form $\sum_i h_i \otimes (t_i/s_i)$, where the h_i belong to $\Gamma(U, \mathscr{F})$, the t_i to $\Gamma(U, \mathscr{O}_X)$, and the s_i to $\Gamma(U, \mathscr{S}_f)$. As by hypothesis the images of the s_i in $\Gamma(f^{-1}(U), \mathscr{O}_{X'})$ are regular, we see that $u \circ f$ is defined at every point of $f^{-1}(U)$; in other words, we have

$$f^{-1}(\mathrm{dom}(u)) \subset \mathrm{dom}(u \circ f). \tag{20.1.11.2}$$

We will see later (20.6.5, (ii)) examples (with $\mathscr{F} = \mathscr{O}_X$) where the two sides of (20.1.11.2) can be different.

Consider in particular the case where $\mathscr{M}_f = \mathscr{M}_X$; then, if $\mathscr{L}$ is an invertible $\mathscr{O}_X$-module, the image in $\mathscr{M}_{X'}(f^*(\mathscr{L}))$, under $\Gamma(\theta'^{\#})$, of a *regular* meromorphic section of $\mathscr{L}$ over X (20.1.8) is a *regular* meromorphic section of $f^*(\mathscr{L})$ over X', as it follows immediately from the definition of its sections, and from the fact that a homomorphism of rings sends an invertible element to an invertible element.

Let $f' : X'' \to X'$ be a second morphism of ringed spaces, and suppose that $\mathscr{M}_f = \mathscr{M}_X$ and $\mathscr{M}_{f'} = \mathscr{M}_{X'}$; then, if we set $f'' = f \circ f'$, we also have $\mathscr{M}_{f''} = \mathscr{M}_X$, and we immediately see that for every meromorphic section u of $\mathscr{F}$ over X, we have $u \circ f'' = (u \circ f) \circ f'$.

Proposition (20.1.12). — *If the morphism $f : X' \to X$ is flat* ($\mathbf{0_I}$, 6.7.1), *then we have $\mathscr{M}_f = \mathscr{M}_X$, and the homomorphism $\phi \mapsto \phi \circ f$ is defined on all of $M(X)$. In addition, if f is a (flat) morphism of locally ringed spaces, then we have* $\mathrm{dom}(\phi \circ f) = f^{-1}(\mathrm{dom}(\phi))$*; if in addition f is surjective (thus faithfully flat), then the homomorphism $\phi \mapsto \phi \circ f$ is injective.*

Proof. The first assertion follows from the fact that, if B is an A-algebra which is a flat A-module, then every element of A not a divisor of 0 in A is not a divisor of 0 in B ($\mathbf{0_I}$, 6.3.4). To prove the other assertions, note that, for every $x' \in X'$, if $x = f(x')$, then $\mathscr{O}_{X',x'}$ is a flat $\mathscr{O}_{X,x}$-module, and as the homomorphism $\mathscr{O}_{X,x} \to \mathscr{O}_{X',x'}$ is *local* by hypothesis, it is injective (($\mathbf{0_I}$, 6.5.1) and ($\mathbf{0_I}$, 6.6.2)); if we set $A = \mathscr{O}_{X,x}$, $B = \mathscr{O}_{X',x'}$, such that A identifies with a subring of B, then $(f^*(\mathscr{M}_X))_{x'}$ is equal to $S^{-1}A \otimes_A B = S^{-1}B$, where S is the set of regular elements of A, $(\mathscr{M}_{X'})_{x'}$ is equal to $T^{-1}B$, where T is the set of regular elements of B, and as we have seen that $S \subset T$, the homomorphism $S^{-1}B \to T^{-1}B$ **IV-4** | 231 is injective; in other words, this proves that the homomorphism (20.1.11.1) $f^*(\mathscr{M}_X) \to \mathscr{M}_{X'}$ is *injective* (hence the last assertion of the statement). The quotient $f^*(\mathscr{M}_X)/\mathscr{O}_{X'}$ identifies with an $\mathscr{O}_{X'}$-submodule of $\mathscr{M}_{X'}/\mathscr{O}_{X'}$, and $(f^*(\mathscr{M}_X)/\mathscr{O}_{X'})_{x'}$ identifies with $(\mathscr{M}_X/\mathscr{O}_X)_x \otimes_{\mathscr{O}_{X,x}} \mathscr{O}_{X',x'}$. Then suppose that $x \notin \mathrm{dom}(\phi)$; the image of ϕ_x in $(\mathscr{M}_X/\mathscr{O}_X)_x$ is therefore $\neq 0$; by faithful flatness, we deduce that it is the same for the image of $(\phi \circ f)_{x'}$, so $x' \notin \mathrm{dom}(\phi \circ f)$, which finishes the proof. □

Remark (20.1.13). — Let X be a *reduced* complex analytic space; then the notion of a meromorphic function on X defined above coincides with the usual notion. Consider on the other hand a prescheme Y, locally of finite type over the field $\mathbf{C}$; we then know that we can associate to Y an analytic space Y^{an} having the same underlying topological space, and the canonical morphism $f : Y^{\mathrm{an}} \to Y$ is *flat* [**?**]; by virtue of (20.1.12), the canonical homomorphism $u \mapsto u \circ f$ from $M(Y)$ to $M(Y^{\mathrm{an}})$ is therefore always defined and is injective; but it is not *surjective* in general. For example, when $Y = \mathbf{V}^r_{\mathbf{C}}$ ($\mathbf{Err_{III}}$, 14) is affine r-space over $\mathbf{C}$, $M(Y)$ canonically identifies with the field $R(Y)$ of rational functions on Y (20.2.13, (i)), while $M(Y^{\mathrm{an}})$ is the field of usual meromorphic functions on $\mathbf{C}^r$. Because of this fact, it is often preferable, in algebraic geometry, to abstain from the terminology introduced in this section, and to use the equivalent terminology of "pseudo-function" which will be defined below.

20.2. Pseudo-morphisms and pseudo-functions The only ringed spaces discussed in this section are *preschemes*.

(20.2.1). Recall (11.10.2) that in a prescheme X, we say that an open U is *schematically dense* if, for every open V of X, the canonical homomorphism $\Gamma(V, \mathscr{O}_X) \to \Gamma(V \cap U, \mathscr{O}_X)$ is *injective*. **IV-4** | 231

Consider two preschemes X, Y, two *schematically dense* opens U, U' of X; we say that two morphisms $u : U \to Y$, $u' : U' \to Y$ are *equivalent* if there exists an open $U'' \subset U \cap U'$, *schematically dense* in X, such that $u|U'' = u'|U''$. As it results immediately from the definition of schematically dense opens that the intersection of two such opens is again one, it is immediate that the preceding relation is indeed an equivalence relation. An equivalence class for this relation is called a *pseudo-morphism* of X in Y, or a *strict rational application* of X in Y.

If S is a prescheme and X, Y two S-preschemes, we call *pseudo-S-morphism* the equivalence class (for the preceding relation) of an *S-morphism* of a schematically dense open of X in Y. A pseudo-morphism is therefore nothing other than a pseudo-S-morphism for $S = \operatorname{Spec}(\mathbf{Z})$.

We denote by $\operatorname{Ps.hom}(X, Y)$ (resp. $\operatorname{Ps.hom}_S(X, Y)$) the set of pseudo-morphisms (resp. pseudo-S-morphisms) of X in Y.

(20.2.2). It follows from the definition recalled above that if U is a schematically dense open in X, then, for every open V of X, $U \cap V$ is schematically dense in V. If two morphisms $u : U \to Y$, $u' : U' \to Y$ of schematically dense opens of X in Y are equivalent, it follows that their restrictions $u|(V \cap U)$ and $u'|(V \cap U')$ are also equivalent morphisms (for the prescheme induced on V); their class is called the *restriction* to V of the pseudo-morphism ω, the class of u, and we denote this pseudo-morphism $\omega|V$. If $W \subset V$ is an open of X, it is clear that $(\omega|V)|W = \omega|W$. This shows that the restriction maps define a *presheaf* of sets $U \mapsto \operatorname{Ps.hom}(U, Y)$; we note that this presheaf *is not* in general a sheaf (cf. (20.2.16)); the associated sheaf is denoted $\mathscr{P}\!s.\mathrm{hom}(X, Y)$. For pseudo-$S$-morphisms, we see that $U \mapsto \operatorname{Ps.hom}_S(U, Y)$ is a presheaf of sets, whose associated sheaf we denote $\mathscr{P}\!s.\mathrm{hom}_S(X, Y)$. **IV-4** | 232

If V is a *schematically dense* open in X, then for every open U schematically dense in X, $U \cap V$ is also schematically dense in X, so the map $\omega \mapsto \omega|V$ is a *bijection* of $\operatorname{Ps.hom}(X, Y)$ (resp. $\operatorname{Ps.hom}_S(X, Y)$) onto $\operatorname{Ps.hom}(V, Y)$ (resp. $\operatorname{Ps.hom}_S(V, Y)$).

(20.2.3). Given a pseudo-S-morphism ω of X in Y, we say that it is *defined* at a point $x \in X$ if there exists an open neighbourhood V of x in X, an open $U \subset V$ *containing x and schematically dense in V*, and finally an S-morphism $u : U \to Y$ whose class in $\operatorname{Ps.hom}_S(V, Y)$ is equal to $\omega|V$ (20.2.2); we call the *domain of definition* of ω and denote $\operatorname{dom}_S(\omega)$ (or simply $\operatorname{dom}(\omega)$ if $S = \operatorname{Spec}(\mathbf{Z})$) the set of $x \in X$ where ω is defined; this is evidently an *open* of X. Moreover, for every open W of X, we have

(20.2.3.1) $$\operatorname{dom}_S(\omega|W) = (\operatorname{dom}_S(\omega)) \cap W$$

by virtue of the property of schematically dense opens recalled in (20.2.2).

Proposition (20.2.4). — *Suppose X, Y are S-preschemes such that the structural morphism $Y \to S$ is separated; then, if ω is a pseudo-S-morphism of X in Y, $\operatorname{dom}_S(\omega)$ is the largest of the schematically dense opens U of X such that there exists an S-morphism $u : U \to Y$ belonging to the class ω.*

Proof. It suffices to prove that if U, U' are two schematically dense opens in X such that two S-morphisms $u : U \to Y$ and $u' : U' \to Y$ are equivalent, then the restrictions of u and u' to $U \cap U'$ are equal. But there exists by hypothesis an open $U'' \subset U \cap U'$, schematically dense in X, in which u and u' coincide; as U'' is also schematically dense in $U \cap U'$, our assertion results from (11.10.1, (d)). ☐

Corollary (20.2.5). — *Let S be an S_0-scheme, and let X, Y be two S-preschemes such that the composite morphism $Y \to S \to S_0$ is separated (which implies, by* (**I**, 5.5.1), *that $Y \to S$ is also separated). For every pseudo-S-morphism ω of X in Y, one then has $\operatorname{dom}_S(\omega) = \operatorname{dom}_{S_0}(\omega)$. In particular, if Y is a schema, one has $\operatorname{dom}_S(\omega) = \operatorname{dom}(\omega)$.*

Proof. Indeed, it is clear that $\operatorname{dom}_S(\omega) \subset \operatorname{dom}_{S_0}(\omega)$ without the hypothesis of separation; by virtue of (20.2.4), it suffices to prove that if $u_0 : U_0 \to Y$ is an S_0-morphism of a schematically dense open U_0 of X in Y such that the composed $v_0 : U_0 \xrightarrow{u_0} Y \to S$ coincides with the restriction of the **IV-4** | 233 structural morphism $w_0 : U_0 \to S$ in an open $U \subset U_0$ schematically dense in X, then one has $v_0 = w_0$. But by virtue of the hypothesis that the morphism $S \to S_0$ is separated, this again results from (11.10.1, , d)). □

Corollary (20.2.6). — *Under the hypotheses of (20.2.4), the presheaf* $U \mapsto \mathrm{Ps.hom}_S(U, Y)$ *is a sheaf.*

Proof. Indeed, let U be an open of X, (U_α) a covering of U by opens of X, and suppose given for each α a pseudo-S-morphism ω_α of U_α in Y, such that for every pair of indices α, β, the restrictions (20.2.2) of ω_α and ω_β to $U_\alpha \cap U_\beta$ are equal; by virtue of (20.2.3.1), this implies that $\mathrm{dom}_S(\omega_\alpha) \cap U_\beta = \mathrm{dom}_S(\omega_\beta) \cap U_\alpha$. Let W be the union of the opens $\mathrm{dom}_S(\omega_\alpha)$, and, for each α, let u_α be the unique S-morphism $\mathrm{dom}_S(\omega_\alpha) \to Y$ belonging to the class ω_α (20.2.4); by virtue of (20.2.4), the restrictions of u_α and u_β to $\mathrm{dom}_S(\omega_\alpha) \cap \mathrm{dom}_S(\omega_\beta)$ are equal, so there exists a morphism $u : W \to Y$ whose restriction to each $\mathrm{dom}_S(\omega_\alpha)$ is equal to u_α; it is clear that W is schematically dense in U, so it defines a pseudo-S-morphism ω of U in Y whose restrictions to the U_α are the ω_α. □

Remark (20.2.7). — We know (11.10.4) that when X is *reduced*, it amounts to the same to say that an open of X is dense in X or that it is schematically dense in X; the notion of pseudo-morphism (resp. pseudo-S-morphism) of X in Y then coincides with that of *rational application* (resp. *S-rational application*) of X in Y (**I**, 7.1.2). In the general case, the notion of pseudo-morphism seems more useful than that of rational application.

(20.2.8). We call *pseudo-function* on X a pseudo-morphism of X in $\mathrm{Spec}(\mathbf{Z}[T])$ (T indeterminate), or, what amounts to the same, an X-pseudo-morphism of X in $X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$; this amounts also to the same (**I**, 3.3.15) as saying that a pseudo-function on X is an *equivalence class of sections of $\mathscr{O}_X$ over schematically dense opens* of X, two sections $g \in \Gamma(U, \mathscr{O}_X)$, $g' \in \Gamma(U', \mathscr{O}_X)$ over such opens being equivalent if there exists an open $U'' \subset U \cap U'$, schematically dense in X, where g and g' coincide. We can apply (20.2.4) with $S = X$ and $Y = X \otimes_{\mathbf{Z}} \mathbf{Z}[T]$; then, for every pseudo-function φ on X, there is a *largest* schematically dense open $\mathrm{dom}(\varphi)$ in X such that there is a section of $\mathscr{O}_X$ over $\mathrm{dom}(\varphi)$ belonging to the class φ. It is clear that the sheaf $\mathscr{P}s.\mathrm{hom}_X(X, X \otimes_{\mathbf{Z}} \mathbf{Z}[T])$ is here a *sheaf of rings*, and even an *$\mathscr{O}_X$-Algebra*, which we denote $\mathscr{M}'_X$; it results from (20.2.6) that, for every open U of X, $\Gamma(U, \mathscr{M}'_X)$ is equal to the set of pseudo-morphisms of U in $\mathrm{Spec}(\mathbf{Z}[T])$; we set $M'(X) = \Gamma(X, \mathscr{M}'_X)$. To say that a section φ of $\mathscr{M}'_X$ over U is *invertible* in the ring $\Gamma(U, \mathscr{M}'_X)$ means that there exists an open U' schematically dense in $\mathrm{dom}(\varphi)$, thus in U, such that, if g is the unique section of $\mathscr{O}_X$ over $\mathrm{dom}(\varphi)$ belonging to the class φ, $g|U'$ is *invertible* in $\Gamma(U', \mathscr{O}_X)$. It results from (**I**, 3.3.15) that, in the canonical correspondence between $\Gamma(V, \mathscr{O}_X)$ and $\mathrm{Hom}(V, \mathbf{Z}[T])$ (V open of X), the invertible elements of $\Gamma(V, \mathscr{O}_X)$ correspond to the morphisms which *factor through* $V \to \mathrm{Spec}(\mathbf{Z}[T, T^{-1}]) \to$ **IV-4** | 234
$\mathrm{Spec}(\mathbf{Z}[T])$. We conclude that the sheaf $\mathscr{M}'^*_X$ of invertible germs of sections of $\mathscr{M}'_X$ is canonically identified with the sheaf $\mathscr{P}s.\mathrm{hom}_X(X, X \otimes_{\mathbf{Z}} \mathbf{Z}[T, T^{-1}])$.

Lemma (20.2.9). — *Let U be an open of X, s a regular element of the ring $\Gamma(U, \mathscr{O}_X)$; then the open U_s of the $x \in U$ such that $s(x) \neq 0$ is schematically dense in U.*

Proof. Indeed, let V be an open of U, t a section of $\mathscr{O}_X$ over V such that $t|(V \cap U_s) = 0$. For every affine open $W \subset V$, there exists an integer n such that $s^n t|W = 0$ (**I**, 1.4.1); but since s is a *regular* element of $\Gamma(U, \mathscr{O}_X)$, this implies $t|W = 0$, hence $t = 0$. □

(20.2.10). Consider a meromorphic function $f \in M(X)$ (20.1.4); then $\mathrm{dom}(f)$ is *schematically dense* in X: indeed, every point of X admits an open neighbourhood U such that there exists a section $s \in \Gamma(U, \mathscr{O}_X)$ which is a regular element of this ring, and for which $s(f|U) \in \Gamma(U, \mathscr{O}_X)$; as $s|U_s$ is invertible, we conclude that $f|U_s \in \Gamma(U_s, \mathscr{O}_X)$, hence $U_s \subset \mathrm{dom}(f)$ by definition (20.1.4), and our assertion results from the lemma (20.2.9). We can thus associate to f the pseudo-function equal to the class of the section of $\mathscr{O}_X$ over $\mathrm{dom}(f)$, restriction of f; operating in the same way replacing X by an arbitrary open of X, we thus obtain a canonical *homomorphism of $\mathscr{O}_X$-Algebras*

(20.2.10.1) $$\mathscr{M}_X \longrightarrow \mathscr{M}'_X$$

which, by restriction, gives evidently a homomorphism of sheaves of abelian groups

(20.2.10.2) $$\mathscr{M}^*_X \longrightarrow \mathscr{M}'^*_X$$

for the sheaves of invertible germs of sections of $\mathscr{M}_X$ and $\mathscr{M}'_X$.

Proposition (20.2.11). — (i) *The canonical homomorphism* (20.2.10.1) *(and hence also* (20.2.10.2)*) is* injective*.*

(ii) *Suppose either that X is locally Noetherian, or that X is reduced and the set of its irreducible components is locally finite. Then the canonical homomorphism* (20.2.10.1) *(and hence also* (20.2.10.2)*) is* bijective*.*

Proof. The questions being local on X, we can restrict to the case where $X = \operatorname{Spec}(A)$ is affine, and show that the canonical homomorphism $M(X) \to M'(X)$ is injective, and that it is bijective in the cases considered in (ii). Taking into account the definition of the sheaf $\mathscr{M}_X$ (20.1.3), we can in fact remark that (20.2.10.1) comes from a homomorphism of *presheaves*

$$\Gamma(U, \mathscr{O}_X)[\Gamma(U, \mathscr{S})^{-1}] \longrightarrow \Gamma(U, \mathscr{M}'_X)$$

and it suffices to show that the latter is injective (resp. bijective under the hypotheses of (ii)). Denoting by S the set of regular elements of A (so that $S^{-1}A$ is the *total ring of fractions* of A), we must therefore consider the canonical homomorphism

$$S^{-1}A \longrightarrow M'(X). \tag{20.2.11.1}$$

We can write $S^{-1}A = \varinjlim A_t$, where t runs through the set of *regular* elements of A, ordered by the relation "t is a divisor of t'" (**0**, 1.4.5); we can also write $A_t = \Gamma(\mathrm{D}(t), \mathscr{O}_X)$. On the other hand, by definition $M'(X) = \varinjlim \Gamma(U, \mathscr{O}_X)$, where U runs through the set of schematically dense opens of X (ordered by the relation $\supset$), and the map (20.2.11.1) is none other than the canonical map coming from the fact that the $\mathrm{D}(t)$ constitute a part of the set of schematically dense opens in X (20.2.9). Note that by definition of a schematically dense open, the homomorphism $\Gamma(U, \mathscr{O}_X) \to \Gamma(U', \mathscr{O}_X)$ for any two such opens $U' \subset U$ is always *injective*, hence the homomorphisms $\Gamma(U, \mathscr{O}_X) \to M'(X)$ are also injective, and we conclude that (20.2.11.1) is injective. To prove that (20.2.11.1) is bijective, it suffices to show that the $\mathrm{D}(t)$ are *cofinal* in the set of schematically dense opens, i.e., that for such an open U, there exists t regular in A such that $\mathrm{D}(t) \subset U$. Now, when X is reduced and the irreducible components of X form a locally finite set, this set is finite since X has been supposed affine, so A has only a finite number of minimal primes $\mathfrak{p}_i$; as A is reduced, the intersection of the $\mathfrak{p}_i$ is zero, and to say that t is regular is equivalent to saying that t does not belong to any of the $\mathfrak{p}_i$; we conclude by the reasoning of (**I**, 7.1.9.1). When A is Noetherian, to say that $U = X - Y$ (where $Y = V(\mathfrak{J})$ is closed in X) is schematically dense means (5.10.2) that Y *does not meet* $\operatorname{Ass}(\mathscr{O}_X)$, and by virtue of Bourbaki, *Alg. comm.*, chap. IV, §1, n° 4, prop. 8, this implies the existence of a $t \in \mathfrak{J}$ such that t is A-regular, hence $U \supset \mathrm{D}(t)$. **IV-4** | 235

We have moreover proved in the course of this demonstration the □

Corollary (20.2.12). — *If $X = \operatorname{Spec}(A)$, where A is Noetherian, or reduced and having only a finite number of minimal prime ideals, the schematically dense opens in X are those which contain an open of the form $\mathrm{D}(t)$, where t is regular in A, and $M(X) = M'(X)$ is the total ring of fractions $S^{-1}A$, where S is the set of regular elements of A.*

Remarks (20.2.13). — (i) Let φ be an element of $M(X)$, φ' its image in $M'(X)$; by definition ((20.1.4) and (20.2.8)) $\operatorname{dom}(\varphi) \subset \operatorname{dom}(\varphi')$; but in fact one has the equality $\operatorname{dom}(\varphi) = \operatorname{dom}(\varphi')$, because there exists a section of $\mathscr{O}_X$ over $\operatorname{dom}(\varphi')$, belonging to the class φ', and the corresponding meromorphic function equals φ by (20.2.11, (i)), hence $\operatorname{dom}(\varphi') \subset \operatorname{dom}(\varphi)$.

(ii) We have already noted that when X is *reduced*, one has $\mathscr{M}'_X = \mathscr{R}(X)$ (sheaf of rational functions on X (**I**, 7.3.2)); if moreover the irreducible components of X form a locally finite set, one has $\mathscr{M}_X = \mathscr{M}'_X = \mathscr{R}(X)$. In general, since every schematically dense open is dense, one has a canonical homomorphism $\mathscr{M}'_X \to \mathscr{R}(X)$, but even when X is locally Noetherian, this homomorphism is not necessarily injective. For example, if $X = \operatorname{Spec}(A)$, where A is a Noetherian ring such that $\operatorname{Ass}(A)$ contains embedded prime ideals (which implies that A is not reduced), then $M(X)$ and $M'(X)$ identify with the total ring of fractions $S^{-1}A$, where **IV-4** | 236
S is the set of regular elements of A, the complement of the union of the ideals $\mathfrak{p} \in \operatorname{Ass}(A)$; on the other hand, $R(X)$ identifies with $Q^{-1}A$, where Q is the complement of the union of the *minimal* prime ideals of A (**I**, 7.1.9), and the canonical homomorphism $A \to Q^{-1}A$

(and *a fortiori* $S^{-1}A \to Q^{-1}A$) is therefore not injective, since there exist nonzero elements of $A - Q$ annihilated by elements of Q (Bourbaki, *Alg. comm.*, chap. IV, §1, n° 1, cor. 2 of prop. 1).

(iii) We note also that when X is locally Noetherian, the $\mathscr{O}_X$-Module $\mathscr{M}'_X$ is *not necessarily quasi-coherent*. Consider for example a local Noetherian ring A of dimension $\geqslant 2$, with maximal ideal $\mathfrak{m} \in \mathrm{Ass}(A)$, and if we set $X = \mathrm{Spec}(A)$, the scheme induced on the open U of X, complementary of $\{\mathfrak{m}\}$, is *integral*. The only regular elements of A are the invertible elements, so $\Gamma(X, \mathscr{M}'_X) = M'(X) = A$; if $\mathscr{M}'_X$ were quasi-coherent, it would therefore be equal to $\mathscr{O}_X$; but as U is an integral scheme, $\mathscr{M}'_X|U = R(U)$ is a simple sheaf (**I**, 7.3.5), whereas $\mathscr{O}_U$ is not a simple sheaf since $\dim(A) \geqslant 2$.

It remains to give an example of a ring A having the above properties. It suffices to consider an integral local ring B of dimension $\geqslant 2$ and residue field k, and to take $A = B \oplus k$ with the law of multiplication $(b, x)(b', x') = (bb', bx' + b'x)$.

(iv) If X is locally Noetherian, it results from (5.10.2) that the schematically dense opens in X are those which *contain* $\mathrm{Ass}(\mathscr{O}_X)$.

(20.2.14). Let X be a prescheme, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent and *strictly torsion-free* (20.1.5), so that $\mathscr{F}$ identifies via (20.1.5.1) with a sub-$\mathscr{O}_X$-Module of $\mathscr{M}_X(\mathscr{F})$. For every meromorphic section u of $\mathscr{F}$ over X, we call the *ideal of denominators* of u the Ideal $\mathscr{J}$ of the section $\overline{u}$, image of u in $\mathscr{M}_X(\mathscr{F})/\mathscr{F}$. The Ideal $\mathscr{J}$ is *quasi-coherent*: indeed, the question being local on X, we can restrict to the case where X is affine, and there exists a section $s \in \Gamma(X, \mathscr{S}(\mathscr{O}_X))$ such that $v = su \in \Gamma(X, \mathscr{F})$; to say that a section $f \in \Gamma(U, \mathscr{O}_X)$ belongs to $\Gamma(U, \mathscr{J})$ means that $f(u|U) \in \Gamma(U, \mathscr{F})$, and since $s|U$ is a regular element of $\Gamma(U, \mathscr{O}_X)$ and $\mathscr{F}$ is strictly torsion-free, the preceding relation is equivalent to $f((su)|U) \in \Gamma(U, s\mathscr{F})$; so if $\overline{v}$ is the section of $\mathscr{F}/s\mathscr{F}$, canonical image of v, we see that $\mathscr{J}$ is the kernel of the homomorphism $\mathscr{O}_X \to \mathscr{F}/s\mathscr{F}$ obtained by multiplication by the section $\overline{v}$. As $\mathscr{F}/s\mathscr{F}$ is quasi-coherent, so is $\mathscr{J}$.

It results immediately from the preceding definition that $\mathrm{dom}(u)$ is the open complement of the closed sub-prescheme of X defined by the Ideal of denominators of u.

Proposition (20.2.15). — *Let $f : X' \to X$ be a morphism, $\mathscr{F}$ an $\mathscr{O}_X$-Module quasi-coherent, φ a section of $\mathscr{M}_f(\mathscr{F})$ over X (20.1.11). Then $f^{-1}(\mathrm{dom}(\varphi))$ is schematically dense in X'.*

Proof. The question being evidently local on X and X', we can suppose $X = \mathrm{Spec}(A)$, $X' = \mathrm{Spec}(A')$ affines, $\mathscr{F} = \widetilde{M}$, and $\varphi = h \otimes (1/s)$, where $h \in M$ and s is a regular element of A whose image s' in A' is a regular element. We know then (20.2.9) that $\mathrm{D}(s')$ is a schematically dense open in X', and it is the inverse image by f of $\mathrm{D}(s)$, which is contained in $\mathrm{dom}(\varphi)$. □

Remark (20.2.16). — When Y is not separated, the presheaf $U \mapsto \mathrm{Ps.hom}_S(U, Y)$ on X is not necessarily a sheaf, even when X is Noetherian, as the following example shows. We take for X the spectrum of a semi-local Noetherian ring A of dimension $\geqslant 2$, having exactly two maximal ideals $\mathfrak{m}', \mathfrak{m}''$ (so that X has exactly two closed points x', x''), such that $\mathfrak{m}'$ and $\mathfrak{m}''$ belong to $\mathrm{Ass}(A)$ **IV-4** | 237
and that the open $U = X - \{x', x''\}$ is *integral*. Note that the only schematically dense opens in X are those containing x' and x'' (20.2.13, (iv)), and hence they are necessarily equal to X. Let $U' = X - \{x'\}$, $U'' = X - \{x''\}$, so that $U = U' \cap U''$. To have a counterexample, it suffices to define two S-morphisms $u' : U' \to Y$, $u'' : U'' \to Y$ (with $S = X$) whose restrictions to U belong to the same pseudo-morphism, but which for every neighbourhood V' of x' in U' and every neighbourhood V'' of x'' in U'', the restrictions of u' and u'' to $V' \cap V''$ are distinct. For this, let us consider a closed irreducible part Z of X, distinct from X; let Y be the X-prescheme obtained by gluing two copies Y', Y'' of X along the open $X - Z$ (**I**, 2.3.1). We take for u' and u'' the restrictions to U' and U'' respectively of the structural isomorphisms $Y' \xrightarrow{\sim} X$, $Y'' \xrightarrow{\sim} X$. Since V' and V'' contain the generic point of Z, the restrictions of u' and u'' to $V' \cap V''$ are distinct; but the restrictions of u' and u'' to $U - (U \cap Z)$ are equal, and this is a dense open in U, hence schematically dense since U is reduced.

It remains therefore only to define A and Z so as to have the preceding properties. Let $X_0 = \mathrm{Spec}(B)$ be an integral affine scheme (for example the affine plane over a field k), Y an irreducible closed subvariety of X_0 (for example a line) containing two distinct closed points x', x'', corresponding to maximal ideals $\mathfrak{n}', \mathfrak{n}''$ of B. Consider the ring $C = B \oplus (B/\mathfrak{n}' \oplus B/\mathfrak{n}'')$ with the multiplication

$(b,z)(b',z') = (bb', bz' + b'z)$. If $X_1 = \mathrm{Spec}(C)$, we have $X_0 = (X_1)_{\mathrm{red}}$ and X_1 is reduced except at the points x', x''. It then suffices to take $A = R^{-1}C$, where R is the complement of the union of the maximal ideals of C at the points x', x'', and for Z the trace of Y on $X = \mathrm{Spec}(A)$.

20.3. Composition of pseudo-morphisms

(20.3.1). Let X, Y, Z be three preschemes, ω a pseudo-morphism of X in Y, $f : Y \to Z$ a morphism. It is clear that if U', U'' are two schematically dense opens in X, $u' : U' \to Y$, $u'' : U'' \to Y$ two morphisms of the class ω, the morphisms fu' and fu'' are equivalent (for the relation defined in (20.2.1)), so, for all morphisms u of the class ω, the morphisms fu belong to a single equivalence class, which we call the pseudo-morphism of X in Z *composed* of f and ω. One has $\mathrm{dom}(f\omega) = \mathrm{dom}(\omega)$. If $g : Z \to T$ is a morphism, it is clear that $g \circ (f\omega) = (g \circ f)\omega$.

(20.3.2). Let us now suppose given a pseudo-S-morphism ω of X in Y, where Y is *separated over* S, so that there exists an S-morphism $u : \mathrm{dom}_S(\omega) \to Y$ of the class ω (20.2.4). Let $f : X' \to X$ be an S-morphism such that the open $V' = f^{-1}(\mathrm{dom}_S(\omega))$ is schematically dense in X'; then we say that the class (for the equivalence relation of (20.2.1)) of the S-morphism $u \circ (f|V')$[5] (a class defined by virtue of the preceding hypothesis) is the pseudo-S-morphism *composed* of ω and f, and we denote it $\omega \circ f$; it is clear that one has

$$f^{-1}(\mathrm{dom}_S(\omega)) \subset \mathrm{dom}_S(\omega \circ f). \tag{20.3.2.1}$$

For $f : X' \to X$ given, we denote by $\mathrm{Ps.\,hom}_S(X,Y)^f$ the set of pseudo-S-morphisms ω of X in **IV-4 | 238**
Y which verify the preceding condition. If ω is such a pseudo-S-morphism, it is clear that for every open V of X,

$$f^{-1}(\mathrm{dom}_S(\omega|V)) = f^{-1}(V \cap \mathrm{dom}_S(\omega)) = f^{-1}(V) \cap f^{-1}(\mathrm{dom}_S(\omega))$$

is schematically dense in $f^{-1}(V)$, so, if $f^V : f^{-1}(V) \to V$ is the restriction of f, the composed $(\omega|V) \circ f^V$ is defined and equal to $(\omega \circ f)|f^{-1}(V)$. One thus defines a canonical restriction map $\mathrm{Ps.\,hom}_S(X,Y)^f \to \mathrm{Ps.\,hom}_S(V,Y)^{f'}$, which allows to define a *presheaf* of sets $V \mapsto \mathrm{Ps.\,hom}_S(V,Y)^{f'}$ on X, sub-presheaf of $V \mapsto \mathrm{Ps.\,hom}_S(V,Y)$, which gives a sheaf of sets that we denote $\mathscr{P}\mathrm{s.hom}_S(X,Y)^f$. Moreover, for every open V of X, we have a map $\omega \mapsto \omega \circ f^V$ in $\mathrm{Ps.\,hom}_S(f^{-1}(V), Y)$, which defines an f-morphism of sheaves of sets

$$\mathscr{P}\mathrm{s.hom}_S(X,Y)^f \longrightarrow \mathscr{P}\mathrm{s.hom}_S(X',Y).$$

(20.3.3). Let now $f' : X'' \to X'$ be an S-morphism such that the open $f'^{-1}(f^{-1}(\mathrm{dom}_S(\omega)))$ is schematically dense in X''; then $\omega \circ (f \circ f')$ is defined and $u \circ f \circ f'$ belongs to this pseudo-S-morphism; moreover, by virtue of (20.3.2.1), $f'^{-1}(\mathrm{dom}_S(\omega \circ f))$ is *a fortiori* schematically dense in X'', so $(\omega \circ f) \circ f'$ is also defined and $u \circ f \circ f'$ belongs to this pseudo-S-morphism, hence $(\omega \circ f) \circ f' = \omega \circ (f \circ f')$. On the other hand, for every S-morphism $g : Y \to Z$, one has $\mathrm{dom}_S(g\omega) = \mathrm{dom}_S(\omega)$ (20.3.1), so $(g\omega) \circ f$ is defined, and $g \circ u \circ f$ belongs to this pseudo-S-morphism, which shows that $(g\omega) \circ f = g \circ (\omega \circ f)$.

(20.3.4). The most important case in the definition (20.3.2) is that where one has $\mathscr{P}\mathrm{s.hom}_S(X,Y)^f = \mathscr{P}\mathrm{s.hom}_S(X,Y)$: it suffices for this that for every open U of X and every open V schematically dense in U, $f^{-1}(V)$ is schematically dense in $f^{-1}(U)$; when this holds, then, for *every* open U of X and *every* pseudo-S-morphism $\omega : U \to Y$, one can define the composed $\omega \circ f^U$, *even if Y is not separated over S*. Indeed, if $u' : U' \to Y$ and $u'' : U'' \to Y$ are two morphisms of the class ω, they coincide in an open U_0 schematically dense in U, so the composed morphisms $f^{-1}(U') \xrightarrow{u'} Y$ and $f^{-1}(U'') \xrightarrow{u''} Y$ coincide in $f^{-1}(U_0)$, which is by hypothesis schematically dense in $f^{-1}(U)$; one can therefore define $\omega \circ f^U$ as the class of any of the morphisms $f^{-1}(U') \xrightarrow{u'} Y$, where u' belongs to ω.

Proposition (20.3.5). — *Let X, X' be two preschemes, $f : X' \to X$ a morphism. Then, in each of the three following cases, for every open U of X and every open V schematically dense in U, $f^{-1}(V)$ is schematically dense in $f^{-1}(U)$:*

[5] The printed text has $f|U'$ here, but the open introduced in the preceding sentence is $V' = f^{-1}(\mathrm{dom}_S(\omega))$.

(i) *f is flat and locally of finite presentation.*
(ii) *f is flat and the underlying space of X is locally Noetherian.*
(iii) *X' is reduced, the set of irreducible components of X' is locally finite, and every irreducible component of X' dominates an irreducible component of X.*

Proof. Assertion (i) results from (11.10.5, (ii), b)); assertion (ii) results from (11.10.5, (ii), a)), since then every open V in U is retrocompact, i.e. the canonical injection $j : V \to U$ is a quasi-compact morphism. Finally, to prove (iii), note that since X' is reduced, it suffices to show that for every open U of X and every open V dense in U, $f^{-1}(V)$ is *dense* in $f^{-1}(U)$. Now, for $f^{-1}(V)$ to be dense in $f^{-1}(U)$, it suffices that $f^{-1}(V)$ contain all the maximal points of X' belonging to $f^{-1}(U)$; the conclusion therefore results from the hypothesis that the image under f of every maximal point of X' belonging to $f^{-1}(U)$ is a maximal point of X belonging to U, hence belongs to V since the set of irreducible components of X is locally finite. ☐ **IV-4** | 239

(20.3.6). Let X, Y be two S-preschemes; suppose that X satisfies one of the two following hypotheses:

a) X is locally Noetherian;
b) X is reduced and the set of its irreducible components is locally finite.

Then, for every $x \in X$, the canonical S-morphism $j : \mathrm{Spec}(\mathscr{O}_{X,x}) \to X$ is flat and satisfies condition (ii) of (20.3.5) in case a), condition (iii) of (20.3.5) in case b). For every pseudo-S-morphism ω of X in Y, the composite $\omega \circ j$ is therefore defined and is a pseudo-S-morphism of $\mathrm{Spec}(\mathscr{O}_{X,x})$ in Y, called the *restriction* of ω to $\mathrm{Spec}(\mathscr{O}_{X,x})$. Note now that if X satisfies condition a) (resp. b)) of (20.3.6), the same holds for every prescheme induced on an open U of X containing x. By passage to the inductive limit, we deduce from the canonical maps $\mathrm{Ps.\,hom}_S(U, Y) \to \mathrm{Ps.\,hom}_S(\mathrm{Spec}(\mathscr{O}_{X,x}), Y)$ thus obtained, a canonical map

(20.3.6.1) $$(\mathscr{P}\!s.\mathrm{hom}_S(X,Y))_x \longrightarrow \mathrm{Ps.\,hom}_S(\mathrm{Spec}(\mathscr{O}_{X,x}), Y)$$

where the left-hand side is the stalk at the point x of the sheaf $\mathscr{P}\!s.\mathrm{hom}_S(X,Y)$, the set of germs at x of pseudo-S-morphisms from open neighbourhoods of x into Y.

Proposition (20.3.7). — *Under the hypotheses of (20.3.6), the canonical map (20.3.6.1) is injective. If moreover Y is locally of finite presentation over S, the map (20.3.6.1) is bijective.*

Proof. By application of the method of (8.1.2, (a)), this proposition will be a particular case of the following: ☐

Proposition (20.3.8). — *The notations being those of* (8.8.1)*, suppose X_α quasi-compact (resp. quasi-compact and quasi-separated) and Y_α locally of finite type (resp. locally of finite presentation) over S_α. Suppose moreover that one of the following conditions is satisfied:*

(i) *The transition morphisms $S_\mu \to S_\lambda$ (for $\lambda \leqslant \mu$) are flat, and the X_λ and X are Noetherian.*
(ii) *The X_λ are reduced, the sets of irreducible components of X and of each X_λ are finite, and, for $\lambda \leqslant \mu$, every irreducible component of X_μ dominates an irreducible component of X_λ.*

Then, the canonical map

(20.3.8.1) $$\varinjlim \mathrm{Ps.\,hom}_{S_\lambda}(X_\lambda, Y_\lambda) \longrightarrow \mathrm{Ps.\,hom}_S(X,Y)$$

is injective (resp. bijective).

Proof. Note first that, in case (i), the morphisms $X_\mu \to X_\lambda$ (for $\lambda \leqslant \mu$) and $X \to X_\lambda$ are flat, so by (20.3.4) and (20.3.5) the canonical maps **IV-4** | 240

$$\mathrm{Ps.\,hom}_{S_\lambda}(X_\lambda, Y_\lambda) \longrightarrow \mathrm{Ps.\,hom}_{S_\mu}(X_\mu, Y_\mu)$$

for $\lambda \leqslant \mu$, as well as the maps

$$\mathrm{Ps.\,hom}_{S_\lambda}(X_\lambda, Y_\lambda) \longrightarrow \mathrm{Ps.\,hom}_S(X,Y),$$

are defined (without any separation hypothesis on the Y_λ or on Y); the same is consequently true of the map (20.3.8.1). The proposition will result from the following lemma.

Lemma (20.3.8.2). — *The notations being those of* (8.8.1)*, suppose X_α quasi-compact, and let U_α be an open in X_α; let U_λ and U be its inverse images in X_λ and X. Suppose that one of the conditions (i), (ii) of* (20.3.8) *is satisfied. Then, for U to be schematically dense in X, it is necessary and sufficient that there exist $\lambda \geqslant \alpha$ such that U_λ is schematically dense in X_λ, and in this case U_μ is schematically dense in X_μ for $\mu \geqslant \lambda$.*

Suppose first that condition (i) is satisfied. Denote by $j_\lambda : U_\lambda \to X_\lambda$ and $j : U \to X$ the canonical injections, and by $\mathscr{J}_\lambda$ the kernel of the canonical homomorphism $\mathscr{O}_{X_\lambda} \to (j_\lambda)_*(\mathscr{O}_{U_\lambda})$. Since the immersion j_λ is quasi-compact and quasi-separated, $(j_\lambda)_*(\mathscr{O}_{U_\lambda})$ is a quasi-coherent $\mathscr{O}_{X_\lambda}$-module; hence $\mathscr{J}_\lambda$ is a quasi-coherent ideal of $\mathscr{O}_{X_\lambda}$, and, since X_λ is Noetherian, $\mathscr{J}_\lambda$ is coherent and therefore of finite type. On the other hand, since the transition morphism $p_{\lambda\mu} : X_\mu \to X_\lambda$ (resp. $p_\lambda : X \to X_\lambda$) is flat, it follows from (2.3.1) that one may identify $\mathscr{J}_\mu$ with $p^*_{\lambda\mu}(\mathscr{J}_\lambda)$ (resp. $\mathscr{J}$ with $p^*_\lambda(\mathscr{J}_\lambda)$). The assertion results then from the definition of a schematically dense open, and of (8.5.8, (ii)).

To establish (20.3.8.2) when condition (ii) is satisfied, we first prove two lemmas.

Lemma (20.3.8.3). — *Under the hypotheses of* (8.2.2)*, let M_α (resp. M) be the set of maximal points of S_α (resp. S). Suppose that for every λ, the set M_λ is finite, and that the M_λ form a projective system of sets. Then M is the projective limit of the M_λ.*

Let us first show that a point $s \in \varprojlim M_\lambda$ is maximal in S: indeed, if s' is a generalisation of s, the image s'_λ of s' in S_λ is a generalisation of the image s_λ of s, hence equal to s_λ for all λ, which implies $s' = s$, since the set of points underlying S is the projective limit of the sets underlying the S_λ (8.2.9). Inversely, let s be a maximal point of S and let us prove that it belongs to $\varprojlim M_\lambda$. Let s_λ be the image of s in S_λ, M'_λ the set of maximal points of S_λ that are generalisations of s_λ; the M'_λ are non-empty finite sets forming a projective system, hence $M' = \varprojlim M'_\lambda$ is non-empty and the map $M' \to M'_\lambda$ is surjective (Bourbaki, *Ens.*, chap. III, 2[e] éd., § 7, n[o] 4, Example I). On the other hand, $\operatorname{Spec}(\mathscr{O}_{S,s}) = \varprojlim \operatorname{Spec}(\mathscr{O}_{S_\lambda,s_\lambda})$ by virtue of (8.2.12) and (8.2.9), hence the points of $\varprojlim M'_\lambda$ are also maximal points of $\operatorname{Spec}(\mathscr{O}_{S,s})$ by the first part of the reasoning. Thus $M' = \varprojlim M'_\lambda$ necessarily reduces to the point s; one concludes that M'_λ reduces to the point s_λ and consequently the s_λ are maximal in the S_λ, which completes the proof of the lemma.

Lemma (20.3.8.4). — *The hypotheses being those of* (20.3.8.3)*, suppose moreover that S_α is quasi-compact, and let U_α be an open of S_α, with inverse images U_λ in S_λ for $\lambda \geqslant \alpha$ and U in S.*[6] *If U_α is dense in S_α, then U_λ is dense in S_λ for $\lambda \geqslant \alpha$ and U is dense in S. Inversely, if U is dense in S and if moreover the set M of maximal points of S is finite, there exists $\lambda \geqslant \alpha$ such that U_λ is dense in S_λ.*

Indeed, since M_α is finite, the hypothesis that U_α is dense in S_α implies that $M_\alpha \subset U_\alpha$, hence $M_\lambda \subset U_\lambda$ for $\lambda \geqslant \alpha$ and $M \subset U$ by virtue of (20.3.8.3), which proves the first assertion. Inversely, suppose M finite and U dense in S, hence $M \subset U$; note that the U_λ are open, hence ind-constructible, and the M_λ finite, hence pro-constructible (1.9.6). The second assertion then follows from (8.3.2).

20.3.8.5. End of the proof of (20.3.8.2)*.* Suppose condition (ii) is verified, and note that X is then *reduced* by virtue of (8.7.1); it amounts to the same to say that U_λ (resp. U) is schematically dense in X_λ (resp. X) or that it is dense, and the conclusion results from (20.3.8.4) applied to X_λ and X. **IV-4** | 241

20.3.8.6. End of the proof of (20.3.8)*.* To show that the map (20.3.8.1) is injective, consider two morphisms u'_λ, u''_λ of a schematically dense open $U_\lambda \subset X_\lambda$ in Y_λ, and suppose that their images u', u'', morphisms of U of X in Y, coincide in a schematically dense open V of U. Under either hypothesis (i) or (ii), one can suppose V *quasi-compact*; otherwise, since X has only a finite number of maximal points and is reduced, it suffices to replace V by a union of a finite number of affine open neighbourhoods of these maximal points (contained in V by hypothesis). Then there exists $\lambda \geqslant \alpha$ such that V is the inverse image of a quasi-compact open V_λ of X_λ (8.2.11), and by (20.3.8.2) we can suppose that V_λ is schematically dense in X_λ. It results from (8.8.2, (i)) that, on taking λ sufficiently large, u'_λ and u''_λ coincide on V_λ, hence belong to the same pseudo-morphism.

[6]The printed statement introduces V_α, V_λ, V but immediately and consistently uses U_α, U_λ, U; the latter notation is used here.

Let us prove finally that the map (20.3.8.1) is surjective. Consider now a morphism u of a schematically dense open U of X in Y; as above, we can suppose U quasi-compact and quasi-separated (U being replaceable, in case (ii), by a union of a finite number of disjoint affine opens). We can furthermore suppose there exists $\lambda \geqslant \alpha$ such that U is the inverse image of a quasi-compact open U_λ of X_λ (8.2.11) which is automatically quasi-separated, and by (20.3.8.2) we can suppose that U_λ is schematically dense in X_λ. Since the Y_λ are supposed locally of finite presentation, it results from (8.8.2, (i)) that there exists λ such that u is the image of a morphism $u_\lambda : U_\lambda \to Y_\lambda$; whence the conclusion.

Remarks (20.3.8.7). — (i) To show that the map (20.3.8.1) is *injective*, it is not necessary, in the hypothesis (i) of (20.3.8), to suppose X Noetherian. Indeed, the lemma (20.3.8.2) does not use this hypothesis. With the notations of (20.3.8.6), let Z_λ be the sub-prescheme of coincidences of u'_λ and u''_λ, and let Z be the sub-prescheme of coincidences of u' and u''; it results from the definition (17.4.5) and from (**I**, 3.3.10.1) that Z is the projective limit of the Z_μ for $\mu \geqslant \lambda$. Now, by hypothesis, Z majorizes a schematically dense open of X; it follows that Z is itself induced on an open of X by virtue of the following lemma:

Lemma (20.3.8.8). — *Let T be a prescheme. Then every sub-prescheme W of T which majorizes a schematically dense open V of T is induced on an open (schematically dense) of T.*

Indeed, the subspace of T underlying W is locally closed, hence open in its closure, hence open in T; the conclusion results from (11.10.1, (c)). **IV-4** | 242

This lemma being established, we conclude that for $\mu \geqslant \lambda$ sufficiently large, Z_μ is induced on an open of X_μ by virtue of (8.6.3); indeed, Z_λ, being a sub-prescheme of a Noetherian prescheme, is of finite presentation over X_λ (1.6.2), and the same is therefore true of Z_μ over X_μ for $\mu \geqslant \lambda$ and of Z over X. One can now apply (20.3.8.2), which shows that for $\mu \geqslant \lambda$ sufficiently large, Z_μ is schematically dense in X_μ, whence the conclusion.

(ii) If, in the hypothesis (i) of (20.3.8), we drop the condition that X is Noetherian, the reasoning of (20.3.8.6) shows yet that the image of (20.3.8.1) is formed of the pseudo-S-morphisms having a representative which is an S-morphism $U \to Y$, where U is schematically dense in X and *quasi-compact and quasi-separated*.

□

Corollary (20.3.9). — *Suppose that either hypothesis a) or b) of (20.3.6) is satisfied by X, and let Y be locally of finite presentation over S. Then, for a pseudo-S-morphism ω of X in Y to be defined at a point x (20.2.3), it is necessary and sufficient that its restriction to* $\operatorname{Spec}(\mathscr{O}_{X,x})$ *be everywhere defined (in other words, that it be an S-morphism of* $\operatorname{Spec}(\mathscr{O}_{X,x})$ *in Y).*

The following result, which will be used in the proof of (20.3.11), uses the theory of faithfully flat descent from Chapter VI. The reader may verify that the results of (20.3) will not be used in that theory.

Lemma (20.3.10). — *Let $f : X' \to X$ be an S-morphism faithfully flat and quasi-compact, $X'' = X' \times_X X'$, p_1 and p_2 the canonical projections of X'' to X', and let Y be an S-prescheme separated over S. Let U be an open of X, $U' = f^{-1}(U)$, $U'' = p_1^{-1}(U') = p_2^{-1}(U')$, and suppose that U'' is schematically dense in X''. Let $g : U \to Y$ be an S-morphism; then, if $g \circ (f|U')$ extends to an S-morphism $X' \to Y$, g extends to an S-morphism $X \to Y$.*

The hypotheses imply that U (resp. U') is schematically dense in X (resp. X') (11.10.5, (i)). Thus, if ω denotes the pseudo-S-morphism given by the class of g, the assertion of (20.3.10) can equivalently be stated as follows: if $\omega \circ f$ is everywhere defined, then so is ω.

Proof. To prove (20.3.10), denote by g' an S-morphism $X' \to Y$ which extends $g \circ (f|U')$, and set $g''_i = g' \circ p_i : X'' \to Y$ $(i = 1, 2)$. If one sets $f'' = f \circ p_1 = f \circ p_2 : X'' \to X$, it is clear that g''_1 and g''_2 coincide on U'' with $g \circ (f''|U'')$. But as Y is separated over S and U'' is schematically dense in X'', we have $g''_1 = g''_2$ (11.10.1, (d)). As f is faithfully flat and quasi-compact, it results from descent theory (chap. VI) that there exists a unique S-morphism $h : X \to Y$ such that $h \circ f = g'$; as the restriction $U' \to U$ of f is a faithfully flat and quasi-compact morphism and $U'' = U' \times_U U'$, the preceding uniqueness result, applied to U instead of X, shows that $h|U = g$, which proves the lemma. □

Proposition (20.3.11). — *Let Y be an S-prescheme separated on S, ω a pseudo-S-morphism of X in Y, $f : X' \to X$ an S-morphism. Suppose that f is flat, and that one of the following conditions is satisfied:*

(i) *f is an open morphism, or surjective and quasi-compact, and* $\mathrm{dom}_S(\omega)$ *contains an open U schematically dense in X and retrocompact in X.*

(ii) *f is locally of finite presentation.*

(iii) *Y is locally of finite presentation over S, and X satisfies one of the conditions a), b) of (20.3.6).*

Then, $f^{-1}(\mathrm{dom}_S(\omega))$ is schematically dense in X', so that $\omega \circ f$ is defined (20.3.2) and one has

(20.3.11.1) $$\mathrm{dom}_S(\omega \circ f) = f^{-1}(\mathrm{dom}_S(\omega)).$$

Proof. Let us first prove that $f^{-1}(\mathrm{dom}_S(\omega))$ is schematically dense in X'. The question being local on X', we can suppose X and X' affines and since f is flat, it suffices to verify, by (11.10.5, (ii), a)), that $\mathrm{dom}_S(\omega)$ contains a retrocompact open U schematically dense in X. This is the hypothesis in case (i), and follows from (20.2.12) in case (iii), since in an affine scheme every open of the form $\mathrm{D}(t)$ is retrocompact; in case (ii), we can directly apply (20.3.5, (i)). **IV-4** | 243

Let us now prove (20.3.11.1), i.e., that, for every point $x' \in \mathrm{dom}_S(\omega \circ f)$, one has $x = f(x') \in \mathrm{dom}_S(\omega)$. Note first that we can restrict to the case where f is *faithfully flat and quasi-compact*. This is already the hypothesis in the second case of (i). In the other cases the question is local on X', so we may suppose X and X' affine, and hence f quasi-compact. In the first case of (i), and in case (ii), f is an open morphism (11.3.1), so, replacing X by the open $f(X')$, we may suppose f surjective, hence faithfully flat. In case (iii), by (20.3.9), it suffices to prove that the restriction of ω to $\mathrm{Spec}(\mathscr{O}_{X,x})$ is everywhere defined, and we can therefore replace X by $\mathrm{Spec}(\mathscr{O}_{X,x})$, X' by $X' \times_X \mathrm{Spec}(\mathscr{O}_{X,x})$ and f by its restriction to this last prescheme, which is a surjective morphism (2.3.4), hence faithfully flat.

Suppose therefore f faithfully flat and quasi-compact; with the notations of the lemma (20.3.10), it suffices to see that U'' is schematically dense in X'', by taking for U an open schematically dense in X, contained in $\mathrm{dom}_S(\omega)$; this will be the case, by virtue of (11.10.5, (ii), a)), if U is taken retrocompact in X (since the morphism $f'' : X'' \to X$ is flat). The existence of such an open U is part of the hypothesis in case (i); in case (iii) it results from (20.2.12) and the fact that in an affine scheme $\mathrm{Spec}(A)$, every open of the form $\mathrm{D}(t)$ is retrocompact. Finally, in case (ii), we take $U = \mathrm{dom}_S(\omega)$ and show directly that U'' is schematically dense in X'': it suffices for this to remark that $f'' : X'' \to X$ is flat and locally of finite presentation and to apply (20.3.5, (i)). □

Corollary (20.3.12). — *Let φ be a pseudo-function on a prescheme X. Then, for every morphism flat and locally of finite presentation $f : X' \to X$, the pseudo-function $\varphi \circ f$ is defined and one has* $\mathrm{dom}(\varphi \circ f) = f^{-1}(\mathrm{dom}(\varphi))$.

Remark (20.3.13). — When X satisfies one of the conditions a), b) of (20.3.6), then we have seen (20.2.11) that the pseudo-functions on X are identified with the meromorphic functions on X. By virtue of (20.1.12) and of (20.2.13, (i)), if one only supposes that the morphism $f : X' \to X$ is *flat*, then, for every pseudo-function φ on X, $\varphi \circ f$ is defined and one has $\mathrm{dom}(\varphi \circ f) = f^{-1}(\mathrm{dom}(\varphi))$. **IV-4** | 244

20.4. Properties of domains of definition of rational functions

(20.4.1). Let X, Y be two S-preschemes, ω a pseudo-S-morphism of X in Y. Let u be an S-morphism $U \to Y$ belonging to the class ω, where U is schematically dense in X, and consider the *graph* Γ_u, sub-prescheme of $U \times_S Y$ (**I**, 5.3.11), and therefore sub-prescheme of $X \times_S Y$. Suppose that this sub-prescheme admits an *adherence* (**I**, 9.5.11) in $X \times_S Y$; if $j : \Gamma_u \to X \times_S Y$ is the canonical injection, this will be the case when the $\mathscr{O}_{X\times_S Y}$-Module $j_*(\mathscr{O}_{\Gamma_u})$ is quasi-coherent; it results from the definition of the class of equivalence (20.2.1) that $j_*(\mathscr{O}_{\Gamma_u})$ does not depend on the representative u considered, hence the adherence of this sub-prescheme is well determined by ω; we say that this closed sub-prescheme is the *graph* of the pseudo-S-morphism ω and we denote it Γ_ω. We note that Γ_ω is defined if there exists a morphism $u : U \to Y$ in the class ω such that U is retrocompact in X and if, moreover, Y is quasi-separated over S; indeed, the injection j is then a quasi-compact and quasi-separated morphism (**I**, 2.2,I.7.4). The first hypothesis is always satisfied when X is locally Noetherian. We note also that when X is *reduced*, so is Γ_u, which is isomorphic to U (**I**, 5.3.11); then

Γ_ω exists and is none other than the *reduced* closed sub-prescheme of $X \times_S Y$ having for underlying space the closure in $X \times_S Y$ of the underlying space of Γ_u (**I**, 5.2.1,I.5.2.2). Note finally that if Y is *separated* over S, then Γ_u is *closed* in $U \times_S Y$ (**I**, 5.4.3), and hence is induced by Γ_ω (whenever the latter exists) on the open $\Gamma_\omega \cap (U \times_S Y)$ (**I**, 9.5.10); by contrast, the prescheme induced by Γ_ω is in general different from Γ_u when Y is not separated. In particular, if $v : X \to Y$ is an *S-morphism*, the graph Γ_ω of the class ω of v may be distinct from the graph Γ_v. In particular also in what follows, when the question is about the graph of a pseudo-S-morphism, we will restrict to the case where *Y is separated over S*.

(20.4.2). Suppose therefore Γ_ω defined and Y separated on S; denote by p and q the restrictions to Γ_ω of the canonical projections.

$$\begin{tikzcd} & \Gamma_\omega \arrow[dl, "p"'] \arrow[dr, "q"] & \\ X & & Y \end{tikzcd}$$

Then, if $U \subset \mathrm{dom}_S(\omega)$, the restriction $p^{-1}(U) \to U$ of p is an isomorphism (**I**, 5.3.11); reciprocally, **IV-4** | 245
if U is an open of X having this property, and if u is the isomorphism reciprocal of the restriction $p^{-1}(U) \to U$ of p, $q \circ u$ is an S-morphism of U in Y which coincides with a morphism of the class ω in $U \cap \mathrm{dom}_S(\omega)$. We conclude that $\mathrm{dom}_S(\omega)$ is *the largest open* U of X such that the restriction $p^{-1}(U) \to U$ of p is an isomorphism; let $v : \mathrm{dom}_S(\omega) \to \Gamma_\omega$ be the S-morphism inverse to the restriction $p^{-1}(\mathrm{dom}_S(\omega)) \to \mathrm{dom}_S(\omega)$ of p. The pseudo-S-morphism from X to Γ_ω represented by v is sometimes denoted p^{-1}; its associated rational map (20.2.13, (ii)) is then *birational*. Since $p^{-1}(\mathrm{dom}_S(\omega))$ is the graph of the S-morphism $u = q \circ v : \mathrm{dom}_S(\omega) \to Y$, it is schematically dense in Γ_ω (11.10.3, (iv)), so ω can be considered as the *composite* $q \circ p^{-1}$ (20.3.2). For every subset M of the underlying space of X, one sometimes sets $\omega(M) = q(p^{-1}(M))$, which amounts to considering ω as a map from X to the set of subsets of Y (or, as some authors say, a "multiform function"). We note that when $x \in \mathrm{dom}_S(\omega)$, $\omega(\{x\})$ is the singleton $\{u(x)\}$; in general, for any $x \in X$, if one denotes by s the image of x in S, via the structural morphism, $\omega(\{x\})$ is a part of the prescheme Y_s, the fibre at the point s of the structural morphism.

(20.4.3). In all the rest of this n°, we will restrict to the case where X is *reduced*, so that there is then an identity between pseudo-S-morphisms and *S-rational applications* (20.2.7). Moreover, the *graph* of the S-rational application ω of X in Y exists and is none other than the *reduced* closed sub-prescheme of $X \times_S Y$ having for underlying space the adherence of the graph of any representative $u : \mathrm{dom}_S(\omega) \to Y$ in the space underlying $X \times_S Y$ (**I**, 5.2.1) and (5.2.2).

Proposition (20.4.4). — *Let X be an S-prescheme locally integral, Y an S-prescheme locally of finite type and separated on S, ω an S-rational application of X in Y, $p : \Gamma_\omega \to X$ the canonical projection. Then, if $x \in X$ is a normal point such that $p^{-1}(x)$ contains an isolated point, ω is defined at the point x.*

Proof. Indeed, the first projection $p_1 : X \times_S Y \to X$ is then a separated and locally of finite type morphism, and so is its restriction $p : \Gamma_\omega \to X$, which is moreover *birational*, and as Γ_ω is reduced and X integral, Γ_ω is *integral*; it results then from ($\mathbf{Err_{IV}}$, 30) that the hypothesis on x implies the existence of an open neighbourhood U of x such that the restriction $p^{-1}(U) \to U$ of p is an isomorphism; whence the conclusion by (20.4.2).

We note that (20.4.4) is the original formulation given by Zariski of his "Main theorem" (up to the fact that it was originally restricted to algebraic schemes over a base field). ☐

Proposition (20.4.5). — *The hypotheses and notations being those of (20.4.4), suppose moreover X normal, and let X' be a reduced prescheme, $f : X' \to X$ a morphism locally of finite type and universally open. Then $\omega \circ f$ is defined, and one has* $\mathrm{dom}_S(\omega \circ f) = f^{-1}(\mathrm{dom}_S(\omega))$ *(in other words, if $x' \in X'$ and $x = f(x')$, then for ω to be defined at x it is necessary and sufficient that $\omega \circ f$ be defined at x').*

Proof. As X' is reduced and $f^{-1}(\mathrm{dom}_S(\omega))$ is everywhere dense in X' by virtue of (2.4.11), the **IV-4** | 246
composite $\omega \circ f$ is defined. To prove that, whenever $\omega \circ f$ is defined at x', ω is defined at x, we may plainly replace X' by an open neighbourhood of x' and hence suppose $\omega \circ f$ everywhere defined; moreover, since $f(X')$ is open in X, we may replace X by $f(X')$ and thus suppose f surjective.

By the hypothesis on f, the morphism

$$f_{(Y)} : X' \times_S Y \longrightarrow X \times_S Y$$

is then open, and consequently

$$\overline{f_{(Y)}^{-1}(M)} = f_{(Y)}^{-1}(\overline{M})$$

for every subset M of $X \times_S Y$. Taking for M the graph Γ_u, where $u : \mathrm{dom}_S(\omega) \to Y$ is the restriction of ω to its domain, it follows from the preceding relation and from (**I**, 3.3.12) that the underlying space of $\Gamma_{\omega \circ f}$ is $f_{(Y)}^{-1}(\Gamma_\omega)$. Since $\Gamma_{\omega \circ f}$ is a reduced prescheme (20.4.1), the X'-prescheme $\Gamma_{\omega \circ f}$ is therefore equal to

$$(\Gamma_\omega \times_X X')_{\mathrm{red}}.$$

But by hypothesis the composite morphism

$$\Gamma_{\omega \circ f} \longrightarrow \Gamma_\omega \times_X X' \xrightarrow{p^{(X')}} X'$$

is an isomorphism, so $p^{(X')}$ is necessarily radicial. Since f is surjective, the same is true of $p : \Gamma_\omega \to X$ (**I**, 3.4.8); hence $p^{-1}(x)$ consists of a single point (**I**, 3.6.4), and (20.4.4) shows that ω is defined at x. □

Proposition (20.4.6). — *Let S be a prescheme, X, Y two S-preschemes; suppose X locally Noetherian, Y locally of finite type on S. Let U be an open schematically dense in X, $h : U \to Y$ an S-morphism, $x \in X - U$ a normal point of X, $h'_x : \mathrm{Spec}(k(x)) \to Y$ an S-morphism. In order that h can be extended to an S-morphism $h' : U' \to Y$, where U' is an open of X containing U and x, and where the composed morphism $\mathrm{Spec}(k(x)) \to U' \xrightarrow{h'} Y$ is the S-morphism given by h'_x, it is necessary and sufficient that the following condition be satisfied:*

> (P) For every spectrum S_1 of a discrete valuation ring, of closed point a and generic point b, and for every morphism $u : S_1 \to X$ such that $u(a) = x$, $u(b) \in U$, the composed morphism $h_1 = h \circ (u|\{b\}) : \{b\} = u^{-1}(U) \to Y$ extends to a morphism $h'_1 : S_1 \to Y$ such that the diagram
>
> $$\begin{tikzcd} \mathrm{Spec}(k(a)) \arrow[r] & S_1 \\ \mathrm{Spec}(k(x)) \arrow[r, "h'_x"'] & Y \arrow[u, "h'_1"'] \end{tikzcd}$$
>
> is commutative.

Moreover, if this condition is satisfied, and if $h'' : U'' \to Y$ is a morphism satisfying the same conditions as h', then there exists an open set containing $U \cup \{x\}$, in which h' and h'' coincide.

Proof. Let us first prove the last assertion; we can suppose h' and h'' defined in the same open IV-4 | 247
$U_0 \supset U \cup \{x\}$. The sub-prescheme Z of coincidences of h' and h'' (17.4.5) contains U and x, so there is an open neighbourhood V of x in U_0 such that the sub-prescheme induced by Z on the open $Z \cap V$ is a *closed* sub-prescheme of the prescheme induced by V on V; as this sub-prescheme $Z \cap V$ majorizes the prescheme induced by V on the open $U \cap V$, and the latter is schematically dense in V, $Z \cap V$ is necessarily equal to V (20.3.8.8), which proves that h' and h'' coincide in $U \cup V$.

The necessity of the condition is evident, let us prove that it is sufficient. By the bijective correspondence between S-morphisms of X in Y and X-sections of $X \times_S Y$ (**I**, 3.3.15), we can restrict to the case where $S = X$ and h is therefore a U-section of Y; we note that then Y is locally Noetherian, and we may plainly (since X is locally integral and locally Noetherian) suppose that $X = S$ is irreducible. Moreover, by virtue of (20.3.7), we can suppose that $X = \mathrm{Spec}(A)$, where A is a local Noetherian ring, integral and integrally closed.

Let $y = h'_x(x)$. For every $x' \in U$, the point y is a specialization of $h(x')$. Indeed, there exists the spectrum S_1 of a discrete valuation ring and a morphism $u : S_1 \to X$ such that $u(a) = x$ and $u(b) =$

x' (**II**, 7.1.9). Applying condition (P), we obtain the assertion at once, since $h(x') = h_1(b) = h'_1(b)$ and $y = h'_1(a)$. If Y' is an affine open neighbourhood of y, it follows that $h(X \cap U) \subset Y'$; we may therefore replace Y by Y' and hence suppose that Y is affine, and thus separated over X. Let ω be the rational X-section of Y to which h belongs, so that its graph has as underlying space the closure of $h(U)$ in Y. Since Y is separated over X, we may apply (20.4.4) to ω; it is enough to prove that, if $p : \Gamma_\omega \to X$ is the canonical projection, then $p^{-1}(x)$ consists of the single point y and $h'_x(x) = y$. Indeed, (20.4.4) then extends h to a section h' over an open U' containing $U \cup \{x\}$, with $h'(x) = y$; and since there is only one X-morphism $\operatorname{Spec}(k(x)) \to Y$ carrying x to y, this proves that h'_x is the composite of h' with $\operatorname{Spec}(k(x)) \to U'$.

Since, for every $x' \in U$, y is a specialization of $h(x')$, we have $y \in p^{-1}(x)$. Now let z be any point of $p^{-1}(x)$. Since Γ_ω is the closure of $h(U)$, and every point of $h(U)$ is adherent to $h(\xi)$, where ξ is the generic point of X, Γ_ω is the closure of $h(\xi)$ in Y. There therefore exist the spectrum S_1 of a discrete valuation ring and a morphism $v : S_1 \to Y$ such that $v(a) = z$ and $v(b) = h(\xi)$ (**II**, 7.1.9); set $u = p \circ v$, so that $u(a) = x$ and $u(b) = \xi$. Applying condition (P) to u, we obtain a morphism $h'_1 : S_1 \to Y$ such that $h'_1(a) = y$ and $h'_1(b) = h(\xi)$. But then $z = y$ by (**II**, 7.2.3), since Y is separated over X and v and h'_1 must coincide, being equal at b. C.Q.F.D. $\square$

Corollary (20.4.7). — *The hypotheses on S, X, Y, U and h being those of (20.4.6), let E be a part of $X - U$ such that X is normal at every point of E, and for each $x \in E$, let $h'_x : \operatorname{Spec}(k(x)) \to Y$ be an S-morphism, such that the condition (P) of (20.4.6) is satisfied. Suppose moreover that the union F of the graphs of the h'_x (identified with parts of $X \times_S Y$) for $x \in E$, is contained in a union of a finite number of opens V_i of $X \times_S Y$ which are* separated *on X (a condition automatically satisfied if Y is* separated *on S, or if X is* quasi-compact *and Y of* finite type *on S). Then there exists an S-morphism $h' : U' \to Y$, where U' is an open of X containing $U \cup E$, such that, for every $x \in E$, the composed*

$$\operatorname{Spec}(k(x)) \longrightarrow U' \xrightarrow{h'} Y$$

is equal to h'_x.

Proof. Note first that, in (20.4.6), if Y is supposed *separated*, there is a largest open $U_0 \supset U$, equal to the domain of the S-rational application corresponding to h, in which h can be extended, and this extension is unique (**I**, 7.2.2); whence the conclusion in this case, by virtue of (20.4.6). In the general case, let E_i be the set of $x \in E$ such that $(x, h'_x(x)) \in V_i$. By virtue of (20.4.6), for every $x \in E$, there is an extension $h^{(x)}$ of h to $U \cup W^{(x)}$, where $W^{(x)}$ is an open neighbourhood of x in X such that the graph of $h^{(x)}|W^{(x)}$ is contained in all the V_i such that $x \in E_i$. Since V_i is separated over X and X is reduced, for two points $x', x'' \in E_i$, the morphisms $h^{(x')}$ and $h^{(x'')}$ coincide on $W^{(x')} \cap W^{(x'')}$, because they coincide on the everywhere dense open $U \cap W^{(x')} \cap W^{(x'')}$. There is therefore a morphism $h_i : U \cup U_i \to Y$ extending h, where U_i is an open of X containing E_i. Moreover, for every pair of indices i, j, the graphs of the restrictions $h_i|(U_i \cap U_j)$ and $h_j|(U_i \cap U_j)$ **IV-4** | 248
are contained in $V_i \cap V_j$; since $V_i \cap V_j$ is separated over X and the preceding morphisms coincide on the everywhere dense open $U \cap U_i \cap U_j$ of $U_i \cap U_j$, they are equal. The morphism h', equal to h on U and to h_i on each U_i, answers the question. $\square$

Remark (20.4.8). — When $E = X - U$, it is clear that if, for every affine open T of X, there exists an S-morphism $h'_T : T \to Y$ extending $h|(U \cap T)$ and such that, for every $x \in T - (U \cap T)$, the composite $\operatorname{Spec}(k(x)) \to T \xrightarrow{h'_T} Y$ is equal to h'_x, then the h'_T are the restrictions of an everywhere-defined S-morphism $h' : X \to Y$ extending h, by the uniqueness assertion in (20.4.6). To prove the existence of h', one is therefore reduced to the case where X is *affine*, and it then suffices that the union F of the graphs of the h'_x, is *quasi-compact* in $X \times_S Y$ so that one can apply (20.4.7). The latter will hold in particular when the h'_x are of the form $\operatorname{Spec}(k(x)) \to Z \xrightarrow{h_0} Y$, where Z is a closed sub-prescheme of X having $X - U$ for underlying space, and h_0 an S-morphism.

Corollary (20.4.9). — *Let S be a locally Noetherian prescheme, X a locally Noetherian prescheme, $f : X \to S$ a flat morphism, and $g : Y \to S$ a morphism locally of finite type. Let U be a dense open of S, $h : f^{-1}(U) \to Y$ an S-morphism, $T = S - U$, Z the reduced closed sub-prescheme of X having $f^{-1}(T)$ as underlying space, and $h_0 : Z \to Y$ an S-morphism. Suppose X normal at all points of Z. In order that there* *exist a (necessarily unique) S-morphism $h' : X \to Y$ extending h and h_0, it is necessary and sufficient that the following condition be satisfied:*

For every spectrum S_1 of a discrete valuation ring, of closed point a and generic point b, and for every morphism $u : S_1 \to S$ such that $u(a) \in T$ and $u(b) \in U$, there exists an S_1-morphism $h'_1 : X_{(S_1)} \to Y_{(S_1)}$ which extends $h_{(S_1)} : f^{-1}_{(S_1)}(b) \to Y_{(S_1)}$ and is such that the diagram

$$\begin{tikzcd} \mathrm{Spec}(k(z_1)) \arrow[r] \arrow[d] & Z_{(S_1)} \arrow[d, "(h_0)_{(S_1)}"] \\ X_{(S_1)} \arrow[r, "h'_1"'] & Y_{(S_1)} \end{tikzcd}$$

is commutative for every $z_1 \in Z_{(S_1)}$.

The hypothesis that f is flat implies indeed that $f^{-1}(U)$ is dense in X (2.3.10)*, and it then suffices to apply* (20.4.8).

Corollary (20.4.10). — *Under the hypotheses of* (20.4.6)*, suppose moreover Y separated and locally quasi-finite on S. Let U be an open schematically dense in X, $h : U \to Y$ an S-morphism, ω the S-rational application corresponding to it, $x \in X - U$ a normal point of X. In order that ω be defined at the point x, it is necessary and sufficient that the following condition be satisfied:* there exists a spectrum S_1 of a discrete valuation ring, of closed point a and generic point b, and a morphism $u : S_1 \to X$ such that $u(a) = x$, $u(b) \in U$, and the S-rational application $\omega \circ u$ is everywhere defined. **IV-4** | 249

Indeed, by hypothesis all the fibres of the projection morphism $X \times_S Y \to X$ consist of isolated points, and to apply (20.4.4) *it suffices to know that the fibre $p^{-1}(x)$ is nonempty. Now, if h_1 is the unique morphism in the class $\omega \circ u$, the unique point of $X \times_S Y$ lying over x and over $h_1(a)$ belongs to Γ_ω, whence the conclusion.*

Proposition (20.4.11). — *Let X be a locally Noetherian S-prescheme, Y an S-prescheme affine over S, U an open of X, and $Z = X - U$. Suppose that* $\mathrm{prof}_Z(\mathscr{O}_X) \geqslant 2$ (5.10.1)*; then every S-morphism $f : U \to Y$ extends uniquely to an S-morphism of X in Y.*

Proof. We can restrict to the case where S and X are affines, and (by virtue of (**I**, 3.3.14)) to the case $S = X$; so one has $X = \mathrm{Spec}(A)$, $Y = \mathrm{Spec}(B)$, B being a finite type A-algebra. As B is a quotient of a polynomial algebra $A[(T_\lambda)]_{\lambda \in L}$, Y is a closed sub-prescheme of $Y' = X[(T_\lambda)]_{\lambda \in L}$. On the other hand, the hypothesis on Z implies that U is schematically dense in X by virtue of (20.2.13, (iv)) and (5.10.2). If we prove that every X-morphism $u : U \to Y$ extends uniquely to an X-morphism $v' : X \to Y'$, it will result that v' will factorise as $X \xrightarrow{v} Y \to Y'$: indeed, the sub-prescheme $v'^{-1}(Y)$ is closed and contains U (**I**, 4.4.1), and hence is equal to X (20.3.8.8). In these conditions, v will be the unique S-morphism of X in Y extending u. But then there is a bijective correspondence between the X-morphisms of an open $U \subset X$ in Y' and the families $(s_\lambda)_{\lambda \in L}$ of sections of $\mathscr{O}_X$ over U (**II**, 1.7.9); the conclusion results therefore from (5.10.5). □

Corollary (20.4.12). — *Let X be an S-prescheme locally Noetherian reduced and satisfying the condition (S_2)* (5.7.2) *(for example a locally Noetherian and* normal *prescheme* (5.8.6)*), Y an S-prescheme affine on S, f an S-rational application of X in Y; then every irreducible component of $X - \mathrm{dom}(f)$ is of codimension 1 in X.*

Proof. This amounts to the same as saying that if $Z^{(2)}$ is the set of $x \in X$ such that $\dim(\mathscr{O}_{X,x}) \geqslant 2$, then, for every closed part $Z \subset Z^{(2)}$ of X, every S-morphism of $X - Z$ in Y extends to an S-morphism of X in Y; but this results from the hypothesis on X (5.7.2) and from (20.4.11). □

20.5. Relative pseudo-morphisms

(20.5.1). Let X, Y be two S-preschemes. It results from the definitions (11.10.8) that the intersection of two opens U, U' of X, *universally schematically dense with respect to* S, possesses this same property. We therefore define an equivalence relation between S-morphisms $u : U \to Y$ by replacing in (20.2.1) "schematically dense" by "universally schematically dense with respect to S". An equivalence class for this relation is called a *pseudo-morphism of* X *in* Y *relatively to* S, and the set of these classes is denoted $\mathrm{Ps.\,hom}_{X/S}(X, Y)$.

(20.5.2). As every open of X universally schematically dense with respect to S is in particular schematically dense, the elements of a pseudo-morphism of X in Y relatively to S are equivalent in the sense of (20.2.1), whence a canonical map **IV-4** | 250

(20.5.2.1) $$\mathrm{Ps.\,hom}_{X/S}(X, Y) \longrightarrow \mathrm{Ps.\,hom}_S(X, Y).$$

Proposition (20.5.3). — *Suppose* Y *separated on* S. *Then:*

(i) *The map* (20.5.2.1) *is injective and identifies* $\mathrm{Ps.\,hom}_{X/S}(X, Y)$ *with the subset of* $\mathrm{Ps.\,hom}_S(X, Y)$ *formed of the pseudo-S-morphisms* ω *such that* $\mathrm{dom}_S(\omega)$ *is universally schematically dense relatively to* S.

(ii) *The presheaf* $U \mapsto \mathrm{Ps.\,hom}_{U/S}(U, Y)$ *on* X *is a sheaf.*

Proof. Assertion (i) is immediate, for two S-morphisms $u : U \to Y$, $u' : U' \to Y$ are equivalent in the sense of (20.2.1) means that they are the restrictions of a single morphism $\mathrm{dom}_S(\omega) \to Y$ (20.2.4), and if U and U' are universally schematically dense with respect to S, the same holds *a fortiori* for $\mathrm{dom}_S(\omega)$. To prove (ii), note that $U \mapsto \mathrm{Ps.\,hom}_S(U, Y)$ is then a sheaf (20.2.6); on the other hand, given an open covering (U_α) of U and a pseudo-S-morphism ω from U to Y, it follows immediately from the definitions (cf. (20.2.1)) that $\mathrm{dom}_S(\omega)$ is universally schematically dense in U relative to S if and only if each $\mathrm{dom}_S(\omega) \cap U_\alpha = \mathrm{dom}_S(\omega | U_\alpha)$ is universally schematically dense in U_α relative to S; whence (ii). $\square$

When Y is separated, we denote by $\mathscr{P}\!s.\mathrm{hom}_{X/S}(X, Y)$ the sub-sheaf

$$U \longmapsto \mathrm{Ps.\,hom}_{U/S}(U, Y)$$

of $\mathscr{P}\!s.\mathrm{hom}_S(X, Y)$.

When X is *flat and locally of finite presentation* over S and Y separated on S, the pseudo-morphisms of X in Y relatively to S are identified with the pseudo-S-morphisms ω such that, for every fibre X_s of the morphism $X \to S$, $\mathrm{dom}_S(\omega) \cap X_s$ is *schematically dense* in X_s (11.10.10).

(20.5.4). A particularly important case is that where Y is separated on S and $Y = S[T] = S \otimes_{\mathbf{Z}} \mathrm{Spec}(\mathbf{Z}[T])$ (T indeterminate). There is then a bijective correspondence between the pseudo-S-morphisms of X in Y and the *pseudo-functions* on X (20.2.8) by the definition of a product of $\mathbf{Z}$-preschemes. The pseudo-morphisms of X in Y relatively to S are identified then, by virtue of (20.5.3), with the *pseudo-functions* φ *on* X *such that* $\mathrm{dom}(\varphi)$ *is universally schematically dense relatively to* S. The sheaf $\mathscr{P}\!s.\mathrm{hom}_{X/S}(X, Y)$ is a *sub-sheaf of rings* of $\mathscr{M}'_X$, that we denote $\mathscr{M}'_{X/S}$.

We then set $\mathrm{Ps.\,hom}_{X/S}(X, Y) = M'(X/S)$ and we say that its elements are the *pseudo-functions on* X *relative to* S.

(20.5.5). Let X, Y, Z be three S-preschemes, $f : Y \to Z$ an S-morphism. We can, in the reasoning of (20.3.1), replace everywhere "schematically dense" by "universally schematically dense relatively to S"; for every pseudo-morphism $\omega \in \mathrm{Ps.\,hom}_{X/S}(X, Y)$, the morphisms $f \circ u$, where u runs through ω, belong therefore to a single equivalence class (for the relation defined in (20.5.1)), which we call the pseudo-morphism from X to Z, relative to S, *composite* of f and ω, and denote $f \circ \omega$. If $g : Z \to T$ is a morphism, it is immediate that $g \circ (f \circ \omega) = (g \circ f) \circ \omega$. **IV-4** | 251

(20.5.6). Suppose Y *separated* on S, and let ω be a pseudo-morphism of X in Y relatively to S. If $f : X' \to X$ is an S-morphism such that $f^{-1}(\mathrm{dom}_S(\omega))$ is universally schematically dense in X' *relative to* S, it follows from (20.3.2) that the pseudo-S-morphism $\omega \circ f$ has a domain universally schematically dense relative to S, and hence, by (20.5.3), may be regarded as a pseudo-morphism *relative to* S. When X' is flat and locally of finite presentation over S, the condition that $f^{-1}(\mathrm{dom}_S(\omega))$

be universally schematically dense in X' relative to S is equivalent to saying that for every $s \in S$, $(f^{-1}(\mathrm{dom}_S(\omega)))_s$ (notation of (11.10.10)) is schematically dense in X'_s. This latter condition will in particular be satisfied if, for every $s \in S$, X'_s and f_s satisfy one of the three conditions (i), (ii), (iii) of (20.3.5).

(20.5.7). Suppose now that X and X' are both S-preschemes *flat and locally of finite presentation* over S, and that $f : X' \to X$ is an S-morphism *flat* (or, what amounts to the same (11.3.10), that for every $s \in S$, $f_s : X'_s \to X_s$ is a flat morphism). Then, for every open U of X and every open $V \subset U$ universally schematically dense in U relatively to S, it results from (11.10.5) and (11.10.9) that $f^{-1}(V)$ is universally schematically dense in $f^{-1}(U)$ relatively to S. For every pseudo-S-morphism ω of X in Y relatively to S, it results from (20.3.4) that the pseudo-S-morphism $\omega \circ f$ is defined and is a pseudo-morphism of X' in Y relatively to S, *even when Y is not supposed separated on S*. We deduce from this that for every S-morphism $g : Y \to Z$, $(g \circ \omega) \circ f$ is still defined and equal to $g \circ (\omega \circ f)$ (with the definitions of (20.5.1)), and that it is therefore a pseudo-morphism *relative* to S.

(20.5.8). Let X be an S-prescheme, $S' \to S$ a morphism, $g : X' \to X$ the canonical projection, $X' = X \times_S S'$. Then, for every open U of X and every open $V \subset U$ universally schematically dense in U relatively to S, $V' = g^{-1}(V)$ is universally schematically dense in $U' = g^{-1}(U)$ relatively to S' by definition (11.10.8). Let ω be a pseudo-S-morphism of X in an S-prescheme Y relatively to S; if u_1, u_2 are S-morphisms of the class ω, defined respectively on opens U_1, U_2 of X, universally schematically dense in X relative to S, it follows from what precedes that $u'_1 = u_1 \circ (g|g^{-1}(U_1))$ and $u'_2 = u_2 \circ (g|g^{-1}(U_2))$ coincide on an open U'_3 universally schematically dense relative to S'. Now, if $Y' = Y_{(S')}$ and $h : Y' \to Y$ is the canonical projection, u'_1 and u'_2 factor canonically as $u'_1 = h \circ v_1$ and $u'_2 = h \circ v_2$, where v_1 and v_2 are S'-morphisms into Y' which coincide on U'_3. Thus, as u_1 runs through the class ω, the corresponding S'-morphisms v_1 belong to a single pseudo-morphism relative to S', called the *inverse image* of ω under the base change $S' \to S$ and denoted $\omega_{(S')}$. It is clear that if $S'' \to S'$ is a second morphism, $(\omega_{(S')})_{(S'')} = \omega_{(S'')}$ (for the composed morphism $S'' \to S' \to S$).

20.6. Relative meromorphic functions

(20.6.1). Let S be a prescheme, X an S-prescheme which is *flat and locally of finite presentation* over S; for every $s \in S$, we denote by X_s the fibre at the point s of the structural morphism $X \to S$. In general, if φ is a meromorphic function on X, it is not possible, for each $s \in S$, to associate to it in a "natural" way a meromorphic function "induced" on X_s (20.1.11). For every open U of X, denote by $T_{X/S}(U)$ the set of sections $t \in \Gamma(U, \mathscr{O}_X)$ such that, for every $s \in S$, the image t_s of t by the canonical homomorphism $\Gamma(U, \mathscr{O}_X) \to \Gamma(U \cap X_s, \mathscr{O}_{X_s})$ is a *regular* section; this also implies, by the equivalence of a) and b) in (11.3.7), that t is itself a *regular* section. It is clear that $U \mapsto T_{X/S}(U)$ is a *sub-sheaf* of the sheaf of sets $\mathscr{S} = \mathscr{S}(\mathscr{O}_X)$ (notation of (20.1.3)), that we denote $\mathscr{T}_{X/S}$; we set IV-4 | 252

$$\mathscr{M}_{X/S} = \mathscr{O}_X[\mathscr{T}_{X/S}^{-1}] \tag{20.6.1.1}$$

(notation of (20.1.1)) and we say that this sheaf of rings is the *sheaf of germs of meromorphic functions on X relative to S*; its sections over X are called *meromorphic functions on X relative to S* and their set is denoted $M(X/S)$. It is clear that $\mathscr{M}_{X/S}$ is a *sub-sheaf* of $\mathscr{M}_X = \mathscr{O}_X[\mathscr{S}^{-1}]$; for every meromorphic function $\varphi \in M(X/S)$ and every $s \in S$, the inverse image of φ by the canonical injection morphism $j_s : X_s \to X$ is then defined (20.1.11), and denoted φ_s.

(20.6.2). Let now $\mathscr{F}$ be an $\mathscr{O}_X$-Module quasi-coherent; we set

$$\mathscr{M}_{X/S}(\mathscr{F}) = \mathscr{F}[\mathscr{T}_{X/S}^{-1}] = \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{M}_{X/S}; \tag{20.6.2.1}$$

the sections of $\mathscr{M}_{X/S}(\mathscr{F})$ are called *meromorphic sections of $\mathscr{F}$ over X, relative to S* and their set is denoted $M(X/S, F)$. The canonical homomorphism $\mathscr{F} \to \mathscr{M}_{X/S}(\mathscr{F})$ is not necessarily injective; when it is, we say that $\mathscr{F}$ is *without torsion relatively to S*: this means that for every open U of X and every section $t \in T_{X/S}(U)$, t is $(\mathscr{F}|U)$-regular; this condition is satisfied *a fortiori* when $\mathscr{F}$ is strictly torsion-free (20.1.5). In this last case, it results immediately from the definitions (20.1.2) that the canonical homomorphism of $\mathscr{O}_X$-Modules

$$\mathscr{M}_{X/S}(\mathscr{F}) \longrightarrow \mathscr{M}_X(\mathscr{F}) \tag{20.6.2.2}$$

is *injective*, so that the meromorphic sections of $\mathscr{F}$ relative to S are meromorphic sections of $\mathscr{F}$ in the sense of (20.1.3).

Proposition (20.6.3). — *The image by the injective homomorphism (20.2.10.1) of the sub-sheaf $\mathscr{M}_{X/S}$ of $\mathscr{M}_X$ is the sub-sheaf $\mathscr{M}'_{X/S}$ of pseudo-functions on X relative to S (i.e. of the pseudo-functions whose domain of definition is universally schematically dense relatively to S (20.5.4)).* **IV-4 | 253**

Proof. We can evidently restrict to proving that the image of $M(X/S)$ by the canonical homomorphism $M(X) \to M'(X)$ is equal to $M'(X/S)$; the proposition is then a consequence of the following more general proposition: □

Proposition (20.6.4). — *Let $\mathscr{F}$ be an $\mathscr{O}_X$-Module quasi-coherent of finite presentation and strictly torsion-free. Then, for a meromorphic section φ of $\mathscr{F}$ over X to be a meromorphic section relative to S, it is necessary and sufficient that* $\mathrm{dom}(\varphi)$ *be universally schematically dense relatively to S.*

Proof. The necessity of the condition results from (20.2.15) applied to each canonical injection $X_s \to X$ ($s \in S$), taking into account (11.10.9). To see that the condition is sufficient, it suffices to prove that for every $x \in X$, there exists a neighbourhood U of x and a section of $\mathscr{M}_{X/S}(\mathscr{F})$ over U whose restriction to $U \cap \mathrm{dom}(\varphi)$ coincides with φ in an open schematically dense of $U \cap \mathrm{dom}(\varphi)$. Consider the ideal of denominators $\mathscr{J}$ of φ (20.2.14), which is quasi-coherent and defines the closed sub-prescheme of X whose underlying space is $X - \mathrm{dom}(\varphi)$. By hypothesis, if s is the image of x in S, $\mathrm{dom}(\varphi) \cap X_s$ is schematically dense in the locally Noetherian prescheme X_s, and hence contains $\mathrm{Ass}(\mathscr{O}_{X_s})$ (20.2.13, (iv)); this implies that the Ideal $\mathscr{J}_x$ has an image in $\mathscr{O}_{X_s,x} = \mathscr{O}_{X,x} \otimes_{\mathscr{O}_{S,s}} k(s)$ which is not contained in any of the prime ideals $\mathfrak{p}_i \in \mathrm{Ass}(\mathscr{O}_{X_s,x})$ (finitely many); hence (Bourbaki, *Alg. comm.*, chap. II, §1, n° 1, prop. 2) there exists an element $t_x \in \mathscr{J}_x$ whose image in $\mathscr{O}_{X_s,x}$ does not belong to any of the $\mathfrak{p}_i$, and is therefore *regular* in this Noetherian ring. Let t be a section of $\mathscr{J}$ over an affine neighbourhood U of x, whose germ at x is t_x; since X is flat and locally of finite presentation over S, we can suppose (11.3.8) that t is a regular section of $\mathscr{O}_X$ over U and that, for every $s' \in S$, the image of t in $\Gamma(U \cap X_{s'}, \mathscr{O}_{X_{s'}})$ is also regular; in other words, $t \in \mathrm{T}_{X/S}(U)$. But then, since $\mathscr{F}$ is strictly torsion-free, $t(\varphi|(U \cap \mathrm{dom}(\varphi)))$ is a section u of $\mathscr{F}$ over $U \cap \mathrm{dom}(\varphi)$; on the other hand, $U \cap \mathrm{dom}(\varphi)$ contains the open U_t of $x' \in U$ where $t(x') \neq 0$, and this last is schematically dense in U (20.2.9). We see that in U_t, φ coincides with the restriction to U_t of the section u/t of $\mathscr{M}_{X/S}(\mathscr{F})$ over U. C.Q.F.D. □

Remarks (20.6.5). — (i) Let φ be a meromorphic function on X relative to S, so that for every $s \in S$, φ_s is a meromorphic function on X_s (20.6.1); by virtue of (20.1.11.1), one has

$$\mathrm{dom}(\varphi) \cap X_s \subset \mathrm{dom}(\varphi_s). \tag{20.6.5.1}$$

But it is worth noting that even when S is the spectrum of a discrete valuation ring A and $X = S[T]$ (T indeterminate), the two sides of (20.6.5.1) are not necessarily equal: for example, if π is a uniformizer of A, it is immediate that $\varphi = \pi/T$ is a meromorphic function on X relative to S, since if a and b are the closed and generic points of S, T is regular in $\Gamma(X_a, \mathscr{O}_{X_a}) = k[T]$ and in $\Gamma(X_b, \mathscr{O}_{X_b}) = K[T]$, k being the residue field and K the field of fractions of A. One has $\mathrm{dom}(\varphi) = \mathrm{D}(T)$ in X, but $\mathrm{dom}(\varphi_a) = X_a$ since $\varphi_a = 0$.

(ii) For a meromorphic function φ relative to S to be *invertible in the ring* $M(X/S)$, it is necessary and sufficient that for every $s \in S$, φ_s be invertible in $M(X_s)$ (in other words, that φ_s be a regular meromorphic function on X_s (20.1.8)). The condition is trivially necessary. Conversely, if it is satisfied, and if x is any point of X, s its image in S, there exists by hypothesis a neighbourhood U of x and two sections u, t of $\mathscr{O}_X$ over U such that $t \in \mathrm{T}_{X/S}(U)$ and $\varphi|U = u/t$; the hypothesis implies that u_s is regular at x in $\Gamma(U \cap X_s, \mathscr{O}_{X_s})$, so u_s is regular at point x. Restricting U, we can therefore suppose, by virtue of (11.3.8), that $u \in \mathrm{T}_{X/S}(U)$, whence the conclusion. **IV-4 | 254**

When φ is invertible in $M(X/S)$, we also say that φ is a *regular meromorphic function relative to* S. We note that $\varphi \in M(X/S)$ can be invertible in $M(X)$ (in other words, be *regular* in the sense of (20.1.8)) without being invertible in $M(X/S)$, as the above example of (i) shows.

(iii) Let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module, and let φ be a regular meromorphic section of $\mathscr{L}$ over X (20.1.8); we say that φ is *regular relatively to* S if, for every open U of X such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_U$, $\varphi|U$ corresponds to an element of $\Gamma(U, \mathscr{M}_{X/S})$ which is *regular relatively to* S (20.1.8); by virtue of (ii), it is immediate that it is necessary and sufficient that, for every $s \in S$, φ_s be a regular meromorphic section (20.1.8) of the invertible $\mathscr{O}_{X_s}$-Module $\mathscr{L}_s = \mathscr{L} \otimes_{\mathscr{O}_X} k(s)$. If φ' is the inverse of φ in $\mathscr{L}^{-1}$ (20.1.10), φ' is then also regular relatively to S, and if $\mathscr{L}_1$ is a second $\mathscr{O}_X$-Module invertible, φ_1 a meromorphic section of $\mathscr{L}_1$ over X, regular relatively to S, then $\varphi \otimes \varphi_1$ is a meromorphic section of $\mathscr{L} \otimes \mathscr{L}_1$ over X, regular relatively to S.

Proposition (20.6.6). — *Let X be an S-prescheme flat and locally of finite presentation over S, and let $\mathscr{F}$ be an $\mathscr{O}_X$-Module locally free of finite type; for every $s \in S$, denote by X_s the fibre at the point s of the structural morphism $f : X \to S$. Let φ be a meromorphic section of $\mathscr{F}$ over X,* relatively to S, *and suppose that φ is defined at all points $x \in X$ such that* $\mathrm{prof}(\mathscr{O}_{X_{f(x)},x}) = 1$. *Then φ is everywhere defined.*

Proof. By hypothesis, $\mathrm{dom}(\varphi) \cap X_s$ is schematically dense in X_s for every $s \in S$, hence contains the points x of X_s such that $\mathrm{prof}(\mathscr{O}_{X_s,x}) = 0$ (5.10.2); the hypothesis therefore signifies that if one sets $Z = X - \mathrm{dom}(\varphi)$, one has $\mathrm{prof}(\mathscr{O}_{X_{f(x)},x}) \geqslant 2$ for all points x of Z. It suffices then to apply (19.9.8). □

(20.6.7). Let X, X' be two S-preschemes flat and locally of finite presentation, $f = (\psi, \theta) : X' \to X$ an S-morphism. For every open U of X, denote by $T_f(U)$ the set of sections $t \in T_{X/S}(U)$ whose image in $\Gamma(f^{-1}(U), \mathscr{O}_{X'})$ belongs to $T_{X'/S}(f^{-1}(U))$; it is immediate that $U \mapsto T_f(U)$ is a sub-sheaf of the sheaf of sets $\mathscr{T}_{X/S}$, that we denote $\mathscr{T}_f$. We set $\mathscr{M}_{X/S,f} = \mathscr{O}_X[\mathscr{T}_f^{-1}]$; this is a sub-sheaf of rings of $\mathscr{M}_{X/S}$, and we canonically obtain from $\theta^\# : \psi^*(\mathscr{O}_X) \to \mathscr{O}_{X'}$ a homomorphism of sheaves of rings $\theta'^\# : \psi^*(\mathscr{M}_{X/S,f}) \to \mathscr{M}_{X'/S}$ extending $\theta^\#$. A meromorphic function φ on X, relative to S, that *is a section of $\mathscr{M}_{X/S,f}$*, then has an inverse image $\Gamma(\theta'^\#)(\varphi)$ which is a meromorphic function on X' relative to S, called the *inverse image of φ by f*, and denoted $\varphi \circ f$ if this causes no confusion. We extend immediately in the same way the definitions of (20.1.11) relative to $\mathscr{O}_X$-Modules.

Proposition (20.6.8). — *With the notations of (20.6.7), if the S-morphism $f : X' \to X$ is* flat, *one has $\mathscr{M}_{X/S,f} = \mathscr{M}_{X/S}$, and the homomorphism $\varphi \mapsto \varphi \circ f$ is defined on all of $M(X/S)$.*

Proof. Indeed, the hypothesis implies, by virtue of (11.3.10), that for every $s \in S$, $f_s : X'_s \to X_s$ is flat; hence, for every section $t \in T_{X/S}(U)$, if t' is its inverse image, t'_s is a *regular* section of $\mathscr{O}_{X'_s}$ over $f^{-1}(U) \cap X'_s$; by virtue of (20.1.12), we conclude that by definition $t' \in T_{X'/S}(f^{-1}(U))$, whence the proposition. □ **IV-4 | 255**

We deduce from this, as in (20.1.12), a canonical homomorphism of $\mathscr{O}_{X'}$-algebras

(20.6.8.1) $$f^*(\mathscr{M}_{X/S}) \longrightarrow \mathscr{M}_{X'/S}.$$

(20.6.9). Consider finally a morphism $S' \to S$, and set $X' = X \times_S S'$, which is flat and locally of finite presentation over S'; let $p : X' \to X$ be the canonical projection. Let U be an open of X, t a section belonging to $T_{X/S}(U)$, t' its image in $\Gamma(p^{-1}(U), \mathscr{O}_{X'})$; for every $s' \in S'$, if s is the image of s', we have $X'_{s'} = X_s \otimes_{k(s)} k(s')$, hence the morphism $X'_{s'} \to X_s$ is flat, and by virtue of (20.1.12) the inverse image $t'_{s'}$ of t_s is regular; this proves that $t' \in T_{X'/S'}(p^{-1}(U))$. We deduce from this, as in (20.1.12), a canonical homomorphism of $\mathscr{O}_{X'}$-Algebras

(20.6.9.1) $$p^*(\mathscr{M}_{X/S}) \longrightarrow \mathscr{M}_{X'/S'}.$$

Using the identification permitted by (20.6.3), this notion of base change for relative meromorphic functions is a particular case of the analogous notion for relative pseudo-morphisms (20.5.8).

§21. Divisors

For the contents of the present section, see the comments in the introduction to ğ20. For the *global* properties of divisors, the reader is referred to the section devoted to them in Chapter V.

21.1. Divisors on a ringed space

(21.1.1). Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{M}_X$ be the sheaf of germs of meromorphic functions on X (20.1.3), and $\mathscr{M}_X^*$ the sheaf (of multiplicative groups) of germs of regular meromorphic functions on X (20.1.8). It is clear that the sheaf $\mathscr{O}_X^*$ (of multiplicative groups) of germs of invertible sections of $\mathscr{O}_X$ is canonically identified with a subsheaf (of multiplicative groups) of $\mathscr{M}_X^*$.

(21.1.2). We call the quotient sheaf (of commutative groups) $\mathscr{M}_X^*/\mathscr{O}_X^*$ the *sheaf of divisors on* X, and denote it by $\mathscr{D}iv_X$. Its sections over X are called the *divisors on* X; they form a commutative group denoted $\mathrm{Div}(X)$. For every section f of $\mathscr{M}_X^*$ over X (that is, every regular meromorphic function on X (20.1.8)), the *divisor of* f, denoted $\mathrm{div}(f)$ (or $\mathrm{div}_X(f)$), is the divisor on X which is the image of f under the canonical homomorphism

$$\Gamma(X, \mathscr{M}_X^*) \longrightarrow \Gamma(X, \mathscr{D}iv_X).$$

The *support* of a divisor D is the closed set of $x \in X$ such that $D_x \neq 0$. We denote it $\mathrm{Supp}(D)$. **IV-4** | 256

For every open U of X, we evidently have $\mathscr{M}_X^*|U = \mathscr{M}_U^*$, $\mathscr{O}_X^*|U = \mathscr{O}_U^*$, hence $\mathscr{D}iv_X|U = \mathscr{D}iv_U$, and consequently the sheaf $\mathscr{D}iv_X$ is equal to the presheaf $U \mapsto \mathrm{Div}(U)$.

When $X = \mathrm{Spec}(A)$ is affine, we write $\mathrm{Div}(A)$ instead of $\mathrm{Div}(\mathrm{Spec}(A))$.

(21.1.3). We will always denote *additively* the group $\mathrm{Div}(X)$ of divisors on X. For two regular meromorphic functions f, g on X, we thus have

$$\mathrm{div}(fg) = \mathrm{div}(f) + \mathrm{div}(g), \tag{21.1.3.1}$$

$$\mathrm{div}(f^{-1}) = -\mathrm{div}(f). \tag{21.1.3.2}$$

By definition, for every regular meromorphic function f on X, we have the equivalence

$$\mathrm{div}(f) = 0 \iff f \in \Gamma(X, \mathscr{O}_X^*), \tag{21.1.3.3}$$

whence, for two regular meromorphic functions f, g on X, we have

$$\mathrm{div}(f) = \mathrm{div}(g) \iff fg^{-1} \in \Gamma(X, \mathscr{O}_X^*). \tag{21.1.3.4}$$

(21.1.4). Now let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module, and let s be a *regular meromorphic section* (20.1.8) of $\mathscr{L}$ over X. Every $x \in X$ admits a neighbourhood U such that $\mathscr{L}|U$ is isomorphic to $\mathscr{O}_U$, hence $\mathscr{M}_X(\mathscr{L})|U$ is isomorphic to $\mathscr{M}_X|U$; by one of these isomorphisms, $s|U$ corresponds to a section $f \in \Gamma(U, \mathscr{M}_X^*)$, and since two isomorphisms differ only by multiplication by an element of $\Gamma(U, \mathscr{O}_X^*)$ ($\mathbf{0_I}$, 5.4.3), the element $\mathrm{div}_U(f)$ of $\Gamma(U, \mathscr{D}iv_X)$ is *independent* of the isomorphism chosen; it is clear that these elements (for U variable) are the restrictions of a section of $\mathscr{D}iv_X$ over X, which we call the *divisor of* s and denote $\mathrm{div}(s)$ (such a divisor is not necessarily of the form $\mathrm{div}(g)$ for a regular meromorphic function g on X; see (21.2.9)). For $\mathscr{L} = \mathscr{O}_X$, the definition of $\mathrm{div}(s)$ coincides with that of (21.1.2). If $\mathscr{L}, \mathscr{L}'$ are two invertible $\mathscr{O}_X$-Modules, s (resp. s') a regular meromorphic section of $\mathscr{L}$ (resp. $\mathscr{L}'$) over X, we immediately have

$$\mathrm{div}(s \otimes s') = \mathrm{div}(s) + \mathrm{div}(s') \tag{21.1.4.1}$$

$$\mathrm{div}(s^{\otimes n}) = n.\mathrm{div}(s) \quad \text{for all} \quad n \in \mathbf{Z} \tag{21.1.4.2}$$

(s^{-1} being the regular meromorphic section of $\mathscr{L}^{-1}$ over X defined in (20.1.10)) and, for two regular meromorphic sections s, s' of $\mathscr{L}$ over X, we have the relation

$$\mathrm{div}(s) = \mathrm{div}(s') \iff s' = ts \quad \text{with} \quad t \in \Gamma(X, \mathscr{O}_X^*). \tag{21.1.4.3}$$

(21.1.5). The sheaf $\mathscr{S}(\mathscr{O}_X)$ (20.1.3) whose sections over an open U of X are the regular elements of $\Gamma(U, \mathscr{O}_X)$ is a *sub-sheaf of monoids* of $\mathscr{M}_X^*$; we can write

$$\mathscr{S}(\mathscr{O}_X) = \mathscr{O}_X \cap \mathscr{M}_X^*. \tag{21.1.5.1}$$

If we denote by $\mathscr{S}(\mathscr{O}_X)^{-1}$ the sheaf whose sections over U are the inverses in $\Gamma(U, \mathscr{M}_X^*)$ of the elements of $\Gamma(U, \mathscr{S}(\mathscr{O}_X))$, it is clear that $\Gamma(U, \mathscr{S}(\mathscr{O}_X)) \cap \Gamma(U, \mathscr{S}(\mathscr{O}_X)^{-1}) = \Gamma(U, \mathscr{O}_X^*)$, hence **IV-4** | 257

$$\mathscr{S}(\mathscr{O}_X) \cap \mathscr{S}(\mathscr{O}_X)^{-1} = \mathscr{O}_X^*. \tag{21.1.5.2}$$

Definition (21.1.6). — The sub-sheaf of sets of $\mathscr{D}iv_X$, the canonical image of the sub-sheaf $\mathscr{S}(\mathscr{O}_X)$ of $\mathscr{M}_X^*$, is denoted $\mathscr{D}iv_X^+$; its sections over X are called *positive divisors* and their set is denoted $\mathrm{Div}^+(X)$.

Since $\mathscr{S}(\mathscr{O}_X)$ is a sheaf of monoids (multiplicative), we have

(21.1.6.1) $$\mathrm{Div}^+(X) + \mathrm{Div}^+(X) \subset \mathrm{Div}^+(X)$$

and on the other hand, by virtue of (21.1.5.2) and (21.1.3.3)

(21.1.6.2) $$\mathrm{Div}^+(X) \cap (-\mathrm{Div}^+(X)) = \{0\}.$$

These two relations show that $\mathrm{Div}^+(X)$ is the set of positive elements for an *order structure* on the group $\mathrm{Div}(X)$, compatible with this group structure; we denote this order relation by $D \leqslant D'$, in other words, we have

(21.1.6.3) $$D \geqslant 0 \iff D \in \mathrm{Div}^+(X).$$

We will always suppose in what follows that $\mathrm{Div}(X)$ is equipped with this order structure.

It is clear that $\mathscr{D}iv_X^+|U = \mathscr{D}iv_U^+$ for every open U of X, and hence that $\Gamma(U, \mathscr{D}iv_X^+) = \mathrm{Div}^+(U)$; thus $\mathscr{D}iv_X^+$ defines on $\mathscr{D}iv_X$ a structure of sheaf of ordered groups. The stalk $(\mathscr{D}iv_X^+)_x$ is a sub-monoid of $(\mathscr{D}iv_X)_x$, consisting of its positive elements. Consequently, for a divisor D on X, to say that $D \geqslant 0$ is equivalent to saying that $D_x \geqslant 0$ for every $x \in X$.

By definition, for every regular meromorphic function f on X, we have the relation

(21.1.6.4) $$\mathrm{div}(f) \geqslant 0 \iff f \in \Gamma(X, \mathscr{S}(\mathscr{O}_X))$$

in other words, $\mathrm{div}(f) \geqslant 0$ means that f is a *regular section* of $\mathscr{O}_X$, or equivalently a section of $\mathscr{O}_X$ *invertible in* $M(X)$.

More generally, given a divisor D on X, the relation $\mathrm{div}(f) \geqslant D$ is equivalent to the following: for every open U of X such that $D|U = \mathrm{div}(g)$, where $g \in \Gamma(U, \mathscr{M}_X^*)$, there exists a regular element t of $\Gamma(U, \mathscr{O}_X)$ such that $f|U = tg$.

(21.1.7). Let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-Module, s a regular meromorphic section of $\mathscr{L}$ over X; we have the relation

(21.1.7.1) $$\mathrm{div}(s) \geqslant 0 \iff s \in \Gamma(X, \mathscr{L}) \cap \Gamma(X, (\mathscr{M}_X(\mathscr{L}))^*)$$

as follows immediately from the definitions (21.1.4) and (21.1.6).

Proposition (21.1.8). — *Let X be a locally Noetherian prescheme, D a divisor on X. Suppose that for every $x \in X$ such that* $\mathrm{prof}(\mathscr{O}_{X,x}) = 1$ *we have $D_x \geqslant 0$ (resp. $D_x = 0$). Then $D \geqslant 0$ (resp. $D = 0$).*

Proof. The question being local on X, we can suppose that $D = \mathrm{div}(f)$, f being a regular meromorphic function on X; the hypothesis means that if $T = X - \mathrm{dom}(f)$, then $\mathrm{prof}(\mathscr{O}_{X,x}) \geqslant 2$ for every $x \in T$ (since $\mathrm{dom}(f)$ contains the maximal points of X). By (5.10.5), the restriction homomorphism $\Gamma(X, \mathscr{O}_X) \to \Gamma(X - T, \mathscr{O}_X)$ is *bijective*, which shows that there exists a section s of $\mathscr{O}_X$ over X such that $f = s|(X - T)$. But by definition of T, this implies $T = \varnothing$, hence $f = s$ and $D \geqslant 0$, and we immediately deduce the conclusion by applying the preceding to $-D$, by virtue of (21.1.6.2). ☐ **IV-4** | 258

Corollary (21.1.9). — *Let X be a locally Noetherian prescheme, D a divisor on X. Let S be the support of D. Then, for every maximal point x of S,* $\mathrm{prof}(\mathscr{O}_{X,x}) = 1$.

Proof. Set $X_1 = \mathrm{Spec}(\mathscr{O}_{X,x})$; taking into account (20.2.11) and (20.3.6), the sheaf $\mathscr{M}_{X_1}^*$ is induced on X_1 by $\mathscr{M}_X^*$, hence we can restrict to the case where $X = X_1$, in which case $x \in S$ and x is the maximal point of S, so necessarily $S = \{x\}$. If $\mathrm{prof}(\mathscr{O}_{X,x}) \neq 1$, we would conclude, by virtue of (21.1.8), that $D = 0$, which contradicts the definition of S. ☐

Proposition (21.1.10). — *Let A be a local Noetherian ring; in order that* $\mathrm{Div}(A) = 0$, *it is necessary and sufficient that* $\mathrm{prof}(A) = 0$, *in other words, that the maximal ideal $\mathfrak{m}$ of A be associated to A (**0**, 16.4.6).*

Proof. Indeed, to say that $\mathrm{Div}(A) = 0$ means that in A every regular element is invertible, or equivalently that all the elements of $\mathfrak{m}$ are divisors of zero, which means that $\mathfrak{m} \in \mathrm{Ass}(A)$ (Bourbaki, *Alg. comm.*, chap. IV, § 1, n° 1, cor. 3 of prop. 2). ☐

21.2. Divisors and invertible fractionary ideals

(21.2.1). Let $(X, \mathscr{O}_X)$ be a ringed space. We call *fractionary Ideal* on X a sub-$\mathscr{O}_X$-Module of the $\mathscr{O}_X$-Module $\mathscr{M}_X$ of germs of meromorphic functions on X. A fractionary Ideal $\mathscr{J}$ on X which is an invertible $\mathscr{O}_X$-Module is called an *invertible fractionary Ideal*.

Proposition (21.2.2). — *For an invertible fractionary Ideal $\mathscr{J}$ on X to be invertible, it is necessary and sufficient that for every $x \in X$, there exists a neighbourhood U of x and a section $f \in \Gamma(U, \mathscr{M}_X^*)$ such that $\mathscr{J}|U = \mathscr{O}_U.f$.*

Proof. The condition is evidently sufficient, since the map $s \mapsto s(f|V)$ of $\Gamma(V, \mathscr{O}_X)$ is evidently bijective for every open $V \subset U$. To see that it is necessary, let us note that there exists by hypothesis a neighbourhood U of x and an isomorphism of $\mathscr{O}_X$-Modules $\mathscr{O}_U \xrightarrow{\sim} \mathscr{J}|U$. If f is the image of the section $1 \in \Gamma(U, \mathscr{O}_X)$ by this isomorphism, we may suppose, after restricting U, that $f = u/s$, where $u \in \Gamma(U, \mathscr{O}_X)$ and $s \in \Gamma(U, \mathscr{S}(\mathscr{O}_X))$, and the isomorphism considered then associates, to every section $v \in \Gamma(V, \mathscr{O}_X)$ (where V is an open contained in U), the section $v(u|V)/(s|V)$; to say that this map is bijective means that $u|V$ is a regular element of $\Gamma(V, \mathscr{O}_X)$, hence $f \in \Gamma(U, \mathscr{M}_X^*)$. □

We note that for every open U of X such that $\mathscr{J}|U = \mathscr{O}_U.f$ with $f \in \Gamma(U, \mathscr{M}_X^*)$, the section f is determined *up to multiplication* by an element of $\Gamma(U, \mathscr{O}_X^*)$, since the multiplication by these elements gives all the automorphisms of the $\mathscr{O}_U$-Module $\mathscr{O}_U$ ($\mathbf{0_I}$, 5.4.3). **IV-4** | 259

Corollary (21.2.3). — *(i) Let $\mathscr{J}$ be an invertible fractionary Ideal; then the invertible $\mathscr{O}_X$-Module $\mathscr{J}^{-1}$ is canonically identified with the fractionary Ideal $\mathscr{J}'$ (the* transporter *of $\mathscr{J}$ into $\mathscr{O}_X$) defined as follows: for every open U of X such that $\mathscr{J}|U = \mathscr{O}_U.f$, where $f \in \Gamma(U, \mathscr{M}_X^*)$, we have $\mathscr{J}'|U = \mathscr{O}_U.f^{-1}$.*

(ii) If $\mathscr{J}_1$ and $\mathscr{J}_2$ are two invertible fractionary Ideals, the canonical map $\mathscr{J}_1 \otimes \mathscr{J}_2 \to \mathscr{J}_1\mathscr{J}_2$ is an isomorphism of $\mathscr{O}_X$-Modules.

Proof. The assertion (ii) follows immediately from (21.2.2). On the other hand, the remark made at the end of (21.2.2) shows that there is indeed a unique fractionary Ideal $\mathscr{J}'$ and a unique isomorphism defined by the condition of the statement; the isomorphism of $\mathscr{J}'$ on $\mathscr{J}^{-1} = \mathscr{H}om_{\mathscr{O}_X}(\mathscr{J}, \mathscr{O}_X)$ is obtained by making correspond, for every section $s(f^{-1}|V)$ of $\Gamma(V, \mathscr{J}')$ (where V is an open contained in U and $s \in \Gamma(V, \mathscr{O}_X)$) the homomorphism $t(f|V) \mapsto st$ of $\Gamma(V, \mathscr{J})$ to $\Gamma(V, \mathscr{O}_X)$. □

By virtue of (21.2.3), (i), we will identify the invertible fractionary ideals $\mathscr{J}'$ and $\mathscr{J}^{-1}$ considered as sub-$\mathscr{O}_X$-Modules of $\mathscr{M}_X$.

(21.2.4). It follows from (21.2.3) that the set $\operatorname{Id.inv}(X)$ of invertible fractionary Ideals on X is equipped with a structure of *commutative group* for the law of composition $(\mathscr{J}_1, \mathscr{J}_2) \mapsto \mathscr{J}_1\mathscr{J}_2$, the neutral element of this group being $\mathscr{O}_X$. It is clear that for every open U of X, we have $(\mathscr{J}_1\mathscr{J}_2)|U = (\mathscr{J}_1|U)(\mathscr{J}_2|U)$, hence the restriction map $\mathscr{J} \mapsto \mathscr{J}|U$ is a homomorphism of groups $\operatorname{Id.inv}(X) \to \operatorname{Id.inv}(U)$; we thus define a *presheaf of commutative groups* $U \mapsto \operatorname{Id.inv}(U)$; it is immediate that in fact this presheaf is a *sheaf of commutative groups*, which we denote $\mathscr{J}d.inv_X$.

(21.2.5). For every regular meromorphic function $f \in \Gamma(X, \mathscr{M}_X^*)$, it follows from (21.2.2) that $\mathscr{J}(f) = \mathscr{O}_X.f$ is an invertible fractionary Ideal, and we evidently have $\mathscr{J}(f_1f_2) = \mathscr{J}(f_1)\mathscr{J}(f_2)$, in other words the map $f \mapsto \mathscr{J}(f)$ is a homomorphism of the commutative group $\Gamma(X, \mathscr{M}_X^*)$ to the commutative group $\operatorname{Id.inv}(X)$. Replacing X by an arbitrary open U of X and noting that the homomorphisms thus obtained are compatible with the restriction operations, we obtain a canonical homomorphism of sheaves of commutative groups:

(21.2.5.1) $$I_0 : \mathscr{M}_X^* \longrightarrow \mathscr{J}d.inv_X.$$

Let us note that if $f \in \Gamma(X, \mathscr{O}_X^*)$, then $\mathscr{J}(f) = \mathscr{O}_X$; we deduce immediately that the homomorphism I_0 factorises as

(21.2.5.2) $$\mathscr{M}_X^* \longrightarrow \mathscr{M}_X^*/\mathscr{O}_X^* = \mathscr{D}iv_X \xrightarrow{I} \mathscr{J}d.inv_X$$

where I is a homomorphism of the sheaf of additive groups $\mathscr{D}iv_X$ to the sheaf of multiplicative groups $\mathscr{I}d.inv_X$; we thus have for every open U of X a homomorphism $\mathscr{I}_U : \mathrm{Div}(U) \to \mathrm{Id.\,inv}(U)$ of commutative groups, such that, for every section $f \in \Gamma(U, \mathscr{M}_X^*)$, we have

(21.2.5.3) $$\mathscr{I}_U(\mathrm{div}_U(f)) = \mathscr{O}_U.f.$$

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We conclude that, for every divisor $D \in \mathrm{Div}(X)$, $\mathscr{I}_X(D)$ is the invertible fractional ideal defined as follows: for every open U of X such that $D|U = \mathrm{div}_U(f)$, where $f \in \Gamma(U, \mathscr{M}_U^*)$, one has $\mathscr{I}_X(D)|U = \mathscr{O}_U.f$. Hence, by virtue of (21.1.6), for every regular meromorphic function $f \in \Gamma(X, \mathscr{M}_X^*)$, we have the relation

(21.2.5.4) $$f \in \Gamma(X, \mathscr{I}_X(D)) \iff \mathrm{div}(f) \geqslant D.$$

Proposition (21.2.6). — *The homomorphism $I : \mathscr{D}iv_X \to \mathscr{I}d.inv_X$ is bijective.*

Proof. We define indeed a homomorphism I'_X of $\mathrm{Id.\,inv}(X)$ to $\mathrm{Div}(X)$ by making correspond to every invertible fractional Ideal $\mathscr{J}$ on X the divisor $I'_X(\mathscr{J})$ defined as follows: for every open U of X such that $\mathscr{J}|U = \mathscr{O}_U.f$, where $f \in \Gamma(U, \mathscr{M}_X^*)$ (21.2.2), we take $I'_X(\mathscr{J})|U = \mathrm{div}_U(f)$; by virtue of the remark following (21.2.2), this definition is independent of the generator f chosen for $\mathscr{J}|U$, and determines a divisor on X. Moreover, this definition immediately shows that the homomorphisms $\mathscr{I}_X$ and I'_X are inverses of one another. Replacing X by an arbitrary open U, we deduce the definition of the isomorphism $I' : \mathscr{I}d.inv_X \to \mathscr{D}iv_X$, inverse of I, whence the proposition. We set $I'_X(\mathscr{J}) = \mathrm{div}(\mathscr{J})$, so that for every regular meromorphic function f on X

(21.2.6.1) $$\mathrm{div}(\mathscr{O}_X.f) = \mathrm{div}(f).$$

□

(21.2.7). We will often identify the sheaves $\mathscr{D}iv_X$ and $\mathscr{I}d.inv_X$ (resp. the groups $\mathrm{Div}(X)$ and $\mathrm{Id.\,inv}(X)$) by means of the isomorphisms I and I' (resp. $\mathscr{I}_X$ and I'_X) above. We note that we have the relation

(21.2.7.1) $$D \geqslant 0 \iff \mathscr{I}_X(D) \subset \mathscr{O}_X \quad \text{for} \quad D \in \mathrm{Div}(X)$$

as follows immediately from the definitions (21.1.6) and (21.1.5.1); in other terms, the image $\mathscr{I}_X(\mathrm{Div}^+(X))$ is the set of *Ideals* of $\mathscr{O}_X$ (also sometimes called *integral Ideals*) which are invertible $\mathscr{O}_X$-Modules: such an Ideal $\mathscr{J}$ is further characterised by the fact that for every $x \in X$, there is a neighbourhood U of x in X such that $\mathscr{J}|U = \mathscr{O}_U.f$, where f is a regular element of $\Gamma(U, \mathscr{O}_X)$. The set $\mathscr{I}_X(\mathrm{Div}^+(X))$ of these Ideals is thus a sub-monoid of $\mathrm{Id.\,inv}(X)$, equal to the set of positive elements for an order relation on $\mathrm{Id.\,inv}(X)$, and it is immediate that this relation is none other than the relation *opposite* to the inclusion; in other words, we have

(21.2.7.2) $$\mathscr{J}_1 \subset \mathscr{J}_2 \iff \mathrm{div}(\mathscr{J}_1) \geqslant \mathrm{div}(\mathscr{J}_2)$$

for $\mathscr{J}_1$, $\mathscr{J}_2$ in $\mathrm{Id.\,inv}(X)$.

(21.2.8). For every divisor D on X, we set

(21.2.8.1) $$\mathscr{O}_X(D) = (\mathscr{I}_X(D))^{-1};$$

$\mathscr{O}_X(D)$ is thus an invertible fractional Ideal, defined as follows: for every open U of X such that $D|U = \mathrm{div}_U(f)$, where $f \in \Gamma(U, \mathscr{M}_X^*)$, $\mathscr{O}_X(D)|U$ is the frac- tionary Ideal $\mathscr{O}_U.f^{-1}$; by virtue of (21.1.6), for every regular meromorphic function f, we have the relation IV-4 | 261

(21.2.8.2) $$f \in \Gamma(X, \mathscr{O}_X(D)) \iff \mathrm{div}(f) \geqslant -D.$$

Moreover, it is clear that we have canonical isomorphisms (21.2.3)

(21.2.8.3) $$\begin{cases} \mathscr{O}_X(0) = \mathscr{O}_X, \quad \mathscr{O}_X(D+D') = \mathscr{O}_X(D)\mathscr{O}_X(D') \xrightarrow{\sim} \mathscr{O}_X(D) \otimes \mathscr{O}_X(D') \\ \mathscr{O}_X(nD) = (\mathscr{O}_X(D))^n \xrightarrow{\sim} (\mathscr{O}_X(D))^{\otimes n} \end{cases}$$

for every integer $n \in \mathbf{Z}$, and for any two divisors D, D' on X.

(21.2.9). Let $\mathscr{J}$ be an invertible fractionary Ideal on X. The canonical injection $\mathscr{J} \hookrightarrow \mathscr{M}_X$ defines by tensorisation a homomorphism of $\mathscr{O}_X$-Modules

$$\mathscr{M}_X(\mathscr{J}) = \mathscr{J} \otimes_{\mathscr{O}_X} \mathscr{M}_X \longrightarrow \mathscr{M}_X \otimes_{\mathscr{O}_X} \mathscr{M}_X = \mathscr{M}_X \tag{21.2.9.1}$$

which is an *isomorphism*: indeed, if U is an open of X such that $\mathscr{J}|U = \mathscr{O}_U.f$, where $f \in \Gamma(U, \mathscr{M}_X^*)$, the homomorphism (21.2.9.1) restricted to U is none other than the isomorphism which, for every section t of $\mathscr{M}_X(\mathscr{J})|U = \mathscr{M}_X|U$ over $V \subset U$, makes correspond the section $t(f|V)^{-1}$ of the same sheaf. In the isomorphism (21.2.9.1), the regular meromorphic sections of $\mathscr{J}$ over X correspond to the regular meromorphic functions on X.

Consider in particular the case where $\mathscr{J} = \mathscr{O}_X(D)$, where D is a divisor on X; we then have a canonical isomorphism

$$\mathscr{M}_X(\mathscr{O}_X(D)) \xrightarrow{\sim} \mathscr{M}_X \tag{21.2.9.2}$$

and we denote by s_D the regular meromorphic section of $\mathscr{O}_X(D)$ over X which corresponds by this isomorphism to the section 1 of $\mathscr{M}_X$. If U is an open of X such that $\mathscr{O}_X(D)|U = \mathscr{O}_U.f^{-1}$, where $f \in \Gamma(U, \mathscr{M}_X^*)$, then $s_D|U = 1$ in $\Gamma(U, \mathscr{M}_X)$; as $D|U = \operatorname{div}_U(f)$, we deduce from (21.1.4) that

$$\operatorname{div}(s_D) = D. \tag{21.2.9.3}$$

On the other hand, we also deduce from the canonical isomorphisms (21.2.8.3) the formulas

$$s_0 = 1, \quad s_{D+D'} = s_D \otimes s_{D'}, \quad s_{nD} = s_D^{\otimes n} \quad (n \in \mathbf{Z}). \tag{21.2.9.4}$$

(21.2.10). Consider, between two pairs $(\mathscr{L}, s)$, $(\mathscr{L}', s')$, where $\mathscr{L}$ and $\mathscr{L}'$ are two invertible $\mathscr{O}_X$-Modules, s (resp. s') a regular meromorphic section of $\mathscr{L}$ (resp. $\mathscr{L}'$) over X, the relation: "there exists an isomorphism $u : \mathscr{L} \xrightarrow{\sim} \mathscr{L}'$ such that $\overline{u}(s) = s'$", where $\overline{u} : \Gamma(X, \mathscr{M}_X(\mathscr{L})) \xrightarrow{\sim} \Gamma(X, \mathscr{M}_X(\mathscr{L}'))$ is the isomorphism derived from u (we note that the isomorphism u satisfying this condition is then determined in a unique way). It is clear that this is an equivalence relation, and since there exists a set of invertible $\mathscr{O}_X$-Modules such that every invertible $\mathscr{O}_X$-Module is isomorphic to an element of this set ($\mathbf{0_I}$, 5.4.7), we can speak of the *set* $D(X)$ of *equivalence classes* of pairs $(\mathscr{L}, s)$ for the above relation. For every invertible $\mathscr{O}_X$-Module $\mathscr{L}$ and every regular mero- morphic section **IV-4** | 262
s of $\mathscr{L}$ over X, we denote by $\operatorname{cl}(\mathscr{L}, s)$ the element of $D(X)$ corresponding to the pair $(\mathscr{L}, s)$. It follows from ($\mathbf{0_I}$, 5.4.3) that if s, s' are two regular meromorphic sections of $\mathscr{L}$ over X, the relation $\operatorname{cl}(\mathscr{L}, s) = \operatorname{cl}(\mathscr{L}, s')$ is equivalent to the existence of a section $t \in \Gamma(X, \mathscr{O}_X^*)$ such that $s' = ts$.

It is immediate that if $(\mathscr{L}, s)$ is equivalent to $(\mathscr{L}_1, s_1)$ and $(\mathscr{L}', s')$ to $(\mathscr{L}'_1, s'_1)$, the pairs $(\mathscr{L} \otimes \mathscr{L}', s \otimes s')$ and $(\mathscr{L}_1 \otimes \mathscr{L}'_1, s_1 \otimes s'_1)$ are equivalent; we thus define in $D(X)$ a law of composition by setting

$$\operatorname{cl}(\mathscr{L}, s)\operatorname{cl}(\mathscr{L}', s') = \operatorname{cl}(\mathscr{L} \otimes \mathscr{L}', s \otimes s'); \tag{21.2.10.1}$$

it is immediate that this is the law of a *commutative group*, whose neutral element is $\operatorname{cl}(\mathscr{O}_X, 1)$ and where the inverse of $\operatorname{cl}(\mathscr{L}, s)$ is $\operatorname{cl}(\mathscr{L}^{-1}, s^{\otimes(-1)})$.

Proposition (21.2.11). — *The maps*

$$D \longmapsto \operatorname{cl}(\mathscr{O}_X(D), s_D), \qquad \operatorname{cl}(\mathscr{L}, s) \longmapsto \operatorname{div}(s) \tag{21.2.11.1}$$

are inverse isomorphisms of $\operatorname{Div}(X)$ *to* $D(X)$ *and of* $D(X)$ *to* $\operatorname{Div}(X)$ *respectively.*

Proof. Taking into account (21.2.8.3), (21.2.9.4), and (21.2.10.1), it suffices to verify that the composites of these two maps are the identities in $\operatorname{Div}(X)$ and $D(X)$ respectively. The first assertion is none other than (21.2.9.3). On the other hand, let $D = \operatorname{div}(s)$, where s is a regular meromorphic section of $\mathscr{L}$ over X, and let U be an open of X such that there exists an isomorphism of $\mathscr{L}|U$ onto $\mathscr{O}_U$, transforming $s|U$ to $f \in \Gamma(U, \mathscr{M}_X^*)$, so that $D|U = \operatorname{div}_U(f)$, $\mathscr{O}_X(D)|U = \mathscr{O}_U.f^{-1}$ and $s_D|U$ is the unit element of $\Gamma(U, \mathscr{M}_X)$. There is thus an isomorphism $v_U : \mathscr{L}|U \to \mathscr{O}_X(D)|U$ such that $\overline{v}_U$ (notation of (21.2.10)) transforms $s|U$ to $f.f^{-1} = 1$; one sees immediately that these isomorphisms are compatible with the restriction operations, and thus define an isomorphism $v : \mathscr{L} \to \mathscr{O}_X(D)$ such that $\overline{v}(s) = s_D$. □

We can *transport* by the first of the isomorphisms (21.2.11.1) the structure of ordered group of $\operatorname{Div}(X)$ to $D(X)$; the elements $\geqslant 0$ of $D(X)$ are thus the classes $\operatorname{cl}(\mathscr{L}, s)$ such that $\operatorname{div}(s) \geqslant 0$, that is to say (21.1.7.1) such that

$$s \in \Gamma(X, \mathscr{L}) \cap \Gamma(X, (\mathscr{M}_X(\mathscr{L}))^*).$$

(21.2.12). Let D be a *positive* divisor on a *prescheme* X; the fractionary Ideal $\mathscr{J}_X(D)$ is an Ideal of $\mathscr{O}_X$, which is an invertible $\mathscr{O}_X$-Module; let $Y(D)$ be the sub-prescheme of X that it defines. For every $x \in Y(D)$, there is by hypothesis a neighbourhood U of x in X and a regular section $t \in \Gamma(U, \mathscr{O}_X)$ such that $\mathscr{J}_X(D)|U = \mathscr{O}_U.t$ (21.2.7); in other terms, the canonical immersion $Y(D) \to X$ is *regular and of codimension* 1 (19.1.4). Conversely, if Y is a sub-prescheme of X, regularly immersed in X and of codimension 1 at every point of Y, there exists *a unique positive divisor* D such that $Y(D) = Y$, since there is a neighbourhood U of x in X such that $Y \cap U$ is defined by an Ideal of the form $\mathscr{O}_U.t$, where t is a regular element of $\Gamma(U, \mathscr{O}_X)$. We note that $\operatorname{Supp}(D) = Y(D)$, since to say that $D_x \neq 0$ means that t_x is not invertible, that is to say $x \in Y(D)$.

21.3. Linear equivalence of divisors

(21.3.1). We say that a divisor D on X is *principal* if it is of the form $\operatorname{div}(f)$, where f is a regular meromorphic function on X; the meromorphic functions f' such that $\operatorname{div}(f') = D$ are then all of the form tf, where $t \in \Gamma(X, \mathscr{O}_X^*)$ (21.1.3.4). The set of principal divisors is a subgroup of $\operatorname{Div}(X)$, denoted $\operatorname{Div.princ}(X)$, isomorphic to $\Gamma(X, \mathscr{M}_X^*)/\Gamma(X, \mathscr{O}_X^*)$. Two divisors D, D' are said to be *linearly equivalent* if $D - D'$ is a principal divisor; the principal divisors are thus the divisors linearly equivalent to 0.

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(21.3.2). Recall ($\mathbf{0_I}$, 5.4.7) that we can speak of the set of equivalence classes of invertible $\mathscr{O}_X$-Modules for the isomorphism relation; we denote this set by $\operatorname{Pic}(X)$, and for every invertible $\mathscr{O}_X$-Module $\mathscr{L}$, we denote by $\operatorname{cl}(\mathscr{L})$ the equivalence class of $\mathscr{O}_X$-Modules isomorphic to $\mathscr{L}$; moreover, $\operatorname{Pic}(X)$ is a commutative group for the multiplication defined by $\operatorname{cl}(\mathscr{L})\operatorname{cl}(\mathscr{L}') = \operatorname{cl}(\mathscr{L} \otimes \mathscr{L}')$. It is clear that the map

(21.3.2.1) $$r : \operatorname{cl}(\mathscr{L}, s) \longmapsto \operatorname{cl}(\mathscr{L})$$

is a homomorphism of the group $D(X)$ (21.2.10) to the group $\operatorname{Pic}(X)$. By composition, we thus deduce a homomorphism

(21.3.2.2) $$l : \operatorname{Div}(X) \xrightarrow{\sim} D(X) \xrightarrow{r} \operatorname{Pic}(X)$$

(also denoted l_X) such that, for every divisor D, we have

(21.3.2.3) $$l(D) = \operatorname{cl}(\mathscr{O}_X(D)).$$

Let us note finally that, if $u : X' \to X$ is a morphism of ringed spaces, $\mathscr{L}_1$, $\mathscr{L}_2$ two invertible isomorphic $\mathscr{O}_X$-Modules, the invertible $\mathscr{O}_{X'}$-Modules $u^*(\mathscr{L}_1)$ and $u^*(\mathscr{L}_2)$ ($\mathbf{0_I}$, 5.4.5) are isomorphic; as moreover, for any two invertible $\mathscr{O}_X$-Modules $\mathscr{L}_1$, $\mathscr{L}_2$, we have $u^*(\mathscr{L}_1 \otimes \mathscr{L}_2) = u^*(\mathscr{L}_1) \otimes u^*(\mathscr{L}_2)$ up to canonical isomorphism ($\mathbf{0_I}$, 4.3.3), we see that the morphism u canonically defines a homomorphism of commutative groups

(21.3.2.4) $$\operatorname{Pic}(u) : \operatorname{Pic}(X) \longrightarrow \operatorname{Pic}(X').$$

Proposition (21.3.3). — *(i) The kernel of the canonical homomorphism* $l : \operatorname{Div}(X) \to \operatorname{Pic}(X)$ *is the subgroup* $\operatorname{Div.princ}(X)$*, in other words, for* $\mathscr{O}_X(D)$ *and* $\mathscr{O}_X(D')$ *to be isomorphic, it is necessary and sufficient that* D *and* D' *be linearly equivalent. There is thus a canonical injective homomorphism*

(21.3.3.1) $$\operatorname{Div}(X)/\operatorname{Div.princ}(X) \longrightarrow \operatorname{Pic}(X)$$

derived from l.

(ii) For an invertible $\mathscr{O}_X$*-Module* $\mathscr{L}$ *to be such that* $\operatorname{cl}(\mathscr{L})$ *is of the form* $l(D)$*, or equivalently that* $\mathscr{L}$ *is isomorphic to an* $\mathscr{O}_X$*-Module of the form* $\mathscr{O}_X(D)$*, it is necessary and sufficient that there exists a regular meromorphic section of* $\mathscr{L}$*.*

Proof. The proposition follows immediately from the definitions and from (21.2.10). □ **IV-4** | 264

Proposition (21.3.4). — *Let X be a prescheme satisfying one of the two following hypotheses:*

a) *X is locally Noetherian and $\operatorname{Ass}(\mathscr{O}_X)$ is contained in an affine open of X.*
b) *X is reduced and the set of its irreducible components is locally finite.*

Then the canonical homomorphism $l : \operatorname{Div}(X) \to \operatorname{Pic}(X)$ is surjective, and gives by passage to the quotient an isomorphism

$$\operatorname{Div}(X) / \operatorname{Div.princ}(X) \xrightarrow{\sim} \operatorname{Pic}(X).$$

Proof. It suffices to show that every invertible $\mathscr{O}_X$-Module $\mathscr{L}$ admits a regular meromorphic section over X (21.3.3). In both cases, it will suffice, thanks to (20.2.11), (ii), to define a section s of $(\mathscr{M}_X(\mathscr{L}))^*$ over a *schematically dense* open U of X, or else, in case *a*), over an open U containing $\operatorname{Ass}(\mathscr{O}_X)$ (20.2.13), (iv). Indeed, let V be an arbitrary open of X such that $\mathscr{L}|V$ is isomorphic to $\mathscr{O}_V$, so that there exists an isomorphism of $\mathscr{M}_X(\mathscr{L})|V$ onto $\mathscr{M}_X|V$, transforming $s|(U \cap V)$ to a section f_V of $\mathscr{M}_X^*$ over $U \cap V$. As $U \cap V$ is schematically dense in V, it follows from (20.2.11) that there exists a unique regular meromorphic function g_V such that $g_V|(U \cap V) = f_V$, and this section corresponds by the isomorphism considered to a regular meromorphic section u_V of $\mathscr{L}$ over V such that $u_V|(U \cap V) = s|(U \cap V)$. Moreover, if V' is a second open of X such that $\mathscr{L}|V'$ is isomorphic to $\mathscr{O}_{V'}$, the restrictions of u_V and $u_{V'}$ to $V \cap V'$ are equal, since they correspond by isomorphism to two meromorphic functions which coincide on an open schematically dense $U \cap V \cap V'$, and we conclude again by (20.2.11); the u_V are thus well the restrictions of a single regular meromorphic section of $\mathscr{L}$ over X.

In case *b*), we take for each of the maximal points x_λ of X an open U_λ containing x_λ, meeting no irreducible component of X distinct from $\overline{\{x_\lambda\}}$, and such that $\mathscr{L}|U_\lambda$ is isomorphic to $\mathscr{O}_{U_\lambda}$; we take for s the section such that $s|U_\lambda$ is the section of $\mathscr{L}|U_\lambda$ which corresponds by the preceding isomorphism to the unit section of $\mathscr{O}_{U_\lambda}$.

In case *a*), we can take U to be an affine open (hence Noetherian), in other words, suppose that $X = \operatorname{Spec}(A)$, where A is Noetherian, and $\mathscr{L} = \widetilde{P}$, where P is a projective A-module of rank 1. If S is the set of regular elements of A, we have $\Gamma(X, \mathscr{M}_X) = S^{-1}A$ (20.2.12) and $\Gamma(X, \mathscr{M}_X(\mathscr{L})) = S^{-1}P$; now S is the set of elements not belonging to any of the ideals associated to A, hence $S^{-1}A$ is a semi-local ring whose maximal ideals come from the maximal elements of $\operatorname{Ass}(A)$, and $S^{-1}P$ is a projective $S^{-1}A$-module of rank 1, hence a *free* $S^{-1}A$-module of rank 1 (Bourbaki, *Alg. comm.*, chap. II, § 5, n° 3, prop. 5); an element forming a base of this $S^{-1}A$-module is thus (20.1.8) a regular meromorphic section of $\mathscr{L}$ over X. ☐

Corollary (21.3.5). — *If X is a Noetherian prescheme such that there exists an ample invertible $\mathscr{O}_X$-Module (**II**, 4.5.3) (for example, a quasi-projective prescheme over the spectrum of a Noetherian ring (**II**, 5.3.1 and 4.6.6)), the canonical homomorphism $l : \operatorname{Div}(X) \to \operatorname{Pic}(X)$ is surjective.*

Proof. Indeed (**II**, 4.5.4), there then exists an affine open neighbourhood of the finite set $\operatorname{Ass}(\mathscr{O}_X)$. ☐

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Remark (21.3.6). — Recall ($\mathbf{0_I}$, 5.4.7) that there is a canonical isomorphism $\pi : \mathrm{H}^1(X, \mathscr{O}_X^*) \xrightarrow{\sim} \operatorname{Pic}(X)$ defined as follows. We start from a 1-cocycle $(c_{\alpha\beta})$ with values in $\mathscr{O}_X^*$ corresponding to a covering (U_α) of X, $c_{\alpha\beta}$ being an element of $\Gamma(U_\alpha \cap U_\beta, \mathscr{O}_X^*)$, and we associate to it the class of the invertible $\mathscr{O}_X$-Module obtained by gluing the $\mathscr{O}_{U_\alpha}$ along the isomorphisms $\mathscr{O}_{U_\alpha}|(U_\alpha \cap U_\beta) \xrightarrow{\sim} \mathscr{O}_{U_\beta}|(U_\alpha \cap U_\beta)$ defined by multiplication by $c_{\alpha\beta}$. On the other hand, we deduce from the exact sequence of sheaves of commutative groups

$$1 \longrightarrow \mathscr{O}_X^* \longrightarrow \mathscr{M}_X^* \longrightarrow \mathscr{D}iv_X \longrightarrow 0$$

the boundary homomorphism of the exact cohomology sequence

(21.3.6.1) $$\partial : \operatorname{Div}(X) \longrightarrow \mathrm{H}^1(X, \mathscr{O}_X^*).$$

Let us show that the composite homomorphism

$$\operatorname{Div}(X) \xrightarrow{\partial} \mathrm{H}^1(X, \mathscr{O}_X^*) \xrightarrow{\pi} \operatorname{Pic}(X)$$

is the homomorphism l defined in (21.3.2.2). Indeed, we must start from a covering (U_α) of X and a divisor D such that $D|U_\alpha = \operatorname{div}_{U_\alpha}(g_\alpha)$, where g_α is a regular meromorphic function over U_α; $\partial(D)$ is the cohomology class of the cocycle $(c_{\alpha\beta})$, where $c_{\alpha\beta} = g_{\alpha|\beta} g_{\beta|\alpha}^{-1}$, $g_{\alpha|\beta}$ denoting the restriction of g_α to $U_\alpha \cap U_\beta$. It is clear that the image by π of this cohomology class is the class of the invertible fractionary Ideal $\mathscr{L}$ such that for every α, $\mathscr{L}|U_\alpha = \mathscr{O}_{U_\alpha}.g_\alpha^{-1}$, which is none other by definition than $\mathscr{O}_X(D)$ (21.2.8).

21.4. Inverse images of divisors

(21.4.1). Let $f : X' \to X$ be a morphism of ringed spaces; we propose to give conditions allowing us to associate to a divisor D on X a divisor D' on X', the *inverse image* of D by f. Let us first note that for every section $t \in \Gamma(X, \mathscr{O}_X^*)$, the image of t by the canonical homomorphism $\Gamma(X, \mathscr{O}_X) \to \Gamma(X', \mathscr{O}_{X'})$ is still invertible, in other words belongs to $\Gamma(X', \mathscr{O}_{X'}^*)$. Consider then D as given by the class of equivalence of a pair $(\mathscr{L}, s)$, where $\mathscr{L}$ is an invertible $\mathscr{O}_X$-Module and s a regular meromorphic section of $\mathscr{L}$ over X (21.2.11). Form the invertible $\mathscr{O}_{X'}$-Module $f^*(\mathscr{L}) = \mathscr{L}'$; to say that the inverse image $s \circ f$ of s by f exists (20.1.11) and is a regular meromorphic section of $\mathscr{L}'$ over X' amounts to saying that the inverse images by f of s and $s^{\otimes(-1)}$ exist, in other words that $s \in \Gamma(X, \mathscr{M}_f(\mathscr{L}))$ and $s^{\otimes(-1)} \in \Gamma(X, \mathscr{M}_f(\mathscr{L}^{-1}))$; the remark made above shows that if $(\mathscr{L}_1, s_1)$ is a pair equivalent to $(\mathscr{L}, s)$ in the sense of (21.2.10), the inverse image $s_1 \circ f$ exists and is a regular meromorphic section of $\mathscr{L}_1' = f^*(\mathscr{L}_1)$ over X', and the pairs $(\mathscr{L}', s \circ f)$ and $(\mathscr{L}_1', s_1 \circ f)$ are equivalent. We can therefore pose the following definition:

Definition (21.4.2). — Given a morphism $f : X' \to X$ of ringed spaces, we say that the *inverse image by* f of a divisor D on X exists if $s_D \in \Gamma(X, \mathscr{M}_f(\mathscr{O}_X(D)))$ and $s_{-D} \in \Gamma(X, \mathscr{M}_f(\mathscr{O}_X(-D)))$ (cf. (20.1.11)). We call then the *inverse image* of D by f and denote $f^*(D)$ the divisor on X' which corresponds canonically (21.2.11) to the class of the pair $(f^*(\mathscr{O}_X(D)), s_D \circ f)$.

It follows immediately from this definition that if D and D' have inverse images by f, so does $-D$ and $D + D'$ (taking into account (21.2.9.4)) and we have $f^*(D + D') = f^*(D) + f^*(D')$. In **IV-4** | 266
other words, the set $\operatorname{Div}^f(X)$ of divisors on X whose inverse image by f exists is a *subgroup* of $\operatorname{Div}(X)$, and the map $D \mapsto f^*(D)$ is an *increasing homomorphism* of the ordered subgroup $\operatorname{Div}^f(X)$ to the ordered group $\operatorname{Div}(X')$, making commutative the diagram

(21.4.2.1)
$$\begin{tikzcd} \operatorname{Div}^f(X) \arrow[r, "l_X"] \arrow[d, "f^*"'] & \operatorname{Pic}(X) \arrow[d, "\operatorname{Pic}(f)"] \\ \operatorname{Div}(X') \arrow[r, "l_{X'}"] & \operatorname{Pic}(X') \end{tikzcd}$$

(21.4.3). The definition (21.4.2) also shows immediately that, for $f^*(D)$ to exist, it is necessary and sufficient that, for every open U of X, the inverse image of $D|U$ by $f^U : f^{-1}(U) \to U$ (the restriction of f) exist. Now, if $D = \operatorname{div}(g)$, where g is a regular meromorphic function on X, to say that the inverse image of s_D by f exists and is a regular meromorphic section of $f^*(\mathscr{O}_X(D))$ means (21.2.9) that the inverse image of g by f exists and is a regular meromorphic function on X'. We deduce also immediately a second description of $\operatorname{Div}^f(X)$ and of $f^*(D)$: we consider the subsheaf of groups of $\mathscr{M}_X^*$, denoted $\mathscr{M}_f^*$, formed of germs of meromorphic functions on an open of X whose inverse image exists and is regular on the inverse image open (20.1.11). Then if $f = (\psi, \theta)$, the canonical homomorphism (20.1.11) $\psi^*(\mathscr{M}_f) \to \mathscr{M}_{X'}$ gives by restriction a homomorphism of sheaves of groups $\psi^*(\mathscr{M}_f^*) \to \mathscr{M}_{X'}^*$. Setting $\mathscr{D}iv'_X = \mathscr{M}_f^* / \mathscr{O}_X^*$, we have $\operatorname{Div}^f(X) = \Gamma(X, \mathscr{D}iv'_X)$, and the map $D \mapsto f^*(D)$ corresponds to the homomorphism of sheaves of groups $\psi^*(\mathscr{M}_f^*)/\psi^*(\mathscr{O}_X^*) \to \mathscr{M}_{X'}^* / \mathscr{O}_{X'}^* = \mathscr{D}iv_{X'}$, derived from the preceding by passage to quotients.

(21.4.4). It follows also from the preceding definitions that if $f' : X'' \to X'$ is a second morphism of ringed spaces, D a divisor on X such that the inverse images $f^*(D)$ and $f'^*(f^*(D))$ exist, then $(f \circ f')^*(D)$ exists and is equal to $f'^*(f^*(D))$.

Proposition (21.4.5). — *Let $f : X' \to X$ be a morphism of ringed spaces. In each of the three following cases, the inverse image by f of every divisor on X is defined:*

(i) *f is flat.*
(ii) *X and X' are locally Noetherian preschemes and $f(\mathrm{Ass}(\mathscr{O}_{X'})) \subset \mathrm{Ass}(\mathscr{O}_X)$.*
(iii) *X and X' are preschemes, the set of irreducible components of X is locally finite, X' is reduced and every irreducible component of X' dominates an irreducible component of X.*

Proof. It suffices indeed to show that in all three cases we have $\mathscr{M}_f = \mathscr{M}_X$; in case (i), this follows from (20.1.12). In case (iii), we can restrict to the case where $X = \mathrm{Spec}(A)$ and $X' = \mathrm{Spec}(A')$ are affine; if $s \in A$ is regular, it does not belong to any minimal prime ideal of A (20.1.3.1), hence the hypothesis implies that its image in A' does not belong to any minimal prime ideal of A', and is therefore a regular element of A' (20.1.3.1). In case (ii) the meromorphic functions on X' are identified with the pseudo-functions on X' (20.2.11), and the hypothesis, combined with (20.2.13), (iv), ensures that the inverse image by f of every schematically dense open in X is a schematically dense open in X'; we conclude therefore by (20.3.12). ∎ **IV-4** | 267

Corollary (21.4.6). — *Let X be a prescheme having one of the following properties:*

(i) *X is locally Noetherian.*
(ii) *X is reduced and the set of its irreducible components is locally finite.*

Then, for every $x \in X$, we have a canonical isomorphism

(21.4.6.1) $$(\mathscr{D}iv_X)_x \xrightarrow{\sim} \mathrm{Div}(\mathscr{O}_{X,x}).$$

Proof. This follows indeed from (20.2.11), (20.3.7), and from the fact that $(\mathscr{O}_X^*)_x$ is identified with the group of invertible elements of the ring $\mathscr{O}_{X,x}$. ∎

(21.4.7). Let X, X' be two preschemes, $f : X' \to X$ a morphism. If D is a *positive* divisor on X such that the inverse image $f^*(D)$ is defined (21.4.2), then the sub-prescheme $Y(f^*(D))$ of X' is none other than the inverse image $f^{-1}(Y(D))$; this follows also from the definitions (21.4.2) and (21.2.12).

Proposition (21.4.8). — *Let X, Y be two preschemes, $f : X \to Y$ a faithfully flat morphism. Then, if D is a divisor on Y such that $f^*(D) \geqslant 0$ (the existence of $f^*(D)$ resulting from (21.4.5)), we have $D \geqslant 0$. In particular, the map $D \mapsto f^*(D)$ of $\mathrm{Div}(Y)$ to $\mathrm{Div}(X)$ is injective.*

Proof. The question being local on Y, we can restrict to the case where $D = \mathrm{div}(w)$, with $w = uv^{-1}$, u and v being two regular sections of $\mathscr{O}_Y$ over Y. By hypothesis we have $u_y\mathscr{O}_{X,x} \subset v_y\mathscr{O}_{X,x}$, hence, for every $x \in X$, setting $y = f(x)$, we have $u_y\mathscr{O}_{Y,y} \subset v_y\mathscr{O}_{Y,y}$; we conclude that $u_y\mathscr{O}_{Y,y} \subset v_y\mathscr{O}_{Y,y}$ by virtue of the hypothesis that $\mathscr{O}_{X,x}$ is a faithfully flat $\mathscr{O}_{Y,y}$-module, and since f is surjective, we have $u\mathscr{O}_Y \subset v\mathscr{O}_Y$, whence $D \geqslant 0$. ∎

21.5. Direct images of divisors

(21.5.1). Let X, X' be two preschemes, $f : X' \to X$ a morphism. In this section, we shall give sufficient conditions for one to associate to a divisor D' on X' a divisor on X, the *direct image* of D' by f. We will restrict ourselves to the case where f is a *finite* morphism (for more general conditions, see the chapter of this Treatise devoted to the theory of intersections).

Lemma (21.5.2). — *Let A be a ring, E an A-module free of finite rank. For an endomorphism u of E to be injective, it is necessary and sufficient that $\det(u)$ be a regular element of A.*

Proof. This is proved in Bourbaki, *Alg.*, chap. III, 3^e^ éd., § 8, n^o^ 2, prop. 3. ∎

(21.5.3). Suppose now that the morphism $f : X' \to X$ is *finite*, and in addition that f satisfies one of the two following properties:

I) f is a finite morphism *locally free*, in other words (18.2.7) the $\mathscr{O}_X$-Module $f_*(\mathscr{O}_{X'})$ is *locally free*.

II) X is a locally Noetherian reduced prescheme, the $\mathscr{R}(X)$-Module $f_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} \mathscr{R}(X)$ is locally free, and for every section $s' \in \Gamma(f^{-1}(U), \mathscr{O}_{X'})$ (U open in X), $\mathrm{N}_{f_*(\mathscr{O}_{X'})|\mathscr{O}_X}(s')$ is a section of $\mathscr{O}_X$ over U (cf. (**II**, 6.5.1)). (We recall that condition (II) is satisfied for every *finite morphism* f when X is a locally Noetherian normal prescheme (*loc. cit.*).) **IV-4** | 268

We know then (**II**, 6.5.5) that for every invertible $\mathscr{O}_{X'}$-Module $\mathscr{L}'$ we define the *norm* $\mathscr{L} = \mathrm{N}_{X'/X}(\mathscr{L}')$ (which we also write $\mathrm{N}(\mathscr{L}')$), which is an invertible $\mathscr{O}_X$-Module. Moreover, for every open U of X and every *regular* section $s' \in \Gamma(f^{-1}(U), \mathscr{O}_{X'})$, the norm $\mathrm{N}_{X'/X}(s') = \mathrm{N}_{f_*(\mathscr{O}_{X'})|\mathscr{O}_X}(s')$ (which we also write $\mathrm{N}(s')$) is a regular element of $\Gamma(U, \mathscr{O}_X)$; we are indeed immediately reduced to the case where $U = X$ is affine, and then the conclusion follows from (21.5.2) in hypothesis (I); on the other hand, in hypothesis (II), the fact that $\mathscr{R}(X)$ is a flat $\mathscr{O}_X$-Module implies that the section $s' \otimes 1$ of $\Gamma(U, f_*(\mathscr{O}_{X'}) \otimes_{\mathscr{O}_X} \mathscr{R}(X))$ is also regular ($\mathbf{0_I}$, 6.1.2), and the conclusion follows again from (21.5.2) applied to the ring $\Gamma(U, \mathscr{R}(X))$, taking into account the definition of the norm (**II**, 6.5.3). This being so, let u' be a section of $\mathscr{L}'$ over X'; the morphism f being affine, every point $x \in X$ admits a neighbourhood U such that $u'|f^{-1}(U)$ is of the form t'/s', where $t' \in \Gamma(f^{-1}(U), \mathscr{L}')$ and s' is a regular section of $\mathscr{O}_{X'}$ over $f^{-1}(U)$; the element $\mathrm{N}(t')/\mathrm{N}(s')$ (where $\mathrm{N}(t')$ is the section of $\mathscr{L}$ defined in (**II**, 6.5.3)) is then a section of $\mathscr{L}$ over U, and from what precedes and the multiplicative properties of the norm (**II**, 6.5.3.1) this section does not depend on the choice of $u'|f^{-1}(U)$ but only on its representation in the form t'/s'; for the same reason, the sections of $\mathscr{L}$ over U thus defined are the restrictions of a single meromorphic section of $\mathscr{L}$ over X, which we call the *norm* of u' and denote $\mathrm{N}_{X'/X}(u')$ (or simply $\mathrm{N}(u')$). The map

(21.5.3.1) $$\mathrm{N}_{X'/X} : \Gamma(X', \mathscr{M}_{X'}(\mathscr{L}')) \longrightarrow \Gamma(X, \mathscr{M}_X(\mathrm{N}_{X'/X}(\mathscr{L}')))$$

extends the norm defined in (**II**, 6.5.5.3); if u' is a *regular* meromorphic section of $\mathscr{L}'$ over X', it follows immediately from what precedes that $\mathrm{N}(u')$ is a regular meromorphic section of $\mathscr{L}$ over X, since (with the same notation) $\mathrm{N}(t')$ is regular if t' is. Finally, if $\mathscr{L}'_1$, $\mathscr{L}'_2$ are two invertible $\mathscr{O}_{X'}$-Modules, s'_1 (resp. s'_2) a regular meromorphic section of $\mathscr{L}'_1$ (resp. $\mathscr{L}'_2$) over X', we have, by virtue of what precedes and of (**II**, 6.5.3.1)

(21.5.3.2) $$\mathrm{N}(s'_1 \otimes s'_2) = \mathrm{N}(s'_1) \otimes \mathrm{N}(s'_2).$$

(21.5.4). Suppose always that f satisfies one of hypotheses I), II) of (21.5.3). If $(\mathscr{L}'_1, s'_1)$, $(\mathscr{L}'_2, s'_2)$ are two pairs each formed of an invertible $\mathscr{O}_{X'}$-Module and a regular meromorphic section of this Module over X', which are moreover *equivalent* in the sense of (21.2.10), then the pairs $(\mathrm{N}(\mathscr{L}'_1), \mathrm{N}(s'_1))$ and $(\mathrm{N}(\mathscr{L}'_2), \mathrm{N}(s'_2))$ are also equivalent, since an isomorphism of invertible $\mathscr{O}_{X'}$-Modules has for norm an isomorphism of their norms (**II**, 6.5.4), and we have already seen above that $\mathrm{N}_{X'/X}$ transforms the sections of $\Gamma(X', \mathscr{O}^*_{X'})$ to sections of $\Gamma(X, \mathscr{O}^*_X)$. We can therefore pose the following definition:

Definition (21.5.5). — Given a finite morphism $f : X' \to X$ of preschemes, satisfying one of conditions I), II) of (21.5.3), we call the *direct image* (or *norm*) of a divisor D' on X' by f, and denote $f_*(D')$ (or $\mathrm{N}_{X'/X}(D')$), the divisor on X which corresponds canonically (21.2.11) to the class of the pair $(\mathrm{N}_{X'/X}(\mathscr{O}_{X'}(D')), \mathrm{N}_{X'/X}(s_{D'}))$.

It follows immediately from this definition, taking into account (21.2.9.4) and (21.5.3.2), that if D'_1, D'_2, D' are divisors on X', we have

(21.5.5.1) $$f_*(D'_1 + D'_2) = f_*(D'_1) + f_*(D'_2)$$

and $D' \geqslant 0$ implies $f_*(D') \geqslant 0$, in other words, $D' \mapsto f_*(D')$ is an *increasing homomorphism* of the ordered group $\mathrm{Div}(X')$ to the ordered group $\mathrm{Div}(X)$. The definition (21.5.5) also shows immediately that for every open U of X, we have $f^U_*(D'|f^{-1}(U)) = f_*(D')|U$ (f^U being the restriction $f^{-1}(U) \to U$ of f), and the homomorphisms f^U_*, for U variable, thus define a *homomorphism of sheaves of ordered groups* **IV-4 | 269**

(21.5.5.2) $$\mathrm{N}_{X'/X} : f_*(\mathscr{D}iv_{X'}) \longrightarrow \mathscr{D}iv_X.$$

Moreover, for every invertible $\mathscr{O}_{X'}$-Module $\mathscr{L}'$ and every regular meromorphic section s' of $\mathscr{L}'$ over $f^{-1}(U)$, we have, from the preceding definitions and (21.1.4)

(21.5.5.3) $$\mathrm{div}_U(\mathrm{N}(s')) = f^U_*(\mathrm{div}_{f^{-1}(U)}(s')).$$

Proposition (21.5.6). — *Let $f : X' \to X$ be a finite morphism locally free and suppose that $f_*(\mathscr{O}_{X'})$ is of constant rank n. Then, for every divisor D on X, $f_*(D)$ is defined and we have*

(21.5.6.1) $$f_*(f^*(D)) = nD.$$

Proof. The first assertion follows from the fact that f is flat (21.4.5), and the second is an immediate consequence of the definitions and of (**II**, 6.5.3.2). □

Proposition (21.5.7). — *Let $f : X' \to X$ be a finite morphism satisfying one of hypotheses I), II) of (21.5.3), $f' : X'' \to X'$ a finite morphism locally free of constant rank n. Then $f'' = f \circ f' : X'' \to X$ satisfies the same hypothesis as f, and for every divisor D'' on X'', we have*

(21.5.7.1)
$$f''_*(D'') = f_*(f'_*(D'')).$$

Proof. Taking into account the definition (21.5.5), it suffices to prove the following: □

Lemma (21.5.7.2). — *Under the hypotheses of (21.5.7), we have a functorial isomorphism*

(21.5.7.3)
$$\mathrm{N}_{X''/X}(\mathscr{L}'') \xrightarrow{\sim} \mathrm{N}_{X'/X}(\mathrm{N}_{X''/X'}(\mathscr{L}''))$$

in the category of invertible $\mathscr{O}_{X''}$-Modules.

Proof. Indeed, taking into account the definition of the norm of a section $\mathscr{L}''$ (**II**, 6.5.3) and of the definition (21.5.5), we will obtain (21.5.7.3), and it suffices to prove that for every section s of $f''_*(\mathscr{O}_{X''}) = f_*(f'_*(\mathscr{O}_{X''}))$ over an open U of X, we have

(21.5.7.4)
$$\mathrm{N}_{f''_*(\mathscr{O}_{X''})|\mathscr{O}_X}(s) = \mathrm{N}_{f_*(\mathscr{O}_{X'})|\mathscr{O}_X}(\mathrm{N}_{f'_*(\mathscr{O}_{X''})|\mathscr{O}_{X'}}(s))$$

which follows from the transitivity of the norm (Bourbaki, *Alg.*, chap. VIII, § 12, n° 2, prop. 7) when **IV-4** | 270
f satisfies hypothesis (I) and A is Noetherian and reduced, one can suppose that A' is an A-module free of rank m, and then A'' is an A-module projective of rank mn, and restricting X to a suitable open, one can suppose that A'' is an A-module free of rank mn; the formula (21.5.7.4) follows then again from the transitivity of the norms. When f satisfies hypothesis (II), A is Noetherian and reduced, its total ring of fractions $A' \otimes_A \mathrm{R}$ is an R-module projective of rank m, hence $A'' \otimes_A \mathrm{R} = A'' \otimes_{A'} \mathrm{R}$ is an R-module projective of rank mn; the proposition also follows then from the transitivity of the norms. □

Proposition (21.5.8). — *Let $f : X' \to X$ be a finite morphism, $g : Y \to X$ a morphism; set $Y' = X' \times_X Y$, $f' = f_{(Y)} : Y' \to Y$, $g' = g_{(X)} : Y' \to X'$. Suppose satisfying one of the following conditions:*

- *(i) f is locally free and g is flat.*
- *(ii) f is locally free, X and Y are locally Noetherian, $g(\mathrm{Ass}(\mathscr{O}_Y)) \subset \mathrm{Ass}(\mathscr{O}_X)$ and $g'(\mathrm{Ass}(\mathscr{O}_{Y'})) \subset \mathrm{Ass}(\mathscr{O}_{X'})$.*
- *(iii) f satisfies hypothesis* II) *of (21.5.3), Y is locally Noetherian, Y and Y' are reduced and every irreducible component of Y (resp. Y') dominates an irreducible component of X (resp. X').*

Then, for every divisor D' on X', $g^*(D')$ is defined, $g^*(f_*(D'))$ is defined and we have

(21.5.8.1)
$$g^*(f_*(D')) = f'_*(g'^*(D')).$$

Proof. Indeed, in all cases, it follows from (**II**, 6.5.8) that there is a functorial isomorphism

$$g^*(\mathrm{N}_{X'/X}(\mathscr{L}')) \xrightarrow{\sim} \mathrm{N}_{Y'/Y}(g'^*(\mathscr{L}'))$$

in the category of invertible $\mathscr{O}_{X'}$-Modules; moreover (**II**, 6.5.4), if s' is the corresponding section of $\mathscr{O}_{X'}$ over $f^{-1}(U)$, s'' the corresponding section of $\mathscr{O}_{Y'}$ over $g'^{-1}(f^{-1}(U))$ (where U is open of X), $\mathrm{N}_{Y'/Y}(s'')$ is a section of $\mathscr{O}_Y$ over U. The formula (21.5.8.1) will thus follow from the definitions, provided one proves that $g^*(D')$ and $g'^*(D')$ are both defined, for arbitrary divisors D' on X' and D on X. As far as D is concerned, this results from the hypotheses made and from (21.4.5). As far as D' is concerned, in case (i) g' is flat, hence the conclusion by (21.4.5); in what concerns D', in case (ii), the conclusion results again from (21.4.5) by the hypotheses; in case (iii), g' is flat, hence the conclusion again by (21.4.5). □

21.6. 1-codimensional cycle associated to a divisor IV-4 | 270

(21.6.1). We call *multiplicity of* Z *at the point* x and denote by $\mathrm{mult}_x(\mathrm{Z})$ the integer n_x (positive or negative). To say that $\mathrm{Z} \geqslant 0$ means that $\mathrm{mult}_x(\mathrm{Z}) \geqslant 0$ for all $x \in \mathrm{X}$. We call the *support* of Z and denote by $\mathrm{Supp}(\mathrm{Z})$ the union of the $\overline{\{x\}}$ such that $\mathrm{mult}_x(\mathrm{Z}) \neq 0$; by definition of $\mathfrak{Z}(\mathrm{X})$, this is a closed part of X, as the union of a locally finite family of closed parts. We call the *dimension* (resp. *codimension*) of Z and denote by $\dim(\mathrm{Z})$ (resp. $\mathrm{codim}(\mathrm{Z})$) the dimension (resp. codimension in X) of $\mathrm{Supp}(\mathrm{Z})$.

(21.6.2). We say that a closed subspace Y of X is *purely of codimension* p (in X) if *each* of its irreducible components is of codimension p in X. We say that a cycle is *purely of codimension* p, or (by abuse of language) is a p*-codimensional cycle*, if its support is purely of codimension p. We denote by $\mathrm{X}^{(p)}$ the set of $x \in \mathrm{X}$ such that $\mathrm{codim}(\overline{\{x\}}, \mathrm{X}) = p$, or, what amounts to the same (5.1.2), $\dim(\mathscr{O}_{\mathrm{X},x}) = p$: the cycles purely of codimension p form a subgroup $\mathfrak{Z}^p(\mathrm{X})$ of $\mathfrak{Z}(\mathrm{X})$, isomorphic to the sub-group of IV-4 | 271
$\mathbf{Z}^{\mathrm{X}^{(p)}}$ consisting of the $(n_x)_{x \in \mathrm{X}^{(p)}}$ such that the set of $\overline{\{x\}}$ (or the set of x) for which $n_x \neq 0$ is locally finite; this sub-group contains the free group $\mathbf{Z}^{(\mathrm{X}^{(p)})}$ and is identical to it when X is Noetherian. We denote by $\mathfrak{Z}^{p+}(\mathrm{X})$ the set of elements $\geqslant 0$ of $\mathfrak{Z}^p(\mathrm{X})$. It is clear that the ordered group $\mathfrak{Z}(\mathrm{X})$ contains the direct sum of the ordered sub-groups $\mathfrak{Z}^p(\mathrm{X})$, and is identical to this direct sum when X is Noetherian; in the latter case, we consider $\mathfrak{Z}(\mathrm{X})$ as *graded* by the $\mathrm{Z}^p(\mathrm{X})$ for $p \in \mathbf{N}$.

(21.6.3). Let $\mathrm{Z} = \sum_{x \in \mathrm{X}} n_x.\overline{\{x\}}$ be a cycle on X, U an open subset of X; we call the *restriction* of Z to U and denote by $\mathrm{Z}|\mathrm{U}$ the cycle $\sum_{x \in \mathrm{U}} n_x.(\mathrm{U} \cap \overline{\{x\}})$ on U; we have $\mathrm{Supp}(\mathrm{Z}|\mathrm{U}) = \mathrm{Supp}(\mathrm{Z}) \cap \mathrm{U}$. It is clear that $\mathrm{Z} \hookrightarrow \mathrm{Z}|\mathrm{U}$ is a homomorphism of ordered groups from $\mathfrak{Z}(\mathrm{X})$ to $\mathfrak{Z}(\mathrm{U})$ (resp. from $\mathfrak{Z}^p(\mathrm{X})$ to $\mathfrak{Z}^p(\mathrm{U})$), and that if V is an open subset contained in U, we have $\mathrm{Z}|\mathrm{V} = (\mathrm{Z}|\mathrm{U})|\mathrm{V}$; the map $\mathrm{U} \hookrightarrow \mathfrak{Z}^+(\mathrm{U})$ (resp. $\mathrm{U} \hookrightarrow \mathfrak{Z}^{p+}(\mathrm{U})$) is therefore a *presheaf* on X of ordered commutative groups. In fact, this presheaf is a *sheaf*, as a direct sum of sheaves $(i_x)_*(\mathbf{Z}_{\overline{x}})$, where x ranges over X (resp. $\mathrm{X}^{(p)}$) and for each $x \in \mathrm{X}$, i_x is the canonical injection $\overline{\{x\}} \to \mathrm{X}$ and $\mathbf{Z}_{\overline{x}}$ is the simple sheaf associated to the constant sheaf $\mathbf{Z}$ on the space $\overline{\{x\}}$: this follows at once from the description of the sections of a direct sum of sheaves of commutative groups (G, II, 2.7). We denote this sheaf by $\mathscr{Z}_{\mathrm{X}}$ (resp. $\mathscr{Z}_{\mathrm{X}}^p$). We denote by $\mathscr{Z}_{\mathrm{X}}^+$ (resp. $\mathscr{Z}_{\mathrm{X}}^{p+}$) the sub-sheaf of monoids $\mathrm{U} \hookrightarrow \mathfrak{Z}^+(\mathrm{U})$ (resp. $\mathrm{U} \hookrightarrow \mathfrak{Z}^{p+}(\mathrm{U})$) of $\mathscr{Z}_{\mathrm{X}}$ IV-4 | 272
(resp. $\mathscr{Z}_{\mathrm{X}}^p$). The sheaf $\mathscr{Z}_{\mathrm{X}}$ is evidently a *direct sum* of sheaves $\mathscr{Z}_{\mathrm{X}}^p$.

Let us note finally that the sheaves $\mathscr{Z}_{\mathrm{X}}^p$ (and therefore also $\mathscr{Z}_{\mathrm{X}}$) are *flasque*. Indeed, suppose given a section Z of $\mathscr{Z}_{\mathrm{X}}^p$ over an open U, so that the set S of $x \in \mathrm{U}$ such that $\mathrm{mult}_x(\mathrm{Z}) \neq 0$ is locally finite *in* U; this set is also locally finite *in* X, since for all $z \in \mathrm{X}$ and every open Noetherian neighbourhood V of z, $\mathrm{U} \cap \mathrm{V}$ is Noetherian, and thus contains only a finite number of points of S. We thus define a section Z' of $\mathscr{Z}_{\mathrm{X}}^p$ over X by setting $\mathrm{Z}' = \sum_{x \in \mathrm{U}} \mathrm{mult}_x(\mathrm{Z}).\overline{\{x\}}$ and since the closure of x in U is $\overline{\{x\}} \cap \mathrm{U}$, we have $\mathrm{Z}'|\mathrm{U} = \mathrm{Z}$, whence our assertion.

(21.6.4). We now propose to define a canonical homomorphism

$$c : \mathscr{D}iv_{\mathrm{X}} \longrightarrow \mathscr{Z}_{\mathrm{X}}^1 \tag{21.6.4.1}$$

of sheaves of commutative groups. It clearly suffices to define a homomorphism of $\mathscr{M}_{\mathrm{X}}^*$ to $\mathscr{Z}_{\mathrm{X}}^1$, which vanishes on $\mathscr{O}_{\mathrm{X}}^*$ (21.1.2); or, by definition, $\mathscr{M}_{\mathrm{X}}^*$ is the *symmetrised* monoid sheaf of $\mathscr{O}_{\mathrm{X}} \cap \mathscr{M}_{\mathrm{X}}^*$, and a homomorphism $\mathscr{M}_{\mathrm{X}}^* \to \mathscr{Z}_{\mathrm{X}}^1$ is uniquely determined by the data of its restriction $\mathscr{S}(\mathscr{O}_{\mathrm{X}}) \to \mathscr{Z}_{\mathrm{X}}^1$, which is a homomorphism of sheaves of *monoids* subject to the sole condition of being zero on $\mathscr{O}_{\mathrm{X}}^*$; it therefore amounts to the same thing to say that, in order to define c, it suffices to define a homomorphism of *sheaves of monoids*

$$c : \mathscr{D}iv_{\mathrm{X}}^+ \longrightarrow \mathscr{Z}_{\mathrm{X}}^1. \tag{21.6.4.2}$$

(21.6.5). Now, we have seen that every positive divisor D on X gives rise to the closed sub-prescheme Y(D) of X, defined by the Ideal $\mathscr{J}_{\mathrm{X}}(\mathrm{D}) \subset \mathscr{O}_{\mathrm{X}}$, and which is regularly immersed and of codimension 1. At each of the maximal points x of Y(D), we have ((19.1.4) and (5.1.2)) $\mathrm{codim}(\overline{\{x\}}, \mathrm{X}) = \dim(\mathscr{O}_{\mathrm{X},x}) = 1$, in other words $x \in \mathrm{X}^{(1)}$; moreover the set of these maximal points is locally finite in X (3.1.6), and $\mathscr{O}_{\mathrm{Y(D)},x}$ is an *Artinian* ring. At every point $x \in \mathrm{X}^{(1)}$ which is not a maximal point of $\mathrm{Y(D)}$, we necessarily have $x \notin \mathrm{Y(D)}$, so $\mathscr{O}_{\mathrm{Y(D)},x} = 0$. Set

(21.6.5.1)
$$\mathrm{cyc(D)} = \sum_{x \in \mathrm{X}^{(1)}} \mathrm{length}(\mathscr{O}_{\mathrm{Y(D)},x}).\overline{\{x\}}$$

which is thus an element of $\mathfrak{Z}^1(\mathrm{X})$.

Proposition (21.6.6). — *The map* $\mathrm{D} \mapsto \mathrm{cyc(D)}$ *is a homomorphism of the monoid* $\mathrm{Div}^+(\mathrm{X})$ *to* $\mathfrak{Z}^1(\mathrm{X})$.

Proof. It amounts to showing that for two positive divisors D, D', we have

$$\mathrm{cyc(D + D')} = \mathrm{cyc(D)} + \mathrm{cyc(D')}.$$

Now, by (21.2.5) $\mathscr{J}_{\mathrm{X}}(\mathrm{D+D'}) = \mathscr{J}_{\mathrm{X}}(\mathrm{D})\mathscr{J}_{\mathrm{X}}(\mathrm{D'})$; it thus amounts to showing that if $x \in \mathrm{X}^{(1)}$, setting $\mathrm{A} = \mathscr{O}_{\mathrm{X},x}$, and if t and t' are two regular elements of A, then $\mathrm{length}(\mathrm{A}/tt'\mathrm{A}) = \mathrm{length}(\mathrm{A}/t\mathrm{A}) + \mathrm{length}(\mathrm{A}/t'\mathrm{A})$; but since t is regular, $t\mathrm{A}/tt'\mathrm{A}$ is isomorphic to $\mathrm{A}/t'\mathrm{A}$, whence the proposition at once. □

It follows at once from the definitions that for every open $\mathrm{U} \subset \mathrm{X}$, we have

$$\mathrm{Y(D|U)} = \mathrm{Y(D)} \cap \mathrm{U},$$

whence $\mathrm{cyc(D|U)} = \mathrm{cyc(D)|U}$, and the homomorphisms $\mathrm{cyc} : \mathrm{Div}^+(\mathrm{U}) \to \mathfrak{Z}^1(\mathrm{U})$ define a homomorphism of sheaves of monoids (21.6.4.2), whence the sought-after homomorphism (21.6.4.1) of sheaves of groups. **IV-4** | 273

We have $\mathrm{Supp(cyc(D))} \subset \mathrm{Supp(D)}$ for every divisor D and

(21.6.6.2).
$$\mathrm{Supp(cyc(D))} = \mathrm{Supp(D)} \quad \text{when} \quad \mathrm{D} \geqslant 0;$$

we have indeed already seen the second relation (21.2.12); when D is arbitrary, the relation $\mathrm{D}_x = 0$ implies the existence of an open neighbourhood U of x such that $\mathrm{D|U} = 0$, whence $\mathrm{cyc(D)|U} = 0$, which proves our assertion.

(21.6.7). The homomorphism (21.6.4.1) gives, by passage to groups of sections over X, a *homomorphism of ordered groups* $\mathrm{Div(X)} \to \mathfrak{Z}^1(\mathrm{X})$, which we will also denote $\mathrm{D} \mapsto \mathrm{cyc(D)}$; the number $\mathrm{mult}_x(\mathrm{cyc(D)})$ for $x \in \mathrm{X}^{(1)}$ is also denoted $\mathrm{mult}_x(\mathrm{D})$ or $\mathrm{mult}_{\overline{\{x\}}}(\mathrm{D})$ and is called the *multiplicity of the divisor* D *at the point* x, or the *multiplicity of the prime cycle* $\overline{\{x\}}$ *in* D; it is a positive or negative integer, equal as we have seen to $\mathrm{length}(\mathscr{O}_{\mathrm{Y(D)},x})$ when D is a *positive* divisor; we thus have by definition

(21.6.7.1)
$$\mathrm{cyc(D)} = \sum_{x \in \mathrm{X}^{(1)}} \mathrm{mult}_x(\mathrm{D}).\overline{\{x\}},$$

and by virtue of the fact that $\mathrm{D} \mapsto \mathrm{cyc(D)}$ is a homomorphism, we have

(21.6.7.2)
$$\mathrm{mult}_x(-\mathrm{D}) = -\,\mathrm{mult}_x(\mathrm{D}), \qquad \mathrm{mult}_x(\mathrm{D+D'}) = \mathrm{mult}_x(\mathrm{D}) + \mathrm{mult}_x(\mathrm{D'})$$

for any two divisors D, D'.

For every regular meromorphic function f on X, we set in particular, for $x \in \mathrm{X}^{(1)}$

(21.6.7.3)
$$o_x(f) = \mathrm{mult}_x(\mathrm{div}(f))$$

and we say that $o_x(f)$ (a positive or negative integer) is the *order of* f *at the point* $x \in \mathrm{X}^{(1)}$. If $f_x \in \mathscr{O}_{\mathrm{X},x}$ (a regular element of this local ring by hypothesis), then we have

(21.6.7.4)
$$o_x(f) = \mathrm{length}(\mathscr{O}_{\mathrm{X},x}/f_x\mathscr{O}_{\mathrm{X},x});$$

for two regular meromorphic functions f, f' on X, we have

(21.6.7.5)
$$o_x(ff') = o_x(f) + o_x(f'), \qquad o_x(f^{-1}) = -o_x(f)$$

for all $x \in \mathrm{X}^{(1)}$. The 1-codimensional cycles

$$\mathrm{Z}^+(f) = \sum_{x \in \mathrm{X}^{(1)}} (o_x(f))^+.\overline{\{x\}}, \qquad \mathrm{Z}^-(f) = \sum_{x \in \mathrm{X}^{(1)}} (o_x(f))^-.\overline{\{x\}}$$

are called respectively the *cycle of zeros* (or *polar cycle*) of f, and we evidently have $\mathrm{cyc}(\mathrm{div}(f)) = \mathrm{Z}^+(f) - \mathrm{Z}^-(f)$. The 1-codimensional cycles of the form $\mathrm{cyc}(\mathrm{div}(f))$ are called *principal* (or *linearly equivalent to zero*) and form a subgroup $\mathfrak{Z}^1_{\mathrm{princ}}(\mathrm{X})$ of $\mathfrak{Z}^1(\mathrm{X})$. The sections over X of the image $c(\mathscr{D}iv_{\mathrm{X}}) \subset \mathscr{Z}^1_{\mathrm{X}}$ are called *locally principal cycles*; such a cycle Z is therefore characterised by the fact **IV-4** | 274
that, for all $y \in \mathrm{X}$, there is an open neighbourhood U of y in X and a regular meromorphic function f on U such that $\mathrm{Z}|\mathrm{U} = \sum_{x \in \mathrm{U} \cap \mathrm{X}^{(1)}} o_x(f)(\overline{\{x\}} \cap \mathrm{U})$. If $\mathrm{Z} = \sum_{x \in \mathrm{X}^{(1)}} n_x . \overline{\{x\}}$, setting $\mathrm{T}_y = \mathrm{Spec}(\mathscr{O}_{\mathrm{X},y})$ and $\mathrm{Z}_y = \sum_{x \in \mathrm{X}^{(1)} \cap \mathrm{T}_y} n_x(\overline{\{x\}} \cap \mathrm{T}_y)$, Z_y is *principal*, otherwise said there exists a regular meromorphic function g on T_y such that $\mathrm{Z}_y = \mathrm{cyc}(\mathrm{div}(g))$. This follows at once from the definition above and from (20.3.7) which establishes a bijective correspondence between the regular meromorphic functions on T_y and the germs of regular meromorphic functions on the open neighbourhoods of y in X when X is locally Noetherian. We express this further by saying that Z_y is principal in saying that Z is *principal at the point* y. The set of points where Z is principal is evidently *open*, by virtue of what precedes.

We deduce from the canonical homomorphism $\mathrm{cyc} : \mathrm{Div}(\mathrm{X}) \to \mathfrak{Z}^1(\mathrm{X})$ a homomorphism $\mathrm{Div}(\mathrm{X}) / \mathrm{Div.\,princ}(\mathrm{X}) \to \mathfrak{Z}^1(\mathrm{X})/\mathfrak{Z}^1_{\mathrm{princ}}(\mathrm{X})$, by virtue of the definition of $\mathfrak{Z}^1_{\mathrm{princ}}(\mathrm{X})$. We say that the group $\mathfrak{Z}^1(\mathrm{X})/\mathfrak{Z}^1_{\mathrm{princ}}(\mathrm{X})$ is the *group of classes of* 1*-codimensional cycles* of X and we denote it $\mathfrak{Cl}(\mathrm{X})$. Two elements of $\mathfrak{Z}^1(\mathrm{X})$ having the same image in $\mathfrak{Cl}(\mathrm{X})$ are called *linearly equivalent*.

(21.6.8). Let us consider in particular the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, where A is a *Noetherian integral and integrally closed* ring. Then $\mathrm{X}^{(1)}$ is the set of *prime ideals of height* 1 of A, and $\mathfrak{Z}^1(\mathrm{X})$ is identified with the *group of divisors* (in the sense of N. Bourbaki) of the Krull ring A (Bourbaki, *Alg. comm.*, chap. VII, § 1, n° 3, cor. du th. 2 et n° 6, th. 3).

On the other hand, the regular meromorphic functions on X are identified with the nonzero elements of the field of fractions K of A, the map $f \mapsto \mathrm{cyc}(\mathrm{div}(f))$ in $\mathrm{M}(\mathrm{X})$ is identified in $\mathfrak{Z}^1(\mathrm{X})$ with the map denoted $f \mapsto \mathrm{div}(f)$ in Bourbaki (*loc. cit.*, § 1, n° 1); $\mathfrak{Z}^1_{\mathrm{princ}}(\mathrm{X})$ is identified with the *group of principal divisors* of A in the sense of Bourbaki, and $\mathfrak{Cl}(\mathrm{X})$ with the *group of classes of divisors* of A in the sense of Bourbaki (*loc. cit.*, § 1, n° 2 et n° 10).

Theorem (21.6.9). — *Let* X *be a locally Noetherian normal prescheme.*

(i) The canonical homomorphism $\mathrm{cyc} : \mathrm{Div}(\mathrm{X}) \to \mathfrak{Z}^1(\mathrm{X})$ *is injective and its image is formed of locally principal cycles.*

(ii) The following conditions are equivalent:

a) *The homomorphism* $\mathrm{cyc} : \mathrm{Div}(\mathrm{X}) \to \mathfrak{Z}^1(\mathrm{X})$ *is bijective.*

b) *Every* 1*-codimensional cycle is locally principal.*

c) *For all* $x \in \mathrm{X}$*, the local ring* $\mathscr{O}_{\mathrm{X},x}$ *is factorial.*

(We then say that X *is a* locally factorial *prescheme.)*

Proof. (i) It suffices to show that

$$\mathrm{cyc}^{-1}(\mathfrak{Z}^{1+}(\mathrm{X})) = \mathrm{Div}^+(\mathrm{X}), \tag{21.6.9.1}$$

since we have $\mathrm{Div}^+(\mathrm{X}) \cap (-\mathrm{Div}^+(\mathrm{X})) = 0$ and $\mathfrak{Z}^{1+}(\mathrm{X}) \cap (-\mathfrak{Z}^{1+}(\mathrm{X})) = 0$. We are thus reduced to proving that if D is a divisor on X such that $\mathrm{mult}_x(\mathrm{D}) \geqslant 0$ for all $x \in \mathrm{X}^{(1)}$, then $\mathrm{D} \geqslant 0$. Now, for all $x \notin \mathrm{X}^{(1)}$, the local ring $\mathscr{O}_{\mathrm{X},x}$ is integral and integrally closed and of **IV-4** | 275
dimension 0 or $\geqslant 2$, therefore on the one hand $\mathrm{prof}(\mathscr{O}_{\mathrm{X},x}) = 0$ or $\mathrm{prof}(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$ (**0**, 16.5.1). On the other hand, for points $x \in \mathrm{X}^{(1)}$, the ring $\mathscr{O}_{\mathrm{X},x}$ is a discrete valuation ring ((**II**, 7.1.6)), whence $\mathrm{prof}(\mathscr{O}_{\mathrm{X},x}) = 1$ (**0**, 16.5.1); the only points of X such that $\mathrm{prof}(\mathscr{O}_{\mathrm{X},x}) = 1$ are thus the points of $\mathrm{X}^{(1)}$, and the conclusion follows from (21.1.8).

(ii) The equivalence of *a)* and *b)* is clear by virtue of (i). From the characterisation of locally principal cycles given in (21.6.7), and the relation between 1-codimensional cycles on Spec(A) and divisors (in the sense of Bourbaki) of the ring A when A is a Noetherian integrally closed ring (21.6.8), condition *b)* is equivalent to saying that for all $x \in \mathrm{X}$, every divisor of the ring $\mathscr{O}_{\mathrm{X},x}$ is principal, in other words that the ring $\mathscr{O}_{\mathrm{X},x}$ is factorial (Bourbaki, *Alg. comm.*, chap. VII, § 3, n° 1), whence the equivalence of *b)* and *c)*.

□

(21.6.9.2). When A is a Noetherian *factorial* ring, it is clear that $X = \mathrm{Spec}(A)$ is locally factorial (Bourbaki, *Alg. comm.*, chap. VII, § 3, nº 3, prop. 3). In this case, if one writes an element $f \neq 0$ of the field of fractions K of A in the form r/s, where r and s are two *coprime* elements of A, whose divisors are well-determined (*loc. cit.*, § 3, nº 3), these divisors are identified respectively with the *divisor of zeros* and the *divisor of poles* of f (21.6.7).

Corollary (21.6.10). — *Let* X *be a locally Noetherian normal prescheme.*

(i) There exists a canonical injective homomorphism

(21.6.10.1) $$\mathrm{Pic}(X) \longrightarrow \mathfrak{Cl}(X).$$

(ii) If X *is locally factorial, the homomorphism* (21.6.10.1) *is bijective, and conversely.*

Proof. We have seen (21.6.7) that the image of $\mathrm{Div.princ}(X)$ by the homomorphism $\mathrm{cyc} : \mathrm{Div}(X) \to \mathfrak{Z}^1(X)$ is $\mathfrak{Z}^1_{\mathrm{princ}}(X)$; we thus deduce from (21.6.9) that the homomorphism $\mathrm{Div}(X)/\mathrm{Div.princ}(X) \to \mathfrak{Z}^1(X)/\mathfrak{Z}^1_{\mathrm{princ}}(X) = \mathfrak{Cl}(X)$ deduced from cyc is injective, and that it is bijective if and only if X is locally factorial. The conclusion follows from the fact that, when X is locally Noetherian and reduced, the canonical homomorphism $\mathrm{Div}(X)/\mathrm{Div.princ}(X) \to \mathrm{Pic}(X)$ (21.3.3.1) is bijective (21.3.4), (ii). □

Corollary (21.6.11). — *Let* X *be a locally Noetherian prescheme which is locally factorial. Then the sheaf* $\mathscr{D}iv_X$ *is flasque, and for every open* U *of* X*, the canonical homomorphism* $\mathrm{Pic}(X) \to \mathrm{Pic}(U)$ *is surjective.*

Proof. Taking into account (21.6.9), (ii), the first assertion is equivalent to saying that the sheaf $\mathscr{Z}^1_X$ is flasque, which was seen above (21.6.3). For every open U of X, the canonical homomorphism $\mathfrak{Z}^1(X) \to \mathfrak{Z}^1(U)$ is thus surjective; as in (21.6.10), the homomorphism $\mathrm{Pic}(X) \to \mathrm{Pic}(U)$ is identified canonically with $\mathfrak{Z}^1(X)/\mathfrak{Z}^1_{\mathrm{princ}}(X) \to \mathfrak{Z}^1(U)/\mathfrak{Z}^1_{\mathrm{princ}}(U)$, it is also surjective. □

Proposition (21.6.12). — *Let* X *be a Noetherian reduced prescheme. Let* $(U_\lambda)_{\lambda \in L}$ *be a decreasing filtered family of opens of* X *satisfying the following conditions:*

1º *If* $Y_\lambda = X - U_\lambda$*, then* $\mathrm{codim}(Y_\lambda, X) \geqslant 2$ *for all* $\lambda \in L$.

2º *For all* $x \in \bigcap_{\lambda \in L} U_\lambda$*, the ring* $\mathscr{O}_{X,x}$ *is factorial.*

Then there exist canonical isomorphisms

(21.6.12.1) $$\varinjlim \mathrm{Div}(U_\lambda) \xrightarrow{\sim} \mathfrak{Z}^1(X), \qquad \varinjlim \mathrm{Pic}(U_\lambda) \xrightarrow{\sim} \mathfrak{Cl}(X)$$

such that, for every open V *of* X*, the diagrams*

(21.6.12.2) $$\begin{tikzcd} \varinjlim \mathrm{Div}(U_\lambda) \arrow[r, "\sim"] \arrow[d] & \mathfrak{Z}^1(X) \arrow[d] \\ \varinjlim \mathrm{Div}(U_\lambda \cap V) \arrow[r, "\sim"] & \mathfrak{Z}^1(V) \end{tikzcd} \qquad \begin{tikzcd} \varinjlim \mathrm{Pic}(U_\lambda) \arrow[r, "\sim"] \arrow[d] & \mathfrak{Cl}(X) \arrow[d] \\ \varinjlim \mathrm{Pic}(U_\lambda \cap V) \arrow[r, "\sim"] & \mathfrak{Cl}(V) \end{tikzcd}$$

are commutative.

Proof. The hypothesis 1º on U_λ implies that for all $\lambda \in L$, we have $X^{(1)} \subset U_\lambda$ (5.1.3.1) and therefore the restriction homomorphism $\mathfrak{Z}^1(X) \to \mathfrak{Z}^1(U_\lambda)$ is bijective and consequently we have an isomorphism $\varinjlim \mathfrak{Z}^1(U_\lambda) \xrightarrow{\sim} \mathfrak{Z}^1(X)$. The canonical homomorphisms $\mathrm{cyc} : \mathrm{Div}(U_\lambda) \to \mathfrak{Z}^1(U_\lambda)$ thus define, by passage to the inductive limit, the first of the canonical homomorphisms (21.6.12.1), and the second is deduced by passage to quotients. Furthermore, it follows from condition 1º that the U_λ are schematically dense in X since X is reduced (11.10.4), and therefore every meromorphic function on X is entirely determined by its restriction to each U_λ; we deduce from this that in the isomorphism $\mathfrak{Z}^1(X) \xrightarrow{\sim} \mathfrak{Z}^1(U_\lambda)$, the image of $\mathfrak{Z}^1_{\mathrm{princ}}(X)$ is $\mathfrak{Z}^1_{\mathrm{princ}}(U_\lambda)$, and therefore we also have a canonical isomorphism $\varinjlim \mathfrak{Cl}(U_\lambda) \xrightarrow{\sim} \mathfrak{Cl}(X)$. The second of the canonical homomorphisms (21.6.12.1) will therefore be an isomorphism if the first is, and it remains to prove this last point, the commutativity of the diagrams (21.6.12.2) being trivial.

Let us first show that the homomorphism $\varinjlim \mathrm{Div}(U_\lambda) \to \mathfrak{Z}^1(X)$ is *injective*. Set $T = \bigcap_\lambda U_\lambda$, and **IV-4** | 276
note that the U_λ form a *fundamental system of neighbourhoods* of T. Indeed, for every open $V \supset T$, $X - V$ is a closed subspace of X, thus a Noetherian space whose every irreducible closed subspace

admits a generic point; as the sets $(X - V) \cap (X - U_\lambda)$ are closed in $X - V$, they form an increasing filtered family and their union is $X - V$; our assertion follows from (**0**, 9.2.4). This being so, it will suffice to show that if $D \in \mathrm{Div}(U_\lambda)$ is such that $\mathrm{cyc}(D) = 0$ in $\mathfrak{Z}^1(U_\lambda)$, then there exists $\mu \geqslant \lambda$ such that $D|U_\mu = 0$. By virtue of what precedes, it will suffice to show that for all $x \in T$, $D_x = 0$; indeed, if this holds, there will be an open neighbourhood $W(x)$ of x in X such that $D|W(x) = 0$, and the union of the $W(x)$ for $x \in T$ contains a U_μ. Taking into account (21.4.6), we are thus reduced to the case where $X = \mathrm{Spec}(\mathscr{O}_{X,x})$ with $x \in T$; but by hypothesis $\mathscr{O}_{X,x}$ is factorial, thus integral and integrally closed, and $\mathrm{Spec}(\mathscr{O}_{X,x})$ is therefore normal; whence the conclusion follows from (21.6.9), (i).

To prove that the homomorphism $\varinjlim \mathrm{Div}(U_\lambda) \to \mathfrak{Z}^1(X)$ is bijective, it will suffice to prove that for all $Z \in \mathfrak{Z}^1(\overline{X})$ and all $x \in T$, there is a neighbourhood $W(x)$ of x in X and a divisor $D^{(x)}$ on $W(x)$ such that $\mathrm{cyc}(D^{(x)}) = Z|W(x)$. Indeed, we can then cover the quasi-compact set T by a finite number of the $W(x_i)$, and by the first part of the proof, there will exist an index λ such that in each of the $W(x_i) \cap W(x_j) \cap U_\lambda$, the restrictions of $D^{(x_i)}$ and $D^{(x_j)}$ coincide; since we can moreover assume that U_λ is contained in the union of the $W(x_i)$, we see that the restrictions $D^{(x_i)}|(W(x_i) \cap U_\lambda)$ are the restrictions of a single divisor $D \in \mathrm{Div}(U_\lambda)$ which will be such that $\mathrm{cyc}(D) = Z|U_\lambda$. We are thus reduced again to the case where $X = \mathrm{Spec}(\mathscr{O}_{X,x})$ with $x \in T$; but as $\mathscr{O}_{X,x}$ is factorial, it is Noetherian, integral and integrally closed, and it suffices to apply (21.6.9), (ii). ☐ **IV-4 | 277**

Corollary (21.6.13). — *Let* A *be a Noetherian local ring which is integral and integrally closed and of dimension* $\geqslant 2$*. Set* $X = \mathrm{Spec}(A)$*, and let* a *be the closed point of* X*,* $U = X - \{a\}$*. For* A *to be factorial, it is necessary and sufficient that* U *is locally factorial and that* $\mathrm{Pic}(U) = 0$*.*

Proof. Indeed, to say that A is factorial is equivalent to saying that $\mathfrak{Cl}(X) = 0$ (Bourbaki, *Alg. comm.*, chap. VII, § 1, n° 4, cor. du th. 2 et § 3, th. 1); it therefore suffices to use the existence of the second isomorphism (21.6.12.1), taking the family (U_λ) restricted to the single open U. ☐

Corollary (21.6.14). — *Let* A *be a Noetherian local ring of dimension* $\geqslant 2$*,* $X = \mathrm{Spec}(A)$*,* a *the closed point of* X*,* $U = X - \{a\}$*. The following conditions are equivalent:*

a) A *is factorial.*
b) $\mathrm{Pic}(U) = 0$ *and* $\mathrm{prof}(A) \geqslant 2$ *(conditions which we will express later* (21.13) *by saying that the ring* A *is* parafactorial*), and* U *is locally factorial.*

Proof. It is clear that if A is factorial, it is normal, whence $\dim(A) \geqslant 2$ (**0**, 16.5.1) and (21.6.13) shows that *a)* implies *b)*. Conversely, if *b)* is satisfied, it suffices to see that A is integrally closed to conclude by (21.6.13) that *b)* implies *a)*. By virtue of the Serre criterion (5.8.6), it suffices to verify for X the conditions (R_1) and (S_2). Now, U being locally factorial satisfies these conditions, and the hypothesis $\mathrm{prof}(A) \geqslant 2$ implies that X also satisfies them. ☐

21.7. Interpretation of positive 1-codimensional cycles in terms of sub-preschemes

(21.7.1). Let X be a locally Noetherian prescheme, $C = \sum_{x \in X^{(1)}} n_x . \overline{\{x\}}$ a positive 1-codimensional cycle (so that $n_x \geqslant 0$ for all $x \in X^{(1)}$, and $n_x = 0$ except at a locally finite set of points). Let $Y(C)$ denote the closed sub-prescheme of X, *closed image* (**I**, 9.5.3) and (9.5.1) of the sub-prescheme *sum* $Y'(C) = \coprod_{x \in X^{(1)}} \mathrm{Spec}(\mathscr{O}_{X,x}/\mathfrak{m}_x^{n_x})$ by the canonical morphism, and let $\mathscr{J}_X(C)$ (or $\mathscr{J}(C)$) denote the Ideal of $\mathscr{O}_X$ defining $Y(C)$.

Proposition (21.7.2). — *Let* X *be a locally Noetherian prescheme satisfying the condition* (R_1) (5.8.2)*. For a closed sub-prescheme* Y *of* X *to be of the form* $Y(C)$*, where* C *is a positive* 1*-codimensional cycle, it is necessary and sufficient that it satisfies the two following conditions:*

(i) Y *is purely of codimension* 1*.* **IV-4 | 278**
(ii) Y *satisfies the condition* (S_1)*, in other words* (5.7.5) *has no associated prime cycle.*

We then have

(21.7.2.1) $$\mathrm{mult}_x(C) = \mathrm{length}\ \mathscr{O}_{Y,x}.$$

The map $C \mapsto Y(C)$ *is a bijection from* $\mathfrak{Z}^{1+}(X)$ *onto the set of closed sub-preschemes of* X *satisfying conditions (i) and (ii).*

Proof. The conditions are necessary (without assuming that X satisfies (R_1)). Indeed, since the question is local on X, we can assume that $X = \operatorname{Spec}(A)$, where A is a Noetherian ring; we then have $Y(C) = \operatorname{Spec}(A/\mathfrak{J})$, where $\mathfrak{J}$ is, by definition (**I**, 9.5.1) the kernel of the canonical homomorphism $A \to \bigoplus A_{\mathfrak{p}_i}/\mathfrak{p}_i^{n_i}A_{\mathfrak{p}_i}$, where the $\mathfrak{p}_i$ are the prime ideals corresponding to points $x \in X^{(1)}$ for which $n_x \geqslant 0$, and we have set $n_i = n_x$. The inverse image $\mathfrak{q}_i$ of $\mathfrak{p}_i^{n_i}A_{\mathfrak{p}_i}$ in A is a $\mathfrak{p}_i$-primary ideal and we have $\mathfrak{J} = \bigcap_i \mathfrak{q}_i$. Moreover, as the $x \in X^{(1)}$ are such that $\overline{\{x\}}$ is of codimension 1, no point of $X^{(1)}$ can be adherent to another point of $X^{(1)}$, so the $\mathfrak{p}_i$ are minimal primes of $A/\mathfrak{J}$ and the set of $\mathfrak{p}_i$ is equal to $\operatorname{Ass}(A/\mathfrak{J})$, which proves conditions (i) and (ii).

The conditions are sufficient; let $\mathscr{J}$ denote the Ideal of $\mathscr{O}_X$ defining Y, and hypothesis (ii) implies that $\operatorname{Ass}(\mathscr{O}_X/\mathscr{J})$ is identical to the set F of maximal points of Y, and hypothesis (i) implies that F is contained in $X^{(1)}$; hence (3.2.6) $\mathscr{O}_X/\mathscr{J}$ is an $\mathscr{O}_X$-Module which is irredundant such that $\operatorname{Ass}(\mathscr{S}(x)) = \{x\}$. Now, since X satisfies (R_1), for each $x \in F$, $\mathscr{O}_{X,x}$ is a discrete valuation ring and so the only primary ideals are the powers $\mathfrak{m}_x^{n_x}$ of the maximal ideal, where the $\mathfrak{p}_i$ correspond to the points of F. We see then that Y is of the form Y(C). □

Corollary (21.7.3). — *Let* X *be a locally Noetherian prescheme such that* X *satisfies* (R_1). *For every positive divisor* D *on* X, *the maximal points of the closed sub-prescheme* Y(D) (21.2.12) *majorise the closed sub-prescheme* Y(cyc(D)) (21.7.1), *and this has the same underlying space; for these two sub-preschemes to be equal, it is necessary and sufficient that* Y(D) *satisfies the condition* (S_1), *or else, for every* $z \in Y(D)$ *distinct from a maximal point of* Y(D), *that* $\operatorname{prof}(\mathscr{O}_{X,z}) \geqslant 2$ *(a condition always satisfied when* X *is* normal*).*

Proof. Indeed, the question is local, and for all $x \in T$, $D_x = 0$ implies the existence of an open neighbourhood U of x such that $D|U = 0$, whence $\operatorname{cyc}(D)|U = 0$, which proves our assertion. We may suppose $X = \operatorname{Spec}(A)$, and that t is a regular section of $\mathscr{O}_X$ over X, such that $\mathscr{J}_X(D) = t\mathscr{O}_X$, meaning that Y(D) satisfies (S_1), at every maximal point x of Y(D), $\mathscr{O}_{X,x}$ is by hypothesis a discrete valuation ring, therefore $t_x\mathscr{O}_{X,x} = \mathfrak{m}_x^{n_x}$, where $n_x = \operatorname{mult}_x(D)$. We can assume $X = \operatorname{Spec}(A)$, and then, if $\mathscr{J}_X(\operatorname{cyc}(D)) = \mathscr{J}'$, it follows from what precedes and from (21.7.1) that $\mathfrak{J}$ and $\mathfrak{J}'$ have the same *non-immersed primary ideals*, whence $\mathfrak{J}' \subset \mathfrak{J}$, since $\mathfrak{J}$ is the intersection of these primary ideals **IV-4** | 279
(21.7.2). This proves that Y(D) majorises Y(cyc(D)) and that these two sub-preschemes are equal if and only if Y(D) has no associated immersed prime cycle. Since for all $x \in Y(D)$, it is a condition on the local ring, and since by hypothesis $\operatorname{prof}(\mathscr{O}_{X,z}) \geqslant 2$ at all non-maximal points $z \in Y(D)$, the condition (S_1) is then also satisfied. □

(21.7.3.1). We see therefore that when X is *normal*, $\operatorname{Div}^+(X)$ is identified canonically with the set of closed sub-preschemes Y of X satisfying conditions (i) and (ii) of (21.7.2) and *regularly immersed* in X.

Proposition (21.7.4). — *Let* X *be a locally Noetherian prescheme, reduced and with isolated points. The following conditions are equivalent:*

- a) *The canonical homomorphism* $c : \mathscr{D}iv_X \to \mathscr{Z}_X^1$ (21.6.4) *is an isomorphism of sheaves of ordered groups, and we have* $\operatorname{Div}(X) = \operatorname{Div.princ}(X)$.
- a′) *The canonical homomorphism* $c : \mathscr{D}iv_X \to \mathscr{Z}_X^1$ *is injective, and every prime* 1*-codimensional cycle on* X *is the image by* cyc *of a positive principal divisor, and the canonical homomorphism* $c : \mathscr{D}iv_X \to \mathscr{Z}_X^1$ *is injective.*
- a″) *For every integral closed sub-prescheme* Y *of* X, *of codimension* 1, *the canonical immersion* $Y \to X$ *is regular.*
- b) X *is normal and the homomorphism* $\operatorname{cyc} : \operatorname{Div}(X) \to \mathfrak{Z}^1(X)$ *is bijective.*
- c) X *is locally factorial and* $\operatorname{Pic}(X) = 0$.
- d) X *is normal, and for every open* U *of* X, *we have* $\operatorname{Pic}(U) = 0$.

Proof. The equivalence of *a), a′), a″), b)* and *c)* follows at once from (21.7.4) and from (21.6.11). Furthermore, it follows at once that every non-empty open U of X also satisfies these conditions, in other words *c)* implies *d)*. It remains to show that *d)* implies *b)*. Now, let us denote by U_λ the open complements of finite unions of sets of the form $\overline{\{x\}}$, where $\dim(\mathscr{O}_{X,x}) \geqslant 2$; it is clear that the U_λ form a decreasing filtered family and that for all $x \in \bigcap_\lambda U_\lambda$, we have $\dim(\mathscr{O}_{X,x}) \leqslant 1$, so $\mathscr{O}_{X,x}$ is a principal ring and therefore one can apply to the family (U_λ) the proposition (21.6.12), which shows that $\mathfrak{Cl}(X)$ is isomorphic to $\varinjlim \mathrm{Pic}(U_\lambda)$, hence $\mathfrak{Cl}(X) = 0$ by virtue of the hypothesis, which proves *b)* by definition. □

Remark (21.7.5). — We cannot replace in (21.7.4) the condition *a′)* by the weaker condition that every prime 1-codimensional cycle on X is the image by cyc of a positive divisor. This is shown by the example where X is the affine scheme defined in (14.1.5) ("the union of a plane and a line that intersects it") *which is not normal*; with the notation introduced in (14.1.5), the prime 1-codimensional cycles of X are those of the plane X_2 and the line X_3; if t_1, t_2, t_3 are the images of T_1, T_2, T_3 in $A/\mathfrak{a}$, the 1-codimensional prime ideals of X are defined by the monogenic prime ideals in $A/\mathfrak{a}$ (P irreducible in $K[T_1, T_2]$) and $t_3 - a$, with $a \neq 0$ in K; these elements are indeed regular elements in $A/\mathfrak{a}$, so every 1-codimensional cycle is indeed the canonical image of a positive divisor.

Corollary (21.7.6). — *Let* X *be a locally Noetherian prescheme, reduced and with isolated points. The following conditions are equivalent:*

a) *The canonical homomorphism $c : \mathscr{D}iv_X \to \mathscr{Z}^1_X$ is an isomorphism of sheaves of ordered groups, and we have* $\mathrm{Div}(X) = \mathrm{Div.princ}(X)$.
b) X *is normal and every* 1*-codimensional cycle on* X *is principal.*
c) X *is locally factorial and* $\mathrm{Pic}(X) = 0$.
d) X *is normal, and for every open* U *of* X*, we have* $\mathrm{Pic}(U) = 0$.

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Remark (21.7.7). — When $X = \mathrm{Spec}(A)$, where A is a Noetherian integral ring, the conditions of (21.7.6) are equivalent to saying that A is a *factorial* ring.

21.8. Divisors and normalization

Lemma (21.8.1). — *Let $f : X' \to X$ be an integral morphism of preschemes. For every $\mathscr{O}_X$-Module locally free $\mathscr{E}'$ of constant rank n and every $x \in X$ such that, if we set $U' = f^{-1}(U)$, $\mathscr{E}'|U'$ is isomorphic to $\mathscr{O}^n_{U'}$.*

Proof. The question being local on X, we can restrict to the case where $X = \mathrm{Spec}(A)$ is affine, in which case $X' = \mathrm{Spec}(A')$, where A′ is an A-algebra that is integral. Then A′ is the inductive limit of its *finite* sub-A-algebras A'_λ. Setting $X'_\lambda = \mathrm{Spec}(A'_\lambda)$ and denoting by $p_\lambda : X' \to X'_\lambda$ the structural morphism, it follows from (8.5.2) and (8.5.5) that there exists an index λ and an $\mathscr{O}_{X'_\lambda}$-Module locally free $\mathscr{E}'_\lambda$ of rank n such that $\mathscr{E}' = p^*_\lambda(\mathscr{E}'_\lambda)$, and it will clearly suffice to prove the lemma for X'_λ and $\mathscr{E}'_\lambda$; in other words, we may assume that the morphism f is *finite*. Set $B = \mathscr{O}_{X,x}$ and let B′ be the ring of the affine scheme $X'_0 = X' \times_X \mathrm{Spec}(\mathscr{O}_{X,x})$; as B is a local ring and B′ is a B-algebra which is finite, B′ is a semi-local ring (Bourbaki, *Alg. comm.*, chap. V, § 2, nº 1, prop. 3); we conclude that the $\mathscr{O}_{X'_0}$-Module locally free $\mathscr{E}'_0 = \mathscr{E}' \otimes_{\mathscr{O}_{X'}} \mathscr{O}_{X,x}$ is isomorphic to $\mathscr{O}^n_{X'_0}$ (Bourbaki, *Alg. comm.*, chap. II, § 5, nº 3, prop. 5). Considering X'_0 as the projective limit of the preschemes induced by X′ on the opens $f^{-1}(U)$, where U ranges over the set of affine open neighbourhoods of x in X, following the method of (8.1.2), *a)*), and applying (8.5.2.5), we obtain the sought conclusion. □ **IV-4 | 281**

Corollary (21.8.2). — *Let $f : X' \to X$ be an integral morphism of preschemes. Then:*

(i) We have $R^1 f_*(\mathscr{O}^*_{X'}) = 0$ (**0**, 12.2.1).
(ii) The group $\mathrm{Pic}(X')$ *is canonically isomorphic to* $H^1(X, f_*(\mathscr{O}^*_{X'}))$.

Proof. (i) (i) is the sheaf associated to the presheaf of commutative groups $U \mapsto H^1(f^{-1}(U), \mathscr{O}^*_{X'})$ on X (*loc. cit.*); the fibre of this sheaf at a point x can be identified with the commutative group $\varinjlim \mathrm{Pic}(f^{-1}(U))$, where U ranges over the open neighbourhoods of x in X, and for every element $\zeta \in \mathrm{Pic}(f^{-1}(U'))$, for every open neighbourhood U′ of x in X, and every element $\zeta \in \mathrm{Pic}(f^{-1}(U'))$, the transition homomorphisms $\mathrm{Pic}(f^{-1}(U')) \to \mathrm{Pic}(f^{-1}(U))$ being defined by (21.3.2.4). Now, for all $x \in X$, every open neighbourhood U′ of x in X, and every element $\zeta \in \mathrm{Pic}(f^{-1}(U'))$, it follows from (21.8.1) that there is an open neighbourhood $U \subset U'$ of x such that the image of ζ in $\mathrm{Pic}(f^{-1}(U))$ at the point x is zero. Therefore the fibre of $R^1 f_*(\mathscr{O}^*_{X'})$ at the point x is zero.

(ii) (ii) The Leray spectral sequence for the composed functor $\mathscr{F}' \mapsto \Gamma(X, f_*(\mathscr{F}'))$ of sheaves of commutative groups on X' (T, 2.4) gives the exact sequence of low degree

$$0 \longrightarrow H^1(X, f_*(\mathscr{O}^*_{X'})) \longrightarrow H^1(X', \mathscr{O}^*_{X'}) \longrightarrow H^0(X, R^1 f_*(\mathscr{O}^*_{X'}))$$

and the conclusion follows from (i) and from the isomorphism of $\mathrm{Pic}(X')$ and $H^1(X', \mathscr{O}^*_{X'})$ (**0**, 5.4.7).

□

Proposition (21.8.3). — *Let $f = (\psi, \theta) : X' \to X$ be an integral morphism of preschemes; suppose that, for every open U of X, the homomorphism $\Gamma(\theta) : \Gamma(U, \mathscr{O}_X) \to \Gamma(U, f_*(\mathscr{O}_{X'}))$ sends regular elements to regular elements, which allows us to canonically extend the homomorphism $\theta : \mathscr{O}_X \to f_*(\mathscr{O}_{X'})$ to a homomorphism of sheaves of rings $\theta' : \mathscr{M}_X \to f_*(\mathscr{M}_{X'})$, whence homomorphisms of sheaves of multiplicative groups $\theta^* : \mathscr{O}^*_X \to f_*(\mathscr{O}^*_{X'})$ and $\theta'' : \mathscr{M}^*_X \to f_*(\mathscr{M}^*_{X'})$, giving by passage to quotients a homomorphism $\theta''' : \mathscr{D}iv_X \to f_*(\mathscr{D}iv_{X'})$. We then have a commutative diagram*

(21.8.3.1)
$$\begin{tikzcd}
1 \arrow[r] & \mathscr{O}^*_X \arrow[r] \arrow[d, "\theta^*"] & \mathscr{M}^*_X \arrow[r] \arrow[d, "\theta''"] & \mathscr{D}iv_X \arrow[r] \arrow[d, "\theta'''"] & 0 \\
1 \arrow[r] & f_*(\mathscr{O}^*_{X'}) \arrow[r] & f_*(\mathscr{M}^*_{X'}) \arrow[r] & f_*(\mathscr{D}iv_{X'}) \arrow[r] & 0
\end{tikzcd}$$

in which the two rows are exact.

Proof. The only point to verify is the exactness of the second row of the diagram, which follows from the application of the exact sequence of cohomology for the functor f_* to the exact sequence of sheaves of commutative groups on X'

$$1 \longrightarrow \mathscr{O}^*_{X'} \longrightarrow \mathscr{M}^*_{X'} \longrightarrow \mathscr{D}iv_{X'} \longrightarrow 0$$

and from (21.8.2). □

Corollary (21.8.4). — *If, in addition to the hypotheses of (21.8.3), the homomorphism θ' is an isomorphism of sheaves of rings, then $\theta^* : \mathscr{O}^*_X \to f_*(\mathscr{O}^*_{X'})$ is injective, $\theta''' : \mathscr{D}iv_X \to f_*(\mathscr{D}iv_{X'})$ is surjective,* $\mathrm{Ker}(\theta''')$ *is isomorphic to* $\mathrm{Coker}(\theta^*)$.

Proof. This is an immediate consequence of the snake diagram (Bourbaki, *Alg. comm.*, chap. I, § 2, n° 4, prop. 2) applied to the diagram (21.8.3.1). □

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Proposition (21.8.5). — *Let X, X' be two locally Noetherian preschemes, $f : X' \to X$ an integral morphism; suppose that there exists a schematically dense open U in X such that $f^{-1}(U)$ is schematically dense in X' (cf. (20.3.5)), and that the morphism $f^{-1}(U) \to U$, restriction of f, is an isomorphism. Then:*

(i) The homomorphism $\theta' : \mathscr{M}_X \to f_*(\mathscr{M}_{X'})$ is defined and bijective, the homomorphism

$$\theta^* : \mathscr{O}^*_X \longrightarrow f_*(\mathscr{O}^*_{X'})$$

is injective, the homomorphism $\theta''' : \mathscr{D}iv_X \to f_*(\mathscr{D}iv_{X'})$ is surjective, $\mathrm{Ker}(\theta''')$ *is isomorphic to $\mathscr{N} =$* $\mathrm{Coker}(\theta^*)$ *and the support of the sheaf of multiplicative groups $\mathscr{N}$ is contained in $S = X - U$.*

(ii) Suppose moreover that the set S is discrete and set, for brevity, $\mathscr{O}'_X = f_*(\mathscr{O}_{X'})$. Then, on a commutative diagram

(21.8.5.1)
$$\begin{tikzcd}
1 \arrow[r] & \prod_{s \in S} \mathscr{O}'^*_{X,s} / \mathscr{O}^*_{X,s} \arrow[r, "j"] \arrow[d] & \mathrm{Div}(X) \arrow[r] \arrow[d, "k_X"] & \mathrm{Div}(X') \arrow[r] \arrow[d, "k_{X'}"] & 0 \\
1 \arrow[r] & (\prod_{s \in S} \mathscr{O}'^*_{X,s} / \mathscr{O}^*_{X,s}) / \mathrm{Im}\, \Gamma(X', \mathscr{O}^*_{X'}) \arrow[r] & \mathrm{Pic}(X) \arrow[r] & \mathrm{Pic}(X') \arrow[r] & 0
\end{tikzcd}$$

where the two rows are exact and where the left vertical arrow is the canonical homomorphism.

Proof. (i) The hypothesis implies that we have a canonical isomorphism $\mathscr{M}'_X \xrightarrow{\sim} f_*(\mathscr{M}_{X'})$ for the sheaves of germs of pseudo-functions (20.2.2), whence the assertion relative to θ', given the existence of the canonical isomorphisms $\mathscr{M}_X \xrightarrow{\sim} \mathscr{M}'_X$ and $\mathscr{M}_{X'} \xrightarrow{\sim} \mathscr{M}'_{X'}$ (20.2.11). The rest of assertion (i) follows from (21.8.4), except what concerns the support of $\mathscr{N}$, which follows directly from the hypothesis on U.

(ii) (ii) Applying the exact sequence of cohomology to the two exact sequences of sheaves of commutative groups

$$1 \longrightarrow \mathscr{N} \longrightarrow \mathscr{D}iv_X \longrightarrow f_*(\mathscr{D}iv_{X'}) \longrightarrow 0 \tag{21.8.5.2}$$

$$1 \longrightarrow \mathscr{O}_X^* \longrightarrow f_*(\mathscr{O}_{X'}^*) \longrightarrow \mathscr{N} \longrightarrow 1 \tag{21.8.5.3}$$

we get respectively the exact sequences

$$1 \longrightarrow \Gamma(X, \mathscr{N}) \xrightarrow{j} \mathrm{Div}(X) \longrightarrow \mathrm{Div}(X') \longrightarrow H^1(X, \mathscr{N})$$

$$\Gamma(X', \mathscr{O}_{X'}^*) \longrightarrow \Gamma(X, \mathscr{N}) \xrightarrow{\partial} H^1(X, \mathscr{O}_X^*)$$

taking into account the canonical isomorphism $\mathrm{Pic}(X') \xrightarrow{\sim} H^1(X, f_*(\mathscr{O}_{X'}^*))$ (21.8.2). Now, **IV-4** | 283
as the support of $\mathscr{N}$ is contained in the discrete set S, closed in X, we have $H^1(X, \mathscr{N}) = H^1(S, \mathscr{N}|S) = \prod_{s \in S} H^1(\{s\}, \mathscr{N}_s) = 0$ by definition of the cohomology (G, II, 4.9.2) and $H^1(S, \mathscr{N}|S) = \prod_{s \in S} H^1(\{s\}, \mathscr{N}_s) = 0$ by definition of the cohomology (G, II, 4.4). Similarly, we have $\Gamma(X, \mathscr{N}) = \prod_{s \in S} \mathscr{N}_s$ and $\mathscr{N}_s = \mathscr{O}_{X,s}^{*''} / \mathscr{O}_{X,s}^*$. By the second exact sequence (21.8.5.3), the image of t by $\partial : \Gamma(X, \mathscr{N}) \to H^1(X, \mathscr{O}_X^*)$ comes from the cocycle obtained as follows: we describe a section t of $\mathscr{N}$ over X, and we describe a covering of X formed of U and of opens U_i such that $U_i \cap S$ reduces to a single point s_i and that $U_i \cap U_j \subset U$ for $i \neq j$, then considering for each i a section t_i of $\mathscr{N}$ over U_i, which may be considered as a section of $\mathscr{M}_{X'}^*$ over $f^{-1}(U_i)$, thus also as a section of $\mathscr{M}_X^*$ over U_i, and finally considering a section u_i of $\mathscr{O}_{X'}^*$ over $f^{-1}(U_i)$, the germ $(t_i)_{s_i}$ comes from a section of $\mathscr{O}_{X'}^*$ over $f^{-1}(U_i)$; it canonically corresponds to this section a section $d_i \in \Gamma(U_i, \mathscr{D}iv_X)$ the restriction of u_i in $U \cap U_i$ to $(u_j|(U_i \cap U_j))(u_i|(U_i \cap U_j))^{-1}$ in $U_i \cap U_j$, whence the expression of $j'(t)$ and the commutativity of the diagram (21.8.5.1).

□

Remark (21.8.6). — (i) The conditions of application of (21.8.5), (i) are in particular satisfied when X is a locally Noetherian prescheme which is *reduced* with only a finite number of irreducible components, X′ is *normalisied* and that X′ is also locally Noetherian ((**II**, 6.3.8)); the $\mathscr{O}_X$-Module $\mathscr{D}iv_X$ is then an *extension* of $f_*(\mathscr{D}iv_{X'})$ by the cokernel $\mathscr{N}$ of $\mathscr{O}_X^* \to f_*(\mathscr{O}_{X'}^*)$, which we can consider as known. If moreover X is an algebraic curve (reduced) over a field k, we are in the conditions of application of (21.8.5), (ii).

(ii) When the conditions of application of (21.8.5), (ii) are satisfied and in addition that the canonical homomorphism globally $\Gamma(X, \mathscr{O}_X) \to \Gamma(X', \mathscr{O}_{X'})$ is *bijective*, we see that the kernels of the surjective homomorphisms $\mathrm{Div}(X) \to \mathrm{Div}(X')$ and $\mathrm{Pic}(X) \to \mathrm{Pic}(X')$ are *isomorphic*.

(iii) When the conditions of application of (21.8.5), (ii) are satisfied, we see that the homomorphism $\mathrm{Div}(X) \to \mathrm{Div}(X')$ cannot be injective (in which case it is bijective) unless $\mathscr{O}_{X,s}^{*''} = \mathscr{O}_{X,s}^*$ for all $s \in S$. For an $s \in S$ such that $\mathscr{O}_{X,s}$ is *local* (which, since $X' = \mathrm{Spec}(\mathscr{O}_X^*)$ (**II**, 1.3), means that there exists no point $s' \in X'$ above s, i.e. the residue fields $\boldsymbol{k}(s)$ and $\boldsymbol{k}(s')$ are equal; moreover, if this condition is satisfied, it is furthermore needed that the multiplicative group $1 + \mathfrak{m}_s$ has for image $1 + \mathfrak{m}_s$, or again $\mathfrak{m}_s = \mathfrak{r}$, which implies that $\mathscr{O}_{X,s}^{*''} \otimes_{\mathscr{O}_{X,s}} \boldsymbol{k}(s)$ is a composed direct of fields isomorphic to $\boldsymbol{k}(s)$ (which, when the morphism f is *finite*, implies in the general case, by (**0**, 23.2.6), the canonical homomorphism $\mathscr{O}_{X,s}^{*''} \to \mathscr{O}_{X,s}^*$ defines, by passage to quotients, a homomorphism of the multiplicative group $(\boldsymbol{k}(s))^*$ into the product of the multiplicative groups $(\boldsymbol{k}(s'_j))^*$, the s'_j being the points (in number > 1) of X′ above s. It is **IV-4** | 284
immediate that this homomorphism cannot be bijective if $\boldsymbol{k}(s)$ and all the $\boldsymbol{k}(s'_j)$ are equal to the field $\mathbf{F}_2$ with 2 elements; moreover, if this condition is satisfied, it is furthermore needed that the multiplicative group $1 + \mathfrak{m}_s$ has for image $1 + \mathfrak{m}_s$, in other words that $\mathfrak{m}_s = \mathfrak{r}$, which

implies that $\mathscr{O}''^{*}_{X,s}$ is *non ramified* at the points s'_j (17.4.1). If for example *no* residue field of X is isomorphic to $\mathbf{F}_2$, the canonical homomorphism $\mathrm{Div}(X) \to \mathrm{Div}(X')$ cannot be bijective if f is an *isomorphism*. In the case where the canonical homomorphism $\Gamma(X, \mathscr{O}_X) \to \Gamma(X', \mathscr{O}_{X'})$ is bijective, we conclude from this and from (ii) that in the preceding considerations, one can replace the homomorphism $\mathrm{Div}(X) \to \mathrm{Div}(X')$ by the homomorphism $\mathrm{Pic}(X) \to \mathrm{Pic}(X')$.

21.9. Divisors on preschemes of dimension 1

(21.9.1). Let X be a topological space, x a point of X, $i_x : \{x\} \to X$ the canonical injection. If $A(x)$ is a commutative group, we can consider the commutative group on the space $\{x\}$ reduced to a single point, and take its direct image sheaf of commutative groups on X (**0**, 3.4.1); it follows at once from the definitions that for every open U of X, $\Gamma(U, (i_x)_*(A(x))) = A(x)$ if $x \in U$, and is reduced to 0 in the contrary case, and for $y \in \overline{\{x\}}$, $((i_x)_*(A(x)))_y = A(x)$, and for $y \notin \overline{\{x\}}$, $((i_x)_*(A(x)))_y = 0$.

If now $\mathscr{F}$ is a sheaf of commutative groups on X and Y a part of X, for every open U of X, we have a canonical homomorphism

$$\Gamma(U, \mathscr{F}) \longrightarrow \prod_{x \in U \cap Y} \mathscr{F}_x = \prod_{x \in Y} \Gamma(U, (i_x)_*(\mathscr{F}_x)) \tag{21.9.1.1}$$

and since these homomorphisms commute with the restriction operators, they define a canonical homomorphism of *sheaves*

$$j_Y : \mathscr{F} \longrightarrow \prod_{x \in Y} (i_x)_*(\mathscr{F}_x). \tag{21.9.1.2}$$

Lemma (21.9.2). — *Let* X *be a locally Noetherian topological space,* X_0 *the set of its closed points,* $\mathscr{F}$ *a sheaf of commutative groups on* X*. The following conditions are equivalent:*

a) *The canonical homomorphism* $j_{X_0} : \mathscr{F} \to \prod_{x \in X_0} (i_x)_*(\mathscr{F}_x)$ *is injective and has for image* $\bigoplus_{x \in X_0} (i_x)_*(\mathscr{F}_x)$.
a′) *There exists a family of commutative groups* $(A(x))_{x \in X_0}$ *such that* $\mathscr{F}$ *is isomorphic to* $\bigoplus_{x \in X_0} (i_x)_*(A(x))$.
b) *Every section of* $\mathscr{F}$ *over an open* U *has a discrete support contained in* X_0.

When this is the case, for all $x \in X_0$*, the group* $A(x)$ *is determined to a unique isomorphism close to* $\mathscr{F}_x$*. Furthermore, the sheaf* $\mathscr{F}$ *is then flasque.*

Proof. As the points of X_0 are closed in X, we have $((i_x)_*(A(x)))_y = 0$ for $y \neq x \in X_0$; this remark **IV-4** | 285
proves the uniqueness assertion relative to the groups $A(x)$, and it is clear moreover that *a)* implies *a′)*. To see that *a′)* implies *b)*, we may restrict to the case where U is Noetherian; then (G, II, 3.10) we have

$$\Gamma(U, \bigoplus_{x \in X_0} (i_x)_*(A(x))) = \bigoplus_{x \in X_0} \Gamma(U, (i_x)_*(A(x))),$$

and thus every section s of $\mathscr{F}$ over U is a direct sum of a *finite* number of sections $s_k \in \Gamma(U, (i_{x_k})_*(A(x_k)))$ $(1 \leqslant k \leqslant n)$ with $x_k \in X_0$. Since x_k is closed in X, the support of s_k is reduced to $\{x_k\}$, so the support of s is contained in the finite set of the x_k, which is evidently discrete. Let us show that *b)* implies *a)*; for every open Noetherian U, every support of a section of $\mathscr{F}$ over U is *finite*. We deduce that for every open U which is discrete and quasi-compact, the image of the homomorphism (21.9.1.1) for $Y = X_0$ is contained in $\bigoplus_{x \in X_0} \Gamma(U, (i_x)_*(\mathscr{F}_x))$ and that this homomorphism is injective, which establishes the hypothesis *b)*.

Finally, to show that $\mathscr{F}$ is flasque, consider a section s of $\mathscr{F}$ over an open U of X; as the support of s is a discrete and closed subspace in X, we extend s to a section s' of $\mathscr{F}$ over X by setting $s'_x = 0$ in $X - U$. ☐

Remark (21.9.3). — (i) In the condition *b)* of (21.9.2), it does not suffice to assume that the support of every section of $\mathscr{F}$ over any open of X is *discrete*; this is shown by the example where we take for X the spectrum of a discrete valuation ring, with a closed point b and a generic point a, and for $\mathscr{F}$ the sheaf of commutative groups such that $\mathscr{F}_b = 0$ and $\mathscr{F}_a = \mathbf{Z}$.

(ii) The conditions of (21.9.2) being supposed satisfied, let E be a discrete part of X_0, and for each $x \in E$, let $a(x)$ be an element of $A(x)$. There then exists a unique section t of X such that $t_x = a(x)$ for all $x \in E$ and that the support of t is contained in E. Indeed, for every

Noetherian open U, $E \cap U$ is finite, and it suffices to take for t the section $\bigoplus_{x\in E}(i_x)_*(A(x))$ whose restriction to each Noetherian open U is the sum of the $a(x)$ for $x \in E \cap U$.

Proposition (21.9.4). — *Let* X *be a locally Noetherian prescheme,* $X^{(1)}$ *the set of points* $x \in X$ *such that* $\dim(\mathscr{O}_{X,x}) \leqslant 1$. *We then have a canonical isomorphism*

$$\mathscr{D}iv_X \xrightarrow{\sim} \bigoplus_{x\in X^{(1)}} (i_x)_*(\mathrm{Div}(\mathscr{O}_{X,x})) \tag{21.9.4.1}$$

and $\mathscr{D}iv_X$ *is flasque.*

Proof. Taking into account the isomorphism (21.4.6.1), the homomorphism (21.9.4.1) is defined by (21.9.1.1): let us prove that it is a bijection. As $\dim(X) \leqslant 1$, the points of $X^{(1)}$ are the *non-isolated closed points* of X. We are thus reduced to proving that: 1º $\mathscr{D}iv_X$ satisfies the condition *b)* of (21.9.2); 2º for every isolated point $x \in X$, we have $(\mathscr{D}iv_X)_x = 0$. The second point follows from the fact that $\mathscr{O}_{X,x}$ is an Artinian ring and that every regular element of $\mathscr{O}_{X,x}$ is therefore invertible. For the 1º, it **IV-4** | 286
suffices to note that for every open U of X and every divisor $D \in \mathrm{Div}(U)$, the maximal points of the support S of D are such that $\mathrm{prof}(\mathscr{O}_{X,x}) \leqslant 1$ (for S is equal to S, and S is therefore discrete, the set of irreducible components of S being locally finite. □

Corollary (21.9.5). — *Let* X *be a locally Noetherian prescheme of dimension* $\leqslant 1$. *For every discrete part* $E \subset X^{(1)}$ *and every family* $(D(x))_{x\in E}$ *with* $D(x) \in \mathrm{Div}(\mathscr{O}_{X,x})$, *there exists a unique divisor* D *on* X *such that the support of* D *is contained in* E *and that* $D_x = D(x)$ *for all* $x \in E$.

Proof. This follows from (21.9.4) and from (21.9.3), (ii). □

Corollary (21.9.6). — *Let* X *be a locally Noetherian prescheme of dimension* $\leqslant 1$. *Every divisor* D *on* X *is of the form* $D' - D''$, *where* D', D'' *are two divisors* $\geqslant 0$ *with supports contained in that of* D.

Proof. By virtue of (21.9.5), it is at once reduced to the case where $X = \mathrm{Spec}(A)$, where A is a Noetherian local ring; it then suffices to observe that M(X) is the total ring of fractions of A, and that a section of $\mathscr{M}_X^*$ over X is written by definition as the quotient b/a of two regular elements of A. □

Corollary (21.9.7). — *Let* X *be a locally Noetherian prescheme of dimension* $\leqslant 1$, *without isolated point, and* U *a dense open in* X. *There then exists a divisor* $D \geqslant 0$ *on* X, *with support contained in* U *and meeting each of the irreducible components of* X.

Proof. We can assume that U is the union of disjoint non-empty opens U_α, each of which is contained in a single irreducible component of X; it suffices then to take in each U_α a point x_α closed in X (there exists one since the unique generic point of U_α cannot be isolated by hypothesis), and to apply (21.9.5) to the discrete set of the x_α. □

The interest of this corollary is that it will allow us to prove that a separated algebraic curve X over a field k is quasi-projective, the divisor $D \geqslant 0$ defined in (21.9.7) being then necessarily *ample* by virtue of the Riemann–Roch theorem for curves (chap. V).

(21.9.8). For locally Noetherian preschemes X of dimension $\leqslant 1$ the proposition (21.9.4) brings back the determination of $\mathscr{D}iv_X$ to the case where $X = \mathrm{Spec}(A)$, A being a Noetherian local ring of dimension 1.

When A is a regular local ring of dimension 1 (in other words, a discrete valuation ring), the group Div(A) is canonically identified with **Z** (21.6.8); by (21.9.2):

Proposition (21.9.9). — *Let* X *be a locally Noetherian regular prescheme of dimension* $\leqslant 1$. *Then the sheaf* $\mathscr{D}iv_X$ *is canonically isomorphic to* $\bigoplus_{x\in X^{(1)}}(i_x)_*(\mathbf{Z}(x))$, *where* $\mathbf{Z}(x)$ *is the additive group* **Z** *considered as a sheaf of groups on the space* $\{x\}$.

(21.9.10). Suppose only that the Noetherian local ring A of dimension 1 is *reduced*; then, if A' is the *normalised* of A (the integral closure of A in its total ring of fractions), A' is Noetherian by virtue of the Krull–Akizuki theorem (Bourbaki, *Alg. comm.*, chap. VII, § 2, nº 5, prop. 5), and we have seen in (21.8.6) how Div(A) can be obtained as an *extension* of $\mathrm{Div}(A')$ and this last group is of the form **IV-4** | 287
$\mathbf{Z}'$.

Proposition (21.9.11). — *Let* X *be a Noetherian prescheme,* X_0 *a closed sub-prescheme of* X *having the following properties:*

1° $\dim(X_0) \leqslant 1$.

2° *For every locally closed part* Y *of* X *such that* $Y \cap X_0$ *is discrete, there exists a part* Y' *of* Y*, closed in* X *and open in* Y*, containing* $Y \cap X_0$*.*

Under these conditions:

(i) Let Z_0 *be the union of the sets* $\overline{\{x\}} \cap X_0$ *when* x *ranges over the part of* $\mathrm{Ass}(\mathscr{O}_X)$ *formed of points such that* $\overline{\{x\}} \cap X_0$ *is finite. Then, for every divisor* D_0 *on* X_0*, whose support does not meet* Z_0*, there exists a divisor* D *on* X *whose inverse image by the injection* $X_0 \to X$ *exists* (21.4) *and is equal to* D_0*; if moreover* $D_0 \geqslant 0$*, we can assume* $D \geqslant 0$*.*

(ii) Suppose in addition that there exists in X_0 *an affine open* U_0 *containing* $(\mathrm{Ass}(\mathscr{O}_{X_0})) \cup Z_0$*. Then the canonical homomorphism* (21.3.2.4) $\mathrm{Pic}(X) \to \mathrm{Pic}(X_0)$ *is surjective.*

Proof. (i) Taking into account (21.9.6), we can restrict to showing the assertion relative to divisors $D_0 \geqslant 0$; the support T of D_0 is a finite discrete and closed set in X_0, that is $T \neq \varnothing$, there is nothing to show. For all $x \in T$, there exists an element $s_x \in \mathscr{O}_{X_0,x}$ which is not a divisor of zero and whose image in $(\mathscr{D}iv_X)_x$ is $(D_0)_x$. There exists an open affine neighbourhood $U^{(x)}$ of x in X and a section $g^{(x)}$ of $\mathscr{O}_X$ over $U^{(x)}$ such that s_x is the image of the germ $(g^{(x)})_x \in \mathscr{O}_{X,x}$; taking $U^{(x)}$ small enough, we can arrange that $g^{(x)}$ is not a zero divisor in $\Gamma(U^{(x)}, \mathscr{O}_X)$, in other words (3.1.9), writing $V^{(x)}$ for the closed part of X defined by $g^{(x)} = 0$, that $V^{(x)} \cap \mathrm{Ass}(\mathscr{O}_X) = \varnothing$. Now, if $z \in \mathrm{Ass}(\mathscr{O}_X)$, the hypothesis $x \notin \overline{\{z\}}$, that is $x \notin \overline{\{z\}} \cap X_0$, either x is not isolated in $\overline{\{z\}} \cap X_0$. Now, if y is a generization of x such that $\overline{\{y\}}$ is of codimension 1 in X, the sub-prescheme reduced Y of X having $\overline{\{y\}}$ for underlying space, $X_0 \cap Y$ being by hypothesis a finite closed part of X and open in Y, containing $Y \cap X_0$, the condition (S_1) is then also satisfied.

We can define a divisor $D'^{(x)} = \mathrm{div}(g^{(x)})$ on $U^{(x)}$ by $D'^{(x)} = \mathrm{div}(g^{(x)})$, and $D'^{(x)}|(X - V^{(x)}) = 0$, which has a meaning since $U^{(x)} \cap (X - V^{(x)})$ is the complement of the zeros of $g^{(x)}$ in $U^{(x)}$; by the hypothesis $x \notin Z_0$ we have $x \notin \overline{\{z\}}$; as $V^{(x)} \cap X_0$ is reduced to $\{x\}$, $V^{(x)}$ is closed in X. On can then define a divisor $D = \sum_{x \in T} D'^{(x)}$, which has a meaning since T is finite.

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(ii) Taking into account the commutative diagram (21.4.2.1), the conclusion will follow if we prove that every $\mathscr{O}_{X_0}$-Module invertible $\mathscr{L}_0$ is isomorphic to an $\mathscr{O}_{X_0}$-Module of the form $\mathscr{O}_{X_0}(D_0)$ (21.2.8), where D_0 is a divisor on X_0 whose support does not meet Z_0. Now, as U_0 is schematically dense in X_0 (20.2.13), (iv), it suffices to prove that there exists a section $t \in \Gamma(U_0, \mathscr{L}_0)$ such that $t(z_0) \neq 0$ at the points of $(\mathrm{Ass}(\mathscr{O}_{X_0})) \cup Z_0$, the conclusion results from ((**II**, 5.1.4)) and, as U_0 is affine, $\mathscr{L}_0$ is ample (**II**, 4.5.4), the conclusion results from (**II**, 5.3.1) and (**II**, 5.5.3).

□

Corollary (21.9.12). — *Let* A *be a Henselian local ring* (18.5.8), $S = \mathrm{Spec}(A)$*,* s_0 *the closed point of* S*,* $f : X \to S$ *a separated morphism of finite presentation, and suppose that the set* $X_0 = f^{-1}(s_0)$ *is of dimension* $\leqslant 1$*. Then, for every closed sub-scheme* X'_0 *of* X*, having* X_0 *as underlying set and of finite presentation over* S*, the canonical homomorphism* (21.3.2.4) $\mathrm{Pic}(X) \to \mathrm{Pic}(X'_0)$ *is surjective.*

Proof. Let us first show that it suffices to prove the corollary when the local ring A is moreover Noetherian. Indeed (18.6.15) we can write $A = \varinjlim A_\lambda$, where the A_λ are local Noetherian and Henselian rings, the homomorphisms $A_\lambda \to A$ being local; there exists in addition an index α and a separated morphism of finite presentation $f_\alpha : X_\alpha \to S_\alpha = \mathrm{Spec}(A_\alpha)$ such that X and f are deduced from X_α and f_α by base change $S \to S_\alpha$ (8.10.5), (v); with the usual notation (8.8.1), the morphisms $f_\lambda : X_\lambda \to S_\lambda$ will be separated for $\lambda \geqslant \alpha$ and we will have $X = X_\lambda \times_{S_\lambda} S$ (8.6.3); since $\dim(X_0) \leqslant 1$ by hypothesis and (4.1.4), we thus have $\dim(X'_{0\lambda}) = \dim(X'_0) \leqslant 1$ by transitivity of the fibres and (4.1.4). It amounts to seeing that if we have proved that the homomorphism $\mathrm{Pic}(X_\lambda) \to \mathrm{Pic}(X'_{0\lambda})$ is surjective for all $\lambda \geqslant \alpha$, we have the same conclusion for X and X'_0. Moreover, we can assume that if $s_{0\alpha}$ is the unique closed point of S_α, there is a closed sub-scheme $X'_{0\alpha}$ of X_α, such that $X'_0 = X'_{0\alpha} \times_{S_\alpha} S$

(8.6.3); since $\dim(X'_0) \leqslant 1$ by hypothesis and $\dim(X'_{0\lambda}) = \dim(X'_0) \leqslant 1$ by transitivity of the fibres and (4.1.4). We evidently have the commutative diagram

$$\begin{tikzcd} \mathrm{Pic}(X_\lambda) \arrow[r] \arrow[d] & \mathrm{Pic}(X) \arrow[d] \\ \mathrm{Pic}(X'_{0\lambda}) \arrow[r] & \mathrm{Pic}(X'_0) \end{tikzcd}$$

For every invertible $\mathscr{O}_{X'_0}$-Module $\mathscr{L}'_0$, there exists $\lambda \geqslant \alpha$ and an invertible $\mathscr{O}_{X'_{0\lambda}}$-Module $\mathscr{L}'_{0\lambda}$ such that $\mathscr{L}'_0$ is deduced by base change $S \to S_\lambda$ (8.5.2) and (8.5.5); it suffices then to take the $\mathscr{O}_X$-Module invertible $\mathscr{L}$ deduced from $\mathscr{L}_\lambda$ by base change $S \to S_\lambda$, and by virtue of the commutativity of the preceding diagram, $\mathrm{cl}(\mathscr{L}'_0)$ will be the image of $\mathrm{cl}(\mathscr{L})$.

Suppose therefore that A is Noetherian, and verify that X and X_0 satisfy the conditions of (21.9.11), (ii). Since by hypothesis $\dim(X'_0) \leqslant 1$; on the other hand, for verifying the condition 2° of (21.9.11), consider a sub-prescheme Y of X having Y as underlying set, the morphism $g : Y \to S$, restriction of f, being quasi-finite at each of the points x_i of $Y \cap X_0$ $(1 \leqslant i \leqslant n)$, we can apply (18.5.11), *c)*) and we see that Y is the *sum* of sub-preschemes *open* $Y_i = \mathrm{Spec}(\mathscr{O}_{Y,x_i})$ $(1 \leqslant i \leqslant n)$ and of a prescheme Y'' such that $Y'' \cap X_0 = \varnothing$, and in addition that the canonical injections $j_i : Y_i \to X$ are such that $f \circ j_i$ is a finite morphism, thus Y_i is *closed* in X; on the question by taking $Y' = \bigcup_{1 \leqslant i \leqslant n} Y_i$.

It remains to verify the supplementary hypothesis of (21.9.11), (ii). Now, X'_0 being a separated curve over the field $\boldsymbol{k}(s_0)$, is a $\boldsymbol{k}(s_0)$-scheme quasi-projective (chap. V) ([1]), thus there exists an $\mathscr{O}_{X'_0}$-Module (**II**, 5.3.1), and this achieves the proof. □

Remark (21.9.13). — Under the conditions of (21.9.12), supposing in addition f proper, the morphism f is even *projective*: indeed, if $\mathscr{L}_0$ is an $\mathscr{O}_{X'_0}$-Module ample, it exists, by virtue of (21.9.12) an $\mathscr{O}_X$-Module invertible $\mathscr{L}$ whose inverse image in X'_0 is isomorphic to $\mathscr{L}_0$, hence is ample and, as every neighbourhood of s_0 in S is necessarily the whole of S, we deduce then from (9.6.4) that $\mathscr{L}$ is an $\mathscr{O}_X$-Module ample, whence the conclusion ((**II**, 5.3.1) and (**II**, 5.5.3)).

21.10. Inverse images and direct images of 1-codimensional cycles

(21.10.1). Let X, X′ be two locally Noetherian preschemes, $f : X' \to X$ a morphism, T a closed rare subset of X contained in $f(X')$, and let Z be a 1-codimensional cycle on X with support contained in T. We propose to define, under suitable hypotheses on f, a 1-codimensional cycle $f^*(Z)$ on X′, the *inverse image* of Z by f. **IV-4** | 289

Let us first suppose that $Z \mapsto f^*(Z)$ is a *homomorphism* of ordered groups from the subgroup of $\mathfrak{Z}^1(X)$ formed by cycles with support contained in T, into the ordered group $\mathfrak{Z}^1(X')$. For this, let

$$Z = \sum_{x \in T \cap X^{(1)}} n_x . \overline{\{x\}} \tag{21.10.1.1}$$

where the family of the $x \in T \cap X^{(1)}$ such that $n_x \neq 0$ is locally finite. For all $x' \in X'^{(1)}$, we define an integer $n_{x'}$ in the following way, setting $x = f(x')$:

1° if $x \notin T$, we set $n_{x'} = 0$;

2° if $x \in X^{(1)}$ and if $\mathscr{O}_{X',x'}$ is a flat $\mathscr{O}_{X,x}$-module, we know (6.1.1) that $\dim(\mathscr{O}_{X',x'}/\mathfrak{m}_x\mathscr{O}_{X',x'}) = 0$, in other words $\mathscr{O}_{X',x'}/\mathfrak{m}_x\mathscr{O}_{X',x'}$ is an $\mathscr{O}_{X',x'}$-module of finite length $\lambda_{x'}$, and we set $n_{x'} = \lambda_{x'} n_x$;

3° if $\mathscr{O}_{X,x}$ is factorial and $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{X',x'})$, we know (21.6.9) that the canonical homomorphism $\mathrm{cyc} : \mathrm{Div}(\mathscr{O}_{X,x}) \to \mathfrak{Z}^1(\mathrm{Spec}(\mathscr{O}_{X,x}))$ is bijective, and on the other hand since $\dim(\mathscr{O}_{X',x'}) = 1$ and $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{X',x'})$, $\mathrm{Ass}(\mathscr{O}_{X',x'})$ consists solely of the maximal points of $\mathrm{Spec}(\mathscr{O}_{X',x'})$, so the hypothesis on f implies that $f(\mathrm{Ass}(\mathscr{O}_{X',x'})) \subset \mathrm{Ass}(\mathscr{O}_{X,x})$ and it follows from (21.4.5), (ii) that the homomorphism $f^* : \mathrm{Div}(\mathscr{O}_{X,x}) \to \mathrm{Div}(\mathscr{O}_{X',x'})$ is defined; finally, x' being the unique closed point of $\mathrm{Spec}(\mathscr{O}_{X',x'})$, $\mathfrak{Z}^1(\mathrm{Spec}(\mathscr{O}_{X',x'}))$ is canonically isomorphic to $\mathbf{Z}$.

We have thus defined a canonical composite homomorphism

$$(21.10.1.2)\qquad \mathfrak{Z}^1(\mathrm{Spec}(\mathscr{O}_{X,x})) \xleftarrow{\mathrm{cyc}^{-1}} \mathrm{Div}(\mathscr{O}_{X,x}) \xrightarrow{f'} \mathrm{Div}(\mathscr{O}_{X',x'}) \xrightarrow{\mathrm{cyc}^0} \mathfrak{Z}^1(\mathrm{Spec}(\mathscr{O}_{X',x'})) \xrightarrow{\sim} \mathbf{Z}$$

If Z_x is the cycle $\sum_{y\in\mathrm{Spec}(\mathscr{O}_{X,x})\cap X^{(1)}} n_y.(\overline{\{y\}}\cap\mathrm{Spec}(\mathscr{O}_{X,x}))$ on $\mathrm{Spec}(\mathscr{O}_{X,x})$, we take for $n_{x'}$ the *image* of Z_x by the homomorphism (21.10.1.2).

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(21.10.2). We propose to show that:

A) When two of the conditions 1°, 2°, 3° of (21.10.1) are *simultaneously* satisfied, the corresponding values of $n_{x'}$ *coincide*.

B) The set of $x'\in X'^{(1)}$ such that $n_{x'}\neq 0$ is *locally finite* in X'.

To prove A), let us first suppose that $x\notin T$ and that x satisfies one of the conditions 2° or 3° of (21.10.1); then $n_x=0$ and if one is in case 2°, we have $n_{x'}=0$; if one is in case 3°, we have $Z_x=0$ since $\mathrm{Supp}(Z)\subset T$, so again $n_{x'}=0$. It remains to consider the case where one is simultaneously in cases 2° and 3°; then, since $\dim(\mathscr{O}_{X,x})=1$, $\mathscr{O}_{X,x}$ is a discrete valuation ring; if t is a uniformiser of this ring, the divisor corresponding to Z_x is $\mathrm{div}(t^{n_x})$ in $\mathrm{Div}(\mathscr{O}_{X,x})$, and its image in $\mathrm{Div}(\mathscr{O}_{X',x'})$ is the divisor of t'^{n_x}, where t' is the image of t. One can evidently restrict to the case where $n_x=1$, and then by the definition (21.6.5.1) the image of Z_x by (21.10.1.1) is the number $\lambda_{x'}$, which finishes the proof of A).

Let us now prove B). Set $T_0=\mathrm{Supp}(Z)\subset T$, $T'_0=f^{-1}(T_0)$; it suffices to prove that the relation $n_{x'}\neq 0$ implies that x' belongs to the set of maximal points of T'_0, this last being locally finite in X'. It is immediate that if x' was not maximal in the closed set T'_0, there would exist a generisation y' of x' in T'_0, distinct from x', and since $x'\in X'^{(1)}$, y' would necessarily be a maximal point of X'; as a result $y=f(y')$ would be a maximal point of X by hypothesis; but this is absurd since $y\in T_0$ and T_0 is purely of codimension 1 in X. **IV-4 | 291**

(21.10.3). We can now set

$$f^*(Z)=\sum_{x'\in X'^{(1)}} n_{x'}\overline{\{x'\}},$$

the sum of the second member having a meaning by virtue of what we proved in (21.10.2); we say that the 1-codimensional cycle $f^*(Z)$ is the *inverse image* of Z by f. It is immediate that the map f^* thus defined is a homomorphism of ordered groups. Furthermore, if U is an open of X, V an open of X' such that $f(V)\subset U$ and $f':V\to U$ the restriction of f, it follows immediately from the definitions that one has

$$(21.10.3.1)\qquad f'^*(Z|U)=f^*(Z)|V.$$

Let us denote by $\Gamma_T(\mathscr{Z}^1_X)$ the largest sub-sheaf of commutative groups of $\mathscr{Z}^1_X$ with support contained in T; it follows from the relation (21.10.3.1) that the maps $\Gamma(U,\Gamma_T(\mathscr{Z}^1_X))\to\Gamma(V,\mathscr{Z}^1_{X'})$ that we have just defined define a homomorphism of sheaves of ordered commutative groups on X'

$$\psi^*(\Gamma_T(\mathscr{Z}^1_X))\longrightarrow\mathscr{Z}^1_{X'}$$

where ψ is the continuous map underlying the morphism f.

Proposition (21.10.4). — *Suppose satisfied the conditions of* (21.10.1). *Then, for every divisor* D *on* X *such that* $\mathrm{Supp}(D)\subset T$ *and such that* $f^*(D)$ *is defined* (21.4.2), *we have*

$$(21.10.4.1)\qquad \mathrm{cyc}(f^*(D))=f^*(\mathrm{cyc}(D)).$$

Proof. The question being local on X, we can restrict to the case where $X=\mathrm{Spec}(A)$ is affine, where $D=\mathrm{div}(t)$, where t is a regular non-invertible element of A, and where the sub-prescheme $Y(D)$ (21.2.12) has only a single maximal point y, so that $\mathrm{cyc}(D)=n_y.\overline{\{y\}}$, where n_y is the length of $\mathscr{O}_{X,y}/t_y\mathscr{O}_{X,y}$ (21.6.5.1). We saw in the proof of (21.10.2), B) that the points $x'\in X'$ such that $\mathrm{mult}_{x'}(f^*(\mathrm{cyc}(D)))\neq 0$ are maximal points of $f^{-1}(\overline{\{y\}})$. If the point x' is in case 3° of (21.10.1), the equality of the multiplicities in x' of the two members of (21.10.4.1) follows from the definition of $n_{x'}$ by means of the homomorphism (21.10.1.1). Suppose on the contrary that x' is in case 2° of (21.10.1), and let $x=f(x')$; since $x\in X^{(1)}\cap\overline{\{y\}}$, we have necessarily $x=y$. Note now that $f^*(D)=\mathrm{div}(t')$, where t' is the image of t in $\Gamma(X', \mathscr{O}_{X'})$, and $Y(f^*(D)) = f^{-1}(Y(D))$; the multiplicity of $f^*(D)$ at the point x' is therefore the length $n_{x'}$ of the $\mathscr{O}_{X',x'}$-module $\mathscr{O}_{X',x'}/t'_{x'}\mathscr{O}_{X',x'} = (\mathscr{O}_{X,y}/t_y\mathscr{O}_{X,y}) \otimes_{\mathscr{O}_{X,y}} \mathscr{O}_{X',x'}$; it follows from (4.7.1) that $n_{x'} = \lambda_{x'}n_y$, so the multiplicities at the point x' of the two members of (21.10.4.1) are again equal, which finishes the proof. □

(21.10.5). Suppose now that $f : X' \to X$ is a morphism of locally Noetherian preschemes, transforming every maximal point of X' into a maximal point of X, and suppose furthermore that for every *closed rare* subset T of X, f satisfies the conditions of (21.10.1); this means that for all $x' \in X'^{(1)}$, either $x = f(x')$ is a maximal point of X, or x' satisfies one of the conditions (ii), (iii) of (21.10.1). Noting that every 1-codimensional cycle with rare support in X is defined for *every* 1-codimensional cycle Z on X; by virtue of (21.10.3.1), we have thus defined a homomorphism of sheaves of ordered commutative groups **IV-4 | 292**

$$\psi^*(\mathscr{Z}^1_X) \longrightarrow \mathscr{Z}^1_{X'}.$$

If furthermore, for every divisor D on X, $f^*(D)$ is defined (21.4.5), the fact that the support of D is rare in X (21.6.6) implies that (21.10.4) is applicable, and we therefore have the formula (21.10.4.1) for every divisor D on X. In particular:

Proposition (21.10.6). — *Let X, X' be two locally Noetherian preschemes, $f : X' \to X$ a flat morphism. Then $f^*(Z)$ is defined for every 1-codimensional cycle Z on X, $f^*(D)$ is defined for every divisor D on X and we have the relation (21.10.4.1).*

Proof. Indeed, if $x' \in X'^{(1)}$ is such that $x = f(x')$ is not maximal, it follows from (6.1.1) that we have necessarily $x \in X^{(1)}$, so one is in case (ii) of (21.10.1). We can therefore apply (21.10.5), taking into account (21.4.5) and (2.3.4). □

Remark (21.10.7). — The existence of $f^*(Z)$ for every 1-codimensional cycle Z on X already follows from the hypothesis that f is *flat at all points x' of X' of codimension* $\leqslant 1$ (i.e. such that $\dim \mathscr{O}_{X',x'} \leqslant 1$); indeed, for every maximal $z' \in X'$, it follows from (6.1.1) that $z = f(z')$ is a maximal point of X since $\dim \mathscr{O}_{X,z} \leqslant \dim \mathscr{O}_{X',z'} = 0$. Moreover, if $x' \in X'^{(1)}$, $x = f(x')$ is maximal or belongs to $X^{(1)}$ by (6.1.1); one can again apply (21.10.5).

Proposition (21.10.8). — *Let X, X', X'' be three locally Noetherian preschemes, $f : X' \to X$, $g : X'' \to X'$ two morphisms; suppose that f (resp. g) is flat at every point of codimension $\leqslant 1$ in X' (resp. X''). Then $f \circ g$ is flat at every point of codimension $\leqslant 1$ in X'' and for every 1-codimensional cycle Z on X, we have $(f \circ g)^*(Z) = g^*(f^*(Z))$.*

Proof. The first assertion follows from (6.1.1) and (2.1.6). The second follows from the fact that f (resp. g, resp. $f \circ g$) transforms maximal points of X (resp. X', resp. X'') into maximal points of X (resp. X, resp. X) and from the fact that, if $x'' = g(x'') \in X''^{(1)}$ and $x = f(x') \in X^{(1)}$ is such that $x' = g(x'') \in X'^{(1)}$, we have

$$\operatorname{length}(\mathscr{O}_{X'',x''}/\mathfrak{m}_x\mathscr{O}_{X'',x''}) = \operatorname{length}(\mathscr{O}_{X'',x''}/\mathfrak{m}_{x'}\mathscr{O}_{X'',x''}) \cdot \operatorname{length}(\mathscr{O}_{X',x'}/\mathfrak{m}_x\mathscr{O}_{X',x'})$$

Indeed, if we set $A = \mathscr{O}_{X,x}$, $A' = \mathscr{O}_{X',x'}$, $A'' = \mathscr{O}_{X'',x''}$, $\mathfrak{m} = \mathfrak{m}_x$, $\mathfrak{m}' = \mathfrak{m}_{x'}$, we have $A''/\mathfrak{m}A'' = (A'/\mathfrak{m}A') \otimes_{A'} A''$, and the formula

$$\operatorname{length}_{A''}(A''/\mathfrak{m}A'') = \operatorname{length}_{A'}(A'/\mathfrak{m}A') \cdot \operatorname{length}_{A''}(A''/\mathfrak{m}'A'')$$

follows from (4.7.1) and the hypothesis of flatness of A'' over A'. □

(21.10.9). Let X be a locally Noetherian prescheme, Λ a commutative group, written additively, and considered as a **Z**-module; we will again denote by **Z** and Λ the simple sheaves on X associated to the constant presheaves respectively equal to **Z** and Λ (**0**, 3.6). We call *sheaf of germs of 1-codimensional cycles with coefficients in* Λ the sheaf of commutative groups $\mathscr{Z}^1_X \otimes_{\mathbf{Z}} \Lambda$. If $\Lambda = \mathbf{Q}$, we say that the sections of $\mathscr{Z}^1_X \otimes_{\mathbf{Z}} \mathbf{Q}$ over X are the 1-*codimensional cycles with rational coefficients*. Since the fibres of $\mathscr{Z}^1_X$ are torsion-free **Z**-modules (21.6.3), the canonical homomorphism $\mathscr{Z}^1_X \to \mathscr{Z}^1_X \otimes_{\mathbf{Z}} \mathbf{Q}$ is *injective*, so that the 1-codimensional cycles are identified with 1-codimensional cycles with rational coefficients. **IV-4 | 293**

(21.10.10). We will now see that under certain conditions, we can enlarge the definition of $f^*(Z)$ given in (21.10.3) for a 1-codimensional cycle Z on X, but to take for $f^*(Z)$ a 1-codimensional cycle *with rational coefficients* on X′. The most general case where we place ourselves is the one where f transforms every maximal point of X′ into a maximal point of X, and where, at every point $x' \in \mathrm{X}'^{(1)}$, we have one of the conditions (i), (ii), (iii) of (21.10.1) *or a fourth condition* (setting $x = f(x')$):

(iv) $x \in \mathrm{X}^{(1)}$, $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{\mathrm{X},x})$ *and furthermore, if we set* $\mathrm{A} = \widehat{\mathscr{O}}_{\mathrm{X},x}$, $\mathrm{A}' = \widehat{\mathscr{O}}_{\mathrm{X}',x'}$, *and if* $\mathrm{K} = \mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K}$ *denotes the total ring of fractions of* A, *then* A′ *is a finite* A*-algebra and* $\mathrm{K}' = \mathrm{A}' \otimes_{\mathrm{A}} \mathrm{K}$ *is a free* K*-module.*

Let us note then $r_{x'}$ the rank of the free K-module K′, $q_{x'}$ the degree of $\boldsymbol{k}(x')$ over $\boldsymbol{k}(x) = \boldsymbol{k}(\mathrm{A})$, and set $\mu_{x'} = r_{x'}/q_{x'}$. For a 1-codimensional cycle with support contained in T, given by (21.10.1.1) and for all $x' \in \mathrm{X}'^{(1)}$, we define $c_{x'} \in \mathbf{Q}$ to equal $n_{x'}$ when one is in one of the cases 1°, 2°, 3° of (21.10.1); but there remains here a fourth possibility:

4° if x' satisfies condition (iv) above, we set $c_{x'} = \mu_{x'} n_x \in \mathbf{Q}$.

(21.10.11). We must still prove that when condition 4° of (21.10.10) is satisfied *at the same time* as one of the conditions 1°, 2°, 3° of (21.10.1), we have $n_{x'} = c_{x'}$. This is evident if $x \notin \mathrm{T}$ since then $n_x = 0$. **IV-4** | 294
To study the other two cases, let us note that $\mathfrak{m}_{x'}\mathscr{O}_{\mathrm{X}',x'}$ is closed for the $\mathfrak{m}_{x'}$-preadic topology, so the completion of $\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x\mathscr{O}_{\mathrm{X}',x'}$ for this last topology is $\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x\mathscr{O}_{\mathrm{X}',x'}$. If one is simultaneously in cases 2° and 4°, $\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x\mathscr{O}_{\mathrm{X}',x'}$ is discrete for the $\mathfrak{m}_{x'}$-preadic topology, so isomorphic to $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$. Since A′ is a finite A-algebra and flat ($\mathbf{0}_{\mathrm{III}}$, 10.2.3), it is a free A-module (*ibid.*, 10.1.3), and the rank of $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$ over $\mathrm{A}/\mathfrak{m}_x\mathrm{A} = \boldsymbol{k}(x)$ is equal to the rank of A′ over A, hence also to the rank of K′ over K on A, hence also $\lambda_{x'}$ of the $\mathscr{O}_{\mathrm{X}',x'}$-module $\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x\mathscr{O}_{\mathrm{X}',x'}$ (or of $\boldsymbol{k}(x')$-module) by the rank $[\boldsymbol{k}(x') : \boldsymbol{k}(x)] = q_{x'}$, which proves the relation $\mu_{x'} = r_{x'}/q_{x'} = \lambda_{x'}$ in this case.

Suppose finally that one is simultaneously in cases 3° and 4°. Then, since $\dim(\mathscr{O}_{\mathrm{X},x}) = 1$, $\mathscr{O}_{\mathrm{X},x}$ is a discrete valuation ring, hence regular. On the other hand, $\mathscr{O}_{\mathrm{X}',x'}$ is of dimension 1, and since $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{\mathrm{X}',x'})$, $\mathscr{O}_{\mathrm{X}',x'}$ is a Cohen-Macaulay ring ($\mathbf{0}$, 16.4.6); finally, since A′ is an A-module of finite type, $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$ is a $\boldsymbol{k}(x)$-vector space of finite rank; since $\mathscr{O}_{\mathrm{X}',x'}/\mathfrak{m}_x\mathscr{O}_{\mathrm{X}',x'}$ is contained in $\mathrm{A}'/\mathfrak{m}_x\mathrm{A}'$, it is also a $\boldsymbol{k}(x)$-vector space of finite rank, hence an Artinian ring. The application of (6.1.5) then shows that $\mathscr{O}_{\mathrm{X}',x'}$ is a flat $\mathscr{O}_{\mathrm{X},x}$-module, so one is also in case 2°, and we conclude by what was seen above.

This being the case, in the case we have considered, we will set

$$f^*(\mathrm{Z}) = \sum_{x' \in \mathrm{X}'^{(1)}} c_{x'}.\overline{\{x'\}}$$

which is thus a 1-codimensional cycle on X′ with *rational* coefficients. We have thus again defined a homomorphism f^* of ordered groups, satisfying (21.10.3.1), and hence a homomorphism of sheaves of commutative ordered groups

$$\psi^*(\mathscr{Z}^1_{\mathrm{X}}) \longrightarrow \mathscr{Z}^1_{\mathrm{X}'} \otimes_{\mathbf{Z}} \mathbf{Q}$$

whence, by tensorisation, a homomorphism of sheaves of $\mathbf{Q}$-vector spaces ordered

$$\psi^*(\mathscr{Z}^1_{\mathrm{X}} \otimes_{\mathbf{Z}} \mathbf{Q}) \longrightarrow \mathscr{Z}^1_{\mathrm{X}'} \otimes_{\mathbf{Z}} \mathbf{Q}. \tag{21.10.11.1}$$

When f satisfies the preceding conditions for every *closed rare* subset T in X, i.e. that for all $x' \in \mathrm{X}'^{(1)}$, either $x = f(x')$ is a maximal point of X, or x' satisfies one of the conditions (ii), (iii) or (iv), $f^*(\mathrm{Z})$ is then defined for *every* 1-codimensional cycle Z on X, and we have defined a homomorphism of sheaves of commutative ordered groups

$$\psi^*(\mathscr{Z}^1_{\mathrm{X}}) \longrightarrow \mathscr{Z}^1_{\mathrm{X}'} \otimes_{\mathbf{Z}} \mathbf{Q}$$

Remark (21.10.12). — When one is in the situation of (21.10.10), one can effectively, for 1-codimensional cycles Z on X, have for $f^*(Z)$ cycles 1-codimensional with *non-integer* coefficients; in other words, the numbers $\mu_{x'}$ can be non-integers. We have an example by taking the integral ring and its integral closure A′ of ($\mathbf{0}$, 6.15.11, (ii)) and (21.10.10), (iv) of Spec(A′) and one has $\mu_{x'} = \frac{1}{2}$.

Lemma (21.10.13). — *Let* A *be a local Noetherian ring of dimension* 1*, t a regular element of* A *belonging to the maximal ideal* $\mathfrak{m}$ *(which implies that* $\mathfrak{m} \notin \mathrm{Ass}(\mathrm{A})$*).*

(i) For every A-module M *of finite type, the kernel* $\mathrm{N}_t(\mathrm{M})$ *and the cokernel* $\mathrm{P}_t(\mathrm{M})$ *of the homothety* $t_{\mathrm{M}} : \mathrm{M} \to \mathrm{M}$ *of ratio t are of finite length. We set* $d_t(\mathrm{M}) = \mathrm{length}\,\mathrm{P}_t(\mathrm{M}) - \mathrm{length}\,\mathrm{N}_t(\mathrm{M})$.

(ii) If $0 \to \mathrm{M}' \to \mathrm{M} \to \mathrm{M}'' \to 0$ *is an exact sequence of A-modules of finite type, we have* $d_t(\mathrm{M}) = d_t(\mathrm{M}') + d_t(\mathrm{M}'')$.

(iii) We have $d_t(\mathrm{M}) \geqslant 0$ *for every A-module* M *of finite type; for* $d_t(\mathrm{M}) = 0$*, it is necessary and sufficient that* M *be of finite length.*

(iv) If K *is the total ring of fractions of* A *and if* $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{K}$ *is a free* K*-module of rank n, we have* $d_t(\mathrm{M}) = n \cdot d_t(\mathrm{A}) = n \cdot \mathrm{length}(\mathrm{A}/t\mathrm{A})$.

(v) If M *satisfies the hypotheses of (iv) and if furthermore t is* M*-regular, we have* $\mathrm{length}(\mathrm{M}/t\mathrm{M}) = n \cdot \mathrm{length}(\mathrm{A}/t\mathrm{A})$.

Proof. (i) $\mathrm{Spec}(\mathrm{A})$ is formed of the point $\mathfrak{m}$ and the minimal prime ideals $\mathfrak{p}_i$; since by hypothesis $t \notin \mathfrak{p}_i$ for all i (Bourbaki, *Alg. comm.*, chap. IV, §1, n° 1, cor. 3 of prop. 2), the image of t in each of the $\mathrm{A}_{\mathfrak{p}_i}$ is invertible, and the supports of the A-modules of finite type $\mathrm{N}_t(\mathrm{M})$ and $\mathrm{P}_t(\mathrm{M})$ are therefore empty or reduced to $\mathfrak{m}$; we conclude from this (**0**, 16.1.10) that these modules are of finite length. **IV-4** | 295

(ii) Since t is regular, we have an exact sequence

$$0 \longrightarrow \mathrm{A} \xrightarrow{t_{\mathrm{A}}} \mathrm{A} \longrightarrow \mathrm{A}/t\mathrm{A} \longrightarrow 0$$

and since $\mathrm{Tor}_i^{\mathrm{A}}(\mathrm{M}, \mathrm{A}) = 0$ for $i \geqslant 1$, the Tor exact sequence gives on one hand the exact sequence

$$0 \longrightarrow \mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/t\mathrm{A}) \longrightarrow \mathrm{M} \xrightarrow{t_{\mathrm{M}}} \mathrm{M} \longrightarrow \mathrm{M}/t\mathrm{M} \longrightarrow 0$$

and on the other hand, for $i \geqslant 2$,

$$0 = \mathrm{Tor}_i^{\mathrm{A}}(\mathrm{M}, \mathrm{A}) \longrightarrow \mathrm{Tor}_i^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/t\mathrm{A}) \longrightarrow \mathrm{Tor}_{i-1}^{\mathrm{A}}(\mathrm{M}, \mathrm{A}) = 0$$

hence $\mathrm{N}_t(\mathrm{M}) = \mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/t\mathrm{A})$ and $\mathrm{Tor}_i^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/t\mathrm{A}) = 0$ for $i \geqslant 2$; the Tor exact sequence gives the exact sequence

$$0 \longrightarrow \mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}', \mathrm{A}/t\mathrm{A}) \longrightarrow \mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}, \mathrm{A}/t\mathrm{A}) \longrightarrow \mathrm{Tor}_1^{\mathrm{A}}(\mathrm{M}'', \mathrm{A}/t\mathrm{A}) \longrightarrow$$
$$\mathrm{M}' \otimes_{\mathrm{A}} (\mathrm{A}/t\mathrm{A}) \longrightarrow \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/t\mathrm{A}) \longrightarrow \mathrm{M}'' \otimes_{\mathrm{A}} (\mathrm{A}/t\mathrm{A}) \longrightarrow 0$$

and this sequence can be written, in view of what precedes

$$0 \longrightarrow \mathrm{N}_t(\mathrm{M}') \longrightarrow \mathrm{N}_t(\mathrm{M}) \longrightarrow \mathrm{N}_t(\mathrm{M}'') \longrightarrow \mathrm{P}_t(\mathrm{M}') \longrightarrow \mathrm{P}_t(\mathrm{M}) \longrightarrow \mathrm{P}_t(\mathrm{M}'') \longrightarrow 0$$

which proves (ii).

(iii) To prove (iii), we note that for M there is a composition series (M_h) of M such that the quotients $\mathrm{M}_h/\mathrm{M}_{h+1}$ are isomorphic to $\mathrm{A}/\mathfrak{m}$ or to one of the $\mathrm{A}/\mathfrak{p}_i$ (Bourbaki, *Alg. comm.*, chap. IV, §1, n° 4, th. 1), and that for M to be of finite length, it is necessary and sufficient that all these quotients be isomorphic to $\mathrm{A}/\mathfrak{m}$. It all comes down therefore (by virtue of (ii)) to proving that $d_t(\mathrm{A}/\mathfrak{m}) = 0$ and $d_t(\mathrm{A}/\mathfrak{p}_i) > 0$. Now, the image of t in $\mathrm{A}/\mathfrak{m}$ being 0, we have $\mathrm{N}_t(\mathrm{A}/\mathfrak{m}) = \mathrm{A}/\mathfrak{m}$ and $\mathrm{P}_t(\mathrm{A}/\mathfrak{m}) = \mathrm{A}/\mathfrak{m}$, whence the first assertion; on the other hand, the image of t in $\mathrm{A}/\mathfrak{p}_i$ is regular, so $\mathrm{N}_t(\mathrm{A}/\mathfrak{p}_i) = 0$ and $\mathrm{P}_t(\mathrm{A}/\mathfrak{p}_i) = \mathrm{A}/(t\mathrm{A} + \mathfrak{p}_i)$ which is not reduced to 0, whence the second assertion.

(iv) There is a basis of $\mathrm{M} \otimes_{\mathrm{A}} \mathrm{K}$ of the form $(x_j/s)_{1 \leqslant j \leqslant n}$, where s is a regular element of A and $x_j \in \mathrm{M}$. Consider the homomorphism $u : \mathrm{A}^n \to \mathrm{M}$ which transforms the element e_j $(1 \leqslant j \leqslant n)$ of the canonical basis of A^n into x_j and let us show that $\mathrm{Ker}(u)$ and $\mathrm{Coker}(u)$ are of *finite length*; indeed, for each i, the image of s in $\mathrm{A}_{\mathfrak{p}_i}$ is invertible, and since $\mathrm{K}_{\mathfrak{p}_i} = \mathrm{A}_{\mathfrak{p}_i}$, the images of the x_j/s in $\mathrm{M}_{\mathfrak{p}_i}$ form a basis of this $\mathrm{A}_{\mathfrak{p}_i}$-module; hence $u_{\mathfrak{p}_i} : \mathrm{A}_{\mathfrak{p}_i}^n \to \mathrm{M}_{\mathfrak{p}_i}$ is bijective. As in (i), we conclude that the supports of $\mathrm{Ker}(u)$ and of $\mathrm{Coker}(u)$ are contained in $\{\mathfrak{m}\}$, so these modules are of finite length. This being so, it follows from (ii) and (iii) that $d_t(\mathrm{M}) = d_t(\mathrm{A}^n) = n \cdot d_t(\mathrm{A}) = n \cdot \mathrm{length}(\mathrm{A}/t\mathrm{A})$.

Finally, it is clear that (v) follows immediately from (iv), since then $\mathrm{N}_t(\mathrm{M}) = 0$. □

This lemma allows us to generalise (21.10.4):

Proposition (21.10.13*). — *Suppose that f satisfies the conditions of* (21.10.10)*. Then, for every divisor* D *on* X *such that* $\mathrm{Supp}(\mathrm{D}) \subset \mathrm{T}$ *and such that* $f^*(\mathrm{D})$ *is defined, we have*

(21.10.13.1) $$\mathrm{cyc}(f^*(\mathrm{D})) = f^*(\mathrm{cyc}(\mathrm{D})).$$

Proof. Reasoning as in (21.10.4), it all amounts to seeing (with the same notation) that if x' is in case 4° of (21.10.10) and if n_y is the length of $\mathscr{O}_{X,y}/t_y\mathscr{O}_{X,y}$, $\mu_{x'}$ being the rational number defined in (21.10.10), then the length $n_{x'}$ of $\mathscr{O}_{X',x'}/t'_{x'}\mathscr{O}_{X',x'}$ is equal to $\mu_{x'}n_y$. Since $\mathscr{O}_{X',x'}/t'_{x'}\mathscr{O}_{X',x'}$ is of finite length, it has the same length as its $\mathfrak{m}_{x'}$-preadic completion $A'/t'_{x'}A'$, which can also be written A'/t_yA'; moreover, since $t'_{x'}$ is regular by hypothesis in $\mathscr{O}_{X',x'}$, it is also regular in A' by flatness ($\mathbf{0}_{\mathrm{I}}$, 6.3.4), and when A' is considered as an A-module, we can also say that t_y is A'-regular. Since A is of dimension 1 and A' is an A-module of finite type such that $A' \otimes_A K$ is a free K-module of rank $r_{x'}$, we can apply (21.10.13), (v) to $M = A'$ and to t_y, and the length of A'/t_yA' as an A'-module is therefore $r_{x'}n_y$. Since $\boldsymbol{k}(x')$ is a $\boldsymbol{k}(x)$-vector space of rank $q_{x'}$, the length of A'/t_yA' as an A'-*module* is therefore $r_{x'}n_y/q_{x'} = \mu_{x'}n_y$, which finishes the proof. ☐ IV-4 | 296

(21.10.14). Let now X, X′ be two locally Noetherian preschemes, $f : X' \to X$ a morphism having the *two* following properties:

a) f is finite;

b) the image by f of every maximal point of X′ is a maximal point of X.

For all $x \in X^{(1)}$, the points $x' \in f^{-1}(x)$ all belong to $X'^{(1)}$, as a consequence of hypothesis *b*) and of the inequality (**0**, 16.3.9.1), the fibre $f^{-1}(x)$ being discrete. Let then

$$Z' = \sum_{x' \in X'^{(1)}} n_{x'}.\overline{\{x'\}}$$

be a 1-codimensional cycle on X′. For all $x \in X^{(1)}$, we define an integer n_x by the formula

$$n_x = \sum_{x' \in f^{-1}(x)} n_{x'} \cdot [\boldsymbol{k}(x') : \boldsymbol{k}(x)]$$

which has a meaning, the points of $f^{-1}(x)$ being in finite number and $\boldsymbol{k}(x')$ being a field of finite degree over $\boldsymbol{k}(x)$ (**I**, 6.4.4). Furthermore, the set of $x \in X^{(1)}$ such that $n_x \neq 0$ is *locally finite* in X, since it is contained in the image by f of the set of $x' \in X'^{(1)}$ such that $n_{x'} \neq 0$, and the conclusion follows from the fact that the morphism f is quasi-compact. We can therefore define a 1-codimensional cycle on X by setting

(21.10.14.1)
$$f_*(Z') = \sum_{x \in X^{(1)}} n_x.\overline{\{x\}}$$

and we say that $f_*(Z')$ is the *direct image* of Z′ by f. It is clear that the map $f_* : \mathfrak{Z}^1(X') \to \mathfrak{Z}^1(X)$ thus defined is a homomorphism of ordered groups. Furthermore, if U is an open of X, $f_U : f^{-1}(U) \to U$ the restriction of f, it follows immediately from the definitions that we have

(21.10.14.2)
$$(f_U)_*(Z'|f^{-1}(U)) = f_*(Z')|U$$

hence, denoting by ψ the continuous map underlying the morphism f, the maps $\Gamma(f^{-1}(U), \mathscr{Z}^1_{X'}) \to \Gamma(U, \mathscr{Z}^1_X)$ that we have just defined define a homomorphism of sheaves of ordered commutative groups on X

$$\psi_*(\mathscr{Z}^1_{X'}) \longrightarrow \mathscr{Z}^1_X.$$

Proposition (21.10.15). — *Let* X, X′, X″ *be three locally Noetherian preschemes,* $f : X' \to X$, $g : X'' \to X'$ *two morphisms satisfying conditions* a) *and* b) *of* (21.10.14)*. Then* $f \circ g$ *satisfies the same conditions, and for every* 1*-codimensional cycle* Z″ *on* X″*, we have* $(f \circ g)_*(Z'') = f_*(g_*(Z''))$. IV-4 | 297

Proof. This follows immediately from the definitions. ☐

Proposition (21.10.16). — *Let* X, X′, X_1 *be three locally Noetherian preschemes,* $f : X' \to X$ *a morphism satisfying conditions* a) *and* b) *of* (21.10.14)*,* $g : X_1 \to X$ *a flat morphism. Set* $X'_1 = X' \times_X X_1$ *(so that* X'_1 *is locally Noetherian) and let* $f_1 : X'_1 \to X_1$ *and* $g_1 : X'_1 \to X'$ *be the canonical projections. Then* f_1 *satisfies conditions* a) *and* b) *of* (21.10.14)*, and for every* 1*-codimensional cycle* Z′ *on* X′*, we have*

(21.10.16.1)
$$g^*(f_*(Z')) = (f_1)_*(g_1^*(Z')).$$

Proof. It is clear that f_1 is finite, and it satisfies condition *b*) of (21.10.14) by virtue of (2.3.7). To prove (21.10.16.1), we are immediately reduced to the case where X, X′ and X_1 are spectra of local Noetherian rings of dimension 1, A, A′ and A_1, A′ being a finite flat A-module. Denoting by x, x' and x_1 the closed points of X, X′ and X_1 respectively, it is a matter of seeing that we have

$$\sum_{x'_1} \lambda_{x'_1}[\boldsymbol{k}(x'_1) : \boldsymbol{k}(x_1)] = \lambda_{x_1}[\boldsymbol{k}(x') : \boldsymbol{k}(x)] \tag{21.10.16.2}$$

where x'_1 ranges over the closed points of X'_1 (i.e. the points lying above both x' and x_1) and where $\lambda_{x'_1} = \operatorname{length}(\mathscr{O}_{X_1,x'_1}/\mathfrak{m}_{x'}\mathscr{O}_{X_1,x'_1})$ and $\lambda_{x_1} = \operatorname{length}(A_1/\mathfrak{m}_x A_1)$. Since we have

$$[\boldsymbol{k}(x') : \boldsymbol{k}(x)]\operatorname{length}_{A'}(A'_1/\mathfrak{m}_{x'}A'_1) = \operatorname{length}_{A_1}(A'_1/\mathfrak{m}_{x'}A'_1)$$

where we have set $A'_1 = A' \otimes_A A_1$. We have thus $A'_1/\mathfrak{m}_{x'}A'_1 = (A'/\mathfrak{m}_{x'}A') \otimes_A A_1$, and as A_1 is a flat A-module, by (4.7.1),

$$\operatorname{length}_{A_1}(A'_1/\mathfrak{m}_{x'}A'_1) = \operatorname{length}_{A_1}(A_1/\mathfrak{m}_x A_1)\operatorname{length}_{A_1}(A'_1/\mathfrak{m}A'_1) = [\boldsymbol{k}(x') : \boldsymbol{k}(x)] \cdot \lambda_{x_1}$$

which finishes the proof. □

Proposition (21.10.17). — *Let* X, X′ *be two locally Noetherian preschemes,* $f : X' \to X$ *a finite locally free morphism. Then* f *satisfies condition* b) *of* (21.10.14)*, and for every divisor* D′ *on* X′*, we have*

$$f_*(\operatorname{cyc}(D')) = \operatorname{cyc}(f_*(D')). \tag{21.10.17.1}$$

Proof. Since f is flat and finite, condition *b*) of (21.10.14) and the relation $f(X'^{(1)}) \subset X^{(1)}$ follow from (6.1.1). The definition (21.10.14.1) shows that we can reduce to the case where $X = \operatorname{Spec}(A) = \operatorname{Spec}(\mathscr{O}_{X,x})$ for an $x \in X^{(1)}$; then $X' = \operatorname{Spec}(A')$, where A′ is a free A-module of finite rank; furthermore, we can suppose that $D' = \operatorname{div}(t')$, where t' is a regular element of A′; we then have $f_*(D') = \operatorname{div}(t)$, where $t = N_{A'/A}(t')$ is a regular element of A (21.5.2). We can restrict to the case where t' is not invertible in A′, the ring A/tA is then of finite length, and $A'/t'A'$ is a direct sum of local Artinian rings A'_i $(1 \leqslant i \leqslant r)$, the residue field of A'_i being $\boldsymbol{k}(x'_i)$, where x'_i $(1 \leqslant i \leqslant r)$ are the points of X′ lying above x. If t'_i $(1 \leqslant i \leqslant r)$ is the image of t' in A'_i, $A'/t'A'$ is a direct sum of the $A'_i/t'_iA'_i$; since the product $\operatorname{length}_{A'_i}(A'_i/t'_iA'_i) \cdot [\boldsymbol{k}(x'_i) : \boldsymbol{k}(x)]$ is equal to $\operatorname{length}_A(A'_i/t'_iA'_i)$, we see that the multiplicity at point x of the first member of (21.10.17.1) is $\operatorname{length}_A(A'/t'A')$, so that the formula to prove reduces to

$$\operatorname{length}_A(A'/t'A') = \operatorname{length}_A(A/tA). \tag{21.10.17.2}$$

This relation follows from the following more general lemma: □

Lemma (21.10.17.3). — *Let* A *be a local Noetherian ring of dimension* 1, M *a free A-module of finite rank,* u *an injective endomorphism of* M*. Then we have* **IV-4** | 298

$$\operatorname{length}_A(\operatorname{Coker} u) = \operatorname{length}_A(A/(\det u)A). \tag{21.10.17.4}$$

Proof. We distinguish several cases.

I) A *is a discrete valuation ring*. Let π be a uniformiser of A, and note $\operatorname{length}_A(A/\pi^k A) = k$; if n is the rank of M and if π^{m_i} $(1 \leqslant i \leqslant n)$ are the invariant factors of u, $\operatorname{Coker}(u)$ is a direct sum of A-modules $A/\pi^{m_i}A$, so has length $m = \sum_{i=1}^n m_i$, and $\det(u)$ is the product of an invertible element and of t^m, whence the conclusion in this case (Bourbaki, *Alg.*, chap. VII, §4, nº 5, cor. 1 of prop. 4).

II) A *is a complete integral ring (of dimension* 1*)*. We know then (**0**, 19.8.8, (ii)) that there is a subring B of A, which is a discrete valuation ring, such that $B \to A$ is a local homomorphism making A a B-module of finite type; since this B-module is evidently torsion-free, it is *free* (Bourbaki, *Alg. comm.*, chap. VI, §3, nº 6, lemma 1). Let M′ denote the set M equipped with its structure of B-module (free), by u' the endomorphism u considered as a B-endomorphism. It follows from I) that

$$\operatorname{length}_B(\operatorname{Coker} u') = \operatorname{length}_B(B/(\det u')B). \tag{21.10.17.5}$$

But $\det u' = N_{A/B}(\det u)$, therefore, by applying (21.10.17.4) to the homothety $x \mapsto (\det u)x$ of the free B-module A, we get

$$\operatorname{length}_B(B/(\det u')B) = \operatorname{length}_B(A/(\det u)A),$$

whence, substituting in (21.10.17.5) and dividing by $[\boldsymbol{k}(\mathrm{A}):\boldsymbol{k}(\mathrm{B})]$, the length of the B-module $\boldsymbol{k}(\mathrm{A})$, we obtain (21.10.17.4) in the case considered.

III) A *is a complete ring*. Let us note that we can further suppose that $\mathfrak{m} \notin \mathrm{Ass}(\mathrm{A})$; indeed, since $\det(u)$ is a regular element of A, if we had $\mathfrak{m} \in \mathrm{Ass}(\mathrm{A})$, then $\det(u)$ would be invertible, u an automorphism of M, and the formula (21.10.17.4) would then be trivial, both members being zero. **IV-4** | 299

In what follows, for an endomorphism v of a module N over a ring R, such that Ker v and Coker v are of *finite length*, we set

$$\chi(\mathrm{N}, v) = \mathrm{length}_{\mathrm{R}}(\mathrm{Ker}(v)) - \mathrm{length}_{\mathrm{R}}(\mathrm{Coker}(v)).$$

We note that the hypothesis on v amounts to saying that the complex

$$\mathrm{K}^0 : 0 \longrightarrow \mathrm{N} \xrightarrow{v} \mathrm{N} \longrightarrow 0$$

has its cohomology modules of finite length and that $\chi(\mathrm{N}, v) = \chi(\mathrm{H}^0(\mathrm{K}^0))$ ($\mathbf{0}_{\mathrm{III}}$, 11.10). We deduce that if N', N, N'' are three R-modules, and if there is a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & \mathrm{N}' \arrow[r, "r"] \arrow[d, "v'"] & \mathrm{N} \arrow[r, "s"] \arrow[d, "v"] & \mathrm{N}'' \arrow[r] \arrow[d, "v''"] & 0 \\
0 \arrow[r] & \mathrm{N}' \arrow[r, "r"'] & \mathrm{N} \arrow[r, "s"'] & \mathrm{N}'' \arrow[r] & 0
\end{tikzcd}$$

whose rows are exact, and if two of the three numbers $\chi(\mathrm{N}, v)$, $\chi(\mathrm{N}', v')$ and $\chi(\mathrm{N}'', v'')$ are defined, then so is the third and one has

(21.10.17.6) $$\chi(\mathrm{N}, v) = \chi(\mathrm{N}', v') + \chi(\mathrm{N}'', v'').$$

This follows indeed immediately from the cohomology exact sequence.

Finally, if N is an R-module of *finite length*, we have $\chi(\mathrm{N}, v) = 0$.

With this notation, we have the following lemma: □

Lemma (21.10.17.7). — *Let* A *be a local Noetherian ring of dimension* 1 *whose maximal ideal* $\mathfrak{m}$ *is such that* $\mathfrak{m} \notin \mathrm{Ass}(\mathrm{A})$; *let* $\mathfrak{p}_i$ *($1 \leqslant i \leqslant n$) be the minimal prime ideals of* A. *Let* M *be a free A-module of finite type, u an endomorphism of* M *such that* $\chi(\mathrm{M}, u)$ *is defined; for each i, set* $\mathrm{M}_i = \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{p}_i)$ *and let* u_i *be the endomorphism* $u \otimes 1_{\mathrm{A}/\mathfrak{p}_i}$ *of* M_i; *then if* $\chi(\mathrm{M}_i, u_i)$ *is defined for each i, we have*

(21.10.17.8) $$\chi(\mathrm{M}, u) = \sum_{i=1}^{n} \mathrm{length}(\mathrm{A}_{\mathfrak{p}_i}) \cdot \chi(\mathrm{M}_i, u_i).$$

Proof. Since $\mathfrak{m} \notin \mathrm{Ass}(\mathrm{A})$, we have a reduced primary decomposition $(0) = \bigcap_{1 \leqslant i \leqslant n} \mathfrak{q}_i$, where the ideal $\mathfrak{q}_i$ is $\mathfrak{p}_i$-primary for $1 \leqslant i \leqslant n$. If we set $\mathrm{M}'_i = \mathrm{M} \otimes_{\mathrm{A}} (\mathrm{A}/\mathfrak{q}_i)$, we have therefore an exact sequence **IV-4** | 300

$$0 \longrightarrow \mathrm{M} \longrightarrow \bigoplus_i \mathrm{M}'_i \longrightarrow \mathrm{M}'' \longrightarrow 0$$

of A-modules, where M'' is of *finite length*: indeed by localising the preceding exact sequence at each of the $\mathfrak{p}_i$, we obtain $\mathrm{M}''_{\mathfrak{p}_i} = 0$, since $(\mathfrak{q}_i)_{\mathfrak{p}_i} = 0$ and $(\mathfrak{q}_j)_{\mathfrak{p}_i} = \mathrm{A}_{\mathfrak{p}_j}$ for $j \neq i$ (Bourbaki, *Alg. comm.*, chap. IV, §2, n° 4, prop. 6), so $(\mathrm{M}'_i)_{\mathfrak{p}_i} = \mathrm{M}_{\mathfrak{p}_i}$ and $(\mathrm{M}'_j)_{\mathfrak{p}_i} = 0$; the support of M'' is therefore reduced to $\mathfrak{m}$, hence M'' is of finite length ($\mathbf{0}$, 16.1.10). If we set $u'_i = u \otimes 1_{\mathrm{A}/\mathfrak{q}_i}$, and if u'' is the endomorphism of M'' deduced from $\bigoplus u'_i$ by passing to quotients, we deduce from (21.10.17.6) that $\chi(\mathrm{M}, u) + \chi(\mathrm{M}'', u'') = \sum_i \chi(\mathrm{M}'_i, u'_i)$, and since M'' is of finite length, $\chi(\mathrm{M}'', u'') = 0$. To prove (21.10.17.8), we can therefore reduce to the case where $n = 1$. We will then denote by $\mathfrak{p}$ the unique minimal prime ideal of A; if $\mathrm{A}_0 = \mathrm{A}/\mathfrak{p}$, $\mathrm{M}_0 = \mathrm{M} \otimes_{\mathrm{A}} \mathrm{A}_0$, $u_0 = u \otimes 1_{\mathrm{A}_0}$, it is a matter of proving that if $\chi(\mathrm{M}_0, u_0)$ is defined, then

(21.10.17.9) $$\chi(\mathrm{M}, u) = \mathrm{length}\, \mathrm{A}_{\mathfrak{p}} \cdot \chi(\mathrm{M}_0, u_0).$$

Let $\mathfrak{n}_j$ ($0 \leqslant j \leqslant r$) be the "$j$-th symbolic power" of $\mathfrak{p}$, the inverse image of $(\mathfrak{p}\mathrm{A}_{\mathfrak{p}})^j$ in A ($1 \leqslant j \leqslant r$), with $\mathfrak{n}_0 = \mathrm{A}$ and $\mathfrak{n}_r = 0$; set

$$\mathrm{M}_j = \mathfrak{n}_j\mathrm{M}/\mathfrak{n}_{j+1}\mathrm{M} \qquad (0 \leqslant j \leqslant r-1),$$

and denote by v_j the endomorphism of M_j deduced from u by restriction to $\mathfrak{n}_j M$ and passage to quotients. The first assertion will imply the second, by applying (21.10.17.6) to each of the exact sequences

$$0 \longrightarrow \mathfrak{n}_j M/\mathfrak{n}_{j+1} M \longrightarrow M/\mathfrak{n}_{j+1} M \longrightarrow M/\mathfrak{n}_j M \longrightarrow 0.$$

and that each of the numbers $\chi(M_j, v_j)$ is defined and one has

(21.10.17.10) $$\chi(M, u) = \sum_j \chi(M_j, v_j).$$

To prove the first assertion, we note that if m is the rank of the free A-module M, M_j is isomorphic to $(\mathfrak{n}_j/\mathfrak{n}_{j+1})^m$, or since $\mathfrak{p}$ annihilates each of the quotients (since $\mathfrak{p} \cdot \mathfrak{n}_j \subset \mathfrak{n}_{j+1}$), M_j is an A_0-module isomorphic to $M_0 \otimes_{A_0} (\mathfrak{n}_j/\mathfrak{n}_{j+1})$. Let us denote by l_j the rank of the A_0-module $\mathfrak{n}_j/\mathfrak{n}_{j+1}$; since the field of fractions K_0 of A_0 is the residue field of $A_\mathfrak{p}$, l_j is also the length of the $A_\mathfrak{p}$-module $(\mathfrak{p}A_\mathfrak{p})^j/(\mathfrak{p}A_\mathfrak{p})^{j+1}$. There is a system of generators of the A_0-module M_j which contains a basis of $M_j \otimes_{A_0} K_0$; hence there is an A_0-homomorphism

$$w_j : M_0^{l_j} \longrightarrow M_j$$

whose localisation at the ideal (0) of A_0 is an isomorphism, so that the support of $\operatorname{Ker}(w_j)$ and of $\operatorname{Coker}(w_j)$ is reduced to the maximal ideal $\mathfrak{m}/\mathfrak{p}$ of A_0; $\operatorname{Ker}(w_j)$ and $\operatorname{Coker}(w_j)$ are therefore A_0-modules of finite length (**0**, 16.1.10). Since by hypothesis $\chi(M_0, u_0)$ is defined, it follows from the same of $\chi(M_0^{l_j}, u_0^{l_j}) = l_j \chi(M_0, u_0)$ and by virtue of (21.10.17.6) and the fact that $\operatorname{Ker}(w_j)$ and $\operatorname{Coker}(w_j)$ are of finite length, we see that $\chi(M_j, v_j)$ is defined and equal to $l_j \chi(M_0, u_0)$; the relation (21.10.17.10) then gives

$$\chi(M, u) = (\sum_j l_j) \chi(M_0, u_0),$$

and by virtue of a preceding remark, $\sum_j l_j$ is none other than the length of $A_\mathfrak{p}$, which finishes the proof of the lemma (21.10.17.7). □

To apply this lemma when A is a complete ring and $\mathfrak{m} \notin \operatorname{Ass}(A)$, we observe that if u is injective, so is each of the u_i (with the notation of the lemma): indeed, $\det u$ is then a regular element of A, so is not contained in any of the $\mathfrak{p}_i$, and its image $\det u_i$ in $A/\mathfrak{p}_i$ is therefore a nonzero element of this integral ring, which proves that u_i is injective. Since $\operatorname{Coker}(u_i)$ is an image of $\operatorname{Coker}(u)$, it is also of finite length and $\chi(M_i, u_i)$ is therefore defined for all i; we have therefore, by the formula (21.10.17.8), $\chi(M, u) + \chi(M'', u'') = \sum_i \chi(M'_i, u'_i)$ and, since M'' is of finite length, $\chi(M'', u'') = 0$. On the other hand, since $\det(u)$ is a regular element of A, it is not contained in any of the $\mathfrak{p}_i$; the ideal $(\det u)A$ is therefore $\mathfrak{m}$-primary and the quotient $A/(\det u)A$ is of finite length. Applying the same reasoning above to the injective homothety $t : \xi \to (\det u)\xi$ of A and to its images $t_i = \det u_i$ in the $A/\mathfrak{p}_i$, we get **IV-4 | 301**

$$\chi(A, t) = \sum_i \operatorname{length}(A_{\mathfrak{p}_i}) \cdot \chi(A/\mathfrak{p}_i, t_i).$$

But all the rings $A/\mathfrak{p}_i$ are integral and complete, and by applying the result of II) to each of them, one obtains again the relation (21.10.17.4) for M and u.

IV) *General case*. Set $A' = \widehat{A}$, $M' = M \otimes_A A'$, $u' = u \otimes 1_{A'}$; we have $\det(u') = \det(u)$, and by flatness, $\operatorname{Coker}(u') = (\operatorname{Coker}(u))_{(A')}$ and $A'/(\det u)A' = (A/(\det u)A)_{(A')}$; since the formula (21.10.17.4) is true for A' and u' by virtue of III), it is also true for A and u.

This finishes the proof of (21.10.17).

Proposition (21.10.18). — *Let* X, X′ *be two locally Noetherian preschemes,* $f : X' \to X$ *a finite morphism, transforming every maximal point of* X′ *into a maximal point of* X, *and satisfying for all* $x' \in X'$ *one of the conditions (ii), (iii), (iv) of* (21.10.10). *Suppose furthermore that there exists an integer n such that, for every maximal point x of* X, $(f_*(\mathscr{O}_{X'}))_x$ *is a free* $\mathscr{O}_{X,x}$*-module of rank n. Then, for every* 1*-codimensional cycle* Z *on* X, *we have*

(21.10.18.1) $$f_*(f^*(Z)) = n \cdot Z$$

("projection formula").

Proof. By virtue of the definitions, we are immediately reduced to the case of a local Noetherian ring A of dimension 1, of closed point x, and where $Z = \overline{\{x\}}$; set $X' = \mathrm{Spec}(A')$ where A' is a finite A-algebra and for every minimal prime ideal $\mathfrak{p}_i$ of A, $A'_{\mathfrak{p}_i}$ is a free $A_{\mathfrak{p}_i}$-module of rank n. Let us show that we can furthermore restrict to the case where A is *complete*; let us make the base change $h : Y \to X$, where $Y = \mathrm{Spec}(B)$, $B = \widehat{A}$, and set $Y' = X' \times_X Y = \mathrm{Spec}(B \otimes_A A')$ and $g = f_{\{Y\}} : Y' \to Y$; the morphism g is then finite, and it satisfies the same conditions as f; if we prove the formula (21.10.18.1) for g and $\overline{\{x\}}$, we verify from (21.10.16.1). We can therefore suppose that A is complete, and hence so is A' which is also decomposed as a direct product of complete local rings; one can therefore also restrict ourselves to the case where A' is local, and verify (21.10.18.1) for each of the conditions (ii), (iii), (iv) of X'. **IV-4** | 302

In case (ii), A' is a flat A-module of finite type, of rank n by hypothesis, hence by definition (21.10.1) and (21.10.3) $f^*(Z) = \lambda_{x'} \cdot \overline{\{x'\}}$, where $\lambda_{x'}$ is the length of the A'-module $A'/\mathfrak{m}_x A'$, hence $f_*(f^*(Z)) = (\lambda_{x'}[\boldsymbol{k}(x') : \boldsymbol{k}(x)]) \cdot \overline{\{x\}}$; but $\lambda_{x'}[\boldsymbol{k}(x') : \boldsymbol{k}(x)]$ is the length of $A'/\mathfrak{m}_x A'$ as A-module, or its rank as $\boldsymbol{k}(x)$-vector space, which is equal to n.

In case (iii), A is a discrete valuation ring, hence regular, and the hypothesis $\mathfrak{m}_{x'} \notin \mathrm{Ass}(\mathscr{O}_{X',x'})$ implies that A' is a Cohen-Macaulay ring (**0**, 16.4.6); since $\dim(A') = \dim(A) = 1$ that $A'/\mathfrak{m}_x A'$ is an Artinian ring, it follows from (6.1.5) that A' is a flat A-module and one is reduced to case (ii).

In case (iv), if K is the total ring of fractions of A, $K' = A' \otimes_A K$ is by hypothesis a free K-module of rank n and by definition (21.10.10), we have $f^*(Z) = (n/[\boldsymbol{k}(x') : \boldsymbol{k}(x)]) \cdot \overline{\{x'\}}$, whence $f_*(f^*(Z)) = n \cdot \overline{\{x\}}$.

We note that the formula (21.10.18.1) is applicable in particular when the morphism f is *finite* and *flat* and such that for every maximal point x of X, $(f_*(\mathscr{O}_{X'}))_x$ is a free $\mathscr{O}_{X,x}$-module of rank n. ☐

Corollary (21.10.19). — *Under the hypotheses of* (21.10.18), *let* D *be a divisor on* X *such that* $f^*(D)$ *is defined* (21.4.5); *then we have*

(21.10.19.1) $$f_*(\mathrm{cyc}(f^*(D))) = n \cdot \mathrm{cyc}(D).$$

Proof. This follows from (21.10.18) and (21.10.13bis). ☐

21.11. Factoriality of regular local rings

Theorem (21.11.1). — (Auslander-Buchsbaum) — *A regular local Noetherian ring is factorial.* **IV-4** | 302

Proof. The proof that follows is due to I. Kaplansky.

Let A be a regular local ring of dimension n; we will reason by induction on n. For $n = 0$, A is a field, and for $n = 1$, a discrete valuation ring, hence principal (and *a fortiori* factorial). Suppose therefore $n \geqslant 2$ and the theorem proved for regular rings of dimension $< n$. Set $X = \mathrm{Spec}(A)$, denote by a the closed point of X and set $U = X - \{a\}$. At every point $y \in U$, we have $\dim(\mathscr{O}_{X,y}) \leqslant n - 1$, and since A is regular, the rings $\mathscr{O}_{X,y}$ are also regular (**0**, 17.3.2), hence by the induction hypothesis they are factorial. Moreover, since $\mathrm{prof}(A) = \dim(A) \geqslant 2$ since A is regular, hence Cohen-Macaulay (**0**, 17.1.3). Using (21.6.14), we are reduced to proving that $\mathrm{Pic}(U) = 0$. Consider therefore an invertible $\mathscr{O}_U$-Module $\mathscr{L}$, and let us prove that it is isomorphic to $\mathscr{O}_U$. It follows from (**I**, 9.4.5) that there exists a *coherent* $\mathscr{O}_X$-Module $\mathscr{F}$ such that $\mathscr{F}|U = \mathscr{L}$. Since A is regular, hence of finite cohomological dimension (**0**, 17.3.1), there exists a *finite* free resolution of $\mathscr{F}$: **IV-4** | 303

$$0 \leftarrow \mathscr{F} \leftarrow \mathscr{O}_X^{n_1} \leftarrow \mathscr{O}_X^{n_2} \leftarrow \ldots \leftarrow \mathscr{O}_X^{n_h} \leftarrow 0$$

(**0**, 17.2.8 and **0**, 17.2.2, (iii)). By restriction to U, we have therefore a finite resolution

(21.11.1.1) $$0 \leftarrow \mathscr{L} \leftarrow \mathscr{O}_U^{n_1} \leftarrow \mathscr{O}_U^{n_2} \leftarrow \ldots \leftarrow \mathscr{O}_U^{n_h} \leftarrow 0.$$

☐

The theorem will follow from the following general considerations. On a ringed space X, let $\mathscr{E}$ be an $\mathscr{O}_X$-Module locally free of finite rank; we will denote by $\Lambda^{\max}\mathscr{E}$ the invertible $\mathscr{O}_X$-Module which, in a neighbourhood of each point of X, is equal to $\Lambda^p\mathscr{E}$, denoting by p the rank of $\mathscr{E}$ in this neighbourhood (which can vary with the connected component). With this notation, we have the

Lemma (21.11.2). — *Let* X *be a ringed space in local rings, and*

$$0 \leftarrow \mathscr{E}_0 \xleftarrow{u_0} \mathscr{E}_1 \leftarrow \ldots \leftarrow \mathscr{E}_h \leftarrow 0$$

an exact sequence of $\mathscr{O}_X$*-Modules locally free of finite rank; then the invertible* $\mathscr{O}_X$*-Module* $\bigotimes_{0\leqslant i\leqslant h}(\Lambda^{\max}\mathscr{E}_i)^{\otimes(-1)^i}$ *is isomorphic to* $\mathscr{O}_X$*.*

Proof. Let us show how this lemma will allow us to conclude the proof of (21.11.1). It suffices to note that, for every integer n, $\Lambda^{\max}(\mathscr{O}_U^n) = \Lambda^n\mathscr{O}_U^n$ is isomorphic to $\mathscr{O}_U$, and on the other hand $\Lambda^{\max}\mathscr{L} = \mathscr{L}$ for every invertible $\mathscr{O}_U$-Module $\mathscr{L}$; the lemma (21.11.2), applied to the exact sequence (21.11.1.1), shows that $\mathscr{L}$ is isomorphic to $\mathscr{O}_U$.

It remains to prove (21.11.2); we proceed by induction on h, the lemma being trivial for $h = 1$. For $h > 1$, $\mathscr{N} = \mathrm{Ker}(u_0)$ is an $\mathscr{O}_X$-Module locally free of finite rank ($\mathbf{0}_{\mathrm{I}}$, 5.5.5), and we have the two exact sequences

$$0 \leftarrow \mathscr{E}_0 \leftarrow \mathscr{E}_1 \leftarrow \mathscr{N} \leftarrow 0$$

$$0 \leftarrow \mathscr{N} \leftarrow \mathscr{E}_2 \leftarrow \ldots \leftarrow \mathscr{E}_h \leftarrow 0$$

By virtue of the induction hypothesis, $(\Lambda^{\max}\mathscr{N}) \otimes (\bigotimes_{2\leqslant i\leqslant h}(\Lambda^{\max}\mathscr{E}_i)^{\otimes(-1)^{i-1}})$ is isomorphic to $\mathscr{O}_X$; it will therefore suffice to define a canonical isomorphism $(\Lambda^{\max}\mathscr{N}) \otimes (\Lambda^{\max}\mathscr{E}_0) \xrightarrow{\sim} \Lambda^{\max}\mathscr{E}_1$ to finish the proof. Now, there exists an open covering (U_α) of X such that in each U_α, $\mathscr{E}_1|U_\alpha$ is a direct sum of $\mathscr{N}|U_\alpha$ and of a free $\mathscr{O}_{U_\alpha}$-Module $\mathscr{M}_\alpha$ ($\mathbf{0}_{\mathrm{I}}$, 5.5.5), whence a canonical isomorphism $v_\alpha : \mathscr{M}_\alpha \xrightarrow{\sim} \mathscr{E}_0|U_\alpha$. Since we have a canonical isomorphism

$$r_\alpha : (\Lambda^{\max}\mathscr{N}|U_\alpha) \otimes (\Lambda^{\max}\mathscr{M}_\alpha) \xrightarrow{\sim} (\Lambda^{\max}\mathscr{E}_1)|U_\alpha$$

we deduce, by means of v_α, an isomorphism

$$u_\alpha : (\Lambda^{\max}\mathscr{N}|U_\alpha) \otimes (\Lambda^{\max}\mathscr{E}_0|U_\alpha) \xrightarrow{\sim} \Lambda^{\max}\mathscr{E}_1|U_\alpha$$

and it is a matter of showing that u_α and u_β coincide in $U_\alpha \cap U_\beta$ for any two indices α, β. Now, if **IV-4** | 304
v'_α and v'_β are the restrictions to $U_\alpha \cap U_\beta$ of v_α and v_β respectively, we have $v_\alpha = v'_\beta \circ w_{\beta\alpha}$, where $w_{\beta\alpha} : \mathscr{M}_\alpha|(U_\alpha \cap U_\beta) \xrightarrow{\sim} \mathscr{M}_\beta|(U_\alpha \cap U_\beta)$ is the "parallel projection" such that for every section $s \in \Gamma(U_\alpha \cap U_\beta, \mathscr{M}_\alpha)$, $w_{\beta\alpha}(s) = s + t$ with $t \in \Gamma(U_\alpha \cap U_\beta, \mathscr{N})$; the identity of u_α and u_β follows immediately from this fact and from the definition of the canonical isomorphism r_α (Bourbaki, *Alg.*, chap. III, 3e ed.). ☐

21.12. The Van der Waerden purity theorem for the ramification locus of a birational morphism

(21.12.1). Let X and U be two preschemes, $f : U \to X$ a quasi-compact and quasi-separated morphism, so that $f_*(\mathscr{O}_U)$ is a quasi-coherent $\mathscr{O}_X$-Algebra (1.7.4). We call *affine envelope* of the X-prescheme U the X-prescheme affine over X

$$U^0 = \mathrm{Aff}(U/X) = \mathrm{Spec}(f_*(\mathscr{O}_U)) = \mathrm{Spec}(\mathscr{A}(U)) \quad (\mathbf{II},\ 1.1.1).$$

If $f^0 : U^0 \to X$ is the structural morphism, we have therefore by definition

$$\mathscr{A}(U^0) = f^0_*(\mathscr{O}_{U^0}) = f_*(\mathscr{O}_U),$$

and at the identity isomorphism of $f_*(\mathscr{O}_U)$, there corresponds by (**II**, 1.2.7) a canonical X-morphism

$$i_U : U \longrightarrow U^0. \tag{21.12.1.1}$$

For i_U to be an *isomorphism*, it is necessary and sufficient that the morphism $f : U \to X$ be *affine*. For every affine X-prescheme V, the map $u \mapsto u \circ i_U$ is a *bijection*

$$\mathrm{Hom}_X(U^0, V) \xrightarrow{\sim} \mathrm{Hom}_X(U, V)$$

functorial in V: this follows from the existence of canonical bijections

$$\mathrm{Hom}_X(U, V) \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_x}(\mathscr{A}(V), \mathscr{A}(U))$$

and $\mathrm{Hom}_X(U^0, V) \xrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_x}(\mathscr{A}(V), \mathscr{A}(U^0))$ (**II**, 1.2.7). We can therefore say that U^0 *represents* the covariant functor $V \mapsto \mathrm{Hom}_X(U, V)$ in the category of X-preschemes affine over X. We deduce from this ($\mathbf{0}_{\mathrm{III}}$, 8.1.7) that for a fixed X, $U \mapsto \mathrm{Aff}(U/X)$ is a covariant functor from the category of X-preschemes quasi-compact and quasi-separated over X, into the category of X-preschemes affine over X; more precisely, if U_1, U_2 are two X-preschemes quasi-compact and quasi-separated over X, to every X-morphism $g : \mathrm{U}_1 \to \mathrm{U}_2$ corresponds the unique X-morphism $g^0 : \mathrm{U}_1^0 \to \mathrm{U}_2^0$ making commutative the diagram

$$\begin{tikzcd} \mathrm{U}_1 \arrow[r, "g"] \arrow[d, "i_{\mathrm{U}_1}"'] & \mathrm{U}_2 \arrow[d, "i_{\mathrm{U}_2}"] \\ \mathrm{U}_1^0 \arrow[r, "g^0"'] & \mathrm{U}_2^0 \end{tikzcd}$$

More generally, consider a commutative diagram **IV-4** | 305

$$\begin{tikzcd} \mathrm{U} \arrow[r, "v"] \arrow[d, "f"'] & \mathrm{U}' \arrow[d, "f'"] \\ \mathrm{X} \arrow[r, "u"'] & \mathrm{X}' \end{tikzcd}$$

where the morphisms f, f' are quasi-compact and quasi-separated and the morphism u *affine*. If $h = u \circ f$, we have $h_*(\mathscr{O}_\mathrm{U}) = u_*(f_*(\mathscr{O}_\mathrm{U})) = u_*(f^0_*(\mathscr{O}_{\mathrm{U}^0}))$ and $u \circ f^0$ is an affine morphism; hence $\mathrm{U}^0 = \mathrm{Aff}(\mathrm{U}/\mathrm{X}')$ (relative to the morphism h), whence a unique X′-morphism $v^0 : \mathrm{U}^0 \to \mathrm{U}'^0$ making commutative the diagram

$$\begin{tikzcd} \mathrm{U} \arrow[r, "v"] \arrow[d, "i_\mathrm{U}"'] & \mathrm{U}' \arrow[d, "i_{\mathrm{U}'}"] \\ \mathrm{U}^0 \arrow[r, "v^0"'] & \mathrm{U}'^0 \end{tikzcd}$$

Proposition (21.12.2). — *Let $f : \mathrm{U} \to \mathrm{X}$ be a quasi-compact and quasi-separated morphism, $h : \mathrm{X}' \to \mathrm{X}$ a flat morphism; set $\mathrm{U}' = \mathrm{U} \times_\mathrm{X} \mathrm{X}'$, $f' = f_{\{\mathrm{X}'\}} : \mathrm{U}' \to \mathrm{X}'$. Then there is a canonical X′-isomorphism*

(21.12.2.1) $$\mathrm{Aff}(\mathrm{U}'/\mathrm{X}') \xrightarrow{\sim} \mathrm{Aff}(\mathrm{U}/\mathrm{X}) \times_\mathrm{X} \mathrm{X}'.$$

Proof. Indeed, we have $\mathrm{Aff}(\mathrm{U}'/\mathrm{X}') = \mathrm{Spec}(f'_*(\mathscr{O}_{\mathrm{U}'}))$, and $\mathrm{Aff}(\mathrm{U}/\mathrm{X}) \times_\mathrm{X} \mathrm{X}' = \mathrm{Spec}(h^*(f_*(\mathscr{O}_\mathrm{U})))$; the isomorphism of the assertion comes from the canonical isomorphism $h^*(f_*(\mathscr{O}_\mathrm{U})) \xrightarrow{\sim} f'_*(\mathscr{O}_{\mathrm{U}'})$ (2.3.1). ☐

Corollary (21.12.3). — *For every quasi-compact and quasi-separated morphism $f : \mathrm{U} \to \mathrm{X}$ and all $x \in \mathrm{X}$, there is a canonical isomorphism up to canonical isomorphism*

$$\mathrm{U}^0 \times_\mathrm{X} \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}) = (\mathrm{U} \times_\mathrm{X} \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}))^0.$$

It also follows from (21.12.2) that, up to canonical isomorphism, for every open V of X

(21.12.4) $$(f^{-1}(\mathrm{V}))^0 = (f^0)^{-1}(\mathrm{V}).$$

(21.12.5). We will consider in particular the case where $f : \mathrm{U} \to \mathrm{X}$ is an *open immersion*, U thus being identified with an open of X. Since the morphism $f^{-1}(\mathrm{U}) \to \mathrm{U}$ is the identity, it follows from (21.12.4) that the morphism $(f^0)^{-1}(\mathrm{U}) \to \mathrm{U}$ restriction of f^0 is an isomorphism, $i_\mathrm{U} : \mathrm{U} \to \mathrm{U}^0$ being thus an *open immersion* allowing us to identify U with an open of U^0. More generally, for an open V of X, the restriction $(f^0)^{-1}(\mathrm{V}) \to \mathrm{V}$ of f^0 is an isomorphism of $(f^0)^{-1}(\mathrm{V})$ onto the prescheme induced on the open $\mathrm{V} \cap \mathrm{U}$ if and only if the open immersion $\mathrm{U} \cap \mathrm{V} \to \mathrm{V}$ is an *affine* morphism. It is clear (**II**, 1.2.1) that the union of these opens V is the *largest* of them, U_1, which contains U; by **IV-4** | 306
virtue of what precedes, U_1 is also the largest open not meeting the set

$$\mathrm{Daf}(\mathrm{U}/\mathrm{X}) = f^0(\mathrm{U}^0) - \mathrm{U}$$

("affinity defect" of the open U relative to X, which is empty if and only if U is affine over X); in other words, the closed set $\mathrm{Z} = \mathrm{X} - \mathrm{U}_1$ is the *closure* of $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$.

We note that for every *flat* morphism $h : \mathrm{X}' \to \mathrm{X}$, if we set $\mathrm{U}' = h^{-1}(\mathrm{U})$, we have

(21.12.5.1) $$\mathrm{Daf}(\mathrm{U}'/\mathrm{X}') = h^{-1}(\mathrm{Daf}(\mathrm{U}/\mathrm{X}))$$

as follows immediately from (21.12.2.1) and from (**I**, 3.4.8). In particular, for every open V of X, we have

(21.12.5.2) $$\mathrm{Daf}((\mathrm{U} \cap \mathrm{V})/\mathrm{V}) = \mathrm{Daf}(\mathrm{U}/\mathrm{X}) \cap \mathrm{V}$$

and for all $x \in \mathrm{X}$,

(21.12.5.3) $$\mathrm{Daf}((\mathrm{U} \cap \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}))/\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})) = \mathrm{Daf}(\mathrm{U}/\mathrm{X}) \cap \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}).$$

We will, when X is locally Noetherian, give precisions on the nature of the set $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$, which will show for example that when U is not everywhere dense in X, $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$ is not any rare closed subset:

Theorem (21.12.6). — *Let* X *be a locally Noetherian prescheme,* U *a non-empty open,* $f : \mathrm{U} \to \mathrm{X}$ *the canonical injection. Then:*

(i) The closure $\mathrm{Z} = \mathrm{X} - \mathrm{U}_1$ *of* $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$ *is of codimension* $\geqslant 2$ *in* X.
(ii) If $\mathrm{T} = \mathrm{X} - \mathrm{U} \supset \mathrm{Z}$ *is of codimension* $\geqslant 2$*, the morphism* $f^0 : \mathrm{U}^0 \to \mathrm{X}$ *is surjective.*

Proof. (i) Let us first show that for every point $x \in \mathrm{Daf}(\mathrm{U}/\mathrm{X})$ we have necessarily $\dim(\mathscr{O}_{\mathrm{X},x}) > 1$. Indeed, x cannot evidently be a maximal point of X, not being contained in $\overline{\mathrm{U}}$; it is therefore a matter of seeing that we cannot have $\dim(\mathscr{O}_{\mathrm{X},x}) = 1$. Now, this relation would imply, by (21.12.5.3), $x \in \mathrm{Daf}(\mathrm{U}^{(a)}/\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x}))$, where $\mathrm{U}^{(a)} = \mathrm{U} \cap \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$. But the only open subsets of $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ are $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ itself and the parts of the (finite) set of maximal points of $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$. Now, the open set reduced to a single maximal point of $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ is affine, by definition of preschemes; we conclude that *all* open subsets of $\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ are affine, hence $\mathrm{Daf}(\mathrm{U}^{(a)}/\mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})) = \varnothing$, contradicting the hypothesis.

To prove (i), it remains to show more, namely that if $\dim(\mathscr{O}_{\mathrm{X},x}) = 1$ is such that $x \in \mathrm{X}$, there exists a neighbourhood open W of x in X such that W does not meet $\mathrm{Daf}(\mathrm{U}/\mathrm{X})$, i.e. such that the canonical injection $f_{\mathrm{W}} : \mathrm{U} \cap \mathrm{W} \to \mathrm{W}$ is affine. But, with the same notations as above, we see that the injection $f^{(a)} : \mathrm{U}^{(a)} \to \mathrm{Spec}(\mathscr{O}_{\mathrm{X},x})$ is affine. We can evidently restrict to the case where X is Noetherian, and x the closed point of X. For this, it suffices to make the change of base $h : \mathrm{X}' = \mathrm{Spec}(\mathrm{A}) \to \mathrm{X}$, where $\mathrm{A} = \widehat{\mathscr{O}}_{\mathrm{X},x}$; if we set $\mathrm{U}' = h^{-1}(\mathrm{U})$, $f' = f_{\{\mathrm{X}'\}}$ the canonical injection $\mathrm{U}' \to \mathrm{X}'$, and since the morphism h is *flat*, it follows from (21.12.2) that if we prove that x belongs to $f'^0(\mathrm{U}'^0)$, then $x \in f^0(\mathrm{U}^0)$. By virtue of (6.1.1), we have made the desired reduction.

Let then X_1 be a closed reduced sub-prescheme of X having as underlying space an irreducible component of X, of *maximal* dimension, among those that contain T, and set $\mathrm{U}_1 = \mathrm{U} \cap \mathrm{X}_1$, $\mathrm{T}_1 = \mathrm{T} \cap \mathrm{X}_1 = \mathrm{X}_1 - \mathrm{U}_1$; we have therefore $\mathrm{codim}(\mathrm{T}_1, \mathrm{X}_1) \geqslant 2$ (**0**, 14.2.1), and the pair $(\mathrm{U}_1, \mathrm{X}_1)$ satisfies the same hypotheses that the pair (U, X) (X_1 being the spectrum of a quotient of A, hence a local Noetherian complete ring). We have on the other hand a commutative diagram **IV-4** | 307

$$\begin{tikzcd} \mathrm{U}_1 \arrow[r, "i"] \arrow[d, "h"'] & \mathrm{U} \arrow[d, "f"] \\ \mathrm{X}_1 \arrow[r, "i"'] & \mathrm{X} \end{tikzcd}$$

where i and j are the canonical injections, i being thus an *affine* morphism; we deduce from (21.12.1) the existence of a morphism $j^0 : \mathrm{U}_1^0 \to \mathrm{U}^0$ making commutative the diagram

$$\begin{tikzcd} \mathrm{U}_1^0 \arrow[r, "j^0"] \arrow[d, "f_1^0"'] & \mathrm{U}^0 \arrow[d, "f^0"] \\ \mathrm{X}_1 \arrow[r, "i"'] & \mathrm{X} \end{tikzcd}$$

and hence, to prove that $x \in f^0(\mathrm{U}^0)$, it suffices to prove that $x \in f_1^0(\mathrm{U}_1^0)$. We can therefore replace X by X_1, and we can hence suppose that the ring A is *integral*. But A is a quotient of a regular Noetherian ring by the Cohen structure theorem (**0**, 19.8.8, (i)) and since it is integral, we can apply (5.11.1) with the family (x_a) reduced to the single maximal point of X; since $\mathrm{codim}(\mathrm{T}, \mathrm{X}) \geqslant 2$ by hypothesis, we see that $f_*(\mathscr{O}_{\mathrm{U}})$ is a *coherent* $\mathscr{O}_{\mathrm{X}}$-Module, hence the morphism $f^0 : \mathrm{U}^0 \to \mathrm{X}$ is *finite*; since this morphism is dominant (U being everywhere dense), it is surjective (**II**, 6.1.10), and hence $x \in f^0(\mathrm{U}^0)$.

(ii) It is a matter of proving that for every point $x \in \mathrm{T}$, we have $x \in f^0(\mathrm{U}^0)$; we will first show that we can reduce to the case where $\mathrm{X} = \mathrm{Spec}(\mathrm{A})$, where A is a *complete* local Noetherian ring, and x the closed point of X. For this, it suffices to make the change of base $h : \mathrm{X}' = \mathrm{Spec}(\mathrm{A}) \to \mathrm{X}$, where $\mathrm{A} = \widehat{\mathscr{O}}_{\mathrm{X},x}$; if we set $\mathrm{U}' = h^{-1}(\mathrm{U})$, $f' = f_{\{X'\}}$ the canonical injection $\mathrm{U}' \to \mathrm{X}'$, and since the morphism h is *flat*, it follows from (21.12.2) that if we prove that x belongs to $f'^0(\mathrm{U}'^0)$, then $x \in f^0(\mathrm{U}^0)$. By virtue of (6.1.1), we have made the desired reduction.

Let then X_1 be a closed reduced sub-prescheme of X having as underlying space an irreducible component of X, of *maximal* dimension, among those that contain T, and set $\mathrm{U}_1 = \mathrm{U} \cap \mathrm{X}_1$, $\mathrm{T}_1 = \mathrm{T} \cap \mathrm{X}_1 = \mathrm{X}_1 - \mathrm{U}_1$; we have therefore $\mathrm{codim}(\mathrm{T}_1, \mathrm{X}_1) \geqslant 2$ (**0**, 14.2.1), and the pair $(\mathrm{U}_1, \mathrm{X}_1)$ satisfies the same hypotheses (the pair (U, X) (X_1 being the spectrum of a quotient of A, hence a local Noetherian complete ring). On the other hand, by the commutative diagram above we see that $x \in f^0(\mathrm{U}^0)$ as we have seen, and we can hence suppose that the ring A is integral. But A is a quotient of a regular Noetherian ring by the Cohen structure theorem (**0**, 19.8.8, (i)) and since it is integral, we can apply (5.11.1) with the family (x_a) reduced to the single maximal point of X; since $\mathrm{codim}(\mathrm{T}, \mathrm{X}) \geqslant 2$ by hypothesis, we see that $f_*(\mathscr{O}_{\mathrm{U}})$ is a *coherent* $\mathscr{O}_{\mathrm{X}}$-Module, hence the morphism $f^0 : \mathrm{U}^0 \to \mathrm{X}$ is *finite*; since this morphism is dominant (U being everywhere dense), it is surjective (**II**, 6.1.10), and hence $x \in f^0(\mathrm{U}^0)$. □

Corollary (21.12.7). — *Let* X *be a locally Noetherian prescheme,* U *a sub-prescheme induced on an open of* X*,* $j : \mathrm{U} \to \mathrm{X}$ *the canonical immersion; suppose that* j *is an affine morphism. Then every irreducible component of* $\mathrm{T} = \mathrm{X} - \mathrm{U}$ *is of codimension* $\leqslant 1$ *(and hence everywhere dense).*

Proof. Suppose indeed that one of the irreducible components T_1 of T is of codimension $\geqslant 2$ in X. Replacing X if needed by a neighbourhood open of the generic point of T_1, we can suppose T irreducible and of codimension $\geqslant 2$. But then the hypothesis that $j : \mathrm{U} \to \mathrm{X}$ is affine implies that **IV-4** | 308
U^0 is identified with U and j^0 with j, in other words $j^0(\mathrm{U}^0) = \mathrm{U}$; but this contradicts the conclusion of (21.12.6) which, under the hypothesis of dimension, says that j^0 must be surjective. □

(21.12.8). Let A be a local Noetherian ring, $\mathrm{Y} = \mathrm{Spec}(\mathrm{A})$, y the unique closed point of Y, $\mathrm{Y}' = \mathrm{Y} - \{y\}$. Consider the following condition:

(W) *For every open* U *of* Y *contained in* Y'*, containing no irreducible component of* Y' *and such that the canonical injection* $\mathrm{U} \to \mathrm{Y}'$ *is affine,* U *is itself an affine open.*

This condition is implied by the following:

(W') *For every closed subset* T *of* Y *whose every irreducible component is of codimension* 1 *and such that for every irreducible component* Y_i *of* Y*,* $\mathrm{Y}_i \cap \mathrm{T} \neq \{y\}$*, the open* $\mathrm{U} = \mathrm{Y} - \mathrm{T}$ *is affine.*

Indeed, if (W') is satisfied and if U is an open of Y satisfying the hypothesis of (W), it follows from (21.12.7) that no irreducible component of $\mathrm{T} = \mathrm{Y} - \mathrm{U}$ can be of codimension $\geqslant 2$; since on the other hand U contains no irreducible component of $\mathrm{Y}_i - \{y\}$ of Y, condition (W') on Y, shows that U is affine.

We note that condition (W') simplifies when Y is *irreducible* and is then equivalent to the following:

(W'') *For every closed irreducible subset* T *of* Y*, of codimension* 1 *in* Y*,* $\mathrm{Y} - \mathrm{T}$ *is an affine open.*

Indeed, it is clear that (W') implies (W'') when Y is irreducible, and the converse is true by considering the irreducible components of T and of Y and noting that the intersection of a finite number of affine open subsets is an affine open (**I**, 5.5.6).

(21.12.9). *Examples.* — If A is a local Noetherian *factorial* ring, it satisfies (W') (and *a fortiori* (W)), since every prime ideal of height 1 is principal (**I**, 1.3.6). But there are local Noetherian rings which are not factorial, for example those of dimension $\leqslant 1$: indeed, as remarked in the proof of (21.12.6), (i), that *all* opens of Y are affine. One can on the other hand prove by using the theory of local duality (chap. III, 3e partie) that every local Noetherian ring of dimension 2 satisfies (W').

The interest of condition (W) resides in the following result:

Proposition (21.12.10). — *Let* X, Y *be two locally Noetherian preschemes,* $g : \mathrm{X} \to \mathrm{Y}$ *a morphism,* y *a closed point of* Y, $\mathrm{X}' = g^{-1}(\mathrm{Y}')$*; suppose that the morphism* $g' : \mathrm{X}' \to \mathrm{Y}'$ *restriction of* g *is an open immersion, and that the local ring* $\mathscr{O}_{\mathrm{Y},y}$ *satisfies the condition* (W) *of* (21.12.8)*. Then for every irreducible* Z *of* $\mathrm{X}_y = g^{-1}(y)$*, either* Z *is of codimension* $\leqslant 1$ *in* X*, or its generic point is isolated in* Z*. If* Z *is locally of finite type over* $\boldsymbol{k}(y)$*, the second alternative implies that* Z *is reduced to a single point.*

Proof. The last assertion of the statement results from the fact that Z is then a prescheme of Jacobson (**I**, 10.4.7), and since the set of closed points of Z is then dense in Z, the generic point of Z cannot be isolated in Z unless Z is reduced to a single point.

Suppose that there exists an irreducible component Z of X_y whose generic point z is not isolated in Z, and such that $\operatorname{codim}(\mathrm{Z}, \mathrm{X}) \geqslant 2$. The question being local on X and Y, suppose X and Y affine; since on the other hand $z \in \mathrm{X}_y$ is the generic point of T, we have $g(\mathrm{T}) \subset \overline{\{y\}} = \{y\}$, i.e. $\mathrm{T} = \mathrm{X}_y$. We are then exactly in the conditions of application of (21.12.10), in other words $\mathrm{T} = \mathrm{X}_y$. Set $\mathrm{X}_i = \overline{\{x_i\}}$. By hypothesis, z is not isolated in X_y, and since z is not a maximal point of X (this would imply $\operatorname{codim}(\mathrm{Z}, \mathrm{X}) = 0$), it follows that there exists x_i in X_y with $x_i \neq z$. Since $\mathrm{X}_i \neq \mathrm{Z}$, there exists X_i a point t_i such that if we set $\mathrm{T}_i = \overline{\{t_i\}}$, we have $\dim(\mathscr{O}_{\mathrm{T}_i,z}) = 1$ (10.5.9); this implies by hypothesis $t_i \notin \mathrm{Z}$, so t_i is not a generisation of z. Replacing X by $\mathrm{X} - \bigcup_i \mathrm{T}_i$, we see that g of $\mathrm{X}' \to \mathrm{X}'_y$ does not contain the images $g(t_i)$, so does not contain y. This being so, since X and Y are affines, the morphism $g : \mathrm{X} \to \mathrm{Y}$ is affine, hence $g' : \mathrm{X}' \to \mathrm{Y}'$ is also an affine morphism of the restriction $g' : \mathrm{X}' \to \mathrm{Y}'$. Set $\mathrm{Y}_1 = \operatorname{Spec}(\mathscr{O}_{\mathrm{Y},y})$, $\mathrm{Y}'_1 = \mathrm{Y}_1 - \{y\} = \mathrm{Y}' \cap \mathrm{Y}_1$, $\mathrm{U}_1 = \mathrm{U} \cap \mathrm{X}_1$, $\mathrm{T}_1 = \mathrm{T} \cap \mathrm{X}_1 = \mathrm{X}_1 - \mathrm{U}_1$; we have therefore $\operatorname{codim}(\mathrm{T}_1, \mathrm{X}_1) \geqslant 2$ (**0**, 14.2.1), and the pair $(\mathrm{U}_1, \mathrm{X}_1)$ satisfies the same hypotheses as the pair (U, X). This implies that U_1 is an open affine in Y_1 by virtue of (W); one would conclude from this that the open $\mathrm{X} - \mathrm{X}_y$ of X, isomorphic to U, would be affine, which contradicts (21.12.7); the proposition is thus proved. □

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Theorem (21.12.12). — (van der Waerden) — *Let* X, Y *be two locally Noetherian integral preschemes,* $g : \mathrm{X} \to \mathrm{Y}$ *a birational morphism and locally of finite type. Suppose furthermore that* Y *is normal and that for all* $y \in \mathrm{Y}$*, the local ring* $\mathscr{O}_{\mathrm{Y},y}$ *satisfies the condition* (W) *of* (21.12.8) *(conditions satisfied in particular when* Y *is* locally factorial (21.12.9)*). If* V *is the largest open of* X *such that the restriction* $g|\mathrm{V} : \mathrm{V} \to \mathrm{Y}$ *is a local isomorphism, all the irreducible components of* $\mathrm{T} = \mathrm{X} - \mathrm{V}$ *are of codimension* 1 *in* X*.* IV-4 | 310

Proof. We note that since g is birational, the open V is not empty; it suffices therefore to prove that a maximal point z of T cannot be isolated in X_y, where $y = g(z)$. But, up to restringing to an open neighbourhood of z, we can restrict to the case $\mathrm{T} = \mathrm{X}_y$; then all the fibres $g^{-1}(y')$ $(y' \in \mathrm{Y})$ would be empty or reduced to a single point, and it would follow from the "Main theorem" (8.12.10) that g would be a local isomorphism, contradicting the hypothesis $z \in \mathrm{T}$. □

Corollary (21.12.13). — *Suppose satisfied the hypotheses of* (21.12.12) *and in addition, suppose that* g *is quasi-finite at each of the points of* $\mathrm{X}^{(1)}$ *(recall that these are the points where* $\dim(\mathscr{O}_{\mathrm{X},x}) = 1$*); if furthermore* g *is separated,* g *is an open immersion.*

Proof. It suffices to prove the first assertion, the second being a consequence of the first and of (**I**, 8.2.8). It all comes down to proving, with the notations of (21.12.12), that $\mathrm{T} = \varnothing$; in the contrary case, a generic point z of an irreducible component of T would belong to $\mathrm{X}^{(1)}$ by virtue of (21.12.12), and by hypothesis it would be isolated in X_y with $y = g(z)$; but as we saw in the proof of (21.12.12) this is not possible. □

Remarks (21.12.14). — (i) The conclusion of (21.12.12) applies when Y is regular and integral, by virtue of the Auslander-Buchsbaum theorem (21.11.1). On the contrary, one knows examples where X and Y are normal algebraic schemes of dimension 3, over an algebraically closed field of any characteristic, and where the conclusion of (21.12.12) is no longer valid.

(ii) The set T of (21.12.12) is also the set of points where g is *ramified*. Indeed, if g is unramified at a point $x \in \mathrm{X}$, it is also unramified in an open neighbourhood of x (17.3.7), hence $g|\mathrm{U}$ is a separated, quasi-finite and birational morphism. Since Y is normal, we conclude from the "Main theorem" (**III**, 4.4.9) that g is an isomorphism local at the point x, hence $x \in \mathrm{X} - \mathrm{T}$. Conversely, g is evidently unramified at every point where it is a local isomorphism. This justifies the title given to this section.

(iii) Without supposing that Y is locally factorial, but instead assuming by contrast that X is intersection complete relative to Y (chap. V), we will see in chap. V that one has a more precise result than (21.12.12), expressing T as a 1-codimensional cycle of an $\mathscr{O}_X$-Module invertible, canonically associated to g. This applies notably when X and Y are both regular, or when g is a S-morphism, X and Y being smooth preschemes over S. **IV-4** | 311

(iv) One may ask whether (21.12.10) admits a converse; in other words, for a local Noetherian ring A, set $Y = \mathrm{Spec}(A)$, y denoting the closed point of Y, and for every locally Noetherian prescheme X and every morphism $g : X \to Y$ such that $g' : X' \to Y'$ is an open immersion, is every irreducible component of X_y of codimension $\leqslant 1$ in X or reduced to a single point, is it true that A satisfies condition (W) of (21.12.8)?

(v) Let Y be an integral *regular* prescheme, X a locally Noetherian *normal* prescheme, $g : X \to Y$ a morphism locally of finite type; suppose furthermore that, if η is the generic point of Y, the fibre X_η is *étale* over $\boldsymbol{k}(\eta)$, and let T′ be the complement of the largest open V′ of X such that $g|V' : V' \to Y$ is *étale*; is it true that all the irreducible components of T′ are of codimension 1? This is indeed the case as well when one moreover supposes that g is *locally quasi-finite* ("purity theorem" of Zariski-Nagata). One can show that the preceding conjecture would be a consequence of the following (and it would even be equivalent): for a regular local ring A contained in a local integral integrally closed ring B, such that B is a finite A-module, the open U of the points of Spec(B) for which Spec(B) is unramified over Spec(A) is affine, or equivalently (or also by the same argument (18.10.1), étale on Spec(A)) is *affine*.

Proposition (21.12.15). — *Let* S *be a prescheme,* $g : X \to S$ *a flat morphism locally of finite presentation, whose fibres* $X_s = g^{-1}(s)$ *are geometrically irreducible* (4.5.2), $h : Y \to S$ *a smooth morphism; denote by* $Y_s = h^{-1}(s)$ *the fibres of this morphism. Let* $f : X \to Y$ *be a proper S-morphism such that, for every maximal point* η *of* S, *the morphism* $f_\eta : X_\eta \to Y_\eta$ *is an isomorphism. Then* f *is an isomorphism.*

Proof. Since h is flat, every maximal point of Y lies above a maximal point of S (2.3.4), hence the union of the Y_η, as η ranges over the set of maximal points of S, is dense in Y; since the f_η are isomorphisms, f is dominant and hence *surjective* since it is proper. We must therefore show that f is an *open immersion*. For every $s \in S$, by (17.9.5), it suffices to prove that for all $s \in S$, $f_s : X_s \to Y_s$ is an open immersion. Since every $s \in S$ is a generisation of a maximal point η, by making the change of base $\mathrm{Spec}(\mathscr{O}_{S,s}) \to S$ we can reduce to the case where S is a local integral scheme of closed point s and generic point η. Since X_s and X_η are geometrically irreducible, the irreducible components of X_s and Y_s are of the same dimension. But by hypothesis X_s is irreducible, and since f_η is an isomorphism, X is irreducible and reduced (**I**, 3.2.1) and by flatness (**I**, 3.3.2), hence X is reduced (*ibid.*, 3.2.1) and by flatness, X is integral. Since $g : X \to S$ is flat, the maximal points of X are contained in X_η (2.3.4) and since f_η is an isomorphism, X is irreducible and the local ring at its generic point is a field; moreover, by flatness (3.3.2), X has no prime cycles of dimension 1, hence X is reduced (*ibid.*, 3.2.1). To prove that f is an open immersion, we note that the assertion is evident at the points of $X^{(1)}$ that belong to X_η; the only point of $X^{(1)}$ belonging to X_s is the generic x of X_s, by virtue of (6.1.1). Now, it follows from (2.4.6) and (14.2.2) that the morphisms g and h are equidimensional; as the irreducible components of X_s and Y_s are all of the same dimension. But by hypothesis X_s is irreducible, and since f_η is surjective, it follows from the same of $f_s : X_s \to Y_s$, and $\dim(f_s^{-1}(y)) = 0$, i.e. f is well *quasi-finite* at the point x, and the conclusion follows from the fact that we can apply the criterion (21.12.13); to see that f is quasi-finite at the points of $X^{(1)}$, we remark that the assertion is evident at those that belong to X_η; the only point of $X^{(1)}$ belonging to X_s is the generic x of X_s, by virtue of (6.1.1). Now, it follows from (2.4.6) and (14.2.2) that the morphisms g and h are equidimensional; as X_η and Y_η are isomorphic, the irreducible components of X_s and of Y_s all have the same dimension. But by hypothesis X_s is irreducible, and since f_η is surjective, hence so is f_s, and $\dim(f_s^{-1}(y)) = 0$, i.e. f is well *quasi-finite* at the point x, and the conclusion follows from (21.12.13). □ **IV-4** | 312

Corollary (21.12.16). — *Let* $g : X \to S$ *be a proper, flat morphism of finite presentation,* $h : Y \to S$ *a smooth, proper morphism of finite presentation,* $f : X \to Y$ *an S-morphism. Suppose that the fibres* $X_s = g^{-1}(s)$ *of* g *are geometrically irreducible. Let* U *be the set of* $s \in S$ *such that* $f_s : X_s \to Y_s$ *is an* *isomorphism. Then* U *is open and closed in* S*, and the morphism* $g^{-1}(\mathrm{U}) \to h^{-1}(\mathrm{U})$*, restriction of* f*, is an isomorphism.*

Proof. The last assertion follows from the first and from (17.9.5). We already know (9.6.1), (xi) that U is locally constructible in S. By virtue of (1.10.1), it suffices to prove that U is stable by *specialisation* and by *generisation*. To prove these assertions, one can, by a change of base $\mathrm{Spec}(\mathscr{O}_{\mathrm{S},s}) \to \mathrm{S}$, reduce to the case where S is a local integral scheme of closed point s and generic point η.

Stability by specialisation follows from (21.12.15) (noting that the projection of the closed point of S is the closed point of S_0) in order to reduce to the case where moreover S is *Noetherian*; but then the conclusion follows from (**III**, 4.6.7, (ii)). **IV-4** | 313

Stability by generisation follows from (21.12.15). □

Remarks (21.12.17). — (i) The conclusion of the proposition (21.12.15) is no longer valid if we eliminate the hypothesis that the fibres X_s are irreducible. Take indeed $\mathrm{S} = \mathrm{Spec}(\mathrm{A})$, where A is a discrete valuation ring, $\mathrm{Y} = \mathbb{P}^1_{\mathrm{S}}$, which is proper and smooth over S (17.3.9). Denote moreover s and η the closed point and the generic point of S; let z be a closed point of Y_s, for example a rational point $k = \boldsymbol{k}(s)$, and let X be the Y-prescheme obtained by blowing up the point z. As in the proof of (15.1.6), if $f : \mathrm{X} \to \mathrm{Y}$ is the structural morphism, $f^{-1}(z)$ in X_s is isomorphic to the complement of Z_1, so $\mathrm{Z}_1 = f^{-1}(z)$ and Z_2, the closure in X_s of the complement of Z_1, are the two irreducible components of X_s. On the other hand, f is evidently proper and $g = h \circ f$ is flat, since X is integral (**II**, 8.1.4) and for every affine open U of X, $\Gamma(\mathrm{U}, \mathscr{O}_{\mathrm{X}})$ is a torsion-free A-module, hence flat since A is a discrete valuation ring (**0**, 6.3.4). It is clear that $f_\eta : \mathrm{X}_\eta \to \mathrm{Y}_\eta$ is an isomorphism, but f_s is not an isomorphism, although $f_s : \mathrm{X}_s \to \mathrm{Y}_s$ is not reduced to a single point for the closed point z, f'' is not an isomorphism. **IV-4** | 313

(ii) The conclusion of (21.12.15) also fails if, under the hypotheses, we suppress the hypothesis that f is *proper*. It suffices for this to see to define S and Y as in (i), but to replace X by the complement X'_2 of Z_2 in the prescheme X defined in (i), f by the restriction $f' : \mathrm{X}' \to \mathrm{Y}$ of f; the morphism $g' : \mathrm{X}' \to \mathrm{S}$, restriction of g, is again flat, and this time X'_2 is geometrically irreducible; X'_2 is moreover an affine scheme isomorphic to the complement of a closed point in $\mathbb{P}^1_k$; since its image under f'_s is not reduced to the closed point z, f' is not a proper morphism and f'_s is not an isomorphism, although f'_η is.

(iii) It is possible that the statement of the proposition (21.12.15) remains valid when we replace the word "isomorphism" by "étale morphism" (cf. (21.12.14), (v)). It will then also be the same of (21.12.16).

21.13. Parafactorial pairs. Parafactorial local rings

Definition (21.13.1). — Let X be a ringed space, Y a closed subset of X, $\mathrm{U} = \mathrm{X} - \mathrm{Y}$. We say that the pair (X, Y) is *parafactorial* if, for every open V of X, the restriction functor $\mathscr{L} \mapsto \mathscr{L}|(\mathrm{U} \cap \mathrm{V})$, from the category of $\mathscr{O}_{\mathrm{V}}$-Modules invertibles into that of $\mathscr{O}_{\mathrm{U}\cap\mathrm{V}}$-Modules invertibles, is an *equivalence of categories*.

To say that the pair (X, Y) is parafactorial means therefore that, for every open V of X, the following conditions are satisfied:

1º the functor $\mathscr{L} \mapsto \mathscr{L}|(\mathrm{U} \cap \mathrm{V})$ is *fully faithful*, in other words, for two $\mathscr{O}_{\mathrm{V}}$-Modules invertibles $\mathscr{L}, \mathscr{L}'$, the map of restriction

$$\mathrm{Hom}_{\mathscr{O}_{\mathrm{V}}}(\mathscr{L}, \mathscr{L}') \longrightarrow \mathrm{Hom}_{\mathscr{O}_{\mathrm{U}\cap\mathrm{V}}}(\mathscr{L}|(\mathrm{U} \cap \mathrm{V}), \mathscr{L}'|(\mathrm{U} \cap \mathrm{V}))$$

is *bijective*;

2º the functor $\mathscr{L} \mapsto \mathscr{L}|(\mathrm{U} \cap \mathrm{V})$ is *essentially surjective*, in other words, for every $\mathscr{O}_{\mathrm{U}\cap\mathrm{V}}$-Module invertible $\mathscr{L}_0$, there exists an $\mathscr{O}_{\mathrm{V}}$-Module invertible $\mathscr{L}$ such that $\mathscr{L}_0$ is isomorphic to $\mathscr{L}|(\mathrm{U} \cap$ **IV-4** | 314
$\mathrm{V})$; this is again expressed by saying that the canonical homomorphism (21.3.2.4)

$$\mathrm{Pic}(\mathrm{V}) \longrightarrow \mathrm{Pic}(\mathrm{U} \cap \mathrm{V})$$

is *surjective*.

Lemma (21.13.2). — *Let* $f : \mathrm{X}' \to \mathrm{X}$ *be a morphism of ringed spaces; for every open* V *of* X*, denote* $f_{\mathrm{V}} : f^{-1}(\mathrm{V}) \to \mathrm{V}$ *the restriction of* f*. The following conditions are equivalent:*

a) *For every open* V *of* X, *the functor* $\mathscr{E} \mapsto f_{\mathrm{V}}^*(\mathscr{E})$ *from the category of* $\mathscr{O}_{\mathrm{V}}$*-Modules locally free of finite rank, into the category of* $\mathscr{O}_{f^{-1}(\mathrm{V})}$*-Modules locally free of finite rank, is faithful (resp. fully faithful).*

a') *For every open* V *of* X, *the functor* $\mathscr{L} \mapsto f_{\mathrm{V}}^*(\mathscr{L})$ *from the category of* $\mathscr{O}_{\mathrm{V}}$*-Modules locally free of rank* 1 *into the category of* $\mathscr{O}_{f^{-1}(\mathrm{V})}$*-Modules locally free of rank* 1 *is faithful (resp. fully faithful).*

b) *For every open* V *of* X, *the canonical homomorphism* ($\mathbf{0}_{\mathrm{I}}$, 4.4.3.2) $\mathscr{E} \to (f_{\mathrm{V}})_*(f_{\mathrm{V}}^*(\mathscr{E}))$ *is injective (resp. an isomorphism) for every* $\mathscr{O}_{\mathrm{V}}$*-Module locally free of finite rank* $\mathscr{E}$.

b') *The canonical homomorphism* $\mathscr{O}_{\mathrm{X}} \to f_*(\mathscr{O}_{\mathrm{X}'})$ *is a monomorphism (resp. an isomorphism) for every* $\mathscr{O}_{\mathrm{V}}$*-Module locally free of finite rank* $\mathscr{E}$.

Suppose that the canonical homomorphism $\mathscr{O}_{\mathrm{X}} \to f_*(\mathscr{O}_{\mathrm{X}'})$ *is bijective. Then, for every* $\mathscr{O}_{\mathrm{X}'}$*-Module locally free of finite rank* $\mathscr{F}$, $\mathscr{F}$ *is of the form* $f^*(\mathscr{E})$, *where* $\mathscr{E}$ *is an* $\mathscr{O}_{\mathrm{X}}$*-Module locally free of finite rank, if and only if the two following conditions are satisfied:*

(i) $f_*(\mathscr{E}')$ *is an* $\mathscr{O}_{\mathrm{X}}$*-Module locally free;*

(ii) *the canonical homomorphism* ($\mathbf{0}_{\mathrm{I}}$, 4.4.3.3) $f^*(f_*(\mathscr{E}')) \to \mathscr{E}'$ *is an isomorphism.*

When these two conditions are satisfied, $\mathscr{E}$ *is isomorphic to* $f_*(\mathscr{E}')$.

Proof. To say that the functor $\mathscr{E} \mapsto (j_{\mathrm{V}})_*(\mathscr{E}|(\mathrm{U} \cap \mathrm{V}))$ is injective (resp. bijective) for every open V of X, the canonical homomorphism $\mathscr{E} \to (j_{\mathrm{V}})_*(j_{\mathrm{V}}^*(\mathscr{E}))$ is injective (resp. bijective); since this must take place in replacing V by an open $\mathrm{W} \subset \mathrm{V}$ and $\mathscr{E}_1$, $\mathscr{E}_2$ by $\mathscr{E}_1|\mathrm{W}$, $\mathscr{E}_2|\mathrm{W}$, and since $\mathrm{Hom}_{\mathscr{O}_{\mathrm{V}}}(\mathscr{E}_1, \mathscr{E}_2|\mathrm{W}) = \Gamma(\mathrm{W}, \mathscr{H}om_{\mathscr{O}_{\mathrm{V}}}(\mathscr{E}_1, \mathscr{E}_2))$, the condition also means that the canonical homomorphism of sheaves **IV-4 | 315**

$$\mathscr{H}om_{\mathscr{O}_{\mathrm{V}}}(\mathscr{E}_1, \mathscr{E}_2) \longrightarrow (f_{\mathrm{V}})_*(f_{\mathrm{V}}^*(\mathscr{H}om_{\mathscr{O}_{\mathrm{V}}}(\mathscr{E}_1, \mathscr{E}_2))) \tag{21.13.2.1}$$

is injective (resp. bijective). But since $\mathscr{E}_1$ and $\mathscr{E}_2$ are locally free of finite rank, $\mathscr{H}om_{\mathscr{O}_{\mathrm{V}}}(\mathscr{E}_1, \mathscr{E}_2)$, isomorphic to $\mathscr{E}_1 \otimes_{\mathscr{O}_{\mathrm{V}}} \mathscr{E}_2$ ($\mathbf{0}_{\mathrm{I}}$, 5.4.2) is also locally free of finite rank; this already shows that b) implies a), and b') is a particular case of a'). It is clear that a') is a particular case of a). Conversely, if b) is a local property, it can, for the verification, be restricted to the case where $\mathscr{E} = \mathscr{O}_{\mathrm{V}}^n$, and this shows that b') implies b).

If the canonical homomorphism $\mathscr{O}_{\mathrm{X}} \to f_*(\mathscr{O}_{\mathrm{X}'})$ is bijective, and if the conditions (i) and (ii) are satisfied, it is clear that $\mathscr{E}' = f^*(\mathscr{E})$ with $\mathscr{E} = f_*(\mathscr{E}')$, up to canonical isomorphism. □

In this paragraph, we will apply the preceding lemma to a canonical injection $j : \mathrm{U} \to \mathrm{X}$, where U is the ringed space induced by X on an *open* of X. With these notations, (21.13.2) translates as:

Corollary (21.13.3). — *Let* X *be a ringed space,* Y *a closed subset of* X, $\mathrm{U} = \mathrm{X} - \mathrm{Y}$, $j : \mathrm{U} \to \mathrm{X}$ *the canonical injection. The following conditions are equivalent:*

a) *For every open* V *of* X, *the functor restriction* $\mathscr{E} \mapsto \mathscr{E}|(\mathrm{U} \cap \mathrm{V})$ *from the category of* $\mathscr{O}_{\mathrm{V}}$*-Modules locally free of finite rank into the category of* $\mathscr{O}_{\mathrm{U} \cap \mathrm{V}}$*-Modules locally free of finite rank is faithful (resp. fully faithful).*

b) *For every open* V *of* X, *the canonical homomorphism* $\mathscr{E} \to (j_{\mathrm{V}})_*((\mathscr{E}|(\mathrm{U} \cap \mathrm{V}))$ *is injective (resp. bijective) for every* $\mathscr{O}_{\mathrm{V}}$*-Module locally free of finite rank* $\mathscr{E}$.

b') *The canonical homomorphism* $\mathscr{O}_{\mathrm{X}} \to j_*(\mathscr{O}_{\mathrm{U}})$ *is injective (resp. bijective).*

Suppose that the canonical homomorphism $\mathscr{O}_{\mathrm{X}} \to j_*(\mathscr{O}_{\mathrm{U}})$ *is bijective. Then, for an* $\mathscr{O}_{\mathrm{U}}$*-Module locally free of finite rank* $\mathscr{F}$ *to be of the form* $\mathscr{E}|\mathrm{U}$, *where* $\mathscr{E}$ *is an* $\mathscr{O}_{\mathrm{X}}$*-Module locally free of finite rank, it is necessary and sufficient that* $j_*(\mathscr{F})$ *is an* $\mathscr{O}_{\mathrm{X}}$*-Module locally free of finite rank, and in this case, we can take* $\mathscr{E} = j_*(\mathscr{F})$.

Lemma (21.13.4). — *Let* X *be a locally Noetherian prescheme,* Y *a closed subset of* X. *For the canonical homomorphism* $\mathscr{O}_{\mathrm{X}} \to j_*(\mathscr{O}_{\mathrm{U}})$ *to be injective (resp. bijective), it suffices that* $\mathrm{prof}_{\mathrm{Y}}(\mathscr{O}_{\mathrm{X}}) \geqslant 1$ *(resp.* $\mathrm{prof}_{\mathrm{Y}}(\mathscr{O}_{\mathrm{X}}) \geqslant 2$*) (in other words, that* $\mathrm{prof}(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$*) for all* $x \in \mathrm{Y}$.

Proof. These assertions are indeed particular cases of (5.10.2) and (5.10.5). □

Proposition (21.13.5). — *Let* X *be a ringed space,* Y *a closed subset of* X, $\mathrm{U} = \mathrm{X} - \mathrm{Y}$, $j : \mathrm{U} \to \mathrm{X}$ *the canonical injection. For the pair* (X, Y) *to be parafactorial, it is necessary and sufficient that the canonical homomorphism* $\mathscr{O}_{\mathrm{X}} \to j_*(\mathscr{O}_{\mathrm{U}})$ *is bijective, and that for every open* V *of* X *and every* $\mathscr{O}_{\mathrm{U} \cap \mathrm{V}}$*-Module invertible* $\mathscr{L}$, $(j_{\mathrm{V}})_*(\mathscr{L})$ *is an* $\mathscr{O}_{\mathrm{V}}$*-Module invertible (notation of (21.13.3)).*

Proof. This is an immediate consequence of the definition (21.13.1) and of (21.13.3). □

Corollary (21.13.6). — *Let* X *be a ringed space,* Y *a closed subset of* X.

(i) If the pair (X,Y) *is parafactorial, then so is* $(\mathrm{W},\mathrm{Y}\cap\mathrm{W})$ *for every open* W *of* X*. Conversely, if* (W_α) *is an open covering of* X *such that each of the pairs* $(\mathrm{W}_\alpha,\mathrm{Y}\cap\mathrm{W}_\alpha)$ *is parafactorial, then the pair* (X,Y) *is parafactorial.*

(ii) If the pair (X,Y) *is parafactorial, then so is* (X,Y') *for every closed subset* Y' *of* Y.

(iii) Suppose that X *is a prescheme, and let* X' *be a prescheme,* $f:\mathrm{X}'\to\mathrm{X}$ *a faithfully flat and quasi-compact morphism; set* $\mathrm{Y}'=f^{-1}(\mathrm{Y})$*. Suppose that the pair* $(\mathrm{X}',\mathrm{Y}')$ *is parafactorial and that the open* $\mathrm{U}=\mathrm{X}-\mathrm{Y}$ *is retrocompact in* X $(\mathbf{0}_{\mathrm{III}}, 9.1.1)$*. Then the pair* (X,Y) *is parafactorial.*

Proof. (i) Since the fact that the canonical homomorphism $\mathscr{O}_\mathrm{X}\to j_*(\mathscr{O}_\mathrm{U})$ is bijective is a local property on X, it all comes down to seeing (by virtue of (21.13.5)) by denoting by j_α the injection canonical $\mathrm{U}\cap\mathrm{W}_\alpha\to\mathrm{W}_\alpha$, that for all α and every $\mathscr{O}_\mathrm{U}$-Module invertible $\mathscr{L}$, $(j_\alpha)_*(\mathscr{L}|(\mathrm{U}\cap\mathrm{W}_\alpha))$ is a $\mathscr{O}_{\mathrm{W}_\alpha}$-Module invertible. But the property of being an $\mathscr{O}_\mathrm{X}$-Module invertible is local on X; since $j_*(\mathscr{L})|\mathrm{W}_\alpha=(j_\alpha)_*(\mathscr{L}|(\mathrm{U}\cap\mathrm{W}_\alpha))$, this proves our assertion. IV-4 | 316

(ii) Set $\mathrm{U}'=\mathrm{X}-\mathrm{Y}'$ and let $j'':\mathrm{U}'\to\mathrm{X}$ be the canonical injection. To say that the canonical homomorphism $\mathscr{O}_\mathrm{X}\to j''_*(\mathscr{O}_{\mathrm{U}'})$ is bijective amounts to saying that for every open V of X, the homomorphism $\Gamma(\mathrm{V},\mathscr{O}_\mathrm{X})\to\Gamma(\mathrm{V}\cap\mathrm{U}',\mathscr{O}_\mathrm{X})$ is bijective; but the homomorphism composed

$$\Gamma(\mathrm{V},\mathscr{O}_\mathrm{X})\longrightarrow\Gamma(\mathrm{V}\cap\mathrm{U}',\mathscr{O}_\mathrm{X})\longrightarrow\Gamma(\mathrm{V}\cap\mathrm{U},\mathscr{O}_\mathrm{X})$$

is bijective by hypothesis (by replacing V by $\mathrm{V}\cap\mathrm{U}'$, we see that $\Gamma(\mathrm{V}\cap\mathrm{U}',\mathscr{O}_\mathrm{X})\to\Gamma(\mathrm{V}\cap\mathrm{U},\mathscr{O}_\mathrm{X})$ is bijective. Noting then that if $j'':\mathrm{U}'\to\mathrm{X}$ is the canonical injection and if $\mathscr{L}'$ is an $\mathscr{O}_\mathrm{U}$-Module invertible, then $j''_*(\mathscr{L}')|\mathrm{U}$ is an $\mathscr{O}_\mathrm{X}$-Module invertible. If we denote by $\mathscr{L}_1$ the $\mathscr{O}_{\mathrm{V}_1}$-Module invertible equal to $\mathscr{L}''$ in W and to $\mathscr{L}'$ in $\mathrm{V}\cap\mathrm{U}'$, on concludes from the fact that $\mathscr{O}_\mathrm{X}\to j_*(\mathscr{O}_\mathrm{U})$ is bijective and from (21.13.2), *b*)) that $j_*(\mathscr{L})|V_1$ is an $\mathscr{O}_\mathrm{X}$-Module invertible. It suffices then to replace X by an open V and X′ by $f^{-1}(\mathrm{V})$ in what precedes.

(iii) Set $\mathrm{U}'=\mathrm{X}'-\mathrm{Y}'=f^{-1}(\mathrm{U})$ and note that since it is a question of preschemes, on can write $\mathrm{U}'=\mathrm{U}\times_\mathrm{X}\mathrm{X}'$; let $f_\mathrm{U}:f^{-1}(\mathrm{U})\to\mathrm{U}$ be the restriction of f and let $j':\mathrm{U}'\to\mathrm{X}'$ be the canonical injection, which we also write $j_{(\mathrm{X}')}$. Let us first show that the canonical homomorphism $\rho:\mathscr{O}_\mathrm{X}\to j_*(\mathscr{O}_\mathrm{U})$ is bijective; for this, it is necessary that for every open V of X, the canonical homomorphism $\mathscr{O}_{\mathrm{X}',x'}\to j'_*(\mathscr{O}_{\mathrm{U}'})$ is bijective; since j is quasi-compact and separated by hypothesis, the morphism f is quasi-compact and separated, hence by (2.3.1), $\mathscr{O}_{\mathrm{X}'}=f^*(\mathscr{O}_\mathrm{X})$, we see that ρ is bijective since $f^*(\rho)$ is bijective and f is faithfully flat (2.2.7). Let then $\mathscr{L}'=f_\mathrm{U}^*(\mathscr{L})$ be a $\mathscr{O}_\mathrm{U}$-Module invertible $\mathscr{L}$. By hypothesis $j'_*(\mathscr{L}')=j'_*(f_\mathrm{U}^*(\mathscr{L}))$ is an $\mathscr{O}_{\mathrm{X}'}$-Module invertible, which is isomorphic to $f^*(j_*(\mathscr{L}))$ by (2.3.1) as $\mathscr{O}_{\mathrm{X}'}=f^*(\mathscr{O}_\mathrm{X})$; we conclude then that ρ is bijective since f is faithfully flat. Since f is faithfully flat and quasi-compact, this implies that $j_*(\mathscr{L})$ is an $\mathscr{O}_\mathrm{X}$-Module locally free (2.5.2). For the proof, it suffices to replace X by an open V and X′ by $f^{-1}(\mathrm{V})$ in what precedes. □

Definition (21.13.7). — We say that a *local ring* A is parafactorial if, setting $\mathrm{X}=\mathrm{Spec}(\mathrm{A})$ and denoting by a the closed point of X, the pair $(\mathrm{X},\{a\})$ is parafactorial.

Proposition (21.13.8). — *The notations being those of* (21.13.7)*, set* $\mathrm{U}=\mathrm{X}-\{a\}$*. For* A *to be parafactorial, it is necessary and sufficient that it satisfies the following conditions:*

(i) The canonical homomorphism $\mathrm{A}=\Gamma(\mathrm{X},\mathscr{O}_\mathrm{X})\to\Gamma(\mathrm{U},\mathscr{O}_\mathrm{X})$ *is bijective.*

(ii) $\mathrm{Pic}(\mathrm{U})=0$.

If furthermore A *is Noetherian, condition (i) is equivalent to* IV-4 | 317

(i bis) $\mathrm{prof}(\mathrm{A})\geqslant 2$.

Proof. Indeed, the only open of X containing a is X itself, and hence every $\mathscr{O}_\mathrm{X}$-Module invertible is isomorphic to $\mathscr{O}_\mathrm{X}$, i.e. $\mathrm{Pic}(\mathrm{X})=0$; the first assertion follows therefore from the definition (21.13.1) and from (21.13.3). The second assertion is nothing but a particular case of (21.13.4). □

(21.13.9). *Examples.* — (i) A local Noetherian parafactorial ring is necessarily of dimension $\geqslant 2$ by virtue of (21.13.8); in other words a local Noetherian ring of dimension $\leqslant 1$ is not parafactorial.

(ii) A local Noetherian *factorial* ring A of dimension $\geqslant 2$ is parafactorial, as follows from (21.13.8) and (21.6.14).

(iii) If A is a local Noetherian ring of dimension $\geqslant 3$ and parafactorial, it is not necessarily normal nor even reduced. Consider a regular local ring B of dimension $\geqslant 3$, and let $\mathrm{A}=\mathrm{D}_\mathrm{B}(\mathrm{B})$ $(\mathbf{0}, 18.2.3)$, isomorphic to $B[T]/(T^2)$; we see immediately that $\mathrm{prof}(A) = \mathrm{prof}(B) \geqslant 3$. To show that A is parafactorial, it suffices therefore, with the notations of (21.13.8), to prove that $\mathrm{Pic}(U) = 0$. Let $\mathfrak{J}$ be the ideal of the augmentation $A \to B$, which is such that $\mathfrak{J}^2 = 0$ and which, as a B-module, is isomorphic to B. Since $B = A/\mathfrak{J}$, $X = \mathrm{Spec}(A)$ and $X_0 = \mathrm{Spec}(B)$ have the same underlying space; if $\mathscr{J} = \widetilde{\mathfrak{J}}$, X_0 is the sub-prescheme defined by $\mathscr{J}$; we will denote U_0 the prescheme induced by X_0 on the open U. For every $z \in \mathfrak{J}$, set $\varphi(z) = 1 + z$; since $(1+z)(1-z) = 1$, φ is the kernel of the surjective canonical homomorphism of groups of multiplicative groups $A^* \to B^*$; in other words, we have an exact sequence of commutative groups

$$0 \longrightarrow \mathfrak{J} \xrightarrow{\varphi} A^* \longrightarrow B^* \longrightarrow 1$$

(the three last groups being multiplicative, the two first being additive). For the same reason, for every $t \in A$, we have the exact sequence

$$0 \longrightarrow \mathfrak{J}_t \xrightarrow{\varphi_t} (A_t)^* \longrightarrow (B_{t_0})^* \longrightarrow 1$$

denoting by t_0 the image of t in B, since $B_t = A_t/\mathfrak{J}_t$; in other words, we have an exact sequence of sheaves of commutative groups on the topological space X

$$0 \longrightarrow \mathscr{J} \xrightarrow{u} \mathscr{O}_X^* \longrightarrow \mathscr{O}_{X_0}^* \longrightarrow 1$$

whence by restriction to the open U, an exact sequence

$$0 \longrightarrow \mathscr{J}|U \longrightarrow \mathscr{O}_U^* \longrightarrow \mathscr{O}_{U_0}^* \longrightarrow 1.$$

By the cohomology exact sequence, we deduce an exact sequence

(21.13.9.1) $$H^1(U, \mathscr{J}) \longrightarrow H^1(U, \mathscr{O}_U^*) \xrightarrow{u} H^1(U, \mathscr{O}_{U_0}^*).$$

But since $\mathfrak{J}$ is a B-module isomorphic to B, we have $H^1(U_0, \mathscr{O}_{U_0}) = H^1(U_0, \mathscr{O}_{U_0})$, and it follows from the chap. III, 3e partie (see also [?, III, Exemple III-1]) that we have $H^1(U_0, \mathscr{O}_{U_0}) = 0$ by reason of the relation $\mathrm{prof}(B) \geqslant 3$. On the other hand, since B is factorial, $H^1(U, \mathscr{O}_{U_0}^*) = \mathrm{Pic}(U_0) = 0$; we obtain therefore from the preceding exact sequence $\mathrm{Pic}(U) = H^1(U, \mathscr{O}_U^*) = 0$. **IV-4** | 318

(iv) There also exist local Noetherian rings of dimension 3 which are integral, integrally closed and parafactorial, but not factorial. Let indeed B be a local Noetherian ring, complete, integral, integrally closed, of dimension $\geqslant 2$ and not factorial (for example the completion of the localised polynomial algebra $k[U, V, W]/(U^2 - UV)$ at the ideal image of $(U) + (V) + (W)$). Then it follows from what we will see below (21.14.2) that the local ring $A = B[[T]]$ of formal power series in T over B is parafactorial, but it is not factorial, without which B would be so by virtue of (21.13.12) below.

(v) One can show that an intersection complete local ring of absolute dimension (19.3.1) of dimension $\geqslant 4$, is parafactorial (cf. [?, XI, 3.13 (i)]).

(vi) We have seen (Remark (ii)) that every local Noetherian factorial ring A of dimension 2 is parafactorial. But there are local Noetherian rings of dimension 2 which are parafactorial and not factorial. One can show that, for an arbitrary local Noetherian ring A of dimension 2 to be factorial and not parafactorial, it is necessary and sufficient that the following three conditions be satisfied:

1º A is a Cohen-Macaulay ring (in other words $\mathrm{prof}(A) = 2$).

2º A is integral and if A' is its integral closure, A' is factorial and is a finite A-algebra.

3º Let $\mathfrak{J}$ be the conductor of A in A', or equivalently the largest ideal of A' contained in A); set $B = A/\mathfrak{J}$, $B' = A'/\mathfrak{J}$; then $\dim(B) = 1$ (which implies $A' \neq A$, i.e. A is not integrally closed), and the canonical map $\mathrm{Div}(B) \to \mathrm{Div}(B')$ (21.4.5) is *surjective*.

One can furthermore show that these conditions imply the following property: 4º the ring B' is also reduced, and the morphism $\mathrm{Spec}(B') \to \mathrm{Spec}(B)$ is bijective.

If we set $X = \mathrm{Spec}(A)$, $X' = \mathrm{Spec}(A')$, $Y = \mathrm{Spec}(B)$, $Y' = \mathrm{Spec}(B')$, so that Y (resp. Y') is defined by the ideal $\mathfrak{J}$ of A (resp. A'), the structural morphism $f : X' \to X$ is an isomorphism on $X' - Y'$ over $X - Y$ since f is *bijective* and in other words, X is a *local* Noetherian *unibranch* prescheme (and in particular A' is a *local* Noetherian ring); in general X is not geometrically unibranch. The space Y, of dimension 1, is formed of the closed point x of X and the maximal points y_i $(1 \leqslant i \leqslant r)$ of Y, and the unique point y_i' of Y' above y_i is also a maximal point of Y'; since B and B' are reduced, we deduce from this that $\mathfrak{m}_{\mathscr{O}_{X',y_i'}} = \mathfrak{m}_{y_i}$; for $z = y_i$ and z' the unique point of $f^{-1}(z)$, hence for all

$z \in \mathrm{U} = \mathrm{X} - (x)$ and all z' the unique point of $f^{-1}(z)$; this relation is also verified obviously for $z \notin \mathrm{Y}$ and z' the unique point of $f^{-1}(z)$ (which is not characteristic 0 (21.1.1)) on U (but *not* étale in general).

We will now limit ourselves to demonstrating that conditions 1°, 2°, 3° above are *sufficient* for A to be parafactorial. Since A′ is factorial by virtue of condition 1° and (21.13.8), it suffices to show that $\mathrm{Pic}(\mathrm{U}) = 0$. Since A′ is factorial, by virtue of 2°, we obtain and we deduce from (21.8.5), (ii) an exact sequence **IV-4** | 319

$$1 \longrightarrow (\prod_{s\in\mathrm{S}} \mathscr{O}^*_{\mathrm{U},s} / \operatorname{Im}\Gamma(\mathrm{U}', \mathscr{O}^*_{\mathrm{U}'})) \longrightarrow \mathrm{Pic}(\mathrm{U}) \longrightarrow \mathrm{Pic}(\mathrm{U}') \longrightarrow 0$$

and it is a matter of showing that the second term of this exact sequence is reduced to the neutral element, i.e. for the set y_i $(1 \leqslant i \leqslant r)$ of A and A′ corresponding to points y_i, y'_i, and we note that $\mathfrak{p}_i \cap \mathrm{A} = \mathfrak{p}_i$; denoting for each i an element inversible a'_i/s_i of $\mathrm{A}'_{\mathfrak{p}_i}$, with $a'_i \in \mathrm{A}'$, $s_i \notin \mathfrak{p}_i$; since we have only to examine that the quotient groups $\mathrm{A}^*_{\mathfrak{p}_i}/\mathrm{A}^*_{\mathfrak{p}_i}$ are the same; we can suppose $s_i = 1$ for all i; let $b'_i \in \mathrm{B}'_i$ be the canonical image of a'_i, so that $b'_i/1$ is an element $\neq 0$ of $\boldsymbol{k}(y'_i) = \mathscr{O}_{v',y'_i}$. The open $\mathrm{U} \cap \mathrm{Y}$ formed by the y_i is an open affine of the form $\mathrm{D}(t)$ in Y, with $t \in \mathfrak{m}$, the maximal ideal of A; it exists therefore an invertible element $b' \in \mathrm{B}'$ such that the canonical images of $b'_i/1$ are the images in $\mathrm{A}'_{\mathfrak{p}_i}/\mathrm{A}_{\mathfrak{p}_i}$ are the same, which finishes the proof.

Proposition (21.13.10). — *Let* X *be a locally Noetherian prescheme,* Y *a closed subset of* X*. For the pair* (X, Y) *to be parafactorial, it is necessary and sufficient that, for all* $y \in \mathrm{Y}$*, the local ring* $\mathscr{O}_{\mathrm{X},y}$ *is parafactorial.*

Proof. Set $\mathrm{U} = \mathrm{X} - \mathrm{Y}$ and let $j : \mathrm{U} \to \mathrm{X}$ be the canonical injection. **IV-4** | 320

Let us first show that if the pair (X, Y) is parafactorial, each of the rings $\mathscr{O}_{\mathrm{X},y}$ $(y \in \mathrm{Y})$ is parafactorial. Taking into account the preceding remark, it all amounts to seeing that if $\mathrm{T}_y = \mathrm{Spec}(\mathscr{O}_{\mathrm{X},y})$ and $\mathrm{U}_y = \mathrm{T}_y - \{y\}$, every $\mathscr{O}_{\mathrm{U}_y}$-Module invertible $\mathscr{L}_0$ is isomorphic to $\mathscr{O}_{\mathrm{U}_y}$. Now, when V ranges over the open neighbourhoods of y in X, it follows from (8.2.13) and (1.2.4.2) that the prescheme U_y is projective limit of the preschemes induced by X on the opens $\mathrm{V} - (\mathrm{V} \cap \overline{\{y\}})$, which are quasi-compact since X is Noetherian. Since the $\mathrm{U}_y = \mathrm{V} - (\mathrm{V} \cap \overline{\{y\}})$ are separated, it follows then from (8.5.2), (ii) and (8.5.5) that there is a neighbourhood open affine V of y in X and an $\mathscr{O}_\mathrm{V}$-Module invertible $\mathscr{L}'_\mathrm{V}$ such that the sheaf induced by $j_*(\mathscr{L})$ on U_y is $\mathscr{L}_0$. But $\mathscr{O}_{\mathrm{X},y}$ is parafactorial, so the pair $(\mathrm{V}, \mathrm{V} \cap \overline{\{y\}})$ is parafactorial by (21.13.6), (i)), so there exists an $\mathscr{O}_\mathrm{V}$-Module invertible $\mathscr{L}_\mathrm{V}$ inducing $\mathscr{L}'_\mathrm{V}$ on U_y. But as T_y is projective limit, by replacing V by a smaller neighbourhood open of y, one can suppose that $\mathscr{L}_\mathrm{V}$ is isomorphic to $\mathscr{O}_\mathrm{V}$, whence $\mathscr{L}_0$ is isomorphic to $\mathscr{O}_{\mathrm{U}_y}$, hence invertible. But this contradicts the definition of V, and concludes the proof of (21.13.10).

Conversely, let us prove that if all the $\mathscr{O}_{\mathrm{X},y}$ $(y \in \mathrm{Y})$ are parafactorial, the pair (X, Y) is parafactorial. This is evidently reduced, taking into account the remark at the beginning, to showing that for every $\mathscr{O}_\mathrm{U}$-Module invertible $\mathscr{L}$, $j_*(\mathscr{L})$ is an $\mathscr{O}_\mathrm{X}$-Module invertible. The set of $x \in \mathrm{X}$ such that the restriction of $j_*(\mathscr{L})$ to a neighbourhood open of x in X is invertible is evidently open in X; it is a matter of showing that $\mathrm{V} = \mathrm{X}$. For this, suppose the contrary and set $\mathrm{Z} = \mathrm{X} - \mathrm{V} \neq \varnothing$. Let y be a *maximal* point of Z; since $\mathrm{Z} \subset \mathrm{Y}$, $\mathscr{O}_{\mathrm{X},y}$ is by hypothesis a parafactorial ring. Replacing X if needed by a neighbourhood open of y, we can suppose that the restriction of $j_*(\mathscr{L})$ to $\mathrm{W} - (\mathrm{W} \cap \overline{\{y\}})$ is equal. If we set $\mathrm{V}_1 = \mathrm{V} \cup \mathrm{W}$ and if we denote by $\mathscr{L}_1$ the $\mathscr{O}_{\mathrm{V}_1}$-Module equal to $j_*(\mathscr{L})|W$ in W, $\mathscr{L}_\mathrm{V}$ in V, we have $\mathrm{U} \subset \mathrm{V}_1$ and $\mathscr{L}_1|\mathrm{U} = \mathscr{L}$. We conclude from the fact that $\mathscr{O}_\mathrm{X} \to j_*(\mathscr{O}_\mathrm{U})$ is bijective and from (21.13.2), *b*)) that $j_*(\mathscr{L})|\mathrm{V}_1$ is an $\mathscr{O}_\mathrm{X}$-Module invertible. But this contradicts the definition of V, and concludes the proof of (21.13.10). ☐

Corollary (21.13.11). — *Let* X *be a locally Noetherian prescheme,* Y *a closed subset of* X, $\mathrm{U} = \mathrm{X} - \mathrm{Y}$. *Suppose that the pair* (X, Y) *is parafactorial and that* U *is locally factorial; in other terms* (21.13.10)*, for all* $x \in \mathrm{X}$*, the local ring* $\mathscr{O}_{\mathrm{X},x}$ *is:* 1° *parafactorial if* $x \in \mathrm{Y}$; 2° *factorial if* $x \notin \mathrm{Y}$*. Then* X *is locally factorial (in other words,* $\mathscr{O}_{\mathrm{X},x}$ *is in fact factorial for* all $x \in \mathrm{X}$*).*

Proof. Suppose indeed the contrary, and let y be a point of Y such that $\mathscr{O}_{\mathrm{X},y}$ is not factorial and has a minimal dimension among all the points of Y having this property. Then, if we set $\mathrm{T}_y = \mathrm{Spec}(\mathscr{O}_{\mathrm{X},y})$, $\mathrm{U}_y = \mathrm{T}_y - \{y\}$, the hypothesis that $\mathscr{O}_{\mathrm{X},y}$ is parafactorial implies $\mathrm{Pic}(\mathrm{U}_y) = 0$ and $\mathrm{prof}(\mathscr{O}_{\mathrm{X},y}) \geqslant 2$

(21.13.8); moreover by the choice of y every U_y is locally factorial, contradicting the hypothesis. But then (21.6.14) proves that $\mathscr{O}_{X,y}$ is factorial, contradicting the hypothesis. □

Proposition (21.13.12). — *Let* A, B *be two local Noetherian rings,* $\rho : A \to B$ *a local homomorphism making* B *a flat A-module. Then, if* B *is factorial, so is* A.

Proof. We already know that A is integral and integrally closed (6.5.4), so we can restrict ourselves by induction on $n = \dim(A)$. Let $X = \mathrm{Spec}(A)$, a the closed point of X, $X' = \mathrm{Spec}(B)$, $f : X' \to X$ the structural morphism, $U = X - \{a\}$, $U' = f^{-1}(U)$. The local rings $\mathscr{O}_{X',x'}$ at points $x' \in f^{-1}(a)$ are factorial and of dimension $\geqslant 2$ (6.1.2), hence parafactorial (21.13.9), (ii)); we conclude (21.13.10) that the pair $(X', f^{-1}(a))$ is parafactorial. By virtue of (21.13.6), (iii), this shows that the ring A is parafactorial, hence $\mathrm{prof}(A) \geqslant 2$ and $\mathrm{Pic}(U) = 0$ (21.13.8). Finally, the local rings of U being of dimension $< n$, the induction hypothesis proves that U is *locally factorial*; the conclusion follows then from (21.6.14). □

(21.13.12.1). We note that the statement of (21.13.12) where one replaces "factorial" by "parafactorial" is no longer exact, as shown by the example where $A = k$ is a field and B the local factorial ring $k[[T_1, T_2]]$. However, it follows from (21.13.6), (iii) that, under the conditions of (21.13.12), if $\mathfrak{m}$ denotes the maximal ideal of A and if $\mathfrak{m}B$ is an *ideal of definition* of B (i.e. $\dim(B/\mathfrak{m}B) = 0$), then, when B is parafactorial, so is A. In particular, if the completion $\widehat{A}$ of a local Noetherian ring is parafactorial, so is A. **IV-4** | 321

Remark (21.13.13). — Part of the definitions and results above relates to more general considerations on the cohomology of sheaves of commutative groups, developed in chap. III, 3$^{\text{e}}$ partie, and which we have given here an exposition independent for the convenience of the reader. Indeed, if X is a ringed space, there is a canonical *equivalence* between the category of the $\mathscr{O}_X$-Modules invertibles and that of *principal homogeneous sheaves under the sheaf of groups* $\mathscr{O}_X^*$ (16.5.15). This equivalence is defined by making correspond to every $\mathscr{O}_X$-Module invertible $\mathscr{L}$ the sheaf $\mathscr{L}^* = \mathscr{I}som_{\mathscr{O}_X}(\mathscr{O}_X, \mathscr{L})$ of germs of isomorphisms of $\mathscr{O}_X$ onto $\mathscr{L}$; it is immediate that $\mathscr{L}^*$ is canonically equipped with a structure of principal homogeneous sheaf under $\mathscr{O}_X^* \cong \mathscr{I}som_{\mathscr{O}_X}(\mathscr{O}_X, \mathscr{O}_X)$. The verification of this equivalence of categories is immediate. This being said, consider in general on a topological space X and a sheaf of groups $\mathscr{G}$, and let Y be a closed subset of X, and let i be an integer such that $0 \leqslant i \leqslant 2$; we say that the pair (X, Y) is *i-pure* for $\mathscr{G}$ if, for every open V of X, the restriction functor $\mathscr{P} \mapsto \mathscr{P}|(U \cap V)$, from the category of principal sheaves homogeneous under the sheaf of groups $\mathscr{G}|V$, into the analogous category under $\mathscr{G}|(U \cap V)$, is *faithful* ($i = 0$), resp. *fully faithful* ($i = 1$), resp. an *equivalence of categories* ($i = 2$). In the case where X is a ringed space and $\mathscr{G} = \mathscr{O}_X^*$, to say that (X, Y) is 0-pure therefore means for $i = 0$, that the canonical homomorphism $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ is *injective*; for $i = 1$, that this homomorphism is *bijective*; and finally, for $i = 2$, that the pair (X, Y) is *parafactorial* (21.13.3).

Returning to the general case, recall that the morphisms of the category of principal sheaves are by definition isomorphisms. To say that the pair (X, Y) is 0-pure means therefore that for every open V of X, the canonical homomorphism $H^0(V, \mathscr{G}) \to H^0(U \cap V, \mathscr{G})$ is injective; to say that the pair (X, Y) is 1-pure means that the canonical homomorphism $H^0(V, \mathscr{G}) \to H^0(U \cap V, \mathscr{G})$ is bijective (and so also the canonical homomorphism $H^1(V, \mathscr{G}) \to H^1(U \cap V, \mathscr{G})$ is injective); finally, one can show that for the pair (X, Y) to be 2-pure, it is necessary and sufficient that the homomorphisms $H^i(V, \mathscr{G}) \to H^i(U \cap V, \mathscr{G})$ be bijective for $i = 0$ and $i = 1$. When $\mathscr{G}$ is a sheaf of commutative groups, by introducing the cohomology sheaves $\mathscr{H}^i_Y(\mathscr{G})$ defined in chap. III, 3$^{\text{e}}$ partie, saying that (X, Y) is i-pure for $i \leqslant 2$ means that $\mathscr{H}^i_Y(\mathscr{G}) = 0$ for $p \leqslant i$; under this form, the notion generalises immediately for every integer i.

Proposition (21.13.14). — *Let* X *be a locally Noetherian* normal *prescheme,* $(U_\lambda)_{\lambda \in L}$ *a decreasing filtered family of opens of* X. *The following conditions are equivalent:*

a) *Every* 1-*codimensional cycle* Z *on* X *such that there exists an index* λ *for which* $Z|U_\lambda$ *is locally principal* (21.6.7), *is locally principal.*
b) *For all* $x \in X$ *such that* $\dim(\mathscr{O}_{X,x}) \geqslant 2$ *and such that* x *does not belong to* $\bigcap_{\lambda \in L} U_\lambda$, *the local ring* $\mathscr{O}_{X,x}$ *is parafactorial.*

Proof. It is clear that the set S' of X generisations of points of S is the intersection of the open neighbourhoods of S. We can restrict to the case where X is Noetherian (the properties a) and b) being local on X); as every 1-codimensional cycle on X which is principal at the points of S is also principal at the points of a neighbourhood open of S, we see that condition a) of (21.13.15) amounts to condition a) of (21.13.14) applied to the family (U_λ) of open neighbourhoods of S. The corollary follows then from the equivalence of a) and b) in (21.13.14). **IV-4** | 322

Let us note on the other hand that b') implies b) by virtue of (21.13.10); conversely, if b') is satisfied and if $x \notin U_\lambda$, then $Y = \{x\}$ is contained in $X - U_\lambda$ and one has $\operatorname{codim}(Y, X) \geqslant 2$, hence $\mathscr{O}_{X,x}$ is parafactorial, which proves the equivalence of a') and b').

Let us prove that a) implies c), i.e. a') implies b''). First we prove that a'') implies b''). Let us note first that b') implies b) by virtue of (21.13.10); conversely, if b') is satisfied and if $x \notin U_\lambda$, then $Y = \{x\}$ is contained in $X - U_\lambda$ and one has $\operatorname{codim}(Y, X) \geqslant 2$; if we note that $U = X - Y$, $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ is bijective (21.13.2), hence condition a) of (21.13.15) amounts to the condition a) of (21.13.14).

a'') For every 1-codimensional cycle Z on X which is principal at the points of S of X, Z is locally principal.

b'') The canonical homomorphism $\operatorname{Pic}(X) \to \operatorname{Pic}(U)$ is surjective.

Let us first prove that a) implies c), since if the 1-codimensional cycle Z on X has its support in one of the sets $X - U_\lambda$, $Z|U_\lambda = 0$, hence $Z|U_\lambda = \operatorname{cyc}(D_\lambda)$, where D_λ is a divisor on U_λ; since X is normal, $Z|U_\lambda$ is locally principal by (21.6.10), (i), and the hypothesis implies that Z is locally principal on X.

Inversely c) implies a), since if Z is a 1-codimensional cycle on X such that $Z|U_\lambda$ is locally principal, the image of $Z|U$ in $\mathfrak{Cl}(U)$ is therefore the image of a unique element $\zeta_0 \in \operatorname{Pic}(U)$ (21.6.11); it follows from (21.6.11) that the image of Z in $\mathfrak{Cl}(X)$ is the image of a unique element $\zeta \in \operatorname{Pic}(X)$, and it is clear then that ζ_0 is the image of ζ. □

Corollary (21.13.15). — *Let X be a locally Noetherian normal prescheme, S a subset of X. The following conditions are equivalent:*

a) *Every 1-codimensional cycle on X which is principal at the points of S is locally principal.* **IV-4** | 323
b) *For all $x \in X$ which is not a generisation of a point of S (in other words, such that $\overline{\{x\}} \cap S = \varnothing$) and such that $\dim(\mathscr{O}_{X,x}) \geqslant 2$, the local ring $\mathscr{O}_{X,x}$ is parafactorial.*

Remark (21.13.16). — Under the general conditions of (21.13.14), suppose moreover $\varinjlim \operatorname{Pic}(U_\lambda) = 0$. Then condition a) of (21.13.14) is also equivalent to the following:

c) *Every 1-codimensional cycle on X whose support is contained in one of the sets $X - U_\lambda$ is locally principal.*

We can restrict to the case where X is irreducible and all the U_λ are non-empty. It is clear that a) implies c), since if the 1-codimensional cycle Z on X has its support in $X - U_\lambda$, $Z|U_\lambda = 0$, hence $Z|U_\lambda$ is locally principal. Inversely c) implies a): let Z be a 1-codimensional cycle on X such that $Z|U_\lambda$ is locally principal; since X is normal, $Z|U_\lambda = \operatorname{cyc}(D_\lambda)$, where D_λ is a divisor on U_λ. If $D_\mu = \operatorname{div}(f_\mu)$, f_μ is a regular rational function on U_μ; the hypothesis implies (by virtue of (21.3.4)) that there is a set $U_\mu \subset U_\lambda$ such that $D_\mu|U_\mu = D_\lambda|U_\mu$ is equivalent to 0. If $D_\mu = \operatorname{div}(f_\mu)$, f_μ may be considered as a regular rational function on X, regular on U_μ; hence $Z' = \operatorname{cyc}(\operatorname{div}(f_\mu))$, we see therefore that $Z'' = Z - Z'$ has its support in $X - U_\mu$; by hypothesis, Z'' is locally principal, whence the conclusion.

The condition $\varinjlim \operatorname{Pic}(U_\lambda) = 0$ will in particular be satisfied if (U_λ) is the family of open neighbourhoods of a point $z \in X$, since then U_λ is isomorphic to $\mathscr{O}_{U_\lambda}$; there exists $U_\mu \subset U_\lambda$ such that $\mathscr{L}_\lambda|U_\mu \cong \mathscr{O}_{U_\mu}$. We see therefore that, in (21.13.15), if $S = \{z\}$, conditions a) and b) are equivalent also to the following:

c) *Every 1-codimensional cycle on X whose support does not contain z is locally principal.*

If moreover $\operatorname{Pic}(X) = 0$, condition a) of (21.13.15) is also equivalent to the following:

c) Every 1-codimensional cycle on X whose support is contained in one of the sets $X - U_\lambda$ is locally principal.

21.14. The Ramanujam-Samuel theorem

Theorem (21.14.1). — (Ramanujam-Samuel) — *Let* A *be a local Noetherian ring with maximal ideal* $\mathfrak{m}$, **IV-4** | 323
such that its completion $\widehat{\mathrm{A}}$ *is integral and integrally closed. Let* B *be a local Noetherian ring and* $\rho : \mathrm{A} \to \mathrm{B}$ *a local homomorphism making* B *a smooth A-algebra (for the preadic topologies* (**0**, 19.3.1)*) and such that the residue field of* B *is finite over that of* A*. Then every* 1*-codimensional cycle on* $\mathrm{Spec(B)}$ *which is principal at the point* $\mathfrak{p} = \mathfrak{m}\mathrm{B}$, *is a principal* 1*-codimensional cycle.*

Proof. If $k = \mathrm{A}/\mathfrak{m}$ is the residue field, $\mathrm{B}/\mathfrak{m}\mathrm{B}$ is a formally smooth k-algebra (for the preadic topolo- **IV-4** | 324
gies), and in particular integral, in other words $\mathfrak{p} = \mathfrak{m}\mathrm{B}$ is indeed a prime ideal of B, which justifies the statement. It evidently all amounts to proving that every prime ideal $\mathfrak{q}$ of B not contained in $\mathfrak{m}\mathrm{B}$ is *principal*.

Let $\widehat{\mathrm{A}}$, $\widehat{\mathrm{B}}$ be the completions of A and B respectively, so that the maximal ideal of $\widehat{\mathrm{A}}$ is $\mathfrak{m}\widehat{\mathrm{A}}$; we know (**0**, 19.3.6) that $\widehat{\mathrm{B}}$ is a formally smooth $\widehat{\mathrm{A}}$-algebra for the adic topologies. Let k' be the residue field of $\widehat{\mathrm{B}}$, a finite extension of k; there is a local homomorphism $\widehat{\mathrm{A}} \to \mathrm{C}$, where C is a local Noetherian ring which is an $\widehat{\mathrm{A}}$-module *finite* (and *flat*) and $\mathrm{C}/\mathfrak{m}\mathrm{C}$ isomorphic to k' ($\mathbf{0}_{\mathrm{III}}$, 10.3.1); we conclude then from (7.5.1), (7.5.3) and (6.5.4), (ii) that C is *integral* and *integrally closed*. Furthermore, $\mathrm{D} = \widehat{\mathrm{B}} \otimes_{\mathbf{Z}} \mathrm{C}$ is a semi-local complete ring, a direct sum of complete local rings whose one, D_0, has for residue field k' (since $k' \otimes_k k'$ is isomorphic to k'). Since D is formally smooth over C for the preadic topologies; by $\mathrm{D}_0\,/\mathrm{m}\mathrm{D}_0$ is a k'-algebra formally smooth of residue field k', which implies that it is a k'-isomorphic algebra of power series $k'[[\mathrm{T}_1, \ldots, \mathrm{T}_n]]$ (**0**, 19.6.4); we conclude, by (**0**, 19.7.5), that D_0 is an algebra of formal power series $\mathrm{C}[[\mathrm{T}_1, \ldots, \mathrm{T}_n]]$, and hence *integral and integrally closed* (Bourbaki, *Alg. comm.*, chap. V, §1, nº 4, prop. 14). Since the morphisms $\mathrm{Spec}(\mathrm{D}_0) \to \mathrm{Spec}(\widehat{\mathrm{B}})$ and $\mathrm{Spec}(\mathrm{D}_0) \to \mathrm{Spec}(\mathrm{C})$ are faithfully flat, we deduce that B and $\widehat{\mathrm{B}}$ are
also *integral and integrally closed*. We will thus prove the theorem in the particular case where A is **IV-4** | 325
complete and $\mathrm{B} = \mathrm{A}[[\mathrm{T}_1, \ldots, \mathrm{T}_n]]$. Let us first show that we can reduce to the case where $n = 1$. Indeed, we proceed by induction on n, and let $f \in \mathfrak{q}$ be an element not belonging to $\mathfrak{p} = \mathfrak{m}\mathrm{B}$; it is a formal power series whose reduced series $\overline{f}_0$ is nonzero; hence, by virtue of the preparation theorem (Bourbaki, *Alg. comm.*, chap. VII, §3, nº 8, prop. 5), for every $f \in \mathfrak{q}$, there exists $g \in \mathrm{B}$ and a polynomial $r \in \mathrm{A}[\mathrm{T}]$ such that $f = gf_0 + r$ and one has thus $r \in \mathfrak{q} \cap \mathrm{A}[\mathrm{T}]$; on the other hand (*loc. cit.*, prop. 6) there exists a distinguished non-constant polynomial $\mathrm{F}_0 \in \mathrm{A}[\mathrm{T}]$ and an invertible element $u \in \mathrm{B}$ such that $f_0 = u\mathrm{F}_0$, hence also $\mathrm{F}_0 \in \mathfrak{q} \cap \mathrm{A}[\mathrm{T}]$; since B is flat over $\mathrm{A}[\mathrm{T}]$ ($\mathbf{0}_{\mathrm{I}}$, 7.3.3), it follows from Bourbaki, *Alg. comm.*, chap. VII, §1, nº 10, prop. 15, that $\mathfrak{q}_1$ is a prime ideal of height 1 in $\mathrm{A}[\mathrm{T}]$. But on the other hand, we have necessarily $\mathfrak{q}_1 \cap \mathrm{A} = 0$; otherwise, $\mathfrak{q}_1 \cap \mathrm{A}$ would necessarily be of height $\geqslant 1$, and it would follow from (5.5.3) that $\mathfrak{q}_1 = (\mathfrak{q}_1 \cap \mathrm{A})\mathrm{A}[\mathrm{T}]$. But then, since $\mathfrak{q}_1 \subset \mathfrak{m}\mathrm{A}[\mathrm{T}]$, we would have $\mathfrak{q}_1 \subset \mathfrak{m}\mathrm{A}[\mathrm{T}]$, contradicting the hypothesis on $\mathfrak{q}$. If K is the field of fractions of A, $\mathfrak{q}_1\mathrm{K}[\mathrm{T}]$ is therefore a prime ideal distinct from 0 and of $\mathrm{K}[\mathrm{T}]$, hence of the form $h \cdot \mathrm{K}[\mathrm{T}]$, where $h(\mathrm{T}) = \mathrm{T}^m + a_1\mathrm{T}^{m-1} + \ldots + a_m$ with $m \geqslant 1$ and the $a_i \in \mathrm{K}$, h being irreducible in $\mathrm{K}[\mathrm{T}]$. But we have seen above that there exists in $\mathfrak{q}_1$ a distinguished non-constant $\mathrm{F}_0 \in \mathrm{A}[\mathrm{T}]$. t is a root of F_0 in an extension of K and hence divides F_0 *in* $\mathrm{K}[\mathrm{T}]$; but since h and F_0 are unitary, the coefficients a_i of h are integers over A (Bourbaki, *Alg. comm.*, chap. V, §1, nº 3, prop. 11), hence belong to A since A is integrally closed.

The statement (21.14.1) is equivalent to the following: □

Corollary (21.14.2). — *Under the hypotheses of* (21.14.1) *on* A, B *and* ρ, *for every prime ideal* $\mathfrak{q}$ *of* B *not contained in* $\mathfrak{p} = \mathfrak{m}\mathrm{B}$ *and such that* $\dim(\mathrm{B}_{\mathfrak{q}}) \geqslant 2$, *the ring* $\mathrm{B}_{\mathfrak{q}}$ *is parafactorial. In particular, if* $\dim(\mathrm{B} \otimes_{\mathrm{A}} k) > 0$ *(i.e.* $\dim \mathrm{B} > \dim \mathrm{A}$*)* ((**0**, 19.7.1) *and* (**0**, 6.1.1))*, the ring* B *is parafactorial.*

Proof. Indeed, we saw in the proof of (21.14.1) that the conditions of the statement imply that B is integral and integrally closed; and the equivalence of the statements (21.14.1) and (21.14.2) follows then from (21.13.15) applied to $\mathrm{X} = \mathrm{Spec(B)}$ and $\mathrm{S} = \{\mathfrak{p}\}$. □

Proposition (21.14.3). — *Let* S *be a normal prescheme,* $f : \mathrm{X} \to \mathrm{S}$ *a smooth morphism.*

(i) *If* S *is locally Noetherian (hence also* X*), then, for all* $x \in \mathrm{X}$ *such that* $\dim(\mathscr{O}_{\mathrm{X},x}) \geqslant 2$ *and such that* x *is not a maximal point of its fibre* $f^{-1}(f(x))$, *the ring* $\mathscr{O}_{\mathrm{X},x}$ *is parafactorial. Every* 1*-codimensional* **IV-4** | 326 *cycle* Z *on* X *such that* Supp(Z) *contains no irreducible component of any fibre* X_s, $f^{-1}(s) = X_s$, *is locally principal.*

(ii) Let Y *be a closed subset of* X *containing no irreducible component of any fibre* X_s, $Y_\eta = Y \cap X_\eta$ *is of codimension* $\geqslant 2$ *in* X_η. *Then the pair* (X, Y) *is parafactorial.*

Proof. We note that the conditions of (ii) are satisfied if Y is a closed subset of X such that, for all $s \in S$, $Y_s = Y \cap X_s$ is of codimension $\geqslant 2$ in X_s. **IV-4** | 327

To prove (21.14.3), let us note first that under the hypotheses of (i), if we set $Y = \overline{\{x\}}$, it suffices to verify the conditions of (ii); and we have $Y \cap X_s$ is of codimension $\geqslant 2$ in X_s for every s by the hypotheses on x.

For the proof of (i), we first show that $\mathscr{O}_{X,x}$ is parafactorial; since X is normal (17.5.7), hence A is integral and integrally closed, and since $\mathscr{O}_{X,x}$ is a formally smooth $\mathscr{O}_{S,s}$-algebra for the preadic topologies (17.5.3); since Y_s contains no irreducible component of X_s, and since Y_s is of codimension $\geqslant 2$ in X_s, the prime ideal of $\mathscr{O}_{X,x}$ corresponding to the point x is not contained in $\mathfrak{m}_s\mathscr{O}_{X,x}$ since Y contains no irreducible component of X_s. Furthermore, $\dim(\mathscr{O}_{X,x}) \geqslant \dim(\mathscr{O}_{S,s}) + 2 \geqslant 2$, so we can apply (21.14.2).

For the proof of (ii), since the conditions and the conclusion are local on S and on X by virtue of (21.13.6), (i), we can restrict to the case where S and X are affine, f is therefore of finite presentation. Since the fibres of f are regular, the hypothesis on Y implies that, for every point $x \in Y$, $\operatorname{prof}(\mathscr{O}_{X_{f(s)},x}) = \dim(\mathscr{O}_{X_{f(s)},x}) \geqslant 1$; it follows from (19.9.8) that the canonical homomorphism $\mathscr{O}_X \to j_*(\mathscr{O}_U)$ is *injective*. Furthermore, replacing Y by a larger closed subset following the method of the proof of (19.9.8), and using (21.13.6), (ii), we can suppose that Y is defined by an ideal of *finite type* of the Picard scheme of X, hence the open $U = X - Y$ is *quasi-compact and quasi-separated* and the closed set Y is *constructible*.

To prove (ii), it suffices, for every $\mathscr{O}_U$-Module invertible $\mathscr{L}$ and every point $x \in Y$, to establish the existence of a neighbourhood open V of x in X such that: 1º the canonical homomorphism $\mathscr{O}_V \to (j_V)_*(\mathscr{O}_{U\cap V})$ is surjective; 2º there exists an $\mathscr{O}_V$-Module invertible $\mathscr{L}'_V$ such that $\mathscr{L}'_V|U \cap V = \mathscr{L}|(U \cap V)$ (and (21.13.3)).

Set $s = f(x)$; we will see that we can restrict to the case where $S = S_0 = \operatorname{Spec}(\mathscr{O}_{S,s})$. Let (S_ν) be a fundamental system of affine open neighbourhoods of s in S, and set $X_\nu = f^{-1}(S_\nu)$, $Y_\nu = Y \cap X_\nu$, $U_\nu = X_\nu - Y_\nu$, $j_\nu : U_\nu \to X_\nu$ being the canonical injection; $X_0 = X \times_S S_0$ is projective limit of the X_ν and, denoting by $Y_0 = Y \cap X_0$ and $U_0 = X_0 - Y_0$, $j_0 : U_0 \to X_0$ the canonical injection, $p_\nu : X_0 \to X_\nu$ being the affine morphism, one has $j_*(\mathscr{O}_U) = p_\nu^*((j_\nu)_*(\mathscr{O}_{U_\nu}))$ (**II**, 1.5.2), and the canonical homomorphism $\rho : \mathscr{O}_X \to j_*(\mathscr{O}_U)$ is none other that $p_\nu^*(\rho_\nu)$, where ρ_ν is the canonical homomorphism. Since by hypothesis ρ_0 is surjective, there exists a sufficiently large ν and $\mathscr{L}_\nu$ such that $\mathscr{L}_0 = p_\nu^*(\mathscr{L}_\nu)$ is invertible. The proof finishes like that of the proposition, when A is a *local* ring: since X is normal and f smooth, S is normal (17.5.7), hence A is integral and integrally closed (17.5.7); since the local rings $\mathscr{O}_{S,s}$ are excellent integrally closed rings (**0**, 7.8.3, (ii)), (iii) and (vi)), A is an inductive limit of its sub-**Z**-algebras *of finite type*; by virtue of (1.8.4.2), there exists an index λ and an A_λ-algebra of finite type B_λ such that $B = B_\lambda \otimes_{A_\lambda} A$. By virtue of (1.8.4.2), there exists an index λ and an A_λ-isomorphism near. Set $S_\lambda = \operatorname{Spec}(A_\lambda)$, $X_\lambda = \operatorname{Spec}(B_\lambda)$; if $\rho_\lambda : X \to X_\lambda$ is the canonical projection, one has $j_*(\mathscr{O}_U) = \rho_\lambda^*((j_\lambda)_*(\mathscr{O}_{U_\lambda}))$; it all amounts to proving that $U = X - Y$ is retrocompact in X. Since the fibres of f are regular, the hypothesis on Y implies that for every point $x \in Y$, $\mathscr{O}_{X,x}$ is parafactorial by (i), and the result then follows from (21.13.10). □

(21.14.4). *Remarks.* — (i) It is possible that the statements (21.14.1) and (21.14.2) remain valid *without hypothesis* on the residue field of B. The statement with this generality would be equivalent, by virtue of (21.13.15) and (21.13.12.1) to the following: let A be a local Noetherian ring, complete, integral, integrally closed, B an arbitrary Noetherian ring which is a formally smooth A-algebra, such that $\dim(B) > \dim(A)$; then B is parafactorial. **IV-4** | 328

(ii) We will see in chap. VI, employing a technique of "finite descent" applied to (21.14.1) (or to (21.14.2)), that the conclusion of (21.14.1) holds valid by replacing the hypothesis that $\widehat{A}$ is *reduced*. If we replace this last condition by $\dim B \geqslant \dim A + 3$, we can even suppress the hypothesis that $\widehat{A}$ is reduced. Similarly, the conclusion of (21.14.3) remains valid by replacing the hypothesis that X is normal by the hypothesis that X is reduced, provided that $\dim(\mathscr{O}_{X,x}) \geqslant \dim(\mathscr{O}_{S,f(x)}) + 2$.

(iii) In chap. III, 3$^{\text{e}}$ partie, we prove that if $f : X \to S$ is a smooth morphism, Y a closed locally constructible subset of X such that, for all $s \in S$, with the notations of (21.14.3) $\operatorname{codim}(Y_s, X_s) \geqslant 3$, then the pair (X, Y) is parafactorial. This conclusion is no longer valid when we only suppose $\operatorname{codim}(Y_s, X_s) \geqslant 2$ for all $s \in S$ and when we do not suppose X reduced. For example, let k be a field, $A = k[T]/(T^2)$, $S = \operatorname{Spec}(A)$ (T_1, T_2 indeterminates), so that $X = \mathbb{V}_S^3$, Y being the "zero section"; if s is the unique closed point of S, on a $X_s = \mathbb{V}_k^2$ and Y_s is the "zero section" of X_s. To see that the pair (X, Y) is not parafactorial, it suffices to show that the ring $B = A[T_1, T_2]_{\mathfrak{m}}$, where $\mathfrak{m}$ is the ideal $(T_1) + (T_2)$ which defines Y, is not parafactorial (21.13.10). Let $B_0 = B_{\text{red}}$, and let U and U_0 be the complementary open subsets of the closed point in $\operatorname{Spec}(B)$ and $\operatorname{Spec}(B_0)$; reasoning as in (21.13.9), we have the exact sequence, extension of (21.13.9.1)

$$\Gamma(U, \mathscr{O}_U)^* \longrightarrow \Gamma(U_0, \mathscr{O}_{U_0})^* \longrightarrow H^1(U_0, \mathscr{O}_{U_0}) \longrightarrow \operatorname{Pic}(U) \longrightarrow \operatorname{Pic}(U_0).$$

Now, one has $\Gamma(U, \mathscr{O}_U) = B$, $\Gamma(U_0, \mathscr{O}_{U_0}) = B_0$ since $\operatorname{prof}(B) = \operatorname{prof}(B_0) = 2$ (5.10.5). On the other hand $\operatorname{Pic}(U_0) = 0$ since B_0 is factorial, and $H^1(U_0, \mathscr{O}_{U_0}) \neq 0$ [**?**, 3, Exemple III-1]; since the homomorphism $B^* \to B_0^*$ is surjective, we conclude that $\operatorname{Pic}(U) \neq 0$, hence that B is not parafactorial.

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(21.14.4bis). (iv) The result of (21.14.3) was first proved by C. Seshadri [**?**] in the particular case where S is an algebraic normal scheme over an algebraically closed field k and $X = S \times_k T$, where T is a k-algebraic prescheme smooth over k. The proof of Seshadri [**?**, pp. 188–189] is of global nature and uses the theory of Picard schemes. It gives moreover (*loc. cit.*) results such as the following (for which one does not currently possess a proof by local methods). Let S, T be two preschemes locally of finite type over a field k, Z a 1-codimensional cycle on X (considered as S-prescheme); suppose the following conditions are satisfied:

1° S and T are geometrically normal over k (6.7.6);

2° For every maximal point η of S, the 1-codimensional cycle Z_η on the fibre X_η, having even multiplicity that Z at every point of $X^{(1)} \cap X_\eta$, is locally principal (in other words, is the image of a divisor of X_η, since X_η is normal);

3° For all $s \in S$, Z is principal at the maximal points of the fibre X_s.

Then Z is locally principal. In other words, X being normal (6.14.1), every 1-codimensional cycle Z on X such that for every maximal $x \in X$ which is not maximal in its fibre X_s and which does not belong to any of the "generic fibres" X_η (which implies $\dim(\mathscr{O}_{X,x}) \geqslant 2$ by virtue of (6.1.1)), the local ring $\mathscr{O}_{X,x}$ is parafactorial, by virtue of (21.12.15) [(1)].

21.15. Relative divisors

(21.15.1). Let S be a prescheme, $f : X \to S$ a *flat* morphism and locally of finite presentation. We defined in (20.6.1) the sheaf of rings $\mathscr{M}_{X/S}$ of the *germs of meromorphic functions on* X *relative to* S, sub-sheaf of $\mathscr{M}_X$; it is clear that the canonical injection $\mathscr{O}_X \hookrightarrow \mathscr{M}_X$ (20.1.4.1) maps $\mathscr{O}_X$ into a sub-sheaf of $\mathscr{M}_{X/S}$, with which we identify it. Let $\mathscr{M}_{X/S}^*$ be the sheaf (in multiplicative groups) of germs of invertible sections of $\mathscr{M}_X^*$ which is thus a sub-sheaf of $\mathscr{M}_X^*$ and contains $\mathscr{O}_X^*$ as sub-sheaf.

Definition (21.15.2). — We call *sheaf of divisors on* X *relative to* S, or *sheaf of divisors on* X *transversal to* f, and denote $\mathscr{D}iv_{X/S}$ the quotient sheaf (of commutative groups) $\mathscr{M}_{X/S}^*/\mathscr{O}_X^*$; the sections of this sheaf over X are called *divisors relative to* S, or *divisors on* X *transversal to* f; they form a commutative group denoted $\operatorname{Div}(X/S)$.

It is clear that $\mathscr{D}iv_{X/S}$ identifies canonically with a sub-sheaf of $\mathscr{D}iv_X$, and hence $\operatorname{Div}(X/S)$ with a subgroup of $\operatorname{Div}(X)$, which we denote also *additively*.

For every open U of X, we have $\mathscr{M}_{X/S}^*|U = \mathscr{M}_{U/S}^*$, hence $\mathscr{D}iv_{X/S}|U = \mathscr{D}iv_{U/S}$, and the sheaf $\mathscr{D}iv_{X/S}$ is therefore equal to the presheaf $U \mapsto \operatorname{Div}(U/S)$. **IV-4 | 330**

Since $\mathscr{M}_{X/S}^*$ is a sub-sheaf of $\mathscr{M}_X^*$, the definitions, notations and formulas relative to divisors of sections of $\mathscr{M}_X$ over X (21.1.3) apply without change to sections of $\mathscr{M}_{X/S}$ over X.

(21.15.3). The structure of sheaf of ordered groups on $\mathscr{D}iv_X$ (21.1.6) induces on the sub-sheaf $\mathscr{D}iv_{X/S}$ a structure of sheaf of ordered groups, for which the sheaf of monoids of germs of positive sections is $\mathscr{D}iv_{X/S}^+ = \mathscr{D}iv_X^+ \cap \mathscr{D}iv_{X/S}$, which we denote $\mathscr{D}iv_{X/S}^+$. We have $\Gamma(X, \mathscr{D}iv_{X/S}^+) = \operatorname{Div}^+(X) \cap$

$\mathrm{Div}(X/S)$; we denote this sub-monoid of $\mathrm{Div}(X/S)$ by $\mathrm{Div}^+(X/S)$, and it consists of elements $\geqslant 0$ with respect to the ordered group structure on $\mathrm{Div}(X/S)$. It follows from (21.5.1) that $\mathscr{D}iv^+_{X/S}$ is the image in $\mathscr{D}iv_{X/S}$ of the sub-sheaf of monoids

$$\mathscr{O}^{\mathrm{reg}}_{X/S} = \mathscr{O}_X \cap \mathscr{M}^*_{X/S} \tag{21.15.3.1}$$

For every open U of X, $\Gamma(U, \mathscr{O}^{\mathrm{reg}}_{X/S})$ is the set of sections t of $\mathscr{O}_X$ over U such that t is regular and that $1/t$ belongs to $\Gamma(U, \mathscr{M}_{X/S})$, which means, with the notations of (20.6.1), that $t \in T_{X/S}(U)$, so that the sheaf $\mathscr{O}^{\mathrm{reg}}_{X/S}$ is none other than the sheaf denoted $\mathscr{T}_{X/S}$ in (20.6.1). We can therefore write

$$\mathscr{D}iv^+_{X/S} = \mathscr{O}^{\mathrm{reg}}_{X/S}/\mathscr{O}^*_X \tag{21.15.3.2}$$

up to canonical isomorphism.

(21.15.3bis). Let $D \in \mathrm{Div}^+(X/S)$ and consider the closed sub-prescheme $Y(D)$ defined by the ideal $\mathscr{J}_X(D)$ of $\mathscr{O}_X$ (21.6.5); by virtue of what precedes, for all $x \in Y(D)$, there is an open neighbourhood U of x in X and a section $t \in T_{X/S}(U)$ such that $\mathscr{J}_X(D)|U = \mathscr{O}_U \cdot t$; since $x \in Y(D)$, the image $t_{f(x)}$ of t in $\Gamma(U \cap X_{f(x)}, \mathscr{O}_{X_{f(x)}})$ belongs to the maximal ideal of $\mathscr{O}_{X_{f(x)},x}$, and furthermore, by definition, for all $s \in S$, the image t_s of t in $\Gamma(U \cap X_s, \mathscr{O}_{X_s})$ is a regular section. We deduce therefore from (11.3.8) and (19.2.4) that the canonical injection $Y(D) \to X$ is *transversely regular relative to* S *and of codimension* 1. The converse being immediate, we see that one can *identify* canonically the positive divisors on X relative to S with the closed sub-preschemes Y of X such that the canonical injection $Y \to X$ is a transversely regular immersion relative to S and of codimension 1. We will ordinarily make this identification.

Proposition (21.15.4). — *Let* D *be a divisor on* X, $\mathscr{J}_X(D)$ *the ideal fractionnaire invertible corresponding* (21.2.8) *and* (21.2.9). *For* $D \in \mathrm{Div}(X/S)$, *it is necessary and sufficient that, for all* $x \in X$, $(\mathscr{J}_X(D) \cap \mathscr{M}^*_{X/S})_x \neq 0$ *(or equivalently, that* $\mathscr{J}_X(D) \subset \mathscr{M}_{X/S}$ *and* $\mathscr{J}_X(D)^{-1} \subset \mathscr{M}_{X/S}$*).*

Proof. Indeed, to say that $D \in \mathrm{Div}(X/S)$ means that for all $x \in X$, there exists a neighbourhood open U of x and a section $f \in \Gamma(U, \mathscr{M}^*_{X/S})$ such that $D|U = \mathrm{div}_U(f)$, where $f \in \Gamma(U, \mathscr{M}^*_X)$; since $\mathscr{J}_X(D)|U$ is the fractional ideal $\mathscr{O}_U \cdot f$, the proposition follows immediately. □

Proposition (21.15.5). — *Let* D *be a divisor on* X, $\mathscr{O}_X(D)$ *the fractional ideal invertible and* s_D *the meromorphic regular section of* $\mathscr{O}_X(D)$ *defined canonically by* D (21.2.8) *and* (21.2.9). *For* $D \in \mathrm{Div}(X/S)$, *it is necessary and sufficient that* $s_D \in \Gamma(X, \mathscr{M}_{X/S}(\mathscr{O}_X(D)))$. **IV-4** | 331

Proof. Indeed, if U is an open of X such that $D|U = \mathrm{div}_U(f)$, where $f \in \Gamma(U, \mathscr{M}^*_X)$, to say that $s_D \in \Gamma(U, \mathscr{M}_{X/S}(\mathscr{O}_X(D)))$ means, by virtue of the definitions (20.6.2), that $f^{-1} \in \Gamma(U, \mathscr{M}_{X/S})$, whence the proposition. □

The interpretation of the divisors on X by means of the classes $\mathrm{cl}(\mathscr{L}, s)$ (21.2.11) allows us therefore to interpret the elements of $\mathrm{Div}(X/S)$ as the pairs $(\mathscr{L}, s)$ (up to isomorphism) such that $\mathscr{L}$ is an $\mathscr{O}_X$-Module invertible and that s is a meromorphic section of $\mathscr{L}$ regular relative to S (20.6.5), (iii)).

Proposition (21.15.6). — *Let* D *be a divisor on* X *relative to* S, *and suppose that for all* $x \in X$ *such that* $\mathrm{prof}(\mathscr{O}_{X_{f(s)},x}) = 1$, *we have* $D_x \geqslant 0$ *(resp.* $D_x = 0$*). Then* $D \geqslant 0$ *(resp.* $D = 0$*).*

Proof. As in (21.1.8), we can restrict to the case where $D = \mathrm{div}(\varphi)$, where φ is a meromorphic regular function relative to S, and it all amounts to seeing that if $D_x \geqslant 0$ everywhere then φ is everywhere defined in X. But this hypothesis means that, if $T = X - \mathrm{dom}(\varphi)$, we have $\mathrm{prof}(\mathscr{O}_{X_{f(s)},x}) \geqslant 2$ for all $x \in T$, and it suffices to conclude by applying (20.6.6). □

(21.15.7). Let X′ be a second S-prescheme flat and locally of finite presentation over S, and let $g : X' \to X$ be an S-morphism. If the S-morphism g is *flat*, we know (21.4.5) that the inverse image by g of every divisor on X is defined; if furthermore $D \in \mathrm{Div}(X/S)$, it follows from the definition (21.15.2) and from (20.6.8) that $g^*(D) \in \mathrm{Div}(X'/S)$.

(21.15.8). Let X′ be a second S-prescheme flat and locally of finite presentation over S, and let $g : X' \to X$ be an S-morphism *finite* and *flat*. We note that g is also necessarily of finite presentation (1.4.3), (1.4.6) and (1.6.3), hence $g_*(\mathscr{O}_{X'})$ is a $\mathscr{O}_X$-Module *locally free* (2.1.12), in other words g is a locally free morphism (18.2.7); for all $s \in S$, the corresponding morphism $g_s : X'_s \to X_s$ is also finite and locally free. We deduce then from (21.5.2) and (20.6.1) that for every open U of X and every section $t' \in \Gamma(g^{-1}(U), \mathscr{T}_{X'/S})$, the norm $N_{X'/X}(t')$ belongs to $\Gamma(U, \mathscr{T}_{X/S})$; the reasoning of (21.5.3) proves then that for every $\mathscr{O}_{X'}$-Module invertible $\mathscr{L}'$ and every meromorphic section u' of $\mathscr{L}'$ above S, the norm $\mathscr{L} = N_{X'/X}(\mathscr{L}')$ is a meromorphic section of X, regular relative to S. The interpretation of the direct image of divisors relatives to S given in (21.15.5) and the definition of a direct image of divisors (21.5.5) prove then that *for every divisor* $D' \in \mathrm{Div}(X'/S)$, we have $g_*(D') \in \mathrm{Div}(X/S)$.

(21.15.9). Consider finally an arbitrary morphism $S' \to S$, and (under the hypotheses of (21.15.1)) set $X' = X \times_S S'$, which is flat and locally of finite presentation over S'; if $p : X' \to X$ is the canonical projection, we have seen (20.6.9) that there is a canonical homomorphism $p^*(\mathscr{M}_{X/S}) \to \mathscr{M}_{X'/S'}$, which transforms every section of $\mathscr{M}_{X/S}$ over U, *regular relative to* S, into a section of $\mathscr{M}_{X'/S'}$ regular relative to S' (20.6.5), (iii); we conclude then from the definition (21.15.2) and the right exactness of the functor p^*, that the preceding homomorphism defines by passage to quotients a canonical homomorphism **IV-4 | 332**

(21.15.9.1) $$\mathrm{Div}(X/S) \longrightarrow \mathrm{Div}(X'/S')$$

which evidently transforms elements of $\mathrm{Div}^+(X/S)$ into elements of $\mathrm{Div}^+(X'/S')$. We also set $\mathrm{Div}(X'/S') = \mathrm{Div}_{X/S}(S')$ (resp. $\mathrm{Div}^+(X'/S') = \mathrm{Div}^+_{X/S}(S')$), and we see immediately that we have thus defined two contravariant functors

$$\mathrm{Div}_{X/S} : \mathbf{Sch}_{/S} \longrightarrow \mathbf{Ab}, \qquad \mathrm{Div}^+_{X/S} : \mathbf{Sch}_{/S} \longrightarrow \mathbf{Ens}$$

from the category of S-preschemes into that of commutative groups (resp. sets). We will see further (chap. VI) the important cases where the functor $\mathrm{Div}^+_{X/S}$ is *representable* ($\mathbf{0}_{\mathrm{III}}$, 8.1.8).

For every divisor $D \in \mathrm{Div}(X/S)$, the image of D by the homomorphism (21.15.9.1) is none other than the *inverse image* $p^*(D)$ (in the sense of (21.4.2)): the existence of this inverse image and the preceding assertions are immediate consequences of (20.6.5), (iii) and (20.6.9).

The elements of $\mathrm{Div}(X'/S')$ are often called, by abuse of language, "*families of divisors on* X *relative to* S, *parametrised by the S-prescheme* S'"; this terminology is used especially when it concerns positive divisors.